Privacy-Preserving Vertical Federated Regression

with Threshold Decryption and Differentially Private Releases

Working draft — protocol v3.1, honest-but-curious model

October 28, 2026

Abstract

We study two-party vertical federated least squares between a researcher R, who holds a cohort with features $\mathbf{X}_R$, and an organization O (for example a national statistical agency), which holds a register with features $\mathbf{X}_O$ and the response $\mathbf{y}$. Records are linked by labeled private set intersection with an oblivious pseudorandom function: R learns which of its cohort members are in O's register and where, and O learns only public sizes. The statistics that depend on O's data are aggregated under two-party threshold CKKS. The secret key is split additively between the parties, so neither can decrypt alone, and O never decrypts at all. Instead, O adds the differential-privacy noise inside its share of each decryption, as an exact discrete Gaussian on the coefficient lattice that also floods the encryption error. R therefore receives only noisy statistics, and O can enforce a privacy budget by withholding its share. R's own Gram block is computed locally and is never noised. Each run makes two releases. The first, of the Gram blocks only, chooses the ridge. The second, whose noise is independent of that choice, gives the estimate, a Wald interval for the ridge target from a parametric bootstrap of the mechanism, and a bias-aware interval for the regression coefficients. We give the exact replace-one sensitivity, zCDP accounting with a tight conversion, simulation-based output privacy against an honest-but-curious researcher, and the statistical guarantees of the two-stage procedure. We evaluate two settings of the protocol, each chosen on synthetic data alone by a pre-registered rule: setting I (inference) keeps the intervals and sizes the stage-1 radius by an exact Monte Carlo quantile; setting P (prediction) gives up the intervals for a smaller ridge, subject to its gate refusing at most 1% of releases on both standardized and block-normalized synthetic features (a first choice, made on standardized features only, was refused on most block-normalized releases). On a synthetic population at $\delta = 10^{-9}$, the Wald intervals under setting I cover 94.5–96.6% in every cell (target 95.5%), with and without the ridge, and 94.5–96.4% on fixed cohorts of five real datasets with the calibrated noise of a real budget. Setting I switches the stage-1 ridge off 35% sooner than the previous setting (3.9k to 53k matched records for $\varepsilon = 8$ to 0.5), and prediction comes within 10% of non-private test error at 2.9k to 38k matched records under I and 2.1k to 26k under P. Once the ridge is off, I's coefficient error is slightly higher: it reaches twice that of non-private least squares at 8.0k, 27k and 88k matched records for $\varepsilon = 8$, 4 and 2. On real data with each party normalizing its own block of every row, prediction comes within 10% of non-private test error on protein at 5.0k and 2.5k matched records for $\varepsilon = 1$ and 2 under setting P (8.2k and 4.1k under I), and P's gate refuses at most 1% of releases on bike, elevators, pol and protein. Against AdaSSP on the joined table at $\varepsilon = 1$, $\delta = 10^{-6}$, v3.1 with block normalization has higher test error on 20 of 29 datasets under either setting, with a median gap of +14% under P and +25% under I (+28% at the previous setting), because its ridge is still larger than that of AdaSSP under

the same guarantee (1.8 times under P and 2.8 times under I, with joint normalization). Against three further baselines, noising R's own block as well (full-record SSP) raises test error by a median of about 1% on the 29 datasets; TukeyEM, which needs no bounds, fails its stability test and releases nothing on 25 of them at $\varepsilon \leq 2$; and the parametric bootstrap of Ferrando et al., applied to our release, gives the same interval for the coefficients as ours once the ridge is off, and a finite one that covers, where ours is unbounded, from about a quarter of the ridge threshold on the synthetic population. End-to-end runs of a full OpenFHE implementation reproduce a plaintext simulation of the mechanism under every setting evaluated; in 450 runs under settings I and P the decrypted noise has its calibrated law, with mean zero to within 0.001 of its standard deviation over 14.7 million values. At 10^6 register rows and 20,000 padded queries a full run takes 2.0 minutes on a 4-core machine with O's OPRF evaluation parallel (O: 91 s and 2.3 GB; 475 MB of traffic; 3.0 minutes with it single-threaded), with the linkage exact in all 78 large runs. On the protein dataset (19,206 matched records, previous setting), after each party normalizes its own block of every row, they come within 2–6% of non-private test error at $\varepsilon \in \{1, 4\}$.

Contents

1 Introduction **7**

2 Related work **8**

3 Setting and notation **12**

4 Threat model and guarantees **13**

5 Building blocks **14**
5.1 Two-party threshold CKKS 14
5.2 Oblivious PRF 15
5.3 Labeled PSI 15
5.4 zCDP and the discrete Gaussian 15

6 The protocol **15**
6.1 Per-party preprocessing 19

7 Output privacy **25**
7.1 Sensitivity under replace-one adjacency 25
7.2 The release mechanism 26
7.3 Calibration 30
7.4 The ledger and composition across runs 33

8 Input privacy, cohort privacy and correctness **36**

9 The statistical procedure **39**
9.1 Estimands 39
9.2 Stage 1: choosing the ridge 39
9.3 Stage 2: estimate and gate 41
9.4 Intervals 41
9.5 When can the ridge be dropped? 42

9.6 Mechanism bias . . . 46
9.7 Status of the statistical guarantees . . . 46

10 Parameters **47**

11 Empirical evaluation **52**
11.1 The cryptographic layer is exact . . . 53
11.2 Calibration under v3.1 . . . 53
11.3 Interval coverage under v3.1 . . . 54
11.4 Interval methods: the parametric bootstrap of Ferrando, Wang and Sheldon . . . 56
11.5 End-to-end runs of the implementation . . . 62
11.6 Large-scale runs . . . 63
11.7 Coefficient accuracy against intersection size . . . 64
11.8 Does the bias correction matter? . . . 71
11.9 Benchmark against AdaSSP . . . 73
11.10 Two further baselines: full-record SSP and TukeyEM . . . 77
11.11 Intersection size on real data . . . 83
11.12 Where the utility goes . . . 85

12 Cost **86**

13 Extensions and limitations **88**

14 Summary and open items **89**

A Mechanism bias and the optional correction **91**
A.1 Noise model . . . 91
A.2 Result . . . 91
A.3 Schur-complement reading . . . 92
A.4 Ridge . . . 92
A.5 Why the ridge must not depend on the estimation noise . . . 93
A.6 Optional zero-budget correction . . . 93

B Proof of Theorem A.1 **93**

C Labeled PSI with an OPRF and record alignment **94**

D Status of the proofs **96**

Items to check

1 The bibliographic details of the entries cited in this section are taken from the literature review's verified reference list, and the fields that list omits (some given names, volumes and pages) are omitted here too. Three are still unverified. (a) UnLynx [31] is not in the verified list. Its authors, title, issue, pages and DOI were confirmed only through a search-engine summary of the PoPETs page, and the description of its noise protocol comes from the Drynx paper, through search-engine extracts; check it against the UnLynx text. (b) Ion et al. [42] is not in the verified list. The title of ePrint 2019/723 and its nine-author list conflict with Google Research's listing of the EuroS&P 2020 paper, which has a different title and six authors; the bibliography gives the ePrint title and the first two authors only. Confirm the published title, the authors and the pages (370–389). (c) The later title and the venue of Hwang et al. [41]: see Check (T1) in Section 7.2. 8

2 (T4) Model (M5) assumes mean-zero subgaussian errors. OpenFHE's floating-point inversion-table sampler was checked to be 3.2^2-subgaussian only numerically. The check is tight: its table gives the variance proxy $10.1761 = 3.19^2$, perturbing every probability by $+2^{-53}$, the granularity of its uniform generator, gives $10.2392 \leq 10.24 = 3.2^2$, and a perturbation of 2^{-52} would fail it (10.32). Its mean is zero only to about 2^{-53}, which changes the bound by a relative term below 2^{-38} at the evaluation sizes. Add the mean term and the numerical check to the proofs, or draw the errors with an exact sampler. 28

3 (T2) Kluczniak and Santato (CiC 2(1), 2025): the description (indistinguishability-based circuit privacy, a KL-divergence analysis, noise exponential in the security parameter, tightness, partial decryptions of multikey and threshold schemes) comes from search-engine extracts. Check the theorem statements and the exact noise condition, and whether their tightness result says anything about a statistical flooding parameter $\kappa = 40$ accounted for as an explicit slack. 28

4 (T3) Guo et al. (USENIX Security 2024) and Checri et al. (CRYPTO 2024): the claims are from abstracts and search-engine extracts. Check which OpenFHE modes they attacked. In OpenFHE 1.5.1, multiparty decryption adds a Gaussian with $\sigma = 2^{20}$ by default, estimate-based flooding under `NOISE_FLOODING_DECRYPT`, and uniform noise under `NOISE_FLOODING_MULTIPARTY`. The given names of Checri et al. and the page numbers of Guo et al. (7447–7461, from an extract) are not verified and are not in the bibliography. 28

5 (T1) Hwang et al. (ePrint 2025/395; the later version's title, used in the bibliography, is "On the (In)security of Approximate Computation Protocols from CKKS", listed as accepted at ASIACRYPT 2026): the definitions and the two-party obstruction were checked against the authors' FHE.org 2026 poster only, and the abstract, the later title and the venue only through search-engine extracts. Confirm against the full text the exact standard and liberal definitions; that the obstruction needs the corrupted party's noise share to be non-negligible next to the rest of the output noise; and the statement for the case of one party with input. Fix the venue in the bibliography once confirmed. 30

6 (T8) Release timing: the padding makes R's observed release latency independent of eta except when a release overruns its deadline. The overrun probability is only measured (0 of 1,000 releases on the idle machine, 4 of 200 under CPU contention), not bounded, and it is not charged to delta. Bound it, for example by a tail bound on the sampler's running time on an isolated core, or charge it to the slack; and pad the flooding of the PSI replies as well. 30
7 Composition: the proof, including the adaptive hybrid and the hockey-stick ledger, is written out in the composition appendix of the proofs (Theorems 4.2 and 5.2, Corollary 5.4), with a self-contained Rényi filter, so the cited filter results are attributions only. Not proved there: that the first rule (runs fixed in advance) is valid when the requests are chosen adaptively, where the proof gives only $\delta(\sup \rho_\Sigma, \varepsilon_L) + (1 + e^{\varepsilon_L}) \sup \Gamma_\Sigma$, which can approach $2\delta_L$; and the cut-off argument for the expected-polynomial-time exact sampler inside the reductions, which is stated but not written out. The per-run results it takes from Part A of the proofs were not re-verified. The bibliographic details of the filter literature were checked only through search-engine extracts: Feldman and Zrnic's theorem number in the NeurIPS version, the pages of Rogers et al. (1921–1929), and the volume and pages of Whitehouse et al. (PMLR 202, 36990–37007). 35
8 G4, second layer: check that OpenFHE 1.5.1's BFVrns (its RNS multiplication, scaling and relinearization) satisfies the BFV model of the proofs' bound on the evaluation noise (Part A, Proposition 7.7), or bound that noise directly for it. The circuit-privacy statements of CLR17 and the role of OPRF preprocessing in CHLR18 were checked only through search-engine extracts; confirm against the full texts, including whether CHLR18 state that OPRF preprocessing removes the need for circuit privacy. 38
9 The switch-off factors 1.2 (setting I and the previous setting) and 1.4 (setting P) remain empirical. A bound needs a tail bound for lambda-min of the noisy stage-1 Gram that uses the spectrum and is much sharper than the comparison bound; at n-off the three smallest eigenvalues of the simulation population's Gram lie within about 2.4 sigma-1 of each other, too close for second-order perturbation. 46
10 AdaSSP's adjacency: the published constants are consistent with add/remove adjacency (sensitivity 1 for each statistic); confirm against the statement in Wang (2018) before citing it. 77
11 TukeyEM: the adjacency (a change of one record changes one of the m fits; the PTR distance is a Hamming distance between the sets of fits) and the PTR noise scale 2/epsilon are taken from the reference code and the PTR threshold, because the paper of Amin et al. was not accessible when this was written. Confirm both against the paper, and the choice of m against its recommendation. 82
12 Full-record SSP: the measured cost assumes that R uploads the products b x-j x-k. The U-side bound of those vectors (at most B-R squared) enters the derivation of B-tot, which was only re-evaluated with the larger entry counts, not re-derived. A secure-aggregation protocol of the kind of Karr et al. or Gascon et al. would have its own cost. 82

13 Full-record SSP: the closed form of the proofs (Part B, Theorem S10c) covers blocks of dimension at least two; for tamielectric, whose block for R has dimension one, it is a conservative bound and the reported value comes from the numerical maximization. Decide whether the baseline code should switch to the enumeration. 82

1 Introduction

Organizations such as Statistics Canada (StatCan) hold administrative and survey datasets that span the full Canadian population. University researchers, by contrast, often hold smaller cohort datasets enriched with domain-specific variables—clinical measurements, survey responses, or behavioural signals—that StatCan does not collect. Neither dataset alone is sufficient to answer certain research questions: a policy researcher studying labour-market outcomes might hold a cohort's health and education records, while StatCan holds their tax filing history, employment insurance claims, and income trajectories. The scientific value of joining these datasets is evident; the barrier to doing so is equally clear. Privacy law, institutional policy, and the practical realities of data governance make direct data sharing infeasible.

The Research Data Centre (RDC) model partially addresses this by letting researchers work with anonymized StatCan data inside a controlled environment. However, it does not permit a researcher to bring their own dataset and perform a record linkage against StatCan records in order to jointly train a statistical model—a process that typically requires six to eight weeks of vetting, human review, and institutional approvals. This overhead is a meaningful bottleneck on the pace and scope of Canadian policy research.

This document proposes a cryptographic protocol for *vertical federated learning* (VFL) between a researcher and StatCan. The term "vertical" refers to the structure of the data: both parties hold overlapping sets of individuals (rows) but non-overlapping sets of variables (columns). The goal is to jointly fit a regression model over the combined column space without either party ever observing the other's raw data. The two datasets are linked by a common identifier, but the linkage itself must remain private: StatCan should not learn which of its records belong to the researcher's cohort.

Scope. We treat the case in which **StatCan holds the response variable y** (e.g. tax-reported income, employment status, benefit receipt), and the researcher wishes to learn whether their cohort-specific covariates predict that outcome, using StatCan's covariates as controls. Both parties are honest-but-curious.

The design in one paragraph. Linkage is labeled private set intersection (PSI) on oblivious-PRF values of the identifiers, after which R knows its matches and their positions in O's secretly shuffled register. R computes its own Gram block exactly and uploads its features, aligned to O's order and zero elsewhere, encrypted under its share of a two-party CKKS key. O converts them to the joint key and multiplies them by its own plaintext columns, which gives encryptions of every statistic that involves its data. A release is a threshold decryption in which O computes its share of the decryption, adds discrete-Gaussian noise to it, and sends it without the part of the ciphertext that would let R remove the noise. O never decrypts anything, R never sees a noiseless statistic that depends on O's data, and every decryption needs O's cooperation, so O can refuse once the budget is spent.

Contributions. Each ingredient of the protocol has a precedent (Section 2): DP noise added at collective decryption, DP noise on partial decryptions as a defence for threshold approximate HE, noisy sufficient statistics for vertical linear regression, and a block that each data owner computes locally. What is new is their combination, made precise in the items below; Section 2

says what we do not claim.

1. **Protocol.** A two-party protocol that combines OPRF-based labeled PSI, a seeded upload under R's key share converted by O, plaintext–ciphertext aggregation, and a release in which O's decryption share carries the DP noise (Section 6, Algorithms 1–7).
2. **Privacy.** The exact replace-one sensitivity of the released statistics; a release whose single noise term, an exact lattice discrete Gaussian in O's decryption share, is exactly ρ_s-zCDP and also floods every encryption error term against a worst-case bound; simulation-based output privacy; composition of runs, fixed or adaptively chosen, under a per-register ledger with a privacy filter; input and cohort privacy (Sections 7 and 8).
3. **Parameters.** The protocol and every theorem are stated for a parameter vector with explicit admissibility constraints; CALIBRATE derives the flooding parameter from the budget and the ledger cap, and the scale and the noise from both (Section 10).
4. **Statistics.** A two-stage ridge whose choice is independent of the noise in the estimation release; two named estimands; a Wald interval from a parametric bootstrap of the mechanism, after Ferrando et al. [28]; a bias-aware interval for the regression coefficients, after Armstrong et al. [3]; and the leading-order mechanism bias with an optional correction (Section 9 and Appendix A).
5. **Preprocessing.** Per-party transforms with public constants that make typical records fill the clipping bounds, which no joint computation is needed for (Section 6.1).
6. **Evaluation.** Exactness of the cryptographic layer, the calibration of the two-release protocol, interval coverage, end-to-end runs of a full implementation on real and synthetic data, coefficient accuracy against the intersection size, benchmarks against AdaSSP, vertical full-record SSP and TukeyEM on 29 UCI datasets, and a comparison with the parametric-bootstrap intervals of Ferrando et al., under two named settings of the protocol chosen on synthetic data by pre-registered rules: one for inference and one for prediction (Section 11).

Specification and proofs. The specification of v3.1 [82] and the companion proofs [83] are in the repository, under `docs/`. References of the form "proofs, Part A" are to the proofs document: Part A [84], Part B [85] and the composition appendix [86]. Appendix D summarizes their status.

2 Related work

[**Check 1:** The bibliographic details of the entries cited in this section are taken from the literature review's verified reference list, and the fields that list omits (some given names, volumes and pages) are omitted here too. Three are still unverified. (a) UnLynx [31] is not in the verified list. Its authors, title, issue, pages and DOI were confirmed only through a search-engine summary of the PoPETs page, and the description of its noise protocol comes from the Drynx paper, through search-engine extracts; check it against the UnLynx text. (b) Ion et al. [42] is not in the verified list. The title of ePrint 2019/723 and its nine-author list conflict with Google Research's listing of the EuroS&P 2020 paper, which has a different title and six authors; the bibliography gives the ePrint title and the first two authors only. Confirm the published title, the authors and the pages (370–389). (c) The later title and the venue of Hwang et al. [41]: see Check (T1) in Section 7.2.]

The protocol draws on secure regression for vertically partitioned data, differentially private regression and inference, threshold homomorphic encryption, and private set intersection. We

review each and then state what is new.

Secure regression on vertically partitioned data. The statistics literature on this problem predates federated learning. Sanil, Karr, Lin and Reiter [70] compute exact global regression coefficients and goodness-of-fit measures for agencies that hold different attributes of the same records, without revealing the attributes. Karr, Lin, Sanil and Reiter [45] compute the off-diagonal blocks of the covariance matrix by secure matrix products between pairs of owners, and each owner computes its own diagonal block locally. They release coefficients, standard errors and diagnostics, use no third party, and assume that the records are already aligned. Our aggregation has the same block structure: R computes $\mathbf{A}$ locally, and only the blocks that involve O's data are computed jointly. Du, Han and Chen [22] give two-party protocols for multivariate linear regression. Hall, Fienberg and Nardi [38] compute linear and ridge regression under homomorphic encryption, release only the final result, and test the method on the Current Population Survey. In cryptography and machine learning, Nikolaenko et al. [62] combine linearly homomorphic encryption with garbled circuits for ridge regression on hundreds of millions of records, and Giacomelli et al. [36] use linearly homomorphic encryption alone. Gascón et al. [35] solve vertical linear regression with garbled circuits, inner-product protocols and a fixed-point conjugate-gradient method. Hardy et al. [39] link records by privacy-preserving entity resolution and train logistic regression under additively homomorphic encryption, with a coordinator as a third party. SecureML [60] trains linear and logistic regression on two servers, and the FATE platform [54] includes PSI and two-party linear regression among its components. Yang et al. [80] and Liu et al. [55] survey federated and vertical federated learning. All of these give the analyst the exact model or exact statistics. None gives the output a differential-privacy guarantee.

Differentially private linear regression. Sufficient-statistics perturbation (SSP) adds noise to $\mathbf{X}^\top\mathbf{X}$, $\mathbf{X}^\top\mathbf{y}$ and $\mathbf{y}^\top\mathbf{y}$ and solves from the noisy statistics. Vu and Slavković [74] add noise to sufficient statistics for tests on clinical-trial data. The functional mechanism of Zhang et al. [81] perturbs the coefficients of the polynomial objective, which for least squares is SSP with Laplace noise. Foulds et al. [29] use Laplace SSP for conjugate exponential-family posteriors. AdaSSP [75] adds a ridge chosen from a private estimate of the smallest eigenvalue of the Gram matrix. Other estimators perturb the objective [14, 46]. TukeyEM [2] fits least squares on disjoint subsets and selects among the fits by the exponential mechanism on approximate Tukey depth; it needs no bounds on the features or the labels. Insufficient statistics perturbation [9] has error linear in the dimension and independent of the condition number for data of bounded leverage. All of these assume one curator who holds the joined table, and they protect the whole record. Our ridge rule borrows AdaSSP's idea of choosing λ from a private eigenvalue bound, but takes the bound from a separate release, so that the estimation noise is independent of λ (Proposition 9.1).

Differentially private inference. Sheffet [71] derives t-values and confidence intervals for differentially private ordinary least squares. Ferrando, Wang and Sheldon [28] form intervals by a parametric bootstrap of the mechanism, which accounts for sampling and privacy noise together, and prove consistency for a least-squares variant. Our Wald interval applies their idea to a ridge target with noise on only part of the statistics (Section 9.4). Evans, King, Schwenzfeier and Thakurta [26] give bias-corrected estimates with valid uncertainty from privacy-protected

data. Brown et al. [10] derive instance-specific intervals from private gradient descent for least squares. Lin, Slavković and Bhoomireddy [53] give a bias-corrected estimator for linear regression with asymptotic intervals under Gaussian DP. Barrientos, Reiter, Machanavajjhala and Chen [5] test whether a regression coefficient would be significant on the confidential data when the analyst sees only synthetic or protected outputs; the test adds Laplace noise to the mean of truncated t-statistics over subsamples. All of these are central methods. Our bias-aware interval for $\boldsymbol{\beta}$ follows Armstrong, Kolesár and Kwon [3], with the ridge bias bounded through a public or a private bound on $\|\boldsymbol{\beta}\|$.

Differential privacy on vertically partitioned data. Dwork and Nissim [24] extend noisy statistical databases to vertically partitioned databases. Xu, Yuan and Wu [79] apply the functional mechanism to vertically partitioned multiparty learning, including linear regression. A server splits the objective into single-party and cross-party terms, the parties perturb the polynomial coefficients, and one round of noise addition and secure aggregation gives the utility of the centralized functional mechanism. Differentially private vertical linear regression from noisy sufficient statistics is therefore not new. Wu et al. [78] let parties that hold disjoint attributes of the same subjects each release differentially private data. They show that a direct Gaussian mechanism suffers from small eigenvalues and give a release from which the regression converges to the non-private solution, without cryptography. Mohammed et al. [59] release differentially private data from two parties with a two-party exponential mechanism. Heinze-Deml, McWilliams and Meinshausen [40] share perturbed random projections of each party's features. Other work adds noise to the intermediate results of iterative training [76, 21]. None of these links records; they assume aligned rows. Krichene et al. [48] treat some features as public and add noise only to the sufficient statistics that involve the private features or the labels. Our guarantee has this shape: R's features are fixed under the adjacency (Definition 4.1), so $\mathbf{A}$ needs no noise.

Two-party and computational differential privacy. McGregor et al. [56] show that information-theoretic two-party differentially private protocols lose accuracy on functions such as the inner product and the Hamming distance. Computational DP [57] asks for privacy only against computationally bounded adversaries. Our guarantee to R is SIM-CDP in this sense (Definition 4.3). It is also one-sided: DP protects O's records from R, while R's data are protected from O by encryption alone (G1 and G2 in Table 2).

Distributed noise and noise inside the encryption. Dwork et al. [25] let the parties generate Gaussian or exponential noise jointly, so that no trusted curator is needed. Shi et al. [72] let an aggregator decrypt only the sum of users' encrypted values, with the randomness distributed among the users so that the sum stays differentially private even if some users are compromised. Kairouz, Liu and Steinke [44] analyse sums of discrete Gaussians under secure aggregation. In these schemes the noise is split so that no single party knows all of it. In v3.1 all of it comes from O, the party whose records it protects; Remark 7.4 discusses what O's knowledge of the noise implies. Stevens et al. [73] use the LWE error of the encryption as DP noise in secure aggregation. Ogilvie [63] uses the CKKS error as DP noise, with ridge regression by gradient descent as the case study. Our release reuses noise in the other direction: the DP noise floods the encryption error.

Threshold homomorphic encryption with noise at decryption. Threshold fully homomorphic encryption goes back to Asharov et al. [4], and Boneh et al. [8] smudge partial decryptions with noise. Mouchet et al. [61] give N-out-of-N multiparty HE from RLWE, with collective key generation and collective key switching; we use their key generation with two parties (Section 5.1). Froelicher, Hubaux and coauthors build federated-analytics systems that attach DP noise to the collective decryption. UnLynx [31] and Drynx [32] encrypt under a collective EC-ElGamal key. For a differentially private answer, the computing nodes shuffle a list of encrypted Laplace noise values in turn and add the first entry of the shuffled list to the encrypted result, before the collective key switch to the querier. SPINDLE [33] trains generalized linear models under RLWE multiparty CKKS and states that collectively generated noise should be added to the result before the distributed key switch. FAMHE [34] lets the parties choose whether the querier receives exact or differentially private results, without specifying a mechanism. In all four the data are horizontally partitioned, and the noise is added under encryption before the key switch, not inside a decryption share. Nor is it used to flood a decryption error; the ElGamal-based systems have none. Kluczniak and Santato [47] supply the cryptographic half of our release. In threshold and multikey approximate HE, they apply a differentially private mechanism to the partial decryptions, with noise sized to the worst-case error growth, and obtain IND-CPA-style security against an adversary who also sees the partial decryptions. Our release uses noise of this kind. What we add is that the same noise is also the zCDP noise of the released statistic, with a simulation-based proof at a concrete flooding parameter (Section 7.2).

Attacks on approximate homomorphic encryption. Li and Micciancio [50] recover the CKKS secret key from decryptions that the adversary sees (IND-CPA^D security). Li, Micciancio, Schultz and Sorrell [51] add Gaussian noise after decryption to restore security, with nearly matching bounds on the noise needed. Guo et al. [37] recover the key in OpenFHE when the flooding noise is set from non-worst-case noise estimates. Checri et al. [15] show that threshold BFV, BGV and CKKS are exposed to such attacks unless smudging noise is added after partial decryption. Hwang et al. [41] show that two-party CKKS protocols in which each party adds its own smudging noise meet only a relaxed simulation notion when both parties have inputs. Section 7.2 shows how v3.1 answers the first three: O's share, the only partial decryption that is sent, carries noise calibrated to a worst-case bound. Remark 7.4 explains why the fourth does not affect the guarantee to R but does apply to O.

Private set intersection and computation on the intersection. Labeled PSI from homomorphic encryption [16, 17, 19] builds on PSI by oblivious polynomial evaluation [30] and on cuckoo hashing [64]. We key the items with the 2HashDH oblivious PRF [43], specified in RFC 9497 [67]. In these protocols the receiver learns the intersection. Circuit-PSI [65, 68] gives the parties secret shares instead, and would hide the match set from R as well (Remark 13.1). Other protocols compute on the intersection without revealing it. The protocol of Ion et al. [42] reveals only the cardinality of the intersection and the sum of the values attached to it. Buddhavarapu et al. [11] give private matching protocols whose outputs are secret-shared, for later computation. Lepoint et al. [49] compute inner products over the intersection from PIR with default, hiding even its size. In v3.1 R learns the match, which lets it compute $\mathbf{A}$ exactly and locally. Lin, Paquette and Kolaczyk [52] study DP linear regression after probabilistic record linkage, under central DP, with error bounds that separate the linkage error, the estimation

error and the cost of privacy. Our linkage is exact on identifiers, and linkage error is outside our model (Section 13).

Statistical agencies. Bogdanov et al. [7] joined about ten million Estonian tax records with about half a million education records under secure multiparty computation for a statistical study. Dasylva et al. [20] link trade data held by different national statistical offices, Statistics Canada among them, by PSI. Reiter, Oganian and Karr [66] propose verification servers, which tell an analyst how closely results from altered public-use data match those from the confidential data; the tests of Barrientos et al. [5] are a differentially private check of this kind. Barrientos, Williams, Snoke and Bowen [6] evaluate DP summary statistics and regression on IRS tax data and the Current Population Survey, and report that accurate DP regression estimates and intervals were hard to obtain. The application setting is therefore not new. None of these works gives a DP release, under the agency's control, on a cohort that the researcher brings.

What is new. Each ingredient has a precedent, and we do not claim them:

- DP noise attached to collective decryption (UnLynx, Drynx, SPINDLE, FAMHE);
- DP noise on partial decryptions as a defence for threshold approximate HE (Kluczniak and Santato);
- noisy sufficient statistics for vertical linear regression (Xu, Yuan and Wu);
- each owner's block computed locally (Karr et al.);
- noise only on the statistics that involve private data (Krichene et al.).

We found no prior protocol that combines them as follows:

(i) vertically partitioned data linked by OPRF-based labeled PSI and aggregated under two-party threshold CKKS;
(ii) a release in which a single party's decryption share carries an exactly sampled discrete Gaussian that is at once the zCDP noise for that party's records and the worst-case flooding noise for the CKKS error, with a simulation-based proof (Theorem 7.2);
(iii) R's block kept exact and local, with the adjacency and the noise covering O's records only;
(iv) a two-stage ridge whose choice is independent of the estimation noise, on which the validity of the Wald and bias-aware intervals rests (Proposition 9.1, Section 9.4);
(v) a per-register ledger that composes runs, fixed or chosen adaptively, under a privacy filter [69], with the flooding slack accounted next to the zCDP cost (Section 7.4).

3 Setting and notation

O restricts its register to a study scope $D_\Phi = \{j : \Phi(j)\}$, where the predicate Φ uses public attributes only (for example province and birth-year band), and pads D_Φ with dummy rows—fresh random identifiers that never match, with $\mathbf{x}_O = \mathbf{0}$ and $y = 0$—to a public size $\bar{n}_O$. R holds $n_R \le n_R^{\max}$ cohort records. Let I be the cohort members that appear in D_Φ and $n = |I|$. Table 1 collects the notation.

Parameters. The protocol, its algorithms and its theorems are stated for a parameter vector $\mathcal{P}$: security parameters (ring dimension, modulus chain, δ_B, the floor $\kappa_{\min}$ on the flooding

parameter, the security level λ_{sec}), privacy parameters (the budget, the splits f_1 and f_y, the slack share ζ, the ledger cap), statistical parameters, the data bounds and preprocessing, the PSI parameters, and the encoding scales Ω_U and Ω_u. Section 10 lists every parameter with its role, the constraint it must satisfy and the value used in our evaluation, which we call the *evaluation setting*; the evaluation settings are not part of the protocol. CALIBRATE derives the flooding parameter κ, the noise, the scales and the size of q_0 from $\mathcal{P}$, or refuses the run (Section 7.3). Numbers in Sections 6–9 and Appendix A are evaluation settings or worked examples and are labelled as such.

The target is a ridge fit on the matched, column-joined data $\mathbf{X}^I = [\dot{\mathbf{X}}_R^I \mid \mathbf{X}_O^I]$ with

$$\mathbf{G} := \mathbf{X}^{I\top}\mathbf{X}^I = \begin{bmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^\top & \mathbf{B} \end{bmatrix}, \qquad \mathbf{c} := \mathbf{X}^{I\top}\mathbf{y}^I = \begin{bmatrix} \mathbf{c}_R \\ \mathbf{c}_O \end{bmatrix}, \tag{1}$$

where $\mathbf{A}$ is R's own block and $\mathbf{B}$, $\mathbf{C}$, $\mathbf{c}_R$, $\mathbf{c}_O$ and $\mathbf{y}^\top\mathbf{y}$ depend on O's data. We write Π_R and Π_O for the orthogonal projectors onto the two coordinate blocks, so $\Pi_R + \Pi_O = \mathbf{I}_p$. The two estimands, the ridge target $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$ and the regression coefficient $\boldsymbol{\beta}$, are defined in Section 9.1.

4 Threat model and guarantees

Both parties are *honest-but-curious*. They follow the protocol, including key generation and R's preparation of its inputs (clipping, the selection vector), and they try to learn what they can from their views. Following the protocol includes drawing every secret, error and coin from its specified distribution with fresh randomness. In particular, the flooding bound of Theorem 7.1, and with it Theorem 7.2, needs R's encryption and key-share errors to be sampled honestly: a researcher who inflated them would push the term "R's errors times O's data" beyond the bound, and that term is correlated with O's data (Remark 7.4; robustness to such a researcher is an extension, Section 13). O's ledger is honest and admits runs one at a time (Section 7.4). Channels are private and authenticated. R is the analyst and receives the output. O receives no output, and its guarantees G1 and G2 below concern its view of the protocol only.

Definition 4.1 (Datasets and adjacency)**.** A dataset is O's padded scope register $D = \{(\mathrm{id}_j, \mathbf{x}_{O,j}, y_j)\}_{j \le \bar{n}_O}$, with the cohort $(\mathrm{id}, \mathbf{X}_R)$, the identifiers and Φ held fixed. $D \sim D'$ if they differ in $(\mathbf{x}_O, y)$ of exactly one row, matched or not (*replace-one* adjacency).

Definition 4.2 (Zero-concentrated DP)**.** A randomized mechanism $\mathcal{M}$ is ρ-zCDP if the Rényi divergence of order a between $\mathcal{M}(D)$ and $\mathcal{M}(D')$ is at most ρa for all $a > 1$ and all $D \sim D'$ [12]. It is (ε, δ)-DP if $\Pr[\mathcal{M}(D) \in S] \le e^\varepsilon \Pr[\mathcal{M}(D') \in S] + \delta$ for all $D \sim D'$ and measurable S.

Definition 4.3 (SIM-CDP and IND-CDP, after [57])**.** A protocol is (ε, δ)-*SIM-CDP for* R *with statistical slack* γ if there exist a mechanism $\mathcal{M}$ that is (ε, δ)-DP with respect to $\sim$ and a probabilistic polynomial-time simulator $\mathcal{S}$ such that, for every D and every PPT distinguisher $\mathcal{A}$,

$$\left|\Pr[\mathcal{A}(\mathrm{view}_{\mathsf{R}}(D)) = 1] - \Pr[\mathcal{A}(\mathcal{S}(\mathcal{M}(D))) = 1]\right| \le \gamma + \mathrm{negl}(\lambda_{\text{sec}}).$$

It is (ε, δ)-*IND-CDP* if $\Pr[\mathcal{A}(\mathrm{view}_{\mathsf{R}}(D)) = 1] \le e^\varepsilon \Pr[\mathcal{A}(\mathrm{view}_{\mathsf{R}}(D')) = 1] + \delta + \mathrm{negl}(\lambda_{\text{sec}})$ for all $D \sim D'$ and every PPT $\mathcal{A}$.

Mironov et al. state these notions for pure ε; the extension to (ε, δ) is the standard one, and the slack γ makes explicit the statistical terms that are not negligible in λ_{sec} at a concrete flooding parameter.

Lemma 4.1 (From simulation to indistinguishability)**.** *(ε, δ)-SIM-CDP with slack γ implies $(\varepsilon, \delta + (1 + e^{\varepsilon})\gamma)$-IND-CDP.*

Proof. Chain $\text{view}(D) \approx \mathcal{S}(\mathcal{M}(D))$ (cost γ), the DP of $\mathcal{S} \circ \mathcal{M}$ (post-processing; factor e^{ε}, cost δ), and $\mathcal{S}(\mathcal{M}(D')) \approx \text{view}(D')$ (cost $e^{\varepsilon}\gamma$). □

Remark 4.1 (What is revealed by design)**.** R learns which of its cohort appear in O's scope, and n. When membership is itself sensitive (e.g. a benefits register, where membership *is* the outcome), this must be disclosed to data stewards. DP does not protect membership or n, nor R's own cohort in a published model, since $\mathbf{A}$ is exact. A membership-private variant is sketched in Remark 13.1.

5 Building blocks

5.1 Two-party threshold CKKS

CKKS [18] encrypts vectors of $N/2$ real numbers. A ciphertext is a pair (c_0, c_1) of polynomials modulo q, and decryption with the secret key s gives

$$c_0 + c_1 s = \mathsf{Ecd}(\mathbf{z}) + e \pmod{q},$$

the encoded vector plus a small error. In the threshold variant the key is split additively, $s = s_R + s_O$, so decryption splits as $(c_0 + c_1 s_O) + c_1 s_R$: the first term needs O and the second needs R.

Key generation. Following Mouchet et al. [61], from a common random polynomial a each party draws its share and an error and publishes one polynomial,

$$p_{0,R} = -a s_R + e_R, \qquad p_{0,O} = -a s_O + e_O, \qquad pk = \big(p_{0,R} + p_{0,O},\ a\big) = (-as + e,\ a).$$

Rotation keys for the slot shifts 2^i are generated the same way: R sends its rotation-key shares to O, which adds its own and keeps the joint rotation keys, since only O evaluates. O sends R only its public-key share $p_{0,O}$. The circuit has depth two and no ciphertext–ciphertext products, so no relinearization or conjugation key is needed.

Upload at half size. R encrypts with its own share symmetrically, using a random a expanded from a short seed: $c_0 = -a s_R + e + \mathsf{Ecd}(\mathbf{v})$, $a = \text{PRG}(\varsigma)$. It sends (c_0, ς). O regenerates a and sets $c_0 \leftarrow c_0 - a s_O$, after which $c_0 + as = \mathsf{Ecd}(\mathbf{v}) + e$: a ciphertext under the joint key.

What O computes. O adds ciphertexts, multiplies them by its own plaintext columns, and rotates slots to sum a vector. None of this needs a key share.

5.2 Oblivious PRF

R obtains $F_k(x) = H_2(x, H_1(x)^k)$ under O's key k without O seeing x and without R learning k (2HashDH [43, 67]). R sends $\alpha = H_1(x)^r$ for a random r; O returns $\beta = \alpha^k$; R unblinds $\beta^{1/r} = H_1(x)^k$ and hashes. α is uniform in the group, so O learns nothing about x. The output is split into an item part $F_k(x)$, used for matching, and a key part $F'_k(x)$, used to open labels. The OPRF makes every membership test an online interaction with O, which matters because identifiers such as social insurance numbers are a small, guessable space.

5.3 Labeled PSI

In labeled PSI [16, 17, 19] the receiver learns which of its items are in the sender's set and, for each match, a label attached by the sender. O hashes its items into bins, interpolates per bin a membership polynomial whose roots are the items and a label polynomial that maps each item to its encrypted label, and evaluates both on R's encrypted, cuckoo-hashed items. Here the label is the row's position $\tau(j)$ in O's shuffled register. Appendix C gives the details.

5.4 zCDP and the discrete Gaussian

A mechanism that adds Gaussian noise of s.d. σ to a statistic with ℓ_2 sensitivity Δ is exactly $\Delta^2/(2\sigma^2)$-zCDP [12]. The multivariate discrete Gaussian on $\mathbb{Z}^N$ is exactly $\Delta^2/(2\sigma^2)$-zCDP for a statistic with values in $\mathbb{Z}^N$, that is, for integral shifts only [13]: if two centres differed by a non-integral vector, the two outputs would lie on disjoint cosets of $\mathbb{Z}^N$. The release of Section 7.2 is centred at an integral vector, so its shifts are integral. zCDP costs add under composition. A ρ-zCDP mechanism is (ε, δ)-DP with $\varepsilon = \rho + 2\sqrt{\rho \ln(1/\delta)}$; the tight conversion of Canonne, Kamath and Steinke [13] allows a larger ρ at the same (ε, δ) (about 27% larger at $\varepsilon = 1$, $\delta = 10^{-9}$). Floating-point samplers leak through their low-order bits [58]; we use an exact integer sampler.

6 The protocol

Figure 1 shows the message flow and Algorithms 1–7 give the pseudocode.

Phase 0: setup (Algorithm 1). The parties agree on the parameter vector $\mathcal{P}$ (Section 10): the scope, the caps, the budget, the ledger cap of the base register, the statistical constants, the PSI parameters and the encoding scales. Each party preprocesses its own columns with public constants (Section 6.1), and the clipping bounds are chosen on the preprocessed public reference data or from domain knowledge. The sensitivities follow from the bounds alone (Section 7.1). O runs Calibrate (Algorithm 7), which derives the flooding parameter, the CKKS scale and the noise from $\mathcal{P}$ and refuses the run if $\mathcal{P}$ is not admissible; O also refuses if the run's full cost would exceed its ledger's cap, and otherwise charges that cost at once (Section 7.4). R repeats Calibrate, whose outputs depend on public values only, and checks O's. The parties run threshold key generation, and O draws a fresh OPRF key, a fresh shuffle τ, and prepares its PSI polynomials.

ChooseBounds (Algorithm 7) takes the 99.5% quantile of each block's norm, $\|\dot{\mathbf{x}}\|$, $\|\mathbf{x}_O\|$ and $|y|$, on the preprocessed public reference data; this is the evaluation setting. Choosing the

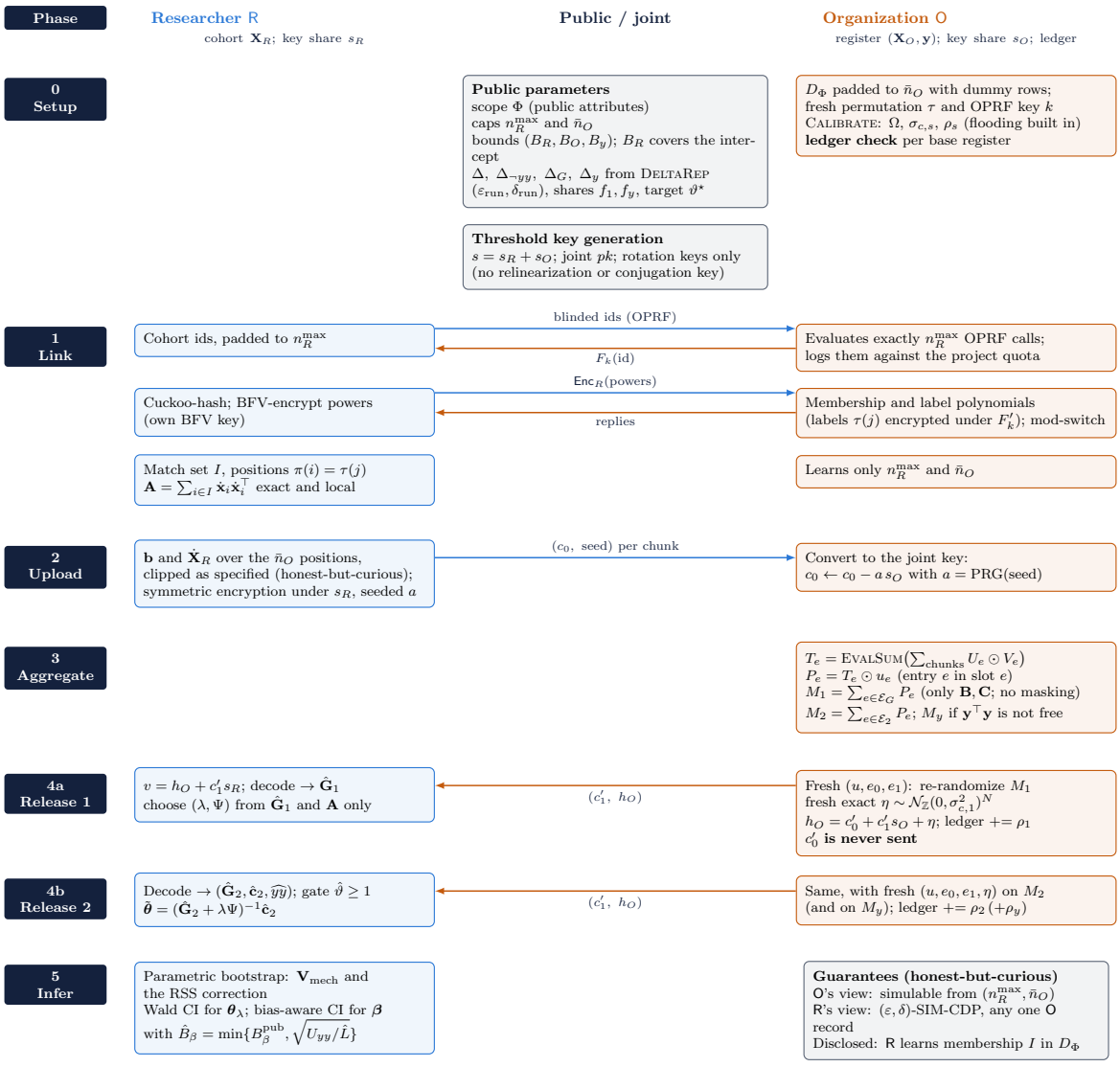


Figure 1: Message flow of protocol v3.1. Blue arrows go from R to O and orange arrows from O to R. O never receives a decryption share, and R never receives c'_0.

quantile levels to minimize the first-order coefficient MSE plus the squared clipping bias is future work.

Phase 1: link (Algorithm 2). R pads its identifiers to $n_R^{\max}$, obtains their OPRF values, and runs labeled PSI. It learns I and the positions $\pi(i) = \tau(j)$, and computes its own block

$$\mathbf{A} = \sum_{i \in I} \dot{\mathbf{x}}_i \dot{\mathbf{x}}_i^\top$$

exactly and locally. $\mathbf{A}$ is never transmitted and never noised: it is a function of $(I, \mathbf{X}_R)$, both known to R, and it does not change between adjacent datasets.

Phase 2: upload (Algorithm 3). R writes the selection vector $\mathbf{b}$ and its features $\dot{\mathbf{X}}_R$ over all $\bar{n}_O$ positions, in τ-order and zero off the match, encrypts them under s_R with seeded randomness, and sends them in chunks of at most $N/2$ slots. O converts each ciphertext to the joint key.

Phase 3: aggregate (Algorithm 4). For each released entry O multiplies one of R's encrypted vectors by one of its plaintext columns and sums:

$$\mathsf{Enc}(B_{jk}) = \textstyle\sum_t \mathsf{Enc}(b_t)\; X_{O,tj} X_{O,tk}, \qquad \mathsf{Enc}(C_{jk}) = \textstyle\sum_t \mathsf{Enc}\Big(\dot{X}_{R,tj}\Big)\; X_{O,tk}, \tag{2}$$

$$\mathsf{Enc}(c_{O,k}) = \textstyle\sum_t \mathsf{Enc}(b_t)\; X_{O,tk}\, y_t, \qquad \mathsf{Enc}(c_{R,j}) = \textstyle\sum_t \mathsf{Enc}\Big(\dot{X}_{R,tj}\Big)\; y_t, \tag{3}$$

and $\mathsf{Enc}\big(\mathbf{y}^\top \mathbf{y}\big) = \sum_t \mathsf{Enc}(b_t)\; y_t^2$. Unmatched positions contribute zero to the values, because $\mathbf{b}$ and $\dot{\mathbf{X}}_R$ are zero there. O then packs the results so that each released entry sits alone in its own slot: M_1 holds only $\mathrm{triu}\,\mathbf{B}$ and $\mathbf{C}$, for stage 1, and M_2 holds every stage-2 entry. A separate M_y holds $\mathbf{y}^\top \mathbf{y}$ when it is not free (Section 7.1).

Phase 4: release (Algorithm 5). For each packed ciphertext $M = (c_0, c_1)$, O re-randomizes it under the joint public key with fresh (u, e_0, e_1), computes its noisy decryption share

$$h_O = c_0' + c_1' s_O + \eta, \qquad \eta \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma_c^2)^N \text{ fresh},$$

and sends (c_1', h_O); the release was paid for when the run was admitted. It never sends c_0'. R computes $v = h_O + c_1' s_R = \mathsf{Ecd}(\mathbf{z}) + e + \eta$ and decodes. Stage 1 releases M_1 with s.d. σ_1, stage 2 releases M_2 with s.d. σ_2, and the releases use independent randomness.

Phase 5: solve and infer (Algorithm 6). From stage 1 and its exact $\mathbf{A}$, R chooses the ridge (λ, Ψ). After stage 2 it checks the gate $\hat{\vartheta} \geq 1$, solves $\tilde{\boldsymbol{\theta}} = (\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1} \hat{\mathbf{c}}_2$, and computes both intervals (Section 9). All of this is post-processing.

Remark 6.1 (Changes relative to earlier drafts)**.** (i) **Keys.** R no longer holds the CKKS secret key. The key is shared, O adds the noise inside its decryption share, and c_0' is never transmitted, which closes a leak of O's key share.

(ii) **Flooding.** The DP noise η itself floods every error term that depends on O's data or secrets, statistically and in ℓ_2. The earlier separately calibrated "error channel" and its cost ρ_{err} are gone.

Algorithm 1 SETUP (Phase 0)

Both: agree on the parameter vector $\mathcal{P}$ (Table 10): the scope Φ (public attributes only), the caps $n_R^{\max}$ and $\bar{n}_O$, the budget $(\varepsilon_{\text{run}}, \delta_{\text{run}})$, the ledger parameters $(\varepsilon_L, \delta_L, \zeta_L, \xi_L, \delta')$, f_1 (and f_y), ζ, $\kappa_{\min}$ (or a fixed κ), δ_B, t_η, $\vartheta^\star$, $\alpha_{\text{CI}}, \alpha_L, \alpha_U$, B_β^{pub}, $\bar{\sigma}^2$, κ_Ψ, η_B, B, the PSI parameters, N, the chain, Ω_U and Ω_u

O: $D_\Phi \leftarrow \{j : \Phi(j)\}$; pad with dummy rows (fresh random identifiers, $\mathbf{x}_O = \mathbf{0}$, $y = 0$) to $\bar{n}_O$; draw a fresh secret permutation τ of $[\bar{n}_O]$

Both: optional: screen the features on public reference data that carries both schemas and drop nearly collinear ones ▷ no budget

R: and **O:** , each on its own columns only: PREPROCESS with constants from public reference data or domain knowledge

Both: $(B_R, B_O, B_y) \leftarrow$ CHOOSEBOUNDS on the preprocessed reference data, or from domain knowledge ▷ B_R covers R's intercept

$(\Delta, \Delta_{\neg yy}) \leftarrow \text{DELTAREP}(B_R, B_O, B_y)$; $\Delta_G \leftarrow \text{DELTAREP}(B_R, B_O, 0)$; $\Delta_y \leftarrow B_y^2$

mode $\leftarrow$ joint if $\Delta - \Delta_{\neg yy} \le 10^{-9}\Delta$, otherwise separate ▷ release $\mathbf{y}^\top\mathbf{y}$ jointly only when it is free

O: $(\kappa, \Omega_V, \Omega_f, \{\sigma_{c,s}, \rho_s\}_s, \mathit{cost}) \leftarrow \text{CALIBRATE}(\mathcal{P})$; refuse if it refuses ▷ R repeats it and checks

O: atomically: **if** the ledger of the base register admits $\mathit{cost} = (\rho_{\text{run}}, \varepsilon'_{\text{run}}, \gamma_L)$ (Section 7.4) **then** charge it in full, for every planned release, **else** refuse ▷ no refund

Both: threshold key generation with a common random a: $s = s_R + s_O$, $pk = (p_0, a)$ with $p_0 = -as + e$, and rotation keys for $\{2^i\}$

(R sends $p_{0,R}$ and its rotation-key shares; O sends only $p_{0,O}$ and keeps the joint rotation keys; no relinearization or conjugation key)

O: fresh OPRF key k; PSI preprocessing on the items $\{F_k(\text{id}_j)\}$ with labels $\tau(j)$ encrypted under keys derived from $F'_k(\text{id}_j)$; bin sizes and partition counts public; log the run against the OPRF quota of (researcher, project)

Algorithm 2 LINK: labeled PSI with an OPRF (Phase 1)

R: pad the cohort identifiers with random dummies to exactly $n_R^{\max}$

R: run the OPRF with O on each padded identifier $\to F_k(\text{id}), F'_k(\text{id})$ ▷ O evaluates exactly $n_R^{\max}$ calls

R: cuckoo-hash the items $F_k(\text{id})$ into the bins; send BFV encryptions of their windowed powers under R's own BFV key

O: evaluate the membership and label polynomials of every bin and partition; mask; SANITIZEBFV; reply ▷ flooding: κ_ψ

R: decrypt $\to$ the match set I and positions $\pi(i) = \tau(j)$ for $i \in I$ (labels opened with keys from $F'_k(\text{id}_i)$)

R: $\mathbf{A} \leftarrow \sum_{i \in I} \dot{\mathbf{x}}_i \dot{\mathbf{x}}_i^\top$, exact and local ▷ never transmitted, never noised

Algorithm 3 UPLOAD (Phase 2)

for $t = 1, \ldots, \bar{n}_O$ **do**

R: $b_t \leftarrow \mathbb{1}[t \in \pi(I)]$; $\dot{\mathbf{x}}_t \leftarrow \big(1,\ \text{clip}(\mathbf{x}_{R,i}; \sqrt{B_R^2 - 1})\big)$ if $t = \pi(i)$, otherwise $\mathbf{0}$

end for

$m \leftarrow \lceil \bar{n}_O / N_s \rceil$ ▷ chunks of at most $N_s = N/2$ slots

for each vector $\mathbf{v} \in \{\mathbf{b}, \dot{\mathbf{x}}_{\cdot 0}, \ldots, \dot{\mathbf{x}}_{\cdot d_R}\}$ and each chunk $h \le m$ **do**

R: draw a seed ς; $a \leftarrow \text{PRG}(\varsigma)$; $c_0 \leftarrow -a\, s_R + e + \mathsf{Ecd}_{\Omega_U}(\mathbf{v}[h])$, with e from the RLWE error distribution; send (c_0, ς)

O: $a \leftarrow \text{PRG}(\varsigma)$; $c_0 \leftarrow c_0 - a\, s_O$ ▷ (c_0, a) now encrypts $\mathbf{v}[h]$ under $s = s_R + s_O$

end for

R encodes at the parameter Ω_U (Section 7.3)

(iii) **Two releases.** The ridge is chosen from a first release instead of from a committed public eigenvalue bound, so it adapts to the data while staying independent of the estimation noise.
(iv) **Accounting.** zCDP with the tight conversion replaces the analytic Gaussian calibration of a single release.
(v) **Linkage.** An OPRF with a fresh key per run replaces a keyed hash with a key known to R, and OPRF queries are padded and counted against a quota.
(vi) **Input checks.** The homomorphic input checks of v3 are removed: they addressed only a malicious researcher, and one could be bypassed. Robustness to a deviating researcher is an extension (Section 13).
(vii) **Bias correction.** The zero-budget correction of Appendix A is now an option, off in our evaluation.
(viii) **Preprocessing.** Each party transforms its own columns in Setup, with public constants (Section 6.1).
(ix) **Parameters.** The protocol is stated for a parameter vector with explicit admissibility constraints (Section 10). The flooding parameter is derived from the budget and the ledger cap, the PSI replies are sanitized with their own flooding parameter, and the ledger has a specified composition rule, which charges a run's full cost at admission.

6.1 Per-party preprocessing

The noise is sized to the clipping bounds, so every record whose norm is well below its bound wastes part of the budget. After standardization, real features are often heavy-tailed. On the protein dataset of Section 11.5, the 99.5% quantile of the squared norm of R's block (with the intercept) is 9.5 times its median, and that of O's block 19 times. Bounds at these quantiles leave typical records using a small part of the budget. Setup therefore includes a preprocessing step, PREPROCESS in Algorithm 7, in which each party transforms its own columns:

(i) standardize each column with public constants, from public reference data or domain knowledge;
(ii) optionally, a per-column tail transform with public constants: winsorizing at $|z| \le c$, a rank-Gaussian transform from public reference quantiles, or a log transform of a skewed variable;
(iii) optionally, *block normalization*: each party scales its own block of every row to unit norm (R excludes the intercept, so $\|\dot{\mathbf{x}}\|^2 = 2$; O gets $\|\mathbf{x}_O\| = 1$). A clipped log of the block norm may be appended as an extra feature to keep the magnitude.

O standardizes y, and may winsorize it, in the same way. The bounds are then chosen on the transformed reference data. For block normalization they are exact: $B_R^2 = 2$, $B_O = 1$.

Proposition 6.1 (Preprocessing preserves the guarantees)**.** *Let each party's transform be a fixed function of its own record and of public constants. Then the protocol run on the transformed data, with bounds that hold on the transformed domain, has the guarantees of Table 2 with respect to replace-one adjacency on the raw data.*

Proof. O's transform maps one raw record to one transformed record, so changing one raw record $(\mathbf{x}_O, y)$ changes exactly one transformed record, which still satisfies the bounds; R's records are fixed under $\sim$. The sensitivity of Section 7.1 therefore bounds the change, and the

rest of the analysis is unchanged. The constants are data-independent, as CALIBRATE's outputs must be. R may standardize with its own constants, since its data are not protected against itself, but it commits to B_R. □

The estimand is the regression on the transformed features. Winsorizing and log transforms keep the coefficients interpretable. Block normalization changes their meaning: each block contributes through its direction only, unless the log-norm feature is appended. It is therefore the choice for prediction rather than for interpreting coefficients.

Why not joint row normalization. Wang's preprocessing for AdaSSP normalizes each whole row $(\mathbf{x}_R, \mathbf{x}_O)$, which needs both parties' columns. We considered two ways to do this here.

- *Exact joint clipping* computes the scale factor $\min\{1, B/\|\mathbf{x}\|\}$ under encryption from R's encrypted row norms and O's plaintext ones. This needs a high-degree polynomial approximation and ciphertext–ciphertext products. Worse, scaling R's part by a factor that depends on O's data makes $\mathbf{A}$ depend on O's data, so $\mathbf{A}$ could no longer be exact.
- *Clipping only O's part to the remaining budget* $B^2 - \|\dot{\mathbf{x}}\|^2$ keeps $\mathbf{A}$ exact and the circuit linear. R assigns each matched record to one of K norm classes and uploads its selection vector and features split by class, still encrypted, and O pre-clips its columns once per class in plaintext, at about K times the upload and aggregation cost. On protein it does not pay. A joint budget alone raises Δ from 87.3 to 102, because the sensitivity is dominated by O's own norm, which enters twice, and records with a small R-norm may then use more of it. Keeping O's separate cap as well lowers Δ only to 82, about 12% fewer records.

The gap is caused by heavy tails under loose bounds, not by the lack of a joint bound, and per-party transforms close most of it. In end-to-end runs on protein, block normalization brings private test error within 2–6% of non-private, against 12–21% with standardization alone (Table 20).

Block normalization is the evaluation's recommended per-party preprocessing (Sections 11.5 and 11.9). Its coefficients describe the regression on the block directions, as noted above. The appended log-norm feature and winsorizing y were not evaluated.

Optional feature screening. Setup may include a screening module, which runs before PREPROCESS (Algorithm 1). On public reference data that carries both parties' schemas, the parties look for features that are nearly collinear, within a block or across the two blocks, and drop one of each such pair. A collinearity across the blocks is invisible to each party in its own data, and it keeps $\lambda_{\min}$ of the joint design small, which delays the switch-off of the ridge (Section 9.5). The module uses only public reference data, so it costs no budget; it needs reference data that carries both schemas, and it helps only as far as that data represents the matched records. On protein, R's feature $x_{R,1}$ is nearly collinear with O's feature $x_{O,5}$ (correlation 0.997 after standardization). Dropping $x_{R,1}$ lowers n_{off} from 691k to 102k at $\varepsilon = 1$, with block normalization at the previous setting and $\delta = 10^{-9}$ (Table 7).

Table 1: Notation. ρ always denotes a zCDP budget, Δ a sensitivity, Ω the CKKS scale and ϑ a validity ratio.

Symbol	**Meaning**
Φ, $\bar{n}_O$, $n_R^{\max}$	study scope (public attributes); padded register size; certified cohort cap
n, I, τ, π	matched records; R's matched cohort members; O's secret shuffle of its padded rows; alignment $\pi(i) = \tau(j)$ for $i \in I$
d_R, d_O, p	features at R and at O; $p = d_R + 1 + d_O$, the $+1$ being R's intercept
$\dot{\mathbf{x}}_i$	R's row with the intercept, $(1, \mathbf{x}_{R,i})$; $\dot{\mathbf{X}}_R$ stacks these rows in τ-order, zero off the match
$\mathbf{b} \in \{0,1\}^{\bar{n}_O}$	selection vector: $b_t = 1$ iff $t \in \pi(I)$
B_R, B_O, B_y	clipping bounds: $\|\dot{\mathbf{x}}\| \le B_R$ (intercept included), $\|\mathbf{x}_O\| \le B_O$, $\|y\| \le B_y$; $r, o, w = B_R^2, B_O^2, B_y^2$
Δ, $\Delta_{\neg yy}$, Δ_G, Δ_y	replace-one ℓ_2 sensitivities in the slot domain: full release with and without $\mathbf{y}^\top\mathbf{y}$; $(\mathrm{triu}\,\mathbf{B}, \mathbf{C})$ only; $\mathbf{y}^\top\mathbf{y}$ alone $(= w)$
$(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$, f_1, f_y	per-run budget; stage-1 share; share of a separate $\mathbf{y}^\top\mathbf{y}$ release
ρ_s, ρ_{run}	zCDP cost of release s; $\rho_{\mathrm{run}} = \sum_s \rho_s$
γ, δ_{DP}, ζ	statistical slack of a run; the part of δ_{run} left to the DP mechanism; the largest share of δ_{run} the slack may take
$(\varepsilon_L, \delta_L)$, ζ_L, ξ_L, δ'; ρ_L, E'_L, Γ_L	ledger cap of a base register, its slack share, its share reserved for smudging, and the per-release slack of smudging; the derived zCDP, smudging and slack capacities
N, $N_s = N/2$, q_0	ring dimension, slots, last-level modulus
Ω_U, Ω_u, Ω_V, Ω_f	encoding scales of R's vectors, of the one-hot masks and of O's data; the exact final scale $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1 q_2)$
Ecd, Dcd	encoding $\mathsf{Ecd}_\Omega(\mathbf{z}) = \lfloor \Omega\, \tau_{\mathrm{can}}^{-1}(\mathbf{z}) \rceil$ and decoding $\mathsf{Dcd}(v) = \tau_{\mathrm{can}}(v)/\Omega_f$, with τ_{can} the canonical embedding
$\sigma_{c,s}$, σ_s	noise scale on the coefficient lattice; the induced slot-domain s.d. $\sigma_s = \sigma_{c,s}\sqrt{N/2}/\Omega_f$
$B_{\mathrm{tot},s}$, δ_B, κ	ℓ_2 bound on all flooded error terms of release s, holding with probability $\ge 1 - \delta_B$; flooding parameter (statistical distance $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ per release), derived by Calibrate
ϑ, $\vartheta^\star$	validity ratio $\lambda_{\min}(\hat{\mathbf{G}} + \lambda\Psi)/(2\sigma_2\sqrt{p})$; its target
$\mathcal{E}$, $\mathcal{E}_G$, $\mathcal{E}_2$	released entries: all; $(\mathrm{triu}\,\mathbf{B}, \mathbf{C})$; stage-2 entries ($\mathbf{y}^\top\mathbf{y}$ only in joint mode)

Table 2: Guarantees of v3.1 against honest-but-curious parties. Correctness assumes honest computation.

	Guarantee	Basis
G1	**R-input privacy**: O's view of the protocol can be simulated from $(n_R^{\max}, \bar{n}_O)$ and the public parameters. It does not cover what R later discloses, which is noiseless to O.	RLWE in s_R; security of the PSI. O never receives a decryption share from R, nor R's decrypted value v_s (Section 8).
G2	**Cohort privacy**: from its view of the protocol, O learns nothing about R's cohort beyond $n_R^{\max}$.	Padded OPRF queries; labeled PSI.
G3	**Output privacy**: R's view of a run is $(\varepsilon_{\text{run}}, \delta_{\text{DP}})$-SIM-CDP for $\sim$ with slack γ, hence $(\varepsilon_{\text{run}}, \delta_{\text{run}})$-IND-CDP. The joint view of all researchers over all runs on a base register, for runs chosen adaptively under the filter, is $(\varepsilon_L, \delta_L)$-IND-CDP for base-register adjacency (Definition 7.1).	Theorem 7.2; Proposition 7.2 (ledger). R samples its errors honestly.
G4	**Disclosed**: R learns I (which cohort members are in D_Φ) and uniformly random positions $\tau(j)$, and nothing else about O's register.	By design (Remark 4.1); OPRF-keyed items and labels and SanitizeBFV (Section 8).

Algorithm 4 Aggregate and pack (O, Phase 3)

Input: entry map $e \mapsto (U_e, V_e)$, with R's ciphertext vector U_e and O's plaintext vector V_e in τ-order:
$\mathbf{B}_{jk}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{x}_{O,j} \odot \mathbf{x}_{O,k})$; $\mathbf{C}_{jk}$: $(\mathsf{Enc}(\dot{\mathbf{x}}_j),\ \mathbf{x}_{O,k})$; $\mathbf{c}_{R,j}$: $(\mathsf{Enc}(\dot{\mathbf{x}}_j),\ \mathbf{y})$;
$\mathbf{c}_{O,k}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{x}_{O,k} \odot \mathbf{y})$; $\mathbf{y}^\top\mathbf{y}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{y} \odot \mathbf{y})$

1: **for** each $e \in \mathcal{E}$ **do**
2: $T_e \leftarrow \textsc{EvalSum}\big(\textsc{Rescale}\big(\sum_{h \le m} U_e[h] \odot \mathsf{Ecd}_{\Omega_V}(V_e[h])\big)\big)$ ▷ sum the chunks first; one EvalSum per entry
3: $P_e \leftarrow \textsc{Rescale}\big(T_e \odot \mathsf{Ecd}_{\Omega_u}(\mathbf{u}_{\text{slot}(e)})\big)$ ▷ $\mathbf{u}_k$: one-hot slot k; entry e sits alone in its slot
4: **end for**
5: $M_1 \leftarrow \sum_{e \in \mathcal{E}_G} P_e$; $M_2 \leftarrow \sum_{e \in \mathcal{E}_2} P_e$; $M_y \leftarrow P_{\mathbf{y}^\top\mathbf{y}}$ if $mode = \mathsf{separate}$ ▷ depth 2; M_1 depends only on triu $\mathbf{B}$ and $\mathbf{C}$

Scales: R's vectors at Ω_U, the masks at Ω_u (parameters), O's data at Ω_V (from Calibrate); the final scale $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1 q_2)$ is tracked exactly, and R decodes with it

Algorithm 5 Release(M, σ_c): one DP release (Phase 4)

1: **O:** sample fresh u from the key distribution and e_0, e_1 from the error distribution
2: **O:** $c_0' \leftarrow c_0 + p_0 u + e_0$; $c_1' \leftarrow c_1 + a u + e_1$ ▷ re-randomize $M = (c_0, c_1)$ under the joint pk
3: **O:** $\eta \leftarrow (\textsc{SampleDG}(\sigma_c))^N$, fresh ▷ exact discrete Gaussian on $\mathbb{Z}^N$
4: **O:** $h_O \leftarrow c_0' + c_1' s_O + \eta \bmod q_0$
5: **O:** send (c_1', h_O); erase c_0', u, e_0, e_1, η ▷ c_0' is never sent; the ledger was charged at admission
6: **R:** $v \leftarrow h_O + c_1' s_R \bmod q_0$; **return** $\mathsf{Dcd}(v)$ restricted to the slots of this release ▷ exact final scale; v never leaves R
7: **R:** before publishing any decoded value, round it to a fixed precision relative to σ_s; never publish v ▷ Remark 8.1; no budget

Algorithm 6 SOLVE and INFER (R, Phase 5)

Input: $\mathbf{A}$, n, p, slot-domain s.d. σ_1, σ_2 (and σ_y); from $\mathcal{P}$: $\vartheta^\star$, $\alpha = \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U$, B_β^{pub}, the variance bound $\bar{\sigma}^2$, κ_Ψ, bootstrap truncation η_B and size B

1: $(\hat{\mathbf{B}}_1, \hat{\mathbf{C}}_1) \leftarrow \textsc{Release}(M_1, \sigma_{c,1})$; $\hat{\mathbf{G}}_1 \leftarrow \begin{pmatrix} \mathbf{A} & \hat{\mathbf{C}}_1 \\ \hat{\mathbf{C}}_1^\top & \hat{\mathbf{B}}_1 \end{pmatrix}$

2: $r_1 \leftarrow \sigma_1\big(2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)}\big)$, or $r_1 \leftarrow \sigma_1 q_{1-\alpha_L}$ (exact quantile) ▷ $\Pr[\lambda_{\max}(\hat{\mathbf{G}}_1 - \mathbf{G}) > r_1] \le \alpha_L$ (one-sided)

3: $\hat{L} \leftarrow \max\{0,\ \lambda_{\min}(\hat{\mathbf{G}}_1) - r_1\}$; $g^\star \leftarrow 2\vartheta^\star\sigma_2\sqrt{p}$

4: $\gamma \leftarrow g^\star + r_1$; $\lambda_I \leftarrow \max\{0,\ \gamma - \lambda_{\min}(\hat{\mathbf{G}}_1)\}$

5: $\lambda_O \leftarrow \max\{0,\ \gamma - \lambda_{\min}(\mathbf{S}_\gamma)\}$ with $\mathbf{S}_\gamma = \hat{\mathbf{B}}_1 - \hat{\mathbf{C}}_1^\top(\mathbf{A} - \gamma\mathbf{I})^{-1}\hat{\mathbf{C}}_1$ if $\lambda_{\min}(\mathbf{A}) > \gamma$, otherwise $\lambda_O \leftarrow \infty$ ▷ closed form: the smallest $\lambda \ge 0$ with $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \ge \gamma$

6: $(\lambda, \Psi) \leftarrow (\lambda_O, \Pi_O)$ if $\lambda_O \le \kappa_\Psi\lambda_I$, otherwise $(\lambda_I, \mathbf{I})$ ▷ uses stage 1 and $\mathbf{A}$ only

7: $(\hat{\mathbf{B}}_2, \hat{\mathbf{C}}_2, \hat{\mathbf{c}}_R, \hat{\mathbf{c}}_O, \widehat{yy}) \leftarrow \textsc{Release}(M_2, \sigma_{c,2})$; if separate, $\widehat{yy} \leftarrow \textsc{Release}(M_y, \sigma_{c,y})$

8: assemble $\hat{\mathbf{G}}_2$, $\hat{\mathbf{c}}_2$; $\hat{\vartheta} \leftarrow \lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi)/(2\sigma_2\sqrt{p})$; **if** $\hat{\vartheta} < 1$ **then return** refused ▷ budget is spent

9: $\tilde{\mathbf{P}} \leftarrow (\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1}$; $\tilde{\boldsymbol{\theta}} \leftarrow \tilde{\mathbf{P}}\hat{\mathbf{c}}_2$ ▷ the bias correction of Appendix A is an option

10: $\widetilde{\mathrm{RSS}} \leftarrow \widehat{yy} - \hat{\mathbf{c}}_2^\top\tilde{\boldsymbol{\theta}} - \lambda\,\tilde{\boldsymbol{\theta}}^\top\Psi\tilde{\boldsymbol{\theta}}$; $\sigma_{yy} \leftarrow \sigma_2$ (joint) or σ_y (separate)

11: **for** $b = 1, \dots, B$ **do** ▷ parametric bootstrap of the mechanism: post-processing

12: draw $\mathbf{E}^*$ (symmetric, s.d. σ_2 on the O-dependent entries, zero on $\mathbf{A}$), $\mathbf{f}^* \sim \mathcal{N}(0, \sigma_2^2\mathbf{I})$, $g^* \sim \mathcal{N}(0, \sigma_{yy}^2)$

13: **keep** the draw only if $\|\tilde{\mathbf{P}}^{1/2}\mathbf{E}^*\tilde{\mathbf{P}}^{1/2}\|_{\mathrm{op}} \le \eta_B$ ▷ truncation: otherwise the moments do not exist

14: $\boldsymbol{\theta}_b^* \leftarrow (\hat{\mathbf{G}}_2 + \mathbf{E}^* + \lambda\Psi)^{-1}(\hat{\mathbf{c}}_2 + \mathbf{f}^*)$; $\mathrm{RSS}_b^* \leftarrow$ the RSS formula at $(\hat{\mathbf{G}}_2 + \mathbf{E}^*, \hat{\mathbf{c}}_2 + \mathbf{f}^*, \widehat{yy} + g^*)$

15: **end for**

16: $\mathbf{V}_{\mathrm{mech}} \leftarrow \widehat{\mathrm{Cov}}_b(\boldsymbol{\theta}_b^*)$; $\mathrm{Corr} \leftarrow \overline{\mathrm{RSS}^*} - \widetilde{\mathrm{RSS}}$; $\tilde{s}^2 \leftarrow \max\{\widetilde{\mathrm{RSS}} - \mathrm{Corr}, 0\}/(n-p)$

17: **CI for the protocol estimand** $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$: $\tilde{\theta}_j \pm z_{1-\alpha_{\mathrm{CI}}/2}\sqrt{(\mathbf{V}_{\mathrm{mech}})_{jj}}$

18: $\mathbf{V} \leftarrow \mathbf{V}_{\mathrm{mech}} + \bar{\sigma}^2\,\tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}}$ ▷ sampling variance from the public bound $\bar{\sigma}^2 \ge \max_i \mathrm{Var}(\tilde{y}_i \mid \mathbf{X})$

19: $\mathbf{V}^{\mathrm{wm}} \leftarrow \mathbf{V}_{\mathrm{mech}} + \tilde{s}^2\,\tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}}$ ▷ sampling variance under the homoskedastic working model

20: $U_{yy} \leftarrow \min\{\widehat{yy} + z_{1-\alpha_U/2}\,\sigma_{yy} + B_y^2\sqrt{n\ln(2/\alpha_U)/2},\ nB_y^2\}$; $\hat{B}_\beta \leftarrow \min\{B_\beta^{\mathrm{pub}}, \sqrt{U_{yy}/\hat{L}}\}$ ▷ ∞ if $\hat{L} = 0$; U_{yy} valid at level α_U

21: $\bar{b}_j \leftarrow \lambda\,\|\Psi\tilde{\mathbf{P}}\mathbf{e}_j\|_2\,\hat{B}_\beta$ ▷ worst-case ridge bias of coordinate j (first order)

22: **CI for** β_j (bias-aware): $\tilde{\theta}_j \pm \mathrm{cv}_{\alpha_{\mathrm{CI}}}\big(\bar{b}_j/\sqrt{V_{jj}}\big)\sqrt{V_{jj}}$, where $\mathrm{cv}_\alpha(t)$ is the $(1-\alpha)$ quantile of $|Z + t|$, $Z \sim \mathcal{N}(0,1)$ ▷ valid whether or not clipping binds

23: **working-model CI for** β_j: the same with $\mathbf{V}^{\mathrm{wm}}$ in place of $\mathbf{V}$ ▷ valid only under the working model

24: **return** $\tilde{\boldsymbol{\theta}}$, the three intervals, each labelled, (λ, Ψ), the shrinkage $-\lambda\tilde{\mathbf{P}}\Psi\tilde{\boldsymbol{\theta}}$, $\hat{\vartheta}$, $\sigma_2\,\mathrm{tr}(\tilde{\mathbf{P}})$, and the (ε, δ) spent

Table 3: What crosses between the parties. All messages travel over a mutually authenticated channel with structured (non-executable) serialization.

R → O	key-generation shares ($p_{0,R}$ and rotation-key shares); $n_R^{\max}$ blinded OPRF elements; BFV encryptions of windowed powers under R's own BFV key; seeded uploads (c_0, ς) of $\mathbf{b}$ and the columns of $\hat{\mathbf{X}}_R$
O → R	public parameters: the agreed $\mathcal{P}$ (including $B_O, B_y, d_O, \bar{n}_O$) and CALIBRATE's outputs $(\kappa, \Omega_V, \Omega_f, q_0, \sigma_{c,s})$, PSI sizes; the public-key share $p_{0,O}$ (not its rotation-key shares); OPRF responses; masked, sanitized PSI replies; per release, (c_1', h_O)
never transmitted	s_R, s_O; $\mathbf{A}$; raw $\mathbf{X}_R$, $\mathbf{X}_O$, $\mathbf{y}$; τ; k; c_0', u, e_0, e_1 and η of every release; R's reconstruction $v_s = h_{O,s} + c_{1,s}'s_R$ of every release (with $h_{O,s}$ and $c_{1,s}'$ it would reveal s_R)

Algorithm 7 Subroutines

function DELTAREP(B_R, B_O, B_y) ▷ replace-one of $(\mathbf{x}_O, y)$, $\dot{\mathbf{x}}$ fixed; $r, o, w = B_R^2, B_O^2, B_y^2$; Proposition 7.1
  if $|r - w| \le 2o$: $\Delta^2 = (2o + r + w)^2/2 + 2rw$; **else if** $r - w \ge 2o$: $\Delta^2 = 4r(o + w)$
  else: exact candidate enumeration (three cubics), rounded up as a safeguard. Return Δ, and $\Delta_{\neg yy}$ from the same maximization without the $(y^2 - y'^2)^2$ term
  ▷ exact suprema for $d_O \ge 2$, valid but conservative for $d_O = 1$; $\Delta_G = (2o + r)/\sqrt{2}$ if $r \le 2o$, otherwise $2\sqrt{ro}$
end function

function CALIBRATE($\mathcal{P}$) ▷ public values only; refuses when a step has no solution
  mode, $n_{\mathrm{rel}} \in \{2, 3\}$ and $\{\Delta_s\}$ as in SETUP; **refuse if** $\log_2(QP) > L_{\max}(N, \lambda_{\mathrm{sec}})$
  $\gamma(k) := n_{\mathrm{rel}}\big(\delta_B + 2^{-k}(1 + 2^{-80}) + 2^{-97}\big)$; $\varepsilon'(k) := \min\{\varepsilon : \delta(\pi 4^{-k}, \varepsilon) \le \delta'\}$ ▷ per run; per release
  unless $\mathcal{P}$ fixes κ: $\kappa \leftarrow \max\{\kappa_{\min}, \kappa_{\mathrm{run}}, \kappa_L\}$ ▷ (8); refuse if none exists
  $\kappa_{\mathrm{run}} = \min\{k : (1 + e^{\varepsilon_{\mathrm{run}}})\gamma(k) \le \zeta\delta_{\mathrm{run}}\}$; $\kappa_L = \min\{k : n_{\mathrm{rel}}\varepsilon'(k) \le (\tilde{\rho}_{\mathrm{run}}/\rho_L)\, E'_L\}$,
  with ρ_L, $E'_L = \xi_L\varepsilon_L/2$ and Γ_L of Section 7.4; **refuse if** $n_{\mathrm{rel}}(\delta_B + \delta') > (\tilde{\rho}_{\mathrm{run}}/\rho_L)\,\Gamma_L$
  $\gamma \leftarrow \gamma(\kappa)$; $\delta_{\mathrm{DP}} \leftarrow \delta_{\mathrm{run}} - (1 + e^{\varepsilon_{\mathrm{run}}})\gamma$; **refuse if** $\delta_{\mathrm{DP}} \le 0$
  $\rho_{\mathrm{run}} \leftarrow \max\{\rho :\ \inf_{a>1} \frac{e^{(a-1)(a\rho - \varepsilon_{\mathrm{run}})}}{a-1}\big(1 - \frac{1}{a}\big)^a \le \delta_{\mathrm{DP}}\}$ ▷ tight zCDP conversion
  $\rho_1 \leftarrow f_1\rho_{\mathrm{run}}$; $\rho_y \leftarrow f_y\rho_{\mathrm{run}}$ (0 in joint mode); $\rho_2 \leftarrow \rho_{\mathrm{run}} - \rho_1 - \rho_y$
  for Ω_f in increasing admissible scales **do** ▷ $\Omega_V = \Omega_f q_1 q_2/(\Omega_U\Omega_u)$, exactly representable
    for each release s: $\widehat{\Delta}_{\mathrm{enc},s} \leftarrow \sqrt{2/N}\,\Omega_f\Delta_s + \Delta_{x,s} + \sqrt{N}$; $\sigma^2_{c,s} \leftarrow \lceil \widehat{\Delta}^2_{\mathrm{enc},s}/(2\rho_s)\rceil$
    if $\sigma_{c,s} \ge \max\{2^\kappa B_{\mathrm{tot},s}(\Omega_f)/\sqrt{2\pi},\ 2^{42}\}$ for every s **then break** ▷ worst-case $B_{\mathrm{tot},s}$
  end for ▷ refuse if no scale floods
  $\sigma_s \leftarrow \sigma_{c,s}\sqrt{N/2}/\Omega_f$; **refuse unless** $q_0 > 2\big(\Omega_f S_{\max} + 2^\kappa B_{\mathrm{tot,max}} + t_\eta\sigma_{c,\max}\big)$
  with $S_{\max} = \bar{n}_O \max(r, o, w, \sqrt{ro}, \sqrt{rw}, \sqrt{ow})$ ▷ no wraparound
  return κ, Ω_V, Ω_f, $\{\sigma^2_{c,s}, \sigma_s, \rho_s\}_s$ and the run's ledger cost $(\rho_{\mathrm{run}},\ n_{\mathrm{rel}}\varepsilon'(\kappa),\ n_{\mathrm{rel}}(\delta_B + \delta'))$
end function

function SAMPLEDG(σ) exact discrete Gaussian on $\mathbb{Z}$ by rejection from a discrete Laplace, with big-integer
  arithmetic, a rational σ^2 and a CSPRNG; never by rounding a floating-point normal
end function

function SANITIZEBFV(ct) ▷ statistical flooding; BFV decryption stays exact
  $ct \leftarrow \mathsf{ModSwitch}_{q'}(ct) \boxplus \mathsf{ModSwitch}_{q'}\big(\mathsf{BFV.Enc}_{pk_B}(\mathbf{0})\big)$; $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^{N_B}, \sigma_\psi}$ ▷ switch both, then add
  return $ct \boxplus (\mathbf{e}, 0)$ ▷ $\sigma_\psi \ge \max\{2^{\kappa_\psi}B_\psi/\sqrt{2\pi}, 2^{42}\}$, $q'/(2t) > t_\eta\sigma_\psi + B_\psi + t$ (Section 8)
end function

function PREPROCESS(own columns, D_{ref}) ▷ each party separately; all constants public
  standardize each column with constants from D_{ref} or domain knowledge (O also standardizes y)
  optional tail transform per column: winsorize at $|z| \le c$, rank-Gaussian from the public quantiles, or log
  optional block normalization: scale the party's block of every row to unit norm (R without the intercept)
end function

function CHOOSEBOUNDS(D_{ref}) the 99.5% quantiles of $\|\dot{\mathbf{x}}\|$, $\|\mathbf{x}_O\|$ and $|y|$ in the preprocessed reference data
  (the evaluation setting; quantile levels that minimize the first-order MSE plus the squared clipping bias are future work)
end function

7 Output privacy

7.1 Sensitivity under replace-one adjacency

Stage 2 releases the stacked statistic

$$\mathbf{z}_2(D) = \big(\mathrm{triu}(\mathbf{B}),\ \mathbf{C},\ \mathbf{c}_R,\ \mathbf{c}_O,\ \mathbf{y}^\top\mathbf{y}\big), \tag{4}$$

without $\mathbf{y}^\top\mathbf{y}$ in separate mode, and stage 1 releases $\mathbf{z}_1(D) = (\mathrm{triu}\,\mathbf{B}, \mathbf{C})$. An unmatched row contributes nothing to these values. Replacing one matched row's $(\mathbf{x}, y)$ by $(\mathbf{x}', y')$, with R's row $\dot{\mathbf{x}}$ fixed, changes $\mathbf{z}_2$ by

$$\big(\mathrm{triu}(\mathbf{x}\mathbf{x}^\top - \mathbf{x}'\mathbf{x}'^\top),\ \dot{\mathbf{x}}(\mathbf{x}-\mathbf{x}')^\top,\ \dot{\mathbf{x}}(y-y'),\ \mathbf{x}y - \mathbf{x}'y',\ y^2 - y'^2\big).$$

Every entry of a release gets noise of the same s.d., so the relevant sensitivity is the ℓ_2 norm of this change, maximized over the bounds.

Proposition 7.1 (Replace-one sensitivity)**.** *With $r = B_R^2$, $o = B_O^2$, $w = B_y^2$ and $\Delta_{\mathrm{add}}^2 = (o+r)(o+w) + w^2$ the add/remove bound, $\Delta_{\mathrm{add}} \le \Delta \le 2\Delta_{\mathrm{add}}$, and*

$$\Delta^2 = \begin{cases} \tfrac{1}{2}(2o + r + w)^2 + 2rw & \text{if } |r - w| \le 2o, \\ 4r(o+w) & \text{if } r - w \ge 2o, \\ \max_{t\in[-B_y, B_y]} F\big(c^*(t), t\big) & \text{if } w - r > 2o, \end{cases} \tag{5}$$

where

$$F(c,t) = 2o^2(1-c^2) + 2ro(1-c) + r(B_y - t)^2 + o(w + t^2) - 2ocB_y t + (w - t^2)^2,$$
$$c^*(t) = \min\big\{1, \max\{-1, -(r + B_y t)/(2o)\}\big\}.$$

For stage 1, $\Delta_G = (2o + r)/\sqrt{2}$ if $r \le 2o$ and $2\sqrt{ro}$ otherwise. These are exact suprema for $d_O \ge 2$ and valid, conservative bounds for $d_O = 1$.

Proof sketch. The lower bound replaces $(\mathbf{x}, y)$ with $(\mathbf{0}, 0)$, the upper bound passes through that dataset with the triangle inequality. At the optimum $\|\dot{\mathbf{x}}\| = B_R$ and $\|\mathbf{x}\| = \|\mathbf{x}'\| = B_O$; the remaining dependence is on $c = \cos\angle(\mathbf{x}, \mathbf{x}')$ and $t = y'$, and maximizing the concave quadratic in c gives $c^*(t)$. In regime 1 the worst case is $y' = -y$ with $c = -(r-w)/(2o)$ and a diagonal change in $\mathbf{B}$; in regime 2 it is $(\mathbf{x}, y) \mapsto (-\mathbf{x}, -y)$. Regime 3 reduces to the roots of three cubics. The full proof is in the companion proofs document (Part B, Theorem S1). □

DeltaRep (Algorithm 7) computes regime 3 by exact enumeration: it evaluates $F(c^*(t), t)$ at $t = \pm B_y$, at the boundaries of the pieces of c^*, and at the real roots of the three cubics $\partial_t F(c^*(t), t) = 0$, one per piece, and takes the largest value. It rounds the result up by a relative 10^{-12} only as a safeguard against floating-point error. The specification's one-dimensional maximization is replaced. The implementation's enumeration agrees with a direct multistart maximization over the vectors ($d_O = 2$) to a relative 10^{-12} in regime 3, for example $\Delta = 26.07$ at $(B_R, B_O, B_y) = (1, 1, 5)$.

Remark 7.1 (When $\mathbf{y}^\top\mathbf{y}$ is free)**.** Releasing $\mathbf{y}^\top\mathbf{y}$ costs nothing exactly when $\Delta = \Delta_{\neg yy}$. This holds whenever $w \le 2(r+o)$ and never when $w > 3(r+o)$; in between it is decided numerically.

It is false in general: for $(B_R, B_O, B_y) = (1, 1, 5)$, $\Delta = 26.07$ with $\mathbf{y}^\top\mathbf{y}$ against 14.14 without. Algorithm 1 therefore releases $\mathbf{y}^\top\mathbf{y}$ in stage 2 only when it is free, and otherwise in its own release with sensitivity $\Delta_y = w$.

Remark 7.2 (Upper triangle or full matrix)**.** For $d_O \geq 2$ the supremum is the same whether the change in $\mathbf{B}$ is measured on its upper triangle or by the full Frobenius norm: the triu maximum is attained only when the change is diagonal. The protocol releases the upper triangle only, which keeps the release and the traffic smaller. Releasing both triangles with independent noise would cost the same worst-case ρ_2 and halve the variance of the averaged off-diagonal entries, at $d_O(d_O - 1)/2$ more released entries per release.

7.2 The release mechanism

For release s, let $\mathbf{z}_s(D) \in \mathbb{R}^{N_s}$ hold the released entries in their slots and zeros elsewhere.

Lemma 7.1 (Sensitivity of the encoded statistic)**.** $\|\tau_{\text{can}}^{-1}(\mathbf{z})\|_2 = \sqrt{2/N}\,\|\mathbf{z}\|_2$. *Hence* $\sup_{D\sim D'} \|\mathsf{Ecd}_\Omega(\mathbf{z}_s(D)) - \mathsf{Ecd}_\Omega(\mathbf{z}_s(D'))\|_2 \leq \sqrt{2/N}\,\Omega\,\Delta_s + \sqrt{N}$, *where* $\sqrt{N}$ *bounds the difference of the two coordinate-wise roundings.*

The centre. The decrypted value differs from $\mathsf{Ecd}_{\Omega_f}(\mathbf{z}_s(D))$ also by deterministic, data-dependent terms, chiefly the spill of the one-hot masks. These terms are part of the query, not hints about secrets, so the mechanism is centred at the noiseless output of the circuit,

$$\hat{\mathbf{z}}_s(D) = \sum_{e\in\mathcal{E}_s} \frac{1}{q_1 q_2}\,\mathrm{Tr}\Big(\sum_{h\leq m} m_{U_e,h}\, m_{V_e,h}\Big) m_{u_e}, \qquad f_s(D) = \lfloor \hat{\mathbf{z}}_s(D) \rceil \in \mathbb{Z}^N,$$

rounded to the coefficient lattice so that the shift $f_s(D) - f_s(D')$ between adjacent datasets is integral, as the discrete Gaussian needs (Section 5.4), where $m_{U_e,h}$, $m_{V_e,h}$ and m_{u_e} are the encodings of R's vectors at Ω_U, of O's vectors at Ω_V and of the mask at Ω_u, and Tr is the trace that EvalSum computes. $f_s(D)$ depends on the data and the public parameters only, not on any key, error or coin. Its sensitivity is, for any encoding scales and for neighbours that differ in a matched or an unmatched row (proofs, Part A, Lemma 4.3; Corollary 3.13 there is the equal-scale case)

$$\sup_{D\sim D'} \|f_s(D) - f_s(D')\|_2 \leq \widehat{\Delta}_{\text{enc},s} := \sqrt{2/N}\,\Omega_f \Delta_s + \Delta_{x,s} + \sqrt{N}, \tag{6}$$

where $\Delta_{x,s}$, an explicit function of N, the encoding scales, q_1, q_2, $|\mathcal{E}_s|$, the bounds and Δ_s, bounds the change of the rounding and spill terms. Ω_f is the exact final scale, tracked from the primes and the encoding scales and never a nominal power of two, and R decodes with it. At the evaluation settings $\Delta_{x,2} \approx 2^{27.6}$ against $\sqrt{2/N}\,\Omega_f\Delta_2 \approx 2^{62.7}$, so centring changes ρ_s by a relative 2^{-34}. Flooding the mask spill instead would need $\log_2 \Omega_f$ of about 67–71 rather than 61.5 at the proofs' example ($\bar{n}_O = 10^6$, $\varepsilon = 1$).

Error decomposition. The value R reconstructs in a release is

$$v_s = h_O + c_1' s_R = c_0' + c_1' s + \eta_s = f_s(D) + e_{\text{tot},s} + \eta_s \pmod{q_0}.$$

$e_{\text{tot},s} \in \mathbb{Z}^N$ collects every term that depends on a key, an error or a coin: R's encryption errors multiplied by O's data, the rescaling and ModDown roundings, the key-switching error (which

involves the errors of O's key shares), the mask spill of these errors, the rounding of the centre, and the fresh-encryption error $e\,u + e_0 + e_1 s$ of the re-randomization.

Lemma 7.2 (Smudging). *For $e \in \mathbb{Z}^N$ and $\eta \sim \mathcal{D}_{\mathbb{Z}^N,\sigma_c}$, $\mathrm{SD}(\eta + e, \eta) \le \|e\|_2/(2\sigma_c)$, and, for σ_c above the smoothing parameter, $\mathrm{SD}(\eta + e, \eta) \le \|e\|_2/(\sqrt{2\pi}\,\sigma_c) + \mathrm{negl}$. If $\sigma_c \ge \max\{2^\kappa B_{\mathrm{tot}}/\sqrt{2\pi},\ 2^{42}\}$ and $\|e\|_2 \le B_{\mathrm{tot}}$, the distance is at most $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$, also given any side information independent of η.*

The floor 2^{42} is what the factor $1 + 2^{-80}$ needs, which holds for $\sigma_c \ge 2.1 \cdot 2^{40}$; earlier drafts stated 2^{20}, at which the proof gives only $1 + 2^{-37.9}$ (proofs, composition appendix, Remark 2.6). The floor never binds at an admissible evaluation setting, where $2^\kappa B_{\mathrm{tot}}/\sqrt{2\pi} \ge 2^{55}$.

Theorem 7.1 (Worst-case error bound). *For the circuit of Algorithm 4 there is an explicit bound $B_{\mathrm{tot},s}$, a function of public parameters only (N, $\bar{n}_O$, d_R, d_O, B_R, B_O, B_y, the primes and the key-switching gadget, Ω_U, Ω_V, Ω_u, δ_B and the error distribution), such that $\Pr[\|e_{\mathrm{tot},s}\|_2 \le B_{\mathrm{tot},s}] \ge 1 - \delta_B$ for every dataset in the clipped domain, the probability being over keys and fresh randomness only (proofs, Part A, Theorem O1). The bound assumes, as model (M5) of the proofs, that every secret and error is drawn independently and honestly from its specified distribution, the errors from a mean-zero σ_e^2-subgaussian distribution with $\sigma_e = 3.2$.*

At the proofs' example ($N = 2^{14}$, $\bar{n}_O = 10^6$, the bounds of the simulation population, all scales and both rescaling primes $2^{52.5}$, centred key switching, $\delta_B = 2^{-50}$), $B_{\mathrm{tot},2} \approx 2^{25.3}$. The error measured in OpenFHE runs at that setting stays 6–8 bits below it.

With OpenFHE's key switching and the evaluation settings of Section 10, $B_{\mathrm{tot},2} \approx 2^{26.6}$. Two properties of OpenFHE 1.5.1, read from its source, change the proofs' centred constants:

- the fast base conversion in ModUp returns digits in $[0, k_j \tilde{Q}_j)$, so the digit factor becomes $\varsigma_\uparrow = 2k_j$;
- ModDown's rounding lies in $(-k_P, 0]$, so the rounding factor becomes $\varsigma_\downarrow = 2k_P$.

Here k_j is the number of primes in digit j and k_P the number of special primes. The dominant terms are the key-switching gadget term ($2^{25.3}$) and the ModDown roundings ($2^{25.7}$). These two properties, and OpenFHE's order of operations (it switches the key before it applies the automorphism), define an abstract model of its key switching, for which the bound is proved, with every other term unchanged (proofs, Part A, Lemma 4.1). That OpenFHE 1.5.1 realizes the model was checked by reading its source again at tag v1.5.1 (`ApproxSwitchCRTBasis`, `ApproxModDown`, the hybrid key switch and `EvalAutomorphism`); this is a statement about a program and is not proved.

Every term of B_{tot} is an analytic bound; none is an estimate or a measurement. The implementation draws its errors with OpenFHE's inversion-table sampler at $\sigma = 3.19$, whose support is $|x| \le 27$. From the sampler's table we checked numerically that it is subgaussian with variance proxy $3.19^2 \le 3.2^2$, as (M5) requires; this is a check, not a proof. Neither party uses OpenFHE's noise estimation, its `NOISE_FLOODING_DECRYPT` mode or its multiparty decryption. O's share is computed directly on the polynomials, and η is the only noise added.

As a sanity check, not part of the guarantee, we measured the whole error $e_{\mathrm{tot},s}$ of releases of the implementation against the exactly computed centre $f_s(D)$. Over 12 runs (24 releases) at $\bar{n}_O \in \{4096, 16384\}$, on the synthetic population and on data at the clipping bounds, its ℓ_2 norm was $2^{17.8}$–$2^{18.8}$, between 7.8 and 8.7 bits below $B_{\mathrm{tot},s}$. The re-randomization term

contributed almost nothing, and data at the clipping bounds did not increase the error. The deterministic part that the centring removes, $\|f_s(D) - \mathsf{Ecd}_{\Omega_f}(\mathbf{z}_s(D))\|_2$, was $2^{21.3}$–$2^{29.6}$; on the extreme data it exceeds $B_{\mathrm{tot},s}$, which is why the mechanism must be centred.

[**Check 2:** (T4) Model (M5) assumes mean-zero subgaussian errors. OpenFHE's floating-point inversion-table sampler was checked to be 3.2^2-subgaussian only numerically. The check is tight: its table gives the variance proxy $10.1761 = 3.19^2$, perturbing every probability by $+2^{-53}$, the granularity of its uniform generator, gives $10.2392 \le 10.24 = 3.2^2$, and a perturbation of 2^{-52} would fail it (10.32). Its mean is zero only to about 2^{-53}, which changes the bound by a relative term below 2^{-38} at the evaluation sizes. Add the mean term and the numerical check to the proofs, or draw the errors with an exact sampler.]

Calibrate must use this worst-case, data-independent bound. A bound estimated from the actual ciphertexts would depend on the data, leak through $\sigma_{c,s}$, and not be a worst-case bound. Recent attacks make the point concrete. Guo, Nabokov, Suvanto and Johansson [37] recover the secret key from a single shared decryption in OpenFHE when the flooding noise is set from non-worst-case noise estimates. Checri, Sirdey, Boudguiga and Bultel [15] show that threshold BFV, BGV and CKKS are exposed to key recovery unless smudging noise is added after partial decryption. In v3.1 the only partial decryption that leaves its owner is O's; it carries η, calibrated to the worst-case bound, and R's is never sent. Kluczniak and Santato [47] show that Gaussian noise exponential in the security parameter, added to the partial decryptions of threshold and multikey approximate HE, gives indistinguishability-based security, and that this noise level is tight. Our release uses noise of the same kind. It differs in two ways: the same noise is the zCDP noise of the released statistic, and we prove a simulation-based guarantee for R's whole view, with an explicit statistical slack at a concrete κ.

[**Check 3:** (T2) Kluczniak and Santato (CiC 2(1), 2025): the description (indistinguishability-based circuit privacy, a KL-divergence analysis, noise exponential in the security parameter, tightness, partial decryptions of multikey and threshold schemes) comes from search-engine extracts. Check the theorem statements and the exact noise condition, and whether their tightness result says anything about a statistical flooding parameter $\kappa = 40$ accounted for as an explicit slack.]

[**Check 4:** (T3) Guo et al. (USENIX Security 2024) and Checri et al. (CRYPTO 2024): the claims are from abstracts and search-engine extracts. Check which OpenFHE modes they attacked. In OpenFHE 1.5.1, multiparty decryption adds a Gaussian with $\sigma = 2^{20}$ by default, estimate-based flooding under `NOISE_FLOODING_DECRYPT`, and uniform noise under `NOISE_FLOODING_MULTIPARTY`. The given names of Checri et al. and the page numbers of Guo et al. (7447–7461, from an extract) are not verified and are not in the bibliography.]

Theorem 7.2 (Output privacy against an honest-but-curious R)**.** *Let the parameter vector satisfy the constraints of Section 10, and let* Calibrate*'s outputs, which depend on public values only, satisfy for every release s*

(i) $\sigma_{c,s} \ge \max\{2^{\kappa} B_{\mathrm{tot},s}/\sqrt{2\pi},\ 2^{42}\}$*, with* $B_{\mathrm{tot},s}$ *from Theorem 7.1;*

(ii) $\rho_s = \widehat{\Delta}^2_{\mathrm{enc},s}/(2\sigma^2_{c,s})$ *and* $\sum_s \rho_s \le \rho_{\mathrm{run}}$*, where* ρ_{run} *is the largest* ρ *whose tight conversion gives* $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$*, with*

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}\big), \qquad \delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - (1 + e^{\varepsilon_{\mathrm{run}}})\gamma > 0.$$

Assume RLWE for the re-randomizer (one sample per release, modulus q_0) and for O's public-key share $p_{0,O}$ (one sample; O's rotation-key shares are never sent), at ring dimension N and

modulus QP with $\log_2(QP) \le L_{\max}(N, \lambda_{\sec})$, and that R samples its errors honestly (Section 4). Then R's view of one run is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP for $\sim$ with statistical slack γ, and therefore $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$-IND-CDP. The ideal mechanism $\mathcal{M}(D) = (f_s(D) + \eta_s)_s$ is a composition of lattice discrete Gaussian mechanisms and is ρ_{run}-zCDP. PSI security is not needed for this guarantee: the identifiers are fixed under $\sim$, so the PSI transcript does not depend on $(\mathbf{x}_O, y)$.

Proof sketch. A sequence of hybrids. H0 is the real view: setup messages, PSI messages, and $\{(c'_{1,s}, h_{O,s})\}_s$. H1 rewrites $h_{O,s} = v_s - c'_{1,s} s_R$, which is identical. H2 replaces v_s by $f_s(D) + \eta_s$; by Lemma 7.2 with side information and Theorem 7.1 this costs at most $\delta_B + 2^{-\kappa}(1+2^{-80}) + 2^{-97}$ per release, γ in total. The side information includes R's random tape, whose errors and key share enter $e_{\mathrm{tot},s}$; Theorem 7.1 needs those errors to be honestly sampled; and γ is a statistical slack, not negligible in $\lambda_{\sec}$. After H2, $(u_s, e_{1,s})$ appear only through $c'_{1,s} = c_{1,s} + a u_s + e_{1,s}$, because $c'_{0,s}$ is never sent. H3 replaces $c'_{1,s}$ by a uniform element (RLWE with secret u_s; fresh u_s per release). H4 simulates the setup messages by plain RLWE for O's public-key share, the only key material O sends. The simulator, given $\mathcal{M}(D)$ and s_R, samples a uniform c_1^* per release and outputs $h^* = \mathcal{M}(D)_s - c_1^* s_R$. The shifts $f_s(D) - f_s(D')$ are integral, so the multivariate discrete Gaussian with sensitivity $\widehat{\Delta}_{\mathrm{enc},s}$ is exactly ρ_s-zCDP [13]; costs add, and the tight conversion gives $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-DP. Lemma 4.1 turns the slack into $\delta_{\mathrm{run}} = \delta_{\mathrm{DP}} + (1 + e^{\varepsilon_{\mathrm{run}}})\gamma$. □

Remark 7.3 (Why the release format matters)**.** If O sent c'_0, or a share $c'_1 s_O$ without η, R would learn $c'_1 s_O$ exactly and could recover s_O, since c'_1 is invertible with overwhelming probability. Re-randomization matters too. The evaluated c_1 is computed from public values and O's plaintext columns, and R chose the seed of a, so without it c_1 could reveal O's data. In a real OpenFHE run of an earlier specification that added the noise as a plaintext scalar, without re-randomizing, R identified the true dataset in 20 of 20 trials, for a matched and for an unmatched record, where an $\varepsilon = 1$ guarantee caps the success rate at $e/(1+e) \approx 73\%$. Reusing η or u across releases would reveal $v_2 - v_1$ exactly, so both are fresh every time. Stage 1 uses its own packed ciphertext M_1, which depends only on triu $\mathbf{B}$ and $\mathbf{C}$, so its error cannot depend on entries released only in stage 2.

Remark 7.4 (Who knows the noise)**.** Hwang et al. [41] show that two-party CKKS protocols in which each party adds its own smudging noise to a shared output are not secure in the standard simulation sense when both parties have inputs. The party that sampled a share of the noise knows it, so its view yields a closer approximation of the exact result than the ideal output reveals. Such protocols meet only a relaxed ("liberal") notion, in which the simulator also receives the exact result. Their remedy is to sample the noise jointly, so that no party knows it.

v3.1 avoids this obstruction for the analyst, at the price of an explicit statistical slack. R adds no noise: its decryption share is never sent, so it needs no flooding of its own, and η is the only noise in the output. R's coins still enter $e_{\mathrm{tot},s}$, through its encryption errors (multiplied by O's data), its key-share errors and s_R. All of this is bounded by $B_{\mathrm{tot},s} \le 2^{-\kappa}\sqrt{2\pi}\,\sigma_{c,s}$ and is absorbed by Lemma 7.2, which allows any side information independent of η, including R's random tape. The simulator of Theorem 7.2 therefore needs only $\mathcal{M}(D)$, never the exact $f_s(D)$. It is also output-consistent, so R's view jointly with its output is simulated in the standard sense, not only the liberal one, when the ideal output is the centred residue of $f_s(D) + \eta_s$ modulo q_0 (proofs, Part A, Corollary 9.3). The price is the slack γ, which is statistical rather

than negligible in λ_{sec}. The deterministic CKKS terms are part of the functionality $f_s(D) + \eta$, not of its noise. Their effect on privacy is accounted for by $\Delta_{x,s}$, so the ideal output is $f_s(D) + \eta$ rather than $\mathsf{Ecd}(\mathbf{z}_s(D)) + \eta$.

Two qualifications follow.

(i) The flooding argument needs R's errors to be sampled honestly. A researcher who inflated its encryption errors would push the term "R's errors times O's data" beyond B_{tot}, and that term is correlated with O's data. This lies outside the honest-but-curious model (Section 13).

(ii) The obstruction applies unchanged to O, which knows η. G1 and G2 are statements about O's view of the protocol, in which O has no output, and they do not extend to R's outputs (Section 8).

[**Check 5:** (T1) Hwang et al. (ePrint 2025/395; the later version's title, used in the bibliography, is "On the (In)security of Approximate Computation Protocols from CKKS", listed as accepted at ASIACRYPT 2026): the definitions and the two-party obstruction were checked against the authors' FHE.org 2026 poster only, and the abstract, the later title and the venue only through search-engine extracts. Confirm against the full text the exact standard and liberal definitions; that the obstruction needs the corrupted party's noise share to be non-negligible next to the rest of the output noise; and the statement for the case of one party with input. Fix the venue in the bibliography once confirmed.]

Release timing. The exact sampler is not constant time, and R observes when each release arrives. O therefore holds every release until a deadline D after it takes up the request, and sends it only then. D is fixed before the first release, from public parameters and the machine alone. In the implementation it is twice the worst of three timed exact samplings of N values at the run's largest σ_c^2, drawn from an independent generator and discarded, plus one re-randomized partial decryption. A release whose computation overruns D is refused, never sent late, and the run's cost stays charged. Outside an overrun, R's observed latency is D plus scheduling jitter, which does not depend on η, so the timing channel adds at most the overrun probability to the statistical distance of the simulation of Theorem 7.2. That probability is measured, not bounded. At the largest σ_c of the 10^6-row runs ($2^{68.5}$), $D = 0.49\,\text{s}$; on the idle machine 1,000 releases took at most $0.68\,D$, with no overrun (overrun probability at most 0.3% at 95% confidence), but with four CPU-bound processes competing for the four cores 4 of 200 overran. The flooding samples of SanitizeBFV in the PSI replies are not padded.

[**Check 6:** (T8) Release timing: the padding makes R's observed release latency independent of eta except when a release overruns its deadline. The overrun probability is only measured (0 of 1,000 releases on the idle machine, 4 of 200 under CPU contention), not bounded, and it is not charged to delta. Bound it, for example by a tail bound on the sampler's running time on an isolated core, or charge it to the slack; and pad the flooding of the PSI replies as well.]

7.3 Calibration

Calibrate (Algorithm 7) derives every quantity the release needs from the parameter vector, using public values only, in five steps.

(1) *Mode and sensitivities.* Δ, $\Delta_{\neg yy}$, Δ_G and Δ_y from the bounds (Proposition 7.1). $\mathbf{y}^\top\mathbf{y}$ joins stage 2 iff it is free (Remark 7.1), so the number of releases is $n_{\text{rel}} = 2$ (joint mode)

or 3 (separate mode).

(2) *Flooding parameter and slack.* κ from (8) below, unless the parameters fix it; $\gamma = n_{\mathrm{rel}}(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97})$ and $\delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - (1 + e^{\varepsilon_{\mathrm{run}}})\gamma$.

(3) *Budget.* ρ_{run}, the largest ρ whose tight conversion gives $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$, split as $\rho_1 = f_1\rho_{\mathrm{run}}$, $\rho_y = f_y\rho_{\mathrm{run}}$ in separate mode, and ρ_2 the rest.

(4) *Scale and noise.* The smallest admissible final scale Ω_f (that is, the smallest Ω_V, given Ω_U and Ω_u) at which $\sigma_{c,s} \geq \max\{2^{\kappa}B_{\mathrm{tot},s}(\Omega_f)/\sqrt{2\pi},\ 2^{42}\}$ for every release, with $\sigma^2_{c,s} = \lceil \widehat{\Delta}^2_{\mathrm{enc},s}/(2\rho_s) \rceil$ an integer, as the exact sampler needs. Rounding up only lowers the cost below ρ_s.

(5) *Modulus and security.* $q_0 > 2(\Omega_f S_{\max} + 2^{\kappa}B_{\mathrm{tot,max}} + t_\eta\,\sigma_{c,\max})$ with $S_{\max} = \bar{n}_O \max(r, o, w, \sqrt{ro}, \sqrt{rw}, \sqrt{ow})$, so that honest inputs never wrap around, and $\log_2(QP) \leq L_{\max}(N, \lambda_{\mathrm{sec}})$, the largest modulus at which RLWE with ring dimension N and ternary secrets has security λ_{sec} [1].

CALIBRATE refuses the run when a step has no solution: no admissible κ, $\delta_{\mathrm{DP}} \leq 0$, no scale that floods, q_0 too small, or a chain above the security bound. When it does not refuse, its outputs are functions of public values only and satisfy hypotheses (i) and (ii) of Theorem 7.2, the no-wraparound condition of Section 8 and, under a ledger cap, the per-run shares of Section 7.4 (proofs, Part A, Proposition 9.1).

The noise. The induced slot-domain noise is, up to a relative term $(\Delta_{x,s}+\sqrt{N})\sqrt{N/2}/(\Omega_f\Delta_s)$ that is below 10^{-10} at the evaluation settings,

$$\sigma_1 = \frac{\Delta_G}{\sqrt{2f_1\rho_{\mathrm{run}}}}, \qquad \sigma_2 = \frac{\Delta}{\sqrt{2(1-f_1)\rho_{\mathrm{run}}}}, \qquad \sqrt{\rho_{\mathrm{run}}} \gtrsim \sqrt{L + \varepsilon_{\mathrm{run}}} - \sqrt{L}, \quad L = \ln(1/\delta_{\mathrm{DP}}), \tag{7}$$

in joint mode ($\Delta_{\neg yy}$ and $1 - f_1 - f_y$ replace Δ and $1 - f_1$ in separate mode). The last inequality, from the simple conversion, gives a planning formula; over $\varepsilon \in [0.1, 8]$ and $\delta \in [10^{-9}, 10^{-6}]$ it is conservative by 6.1–33.4% in σ_2 (Section 11.2).

The flooding parameter. The same η is the DP noise and floods $e_{\mathrm{tot},s}$, and the flooding enters the guarantee only through the slack γ, which Lemma 4.1 converts with the factor $1 + e^{\varepsilon}$. The flooding parameter is therefore a function of the budget, not a constant. Unless the parameters fix it, CALIBRATE sets

$$\kappa = \max\{\kappa_{\min},\ \kappa_{\mathrm{run}},\ \kappa_L\}, \tag{8}$$

where κ_{run} is the smallest integer with $(1 + e^{\varepsilon_{\mathrm{run}}})\gamma \leq \zeta\,\delta_{\mathrm{run}}$, and κ_L is the smallest integer with $n_{\mathrm{rel}}\varepsilon'(\kappa) \leq (\tilde{\rho}_{\mathrm{run}}/\rho_L)\,E'_L$, this run's share of the ledger's smudging budget, with $\varepsilon'(\kappa)$ the smudging cost of one release (Section 7.4, Remark 7.5); $\tilde{\rho}_{\mathrm{run}}$ is the budget at $\delta_{\mathrm{DP}} = (1 - \zeta)\delta_{\mathrm{run}}$, a lower bound on ρ_{run} once $\kappa \geq \kappa_{\mathrm{run}}$. The three terms have separate jobs.

- κ_{run} makes the per-run statement hold with the slack taking at most a share ζ of δ_{run}. γ falls with κ to the floor $n_{\mathrm{rel}}(\delta_B + 2^{-97})$, so κ_{run} exists iff $n_{\mathrm{rel}}(1 + e^{\varepsilon_{\mathrm{run}}})(\delta_B + 2^{-97}) < \zeta\delta_{\mathrm{run}}$; otherwise δ_B must be lowered.
- κ_L makes the ledger run out of zCDP budget before it runs out of smudging budget, whatever the order and the sizes of the runs, provided every run on the register uses $\kappa \geq \kappa_L$. Since the smudging step is accounted in hockey-stick divergence, its $2^{-\kappa}$ part is not multiplied by e^{ε_L}, and κ_L is far below $\kappa_{\min}$ at the evaluation settings.

- $\kappa_{\min}$ is a statistical security floor. γ also bounds what R can learn beyond $\mathcal{M}(D)$, including about O's key share, so it should not grow with a generous δ_{run}.

ζ trades budget for bits. Shrinking δ_{DP} from δ_{run} to $(1-\zeta)\delta_{\text{run}}$ lowers ρ_{run} by a relative $\zeta/\sqrt{L(L+\varepsilon_{\text{run}})}$ to first order (simple conversion, $L = \ln(1/\delta_{\text{run}})$), so σ_2 rises by half that, while each unit of κ adds one bit to Ω_f and to q_0. At $\delta_{\text{run}} = 10^{-9}$ and $\varepsilon_{\text{run}} \in \{1, 4, 8\}$ the tight conversion loses 0.46–0.59% of ρ_{run} at $\zeta = 0.1$ and 0.04–0.06% at $\zeta = 0.01$, and $\zeta = 0.01$ raises κ by 3 bits at $\varepsilon_{\text{run}} = 4$ and 5 at $\varepsilon_{\text{run}} = 8$ (by at most 1 where the floor $\kappa_{\min} = 40$ binds). Table 4 gives κ and the resulting scale at the previous setting ($f_1 = 0.1$); at $f_1 = 0.25$ (setting I) the scale is about 0.13 bits lower and κ is the same.

Table 4: Example at the previous setting ($f_1 = 0.1$): the flooding parameter from (8) and the final scale CALIBRATE chooses, simulation population ($p = 11$, joint mode, $n_{\text{rel}} = 2$), $\delta_{\text{run}} = 10^{-9}$, $\zeta = 0.1$, $\kappa_{\min} = 40$, $\Omega_U = 2^{68}$, $N = 2^{14}$, $q_0 \approx 2^{119}$, $\log_2(QP) = 357 \le 438$. Left: a single-run ledger. Right: ledger cap $(\varepsilon_L, \delta_L) = (10, 10^{-6})$ with $\zeta_L = 0.1$, $\xi_L = 10^{-3}$ and $\delta' = 2^{-100}$; κ_L is below $\kappa_{\min}$, so κ and the scale equal the single-run values (the statistical-distance accounting would give $\kappa_L = 48, 46, 44, 42, 40$). All values were computed by the implementation's CALIBRATE. Every row is feasible. At a fixed $\kappa = 40$, $\varepsilon_{\text{run}} = 8$ is not: the slack exceeds δ_{run}.

	single-run ledger						cap $(10, 10^{-6})$		
ε_{run}	κ_{run}	κ	slack (% of δ)	$\log_2 \Omega_f$ $\bar{n}_O = 4{,}096$	10^6	$\log_2 q_0$ needed, 10^6	κ_L	κ	$\log_2 \Omega_f$ 10^6
0.5	36	40	0.48	61.78	61.78	88.0	22	40	61.78
1	37	40	0.68	62.75	62.75	89.0	21	40	62.75
2	38	40	1.53	63.69	63.69	89.9	19	40	63.69
4	41	41	5.07	65.62	65.66	91.9	17	41	65.66
8	46	46	9.00	71.56	71.94	98.2	15	46	71.94

The scale. Since $\sigma_{c,s} = \sigma_s \Omega_f \sqrt{2/N}$ and σ_s is fixed by the budget, O meets the flooding condition by raising Ω_f; stage 2 binds:

$$\log_2 \Omega_f \approx \kappa + \log_2 B_{\text{tot},2} + \log_2 \sqrt{N/2} - \log_2 \sqrt{2\pi} - \log_2 \sigma_2.$$

Example: for the simulation population at $\varepsilon = 1$, $\delta = 10^{-9}$, $N = 2^{14}$ and $\kappa = 40$, the proofs' bound gives about $40 + 25.3 + 6.5 - 1.3 - 9.0 \approx 61.5$, and the bound for OpenFHE's key switching 62.75 (Table 4).

Encoding scales. We use the circuit as written (Algorithm 4) with separate encoding scales: Ω_U and Ω_u are parameters, CALIBRATE chooses Ω_V, and $\Omega_f = \Omega_U \Omega_V \Omega_u / (q_1 q_2)$ follows and is tracked exactly. The final scale is therefore not tied to the size of a rescaling prime, which a 64-bit implementation keeps below 2^{60}. The constraint on Ω_U comes from B_{tot}: the term from R's encryption errors grows like $(\Omega_f/\Omega_U)\sqrt{\bar{n}_O}$. With $\Omega_U \ge \Omega_f$ it stays below the other terms at every register size; with $\Omega_U < \Omega_f$ the required scale grows with $\bar{n}_O$, and for large enough κ no scale floods, because B_{tot} then grows as fast as Ω_f, and CALIBRATE refuses. In the evaluation $\Omega_U = 2^{68}$ and $\Omega_u = q_1$. The required scale is then independent of $\bar{n}_O \in [4096, 10^6]$ to within 0.05 bits for $\varepsilon \le 4$, and grows by 0.4 bits at $\varepsilon = 8$, where $\Omega_f > \Omega_U$ (Table 4). With

$\Omega_U = q_2 \approx 2^{59}$ instead, the required scale reaches $2^{70.3}$ at $\bar{n}_O = 10^6$, $\varepsilon = 4$ and $\kappa = 40$. The proof review's restructured circuit, which needs about $2^{53.7}$, is not used: its proof is incomplete, and with decoupled scales the larger scale costs nothing.

7.4 The ledger and composition across runs

O keeps one ledger per base register, shared across all researchers and scopes. It admits a run only if the run's full cost fits under the cap, and it produces decryption shares only within admitted runs. Because no decryption is possible without O's share, the cap is enforced technically, not by policy. A refused stage-2 gate (Section 9.3) still spends the budget. The OPRF quota limits membership look-ups across sessions as a matter of policy.

Parameters and state. The ledger's parameters are four:

- a cap $(\varepsilon_L, \delta_L)$;
- the share $\zeta_L \in (0,1)$ of δ_L reserved for statistical slack;
- the share $\xi_L \in (0,1)$ of ε_L reserved for the smudging of the encryption error;
- a per-release constant $\delta' \in (0,1)$.

From them,

$$E'_L = \xi_L \varepsilon_L / 2, \qquad \rho_L = \max\{\rho : \delta(\rho, (1-\xi_L)\varepsilon_L) \le e^{-E'_L}(1-\zeta_L)\delta_L\}, \qquad \Gamma_L = \frac{\zeta_L \delta_L}{1 + e^{\varepsilon_L - E'_L}},$$

where $\delta(\rho, \varepsilon)$ is the tight conversion. A run with n_{rel} releases at flooding parameter κ costs three amounts:

- its zCDP cost ρ_{run};
- a smudging cost $\varepsilon'_{\mathrm{run}} = n_{\mathrm{rel}} \varepsilon'(\kappa)$, where $\varepsilon'(\kappa)$ is the smallest ε with $\delta(\pi 4^{-\kappa}, \varepsilon) \le \delta'$, at most $2\sqrt{\pi \ln(1/\delta')}\, 2^{-\kappa} + \pi 4^{-\kappa}$;
- a slack $\gamma_L = n_{\mathrm{rel}}(\delta_B + \delta')$.

The three costs come from the hockey-stick form of the smudging step (Remark 7.5). The state is their three sums $(\rho_\Sigma, E_\Sigma, \Gamma_\Sigma)$ over the runs admitted on the register. A run is charged its full cost, for all its planned releases, at the moment it is admitted, and it is not refunded if it is aborted or a release is never requested. Refunding the zCDP cost and the $2^{-\kappa}$-dependent smudging cost of releases not made would be sound, but the δ_B part must stay charged, because R's decision to continue may depend on its own key share and encryption errors, which enter e_{tot} (proofs, composition appendix, Remark 4.3); the composition statement below is proved for the rule without refunds. Requests are processed one at a time, and a run's cost is reserved atomically when it is admitted, so concurrent runs cannot jointly exceed the cap. The capacities are rounded down and the costs up. The evaluation uses $\xi_L = 10^{-3}$ and $\delta' = 2^{-100}$.

Admission. Write $\rho^+ = \rho_\Sigma + \rho_{\mathrm{run}}$, $E^+ = E_\Sigma + \varepsilon'_{\mathrm{run}}$ and $\Gamma^+ = \Gamma_\Sigma + \gamma_L$.

- *Runs fixed in advance*, with the parameters and inputs of every run fixed before the first run and independent of any output: admit iff $e^{E^+}\delta(\rho^+, \varepsilon_L - 2E^+) + (1 + e^{\varepsilon_L - E^+})\Gamma^+ \le \delta_L$.
- *Runs chosen after seeing earlier results*: admit iff $\rho^+ \le \rho_L$, $E^+ \le E'_L$ and $\Gamma^+ \le \Gamma_L$. This is a privacy filter in the sense of Rogers et al. [69], and it is the rule for a shared register, where later runs are rarely fixed in advance. With κ_L in (8) for every run, the

smudging condition holds whenever the zCDP condition does. The slack condition does too, provided $n_{\mathrm{rel}}(\delta_B + \delta') \le (\tilde{\rho}_{\mathrm{run}}/\rho_L)\Gamma_L$, a constraint on δ_B that does not involve κ.

Either rule makes the joint view $(\varepsilon_L, \delta_L)$-IND-CDP (Proposition 7.2; proofs, composition appendix, Corollary 5.4). A register on which a single run is allowed is covered by Theorem 7.2 alone, with cap $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$, which that run fills exactly; κ_L plays no role there. The earlier accounting, which measures the smudging step in statistical distance, has two sums, $\Gamma_L = \zeta_L \delta_L/(1 + e^{\varepsilon_L})$ and the raw slack $\gamma = n_{\mathrm{rel}}(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97})$ per run; it is also proved (Theorem 4.2 there) and remains an option of the implementation, but it multiplies the $2^{-\kappa}$ part by e^{ε_L}.

Definition 7.1 (Base-register adjacency)**.** A base register is $\mathfrak{D} = \{(\mathrm{id}_j, \mathbf{a}_j, \mathbf{x}_j, y_j)\}_j$, with identifiers, public attributes, all of O's feature columns and the response. $\mathfrak{D} \sim_{\mathrm{b}} \mathfrak{D}'$ if they have the same identifiers and public attributes and differ in $(\mathbf{x}_j, y_j)$ of exactly one row. Scopes use public attributes only, and preprocessing acts row by row with public constants. So for every run, the padded scope registers of $\mathfrak{D}$ and $\mathfrak{D}'$ either coincide or are adjacent in the sense of Definition 4.1, whatever the scope, the model and the cohort (proofs, composition appendix, Lemma 1.3).

Proposition 7.2 (Composition across runs)**.** *Let all researchers using one base register act as one coalition, possibly colluding. The coalition chooses each run's parameters (scope, model, budget, flooding parameter, ring parameters from a fixed finite menu) and each cohort as a function of the joint view of earlier runs. It follows the protocol within each run, and it may interleave and abort runs. Assume that:*

- *every admitted run draws fresh CKKS key material and fresh release coins, and satisfies the hypotheses of Theorem 7.2;*
- *the register is static;*
- *RLWE holds, as in Theorem 7.2, for every ring parameter set of the menu, against non-uniform adversaries;*
- *O's ledger is honest, admits runs one at a time, and charges a run's full cost at admission.*

Then the joint view of all researchers is $(\varepsilon_L, \delta_L)$-IND-CDP for $\sim_{\mathrm{b}}$ in two cases: under the second (filter) rule, for every PPT coalition; and under the first rule, when the runs are fixed in advance.

More precisely, the joint view is simulated, with statistical slack $\mathbb{E}[\sum_{k \text{ admitted}} \gamma_k] + \mathrm{negl}(\lambda_{\mathrm{sec}})$, by an ideal experiment. This ideal experiment is a post-processing of an adaptive ρ^*-zCDP mechanism whenever $\sum_{k \text{ admitted}} \rho_k \le \rho^*$ in every execution (proofs, composition appendix, Theorem 4.2). With the smudging step measured in hockey-stick divergence, the same holds with an $(\varepsilon'(\kappa), \delta')$ step per release in place of the slack $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ (Theorem 5.2 there), which gives the ledger of this section (Corollary 5.4 there).

Proof sketch. The full proof is in the composition appendix of the proofs (Theorem 4.2). It has four steps.

Statistical hybrid. Write O's share as $v - c_1' s_R$ with $v = f_s(D) + e_{\mathrm{tot},s} + \eta_s$. Remove $e_{\mathrm{tot},s}$ from every release of every run in one hybrid, before any computational step. A hybrid lemma for adaptive processes bounds the statistical distance by a single expectation under the real

experiment: of $\sum_{\text{made}} 2^{-\kappa}(1+2^{-80})+2^{-97}$ plus δ_B for every planned release of every admitted run. That expectation is at most $\mathbb{E}[\sum_{k \text{ admitted}} \gamma_k]$, which each admission rule bounds pointwise.

Computational hybrids, run by run. In admission order, replace each run's re-randomized c_1' and then O's key shares by uniform elements. The reduction for a step has D as advice. It plays earlier runs ideally and later runs honestly, and it computes the adaptively chosen parameters by running the coalition, which is PPT, together with CALIBRATE and the ledger, which are deterministic and efficient. It announces the step's ring parameters before receiving its RLWE challenge, at a loss of the menu's size. The number of steps is at most $(n_{\text{rel}}+1)$ times the coalition's running time; under the filter it is also at most $(n_{\text{rel}}+1)\Gamma_L/\min_k \gamma_k$.

Ideal composition. The ideal experiment releases $f_s(D)+\eta_s$ through lattice Gaussian steps whose dimension, centre and σ depend on the history, interleaved with public steps of cost 0: refusals, coins, the PSI, and uniform key shares and c_1'. A Rényi filter, proved there by the supermartingale argument behind the filters of Rogers et al. [69] and Feldman and Zrnic [27], makes it $(a, a\rho^*)$-Rényi DP for every $a>1$ when $\sum \rho_k \le \rho^*$ pointwise. Refusals and the stage-2 gate are coordinates of the view, not events conditioned on.

Conversion. Chaining as in Lemma 4.1 gives $(\varepsilon, \delta(\rho^*,\varepsilon)+(1+e^{\varepsilon})\Gamma^*)$ for every ε, where Γ^* bounds $\sum_{\text{admitted}} \gamma_k$ pointwise. Under the statistical-distance filter $(\rho^*,\Gamma^*)=(\rho_L,\Gamma_L)$ with $\Gamma_L=\zeta_L\delta_L/(1+e^{\varepsilon_L})$, and the δ budget is used exactly. For runs fixed in advance, the admitted set is deterministic, and the first rule at the last admission gives the same bound. With the hockey-stick ledger, the statistical step is the $(\varepsilon'(\kappa),\delta')$ step of Remark 7.5, applied before every computational step so that the computational losses only add, and the conversion is Corollary 5.4 there: $\varepsilon=(1-\xi_L)\varepsilon_L$, $E^*=E_L'$, and $e^{E_L'}\delta(\rho_L,\varepsilon)+(1+e^{\varepsilon_L-E_L'})\Gamma_L \le \delta_L$. □

[**Check 7:** Composition: the proof, including the adaptive hybrid and the hockey-stick ledger, is written out in the composition appendix of the proofs (Theorems 4.2 and 5.2, Corollary 5.4), with a self-contained Rényi filter, so the cited filter results are attributions only. Not proved there: that the first rule (runs fixed in advance) is valid when the requests are chosen adaptively, where the proof gives only $\delta(\sup\rho_\Sigma,\varepsilon_L)+(1+e^{\varepsilon_L})\sup\Gamma_\Sigma$, which can approach $2\delta_L$; and the cut-off argument for the expected-polynomial-time exact sampler inside the reductions, which is stated but not written out. The per-run results it takes from Part A of the proofs were not re-verified. The bibliographic details of the filter literature were checked only through search-engine extracts: Feldman and Zrnic's theorem number in the NeurIPS version, the pages of Rogers et al. (1921–1929), and the volume and pages of Whitehouse et al. (PMLR 202, 36990–37007).]

Two consequences. First, a statement over many runs multiplies the slack by $1+e^{\varepsilon_L-E_L'}$, not $1+e^{\varepsilon_{\text{run}}}$. The smudging step's $2^{-\kappa}$ part, however, enters only through the small budget E_L'. At $(\varepsilon_L,\delta_L,\zeta_L,\xi_L)=(10,10^{-6},0.1,10^{-3})$, for example, κ_L is 22, 21, 19, 17 and 15 for $\varepsilon_{\text{run}}=0.5,1,2,4,8$. That is below $\kappa_{\min}=40$, so the ledger does not raise κ (Table 4). The statistical-distance accounting would give 48, 46, 44, 42 and 40, which cost 8.1, 6.0, 4.0, 1.0 and 0 bits of Ω_f at $\bar{n}_O=10^6$. The reserved share costs 0.2% of ρ_L. δ_B is still multiplied by $e^{\varepsilon_L-E_L'}$: at $\varepsilon_L=20$ the floor $n_{\text{rel}}\delta_B$ alone exceeds every run's share unless δ_B is lowered well below 2^{-50}. Second, the filter loses nothing against non-adaptive composition, because zCDP composes by summing a single parameter. For (ε,δ)-DP, filters that match advanced composition, and a filter for approximate zCDP, are given by Whitehouse et al. [77].

Remark 7.5 (The smudging step in hockey-stick divergence)**.** Let $e\in\mathbb{Z}^N$ with $\|e\|_2 \le B_{\text{tot}}$ and $\sigma_c \ge 2^{\kappa}B_{\text{tot}}/\sqrt{2\pi}$. Then $\eta+e$ and η are $(\varepsilon'(\kappa),\delta')$-indistinguishable, with $\varepsilon'(\kappa) \le$

$2\sqrt{\pi \ln(1/\delta')}\, 2^{-\kappa} + \pi 4^{-\kappa}$. For $\delta' = 2^{-100}$ this is at most $29.52 \cdot 2^{-\kappa} + \pi 4^{-\kappa}$, or $2.7 \cdot 10^{-11}$ at $\kappa = 40$, where the tight value is $2.0 \cdot 10^{-11}$. The bound follows from the Rényi bound for integer shifts and the conversion, and it needs no smoothing condition and no floor on σ_c (proofs, composition appendix, Lemma 5.1). The ledger uses this form across runs (Theorem 5.2 there). Theorem 7.2 keeps the statistical-distance form for a single run. With the hockey-stick form there too, κ_{run} would be 18, 17, 16, 15 and 14 at a reserved share $\xi = 10^{-3}$ of ε_{run}, and $\kappa = \kappa_{\min}$ in every row of Table 4.

8 Input privacy, cohort privacy and correctness

G1. O sees R's seeded uploads $c_0 = -as_R + e + \mathsf{Ecd}(\mathbf{v})$, which are pseudorandom under RLWE in s_R (with circular security covering R's rotation-key shares, and the PRG modelled as a random oracle); the PSI messages, which are BFV ciphertexts under R's own key and blinded OPRF elements; and R's key-generation shares, which are RLWE samples. O never receives a decryption share from R, so its view can be simulated from $(n_R^{\max}, \bar{n}_O)$, up to a slack $p_{\text{cuckoo}} + p_{\text{false}}$: the failure probability of cuckoo hashing (at most $2^{-49.1}$ with $n_R^{\max} = 2{,}000$ items, $h = 3$ hash functions and 2^{14} bins, and at most 2^{-40} up to a load of 0.64, for an insertion that finds an assignment whenever one exists, as the implementation's random walk with an augmenting-path fallback does; proofs, Part A, Lemma 7.5; above 10,472 items R splits its queries into the fewest balanced tables whose summed bound is at most δ_{psi}: two tables and $2^{-43.4}$ at 20,000, ten and $2^{-41.1}$ at 100,000), and the probability of a false match, most of which R detects and answers with an abort (Appendix C). O observes the abort, and its probability can depend on the cohort, so it belongs in the slack.

G1 and G2 concern O's view of the protocol. O receives no output, and its view includes η, so the guarantees do not extend to anything R later discloses (Remark 7.4).

- *Disclosed results are noiseless to O.* A statistic or model that R computes from the releases and publishes is, to O, a function of $f_s(D) + e_{\text{tot},s}$. The DP noise protects O's records from R and from third parties; it never protects R's data from O. This matters beyond what Remark 4.1 already discloses: $\mathbf{A}$ is exact in a published model, and $\mathbf{C}$ and $\mathbf{c}_R$ become exact to O.
- *v_s must stay with R.* O knows $h_{O,s}$ and $c'_{1,s}$, so v_s would give it $c'_{1,s}s_R$. Since $c'_{1,s}$ is invertible with overwhelming probability, that gives s_R, and with it R's uploads: G1 and G2 would both fail. Disclosing v_s is equivalent to sending R's decryption share, which would need flooding by R (the client-side noise of the protocols analysed by Hwang et al. [41]). v_s is therefore never transmitted (Table 3).
- *Decoded values at double precision are not a key leak.* Rounding $\mathsf{Dcd}(v_s)$ to double precision leaves an error of about 2^{19} per slot, or about 2^{12} per coefficient of v_s, because $|\tau(\eta)_k| \approx \sqrt{N}\sigma_c$. Even all slots at that precision would therefore give O only a rounding-type RLWE sample of s_R with that error at modulus $q_0 \approx 2^{119}$, far inside the secure region for $N = 2^{14}$. Only v_s itself, or a decoding of equal precision, is dangerous. R still rounds every published value (Remark 8.1); rounding is free.

A guarantee for O in the standard sense, jointly with R's output, would need noise that O does not know either. Hwang et al. do this by sampling the noise jointly; here R must not know it either, because of DP. That is a protocol change we do not make.

Remark 8.1 (Publishing guidance)**.** Before R publishes any decoded value of release s, it rounds $\mathsf{Dcd}(v_s)$ to a fixed precision relative to σ_s, for example to a multiple of $10^{\lfloor \log_{10} \sigma_s \rfloor - 2}$, two decimal digits below the leading digit of σ_s (Algorithm 5). Rounding is post-processing and costs no budget. R never publishes v_s. The implementation's `run_R` returns the full decoded slot vectors for local use only; `round_published` in `implementation/vfr/party_r.py` does the rounding.

G2. O evaluates exactly $n_R^{\max}$ OPRF calls on blinded inputs and answers padded PSI queries, so it learns nothing about the cohort or n_R, up to the same slack $p_{\text{cuckoo}} + p_{\text{false}}$ for R's observable aborts.

G4. R learns I and $\pi|_I$ from linkage. That it learns nothing else about O's register needs the labeled PSI to be secure against a semi-honest receiver. Chen, Laine and Rindal prove their PSI from homomorphic encryption secure in the semi-honest model assuming the scheme is IND-CPA secure and circuit private, and note that statistical function privacy, obtained in the leveled setting by re-randomization and noise flooding, suffices [16]. Chen, Huang, Laine and Rindal add OPRF preprocessing, which strengthens the security model [17]. Two layers of the protocol address what the noise of a reply could reveal.

(a) *OPRF-keyed items and labels.* Every item O inserts is $F_k(\mathrm{id}_j)$ under a fresh key, and every label is $\tau(j)$ masked by a key derived from $F'_k(\mathrm{id}_j)$. R learns F_k only at its $n_R^{\max}$ queries, so the item and the label of every row outside I are pseudorandom to it; τ is a uniform permutation, and the masks $\mathbf{r}_1$, $\mathbf{r}_2$ are fresh. A reply, noise included, is a deterministic function of R's ciphertexts and of these values. A simulator that knows $(I, \pi|_I)$, the public sizes and R's view of the OPRF can therefore run O's evaluation on fresh random items and labels outside I and obtain replies indistinguishable from the real ones. In the random-oracle model for H_1, H_2 and the derived hashes, this holds under the computational Diffie–Hellman assumption in the OPRF group, with distinguishing advantage at most $q_{H_2}\mathrm{Adv}_{\mathrm{CDH}}$ plus a negligible collision term, for every input (proofs, Part A, Theorem 7.3). Even without circuit privacy, then, what the noise reveals about O's register beyond $(I, \pi|_I)$ is pseudorandom. When a bin has more than one partition ($\alpha > 1$; $\alpha = 5$ at $\bar{n}_O = 10^6$ with the PSI parameters of Section 10), R also learns which partition flags its match. That index depends on the loads and the insertion order of O's items in the bin, and the simulator reproduces it by running O's preprocessing, partition placement included, on the random items.

(b) SanitizeBFV (Algorithm 7) is part of the protocol nevertheless, so that G4 does not rest on the pseudorandomness of the evaluation's inputs alone. It switches each reply and a fresh encryption of zero under R's BFV key to the reply modulus q', adds them, and floods the noise with an exact discrete Gaussian $\mathcal{D}_{\mathbb{Z}^{N_B},\sigma_\psi}$. After the switch, the error of a reply with plaintext m is $\varphi(m) + w$, where $\varphi(m)$ is an integral function of m with $\|\varphi(m)\|_\infty \le t/2 + 1$, from the change of scale $\lfloor q_B/t \rfloor \to \lfloor q'/t \rfloor$, and w collects the rounding terms of the switches and the scaled evaluation and fresh-encryption noise. Let B_ψ bound $\|w\|_2$ with probability at least $1 - \delta_{B,\psi}$, for every input of O. The requirements are

$$\sigma_\psi \ge \max\{2^{\kappa_\psi} B_\psi/\sqrt{2\pi},\ 2^{42}\} \text{ (flooding)}, \qquad q'/(2t) > t_\eta\, \sigma_\psi + B_\psi + t \text{ (exact decryption)},$$

where the term t covers $\varphi(m)$. Then, as in hybrids H2 and H3 of Theorem 7.2, each reply

ciphertext is within statistical distance $\delta_{B,\psi} + 2^{-\kappa_\psi}(1+2^{-80}) + 2^{-97}$, plus a computational term (RLWE in the randomness of the fresh encryption), of a ciphertext computed from its plaintext and fresh randomness alone. Over the n_ψ reply ciphertexts of a run the distances add. The plaintexts themselves are 0 or uniform, and a label or uniform, because of the masks.

The argument holds for either order of the switch and the addition. Flooding needs only a worst-case bound on w: adding the fresh encryption before the switch leaves one rounding term, and switching both and then adding leaves two, which B_ψ must then cover. Re-randomization needs the switched c_1' to be uniform: under RLWE the fresh part $a_B u' + e_1'$ is uniform modulo q_B, and rounding from $\mathbb{Z}_{q_B}$ to $\mathbb{Z}_{q'}$ maps the uniform distribution to the uniform distribution because $q' \mid q_B$, whether it is switched alone or together with the reply. Algorithm 7 uses the implementation's order, switching both and then adding.

Neither layer is charged to δ: the PSI transcript does not depend on $(\mathbf{x}_O, y)$, so it lies outside the DP guarantee, and κ_ψ is a security parameter of G4 alone. A modulus switch alone, as in the specification, is heuristic. Flooding protects against a semi-honest receiver, whose ciphertexts are fresh; a malicious receiver could send noisier ones, which is outside our model.

At the evaluation settings the implementation uses $\kappa_\psi = 40$ and $\sigma_\psi = 2^{\kappa_\psi} B_{\mathrm{err}} \approx 2^{61}$, with $B_{\mathrm{err}} = 2(\sqrt{N_B}/2 + N_B^{3/2}/2) + 1 \approx 2^{21}$: the worst-case rounding terms of the two switches for a ternary key, plus 1. This σ_ψ meets the flooding requirement for every $B_\psi \le \sqrt{2\pi}\, B_{\mathrm{err}} \approx 2^{22.3}$, which covers the rounding terms in the worst case, the rounding of $\varphi(m)$ included. With q' of two 60-bit limbs, $q'/(2t) \approx 2^{87}$ against $t_\eta \sigma_\psi + B_\psi + t \approx 2^{64.3}$ at $t_\eta = 10$. In the textbook BFV model a worst-case bound covers the switched evaluation noise as well: for every input of O its ℓ_2 norm is at most 2^{-31} after the switch (product trees of depth 3, 65 coefficients per polynomial, q_B of seven limbs and q' of two; proofs, Part A, Proposition 7.7 and Corollary 7.8), against 2^{-72} measured. Whether OpenFHE's BFVrns, whose multiplication and scaling are RNS approximations, satisfies that model was not checked, so for the implementation the $+1$ still rests on the measurement.

[**Check 8:** G4, second layer: check that OpenFHE 1.5.1's BFVrns (its RNS multiplication, scaling and relinearization) satisfies the BFV model of the proofs' bound on the evaluation noise (Part A, Proposition 7.7), or bound that noise directly for it. The circuit-privacy statements of CLR17 and the role of OPRF preprocessing in CHLR18 were checked only through search-engine extracts; confirm against the full texts, including whether CHLR18 state that OPRF preprocessing removes the need for circuit privacy.]

Correctness. With q_0 from Calibrate, honest inputs never wrap around; the failure probability is at most $\delta_B + 2Ne^{-t_\eta^2/2}$ per release, a correctness event that is not charged to δ. No condition is needed at the earlier levels of the circuit: the error decomposition is an identity of congruences whatever the size of the intermediate plaintexts, so only the final reduction modulo q_0 can wrap around (proofs, Part A, Proposition 8.1). The deterministic offset of the centre from $\Omega_f \tau_{\mathrm{can}}^{-1}(\mathbf{z}_s(D))$ has a proved public bound, $2^{30.3}$ in the coefficient domain at $\bar{n}_O = 10^6$ with $n_R^{\max} = 2{,}000$ (Proposition 4.5 there), far inside the slack of Calibrate's condition. With the noise switched off, decryption reproduces the plaintext statistics up to the CKKS approximation error (Section 11.1).

9 The statistical procedure

9.1 Estimands

The protocol targets two quantities.

- The **ridge target** $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$, a fixed function of the matched data once (λ, Ψ) is chosen. Only the mechanism noise enters its uncertainty.
- The **regression coefficient** $\boldsymbol{\beta}$: the coefficient of the working linear model for the clipped data, $\mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}] = \tilde{\mathbf{X}}\boldsymbol{\beta}$ with conditionally independent, homoskedastic errors, where $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{y}}$ are the clipped design and response. Its sample analogue is $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$. Its uncertainty includes sampling variance, and $\boldsymbol{\theta}_\lambda$ is biased for it when $\lambda > 0$.

The working model holds for the clipped data when the bounds are true bounds, so that clipping never binds.

Remark 9.1 (The projection coefficient)**.** Without the working model, the natural target is the projection coefficient $\boldsymbol{\beta}^* = \mathbf{G}^{-1}\tilde{\mathbf{X}}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}]$ of the clipped data, which equals $\boldsymbol{\beta}$ when the model holds. The bound on $\|\boldsymbol{\beta}\|$ that the bias-aware interval uses holds for $\boldsymbol{\beta}^*$ with no model assumed (Table 33), and the Wald interval for $\boldsymbol{\theta}_\lambda$ does not involve $\boldsymbol{\beta}$ at all. The homoskedastic sampling variance of the bias-aware interval, however, does not hold for $\boldsymbol{\beta}^*$ in general.

When clipping binds, the clipped errors are heteroskedastic even if the errors before clipping are homoskedastic. A data-driven residual variance then estimates an average of the conditional variances, not their maximum, and a sandwich variance would need per-record residuals of O's data, that is, a further release. The protocol therefore takes the sampling variance of the bias-aware interval from a public bound $\bar{\sigma}^2 \geq \max_i \mathrm{Var}(\tilde{y}_i \mid \mathbf{X})$, a parameter fixed in Setup before the data are used (Table 10). With it the interval is valid for the projection coefficient, to the same order as under the working model, whether or not clipping binds (proofs, Part B, Proposition S10b). Two choices are admissible: $\bar{\sigma}^2 = B_y^2$, the default, which holds with no model because $|\tilde{y}_i| \leq B_y$; and, when only $\mathbf{y}$ is clipped and the errors before clipping have conditional variance at most σ_0^2, any declared $\bar{\sigma}^2 \geq \sigma_0^2$, since clipping is 1-Lipschitz and cannot increase a variance. Neither party can check the second kind of bound, so it rests on the agreement. The interval with the data-driven $\tilde{s}^2$ is still reported, but only as valid under the working model (Section 9.4).

9.2 Stage 1: choosing the ridge

Stage 1 releases $\hat{\mathbf{B}}_1$ and $\hat{\mathbf{C}}_1$ with s.d. σ_1, and R forms $\hat{\mathbf{G}}_1$ with its exact $\mathbf{A}$. The rule needs a radius r_1 with

$$\Pr\big[\lambda_{\max}(\hat{\mathbf{G}}_1 - \mathbf{G}) > r_1\big] \leq \alpha_L.$$

Then $\hat{L} = \max\{0, \lambda_{\min}(\hat{\mathbf{G}}_1) - r_1\} \leq \lambda_{\min}(\mathbf{G})$ with probability at least $1 - \alpha_L$. This one-sided event is the only one the procedure uses. The protocol allows two radii.

- *Union bound:* $r_1 = \sigma_1(2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)})$. The name is historical: the radius comes from a Gaussian comparison bound, $\mathbb{E}\lambda_{\max}(\mathbf{E}_1) \leq 2\sigma_1\sqrt{p}$, plus concentration of $\lambda_{\max}$ around its mean, not from a union bound. It holds for sub-Gaussian noise, and it is the radius the proofs cover. At $p = 11$ and $\alpha_L = 0.0025$ it is about 1.55 times conservative.
- *Exact quantile:* $r_1 = \sigma_1 q_{1-\alpha_L}$, where $q_{1-\alpha_L}$ is the $(1 - \alpha_L)$ quantile of $\lambda_{\max}(\mathbf{E})$ and $\mathbf{E}$ is the stage-1 noise matrix with unit s.d.: symmetric, zero on $\mathbf{A}$, and independent $\mathcal{N}(0, 1)$

on the released entries of triu $\mathbf{B}$ and $\mathbf{C}$. Its law depends only on (d_R, d_O), so $q_{1-\alpha_L}$ is computed offline by Monte Carlo from public quantities and costs no privacy. The implementation takes an upper 99% confidence bound on the quantile from draws with at least 2,500 expected exceedances (10^6 draws at $\alpha_L = 0.0025$). At $p = 11$ this gives $q_{0.9975} = 7.42$, against 11.53 for the union bound.

A two-sided statement needs $\ln(2/\alpha_L)$ in the union bound and the $(1 - \alpha_L/2)$ quantile in the exact radius. The Monte Carlo quantile is computed for Gaussian slot noise with the calibrated s.d., while the encrypted noise is a discrete Gaussian on the coefficient lattice, each slot value being a sum of N independent terms of it. A coupling settles the difference. Each release's lattice noise η can be coupled with a continuous Gaussian $\mathbf{Z}$ of s.d. $\sigma_{c,s}$ so that $\eta = \lfloor \mathbf{Z} \rceil$, except with probability at most $N\,\mathrm{TV}_1(\sigma_{c,s})$, about 10^{-37} at the evaluation setting ($N = 2^{14}$, $\varepsilon = 1$, previous setting), where the total-variation constant $\mathrm{TV}_1(\sigma) \le 0.0202\,\sigma^{-2} + 0.0315\,\sigma^{-3} + e^{-2\pi^2\sigma^2}$ of one coordinate is proved (proofs, Part B, Lemma S0); the image of $\mathbf{Z}$ in the slots is exactly Gaussian with s.d. σ_s. On that event, and on the events that $\|e_{\mathrm{tot},s}\|_2 \le B_{\mathrm{tot},s}$ and that nothing wraps around, the slot noise differs from the Gaussian one by a vector of ℓ_2 norm at most $\bar{w}_s$, which collects the rounding of $\mathbf{Z}$, the flooded error and the decoding offset, whose public bound is proved (proofs, Part A, Proposition 4.5; Part B, Proposition S7). $\bar{w}_s$ is between about 2×10^{-8} ($\bar{n}_O = 4{,}096$) and 5×10^{-6} ($\bar{n}_O = 10^6$), against $\sigma_1 \ge 128$ in every setting of Table 5. The exact radius therefore holds for the protocol with r_1 enlarged by $\sqrt{2}\,\bar{w}_1$ and the failure probability enlarged by the probability of those events; the implementation uses the unenlarged r_1, a relative difference below 10^{-8}. With Monte Carlo draws independent of the release, the order-statistic radius also has an exact level: $(M - j)/(M + 1) = 0.002385$ at $p = 11$, with $M = 10^6$ draws and the order statistic $j + 1 = 997{,}616$.

With the target $g^\star = 2\vartheta^\star\sigma_2\sqrt{p}$, the full ridge

$$\lambda_I = \max\{0,\ g^\star + r_1 - \lambda_{\min}(\hat{\mathbf{G}}_1)\}$$

guarantees $\lambda_{\min}(\mathbf{G} + \lambda_I\mathbf{I}) \ge g^\star$ on that event. The O-block ridge λ_O is the smallest $\lambda \ge 0$ with $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \ge \gamma$, where $\gamma = g^\star + r_1$. It is finite iff $\lambda_{\min}(\mathbf{A}) > \gamma$, and then it has the closed form

$$\lambda_O = \max\bigl\{0,\ \gamma - \lambda_{\min}(\mathbf{S}_\gamma)\bigr\}, \qquad \mathbf{S}_\gamma = \hat{\mathbf{B}}_1 - \hat{\mathbf{C}}_1^\top(\mathbf{A} - \gamma\mathbf{I})^{-1}\hat{\mathbf{C}}_1,$$

because $\hat{\mathbf{G}}_1 + \lambda\Pi_O - \gamma\mathbf{I} \succeq \mathbf{0}$ iff its Schur complement $\mathbf{S}_\gamma + (\lambda - \gamma)\mathbf{I}$ is positive semidefinite, the block $\mathbf{A} - \gamma\mathbf{I}$ being positive definite. No bisection is needed. The Π_O ridge can never certify a validity ratio above $\lambda_{\min}(\mathbf{A})/(2\sigma_2\sqrt{p})$. The rule takes $(\lambda, \Psi) = (\lambda_O, \Pi_O)$ if $\lambda_O \le \kappa_\Psi\lambda_I$, for a parameter $\kappa_\Psi \ge 0$, and $(\lambda_I, \mathbf{I})$ otherwise. Shrinking only O's block leaves R's coefficients unshrunk when R's own block is well conditioned.

Proposition 9.1 (Independence)**.** (λ, Ψ) *is a function of the stage-1 release and of* $\mathbf{A}$ *only, and is therefore independent of the stage-2 noise.*

This is what makes the bias theory of Appendix A and the intervals below valid conditionally on (λ, Ψ). Choosing λ from the same release that is used for estimation would make λ a function of the noise; Appendix A.5 shows what goes wrong.

Setting I keeps $\vartheta^\star = 2.5$ and $\kappa_\Psi = 2$ from simulations of earlier variants of the rule, and setting P keeps $\kappa_\Psi = 2$ as well. Neither value was re-tuned under the two-stage rule with the exact radius; this is a limitation of both settings (Section 10).

9.3 Stage 2: estimate and gate

Stage 2 releases every entry with s.d. σ_2. R assembles $\hat{\mathbf{G}}_2$ and $\hat{\mathbf{c}}_2$, computes the validity ratio $\hat{\vartheta} = \lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi)/(2\sigma_2\sqrt{p})$, and releases nothing if $\hat{\vartheta} < 1$. Otherwise

$$\tilde{\boldsymbol{\theta}} = \tilde{\mathbf{P}}\hat{\mathbf{c}}_2, \qquad \tilde{\mathbf{P}} = (\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1}.$$

The gate is post-processing; a refused release still spends the budget. Coverage is unconditional: every coverage statement in this paper bounds the probability, over all releases, that R reports an interval that misses its target. A refused release reports no estimate and no interval, so it makes no claim and cannot miss, and it still spends the budget. For $\vartheta^\star > 2$ the probability of refusal is at most $\alpha_L + \exp(-p(\vartheta^\star - 2)^2)$ (proofs, Part B, Lemma S3). For $\vartheta^\star \le 2$, as in the prediction setting P of Section 10 ($\vartheta^\star = 1.5$), there is no such bound. With $\Psi = \mathbf{I}$ the ridge sets $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\mathbf{I}) = g^\star + r_1$, so the gate passes iff

$$r_1 + \big[\lambda_{\min}(\hat{\mathbf{G}}_2) - \lambda_{\min}(\hat{\mathbf{G}}_1)\big] \ge (1 - \vartheta^\star)\, 2\sigma_2\sqrt{p}.$$

The bracket is the difference of the downward shifts that the two noise levels cause in $\lambda_{\min}$; on average it is negative when $\sigma_2 > \sigma_1$. With $\vartheta^\star < 1$ the right side is positive, so the gate passes only when r_1 is large against $\sigma_2\sqrt{p}$, which depends on Δ_G/Δ and f_1 through (7). Block normalization makes Δ_G small against Δ, and a first choice of the prediction setting with $\vartheta^\star = 0.5$ was refused on most block-normalized releases (Sections 11.9 and 11.11). With $\vartheta^\star > 1$ the right side is negative: at $\vartheta^\star = 1.5$ the gate refuses only if the stage-2 noise lowers $\lambda_{\min}$ by more than $r_1 + \sigma_2\sqrt{p}$ beyond what the stage-1 noise did. Setting P was chosen under an explicit bound on the refusal rate. It refuses at most 1% of releases in every simulated cell, including $\sigma_1 = 0.46\,\sigma_2$, and at most 7.2% on any of 29 real datasets (Sections 11.7 and 11.9).

9.4 Intervals

Mechanism variance. R knows the noise distribution, so it estimates the covariance of $\tilde{\boldsymbol{\theta}}$ due to the mechanism by a parametric bootstrap around its own release: symmetric noise $\mathbf{E}^*$ with s.d. σ_2 on the O-dependent entries of $\hat{\mathbf{G}}_2$ and zero on $\mathbf{A}$, noise $\mathbf{f}^* \sim \mathcal{N}(0, \sigma_2^2\mathbf{I})$ on $\hat{\mathbf{c}}_2$, and $\boldsymbol{\theta}_b^* = (\hat{\mathbf{G}}_2 + \mathbf{E}_b^* + \lambda\Psi)^{-1}(\hat{\mathbf{c}}_2 + \mathbf{f}_b^*)$. A draw is kept only if $\|\tilde{\mathbf{P}}^{1/2}\mathbf{E}^*\tilde{\mathbf{P}}^{1/2}\|_{\mathrm{op}} \le \eta_B$, for a truncation parameter $\eta_B \in (0,1)$; without truncation the bootstrap moments do not exist. With B draws the Monte Carlo error of $\mathbf{V}_{\mathrm{mech}}$ is of relative order $B^{-1/2}$. Antithetic pairs $(\pm\mathbf{E}^*, \pm\mathbf{f}^*)$ reduce Monte Carlo error. $\mathbf{V}_{\mathrm{mech}}$ is the sample covariance of the kept draws. To first order, $\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_\lambda \approx \tilde{\mathbf{P}}(\mathbf{f} - \mathbf{E}\boldsymbol{\theta}_\lambda)$, so

$$\mathbf{V}_{\mathrm{mech}} \approx \sigma_2^2\,\tilde{\mathbf{P}}\big(\mathbf{I} + K(\tilde{\boldsymbol{\theta}})\big)\tilde{\mathbf{P}}, \qquad K(\boldsymbol{\theta})_{ii} = \textstyle\sum_j \chi_{ij}\theta_j^2, \quad K(\boldsymbol{\theta})_{ik} = \chi_{ik}\theta_i\theta_k \ (i \ne k),$$

with $\chi_{ij} = 0$ iff i and j both lie in R's block. R may use this closed form instead of the bootstrap.

Wald interval for $\boldsymbol{\theta}_\lambda$. The error level is split as $\alpha = \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U$, a parameter. Setting I uses $\alpha = 0.05$ with $\alpha_L = \alpha_U = \alpha/20$, so $\alpha_{\mathrm{CI}} = 0.045$ and $z_{1-\alpha_{\mathrm{CI}}/2} = 2.005$. Setting P computes no intervals; its $\alpha_L = 0.75$ only sizes r_1. Both intervals are computed only for releases that pass the gate, and their coverage is unconditional in the sense of Section 9.3: a refused release makes no claim. The interval is

$$\tilde{\theta}_j \pm z_{1-\alpha_{\mathrm{CI}}/2}\sqrt{(\mathbf{V}_{\mathrm{mech}})_{jj}}.$$

Residual scale. $\widetilde{\mathrm{RSS}} = \widehat{yy} - \hat{\mathbf{c}}_2^\top \tilde{\boldsymbol{\theta}} - \lambda \tilde{\boldsymbol{\theta}}^\top \Psi \tilde{\boldsymbol{\theta}}$ is biased by the noise at order σ_2^2. The same bootstrap gives $\mathrm{Corr} = \overline{\mathrm{RSS}^*} - \widetilde{\mathrm{RSS}}$ and $\tilde{s}^2 = \max\{\widetilde{\mathrm{RSS}} - \mathrm{Corr}, 0\}/(n-p)$, which removes the $O(\sigma_2^2)$ bias up to $O(\sigma_2^4)$.

Bias-aware interval for $\boldsymbol{\beta}$. The total variance adds the sampling term from the public variance bound of Section 9.1, $\mathbf{V} = \mathbf{V}_{\mathrm{mech}} + \bar{\sigma}^2 \tilde{\mathbf{P}} \hat{\mathbf{G}}_2 \tilde{\mathbf{P}}$. The ridge bias of coordinate j is at most $\bar{b}_j = \lambda \|\Psi \tilde{\mathbf{P}} \mathbf{e}_j\|_2 \, \hat{B}_\beta$, where $\hat{B}_\beta = \min\{B_\beta^{\mathrm{pub}}, \sqrt{U_{yy}/\hat{L}}\}$ bounds $\|\boldsymbol{\beta}\|$ from a public a-priori bound and a private one,

$$U_{yy} = \min\Big\{\widehat{yy} + z_{1-\alpha_U/2}\, \sigma_{yy} + B_y^2 \sqrt{n \ln(2/\alpha_U)/2},\ n B_y^2\Big\},$$

with σ_{yy} the s.d. of the release that carries $\mathbf{y}^\top \mathbf{y}$. If the y_i are conditionally independent given the design, $U_{yy} \geq \mathbb{E}[\mathbf{y}^\top \mathbf{y} \mid \mathbf{X}]$ with probability at least $1 - \alpha_U$: Hoeffding's inequality bounds the sampling deviation of $\mathbf{y}^\top \mathbf{y}$, and the Gaussian tail the release noise, each at level $\alpha_U/2$, and the cap holds deterministically. (The specification's form spent α_U twice and left σ_{yy} undefined.) The interval, following Armstrong, Kolesár and Kwon [3], is

$$\tilde{\theta}_j \pm \mathrm{cv}_{\alpha_{\mathrm{CI}}}\big(\bar{b}_j / \sqrt{V_{jj}}\big) \sqrt{V_{jj}},$$

where $\mathrm{cv}_\alpha(t)$ is the $(1-\alpha)$ quantile of $|Z + t|$ for $Z \sim \mathcal{N}(0,1)$. The private bound is usually infinite while the ridge is on ($\hat{L} = 0$), so without a public B_β^{pub} the interval for $\boldsymbol{\beta}$ is unbounded exactly when the ridge is needed. This is the honest answer: the data cannot rule out large coefficients in the directions the ridge suppresses.

Working-model interval for $\boldsymbol{\beta}$. R also reports the same interval with $\mathbf{V}^{\mathrm{wm}} = \mathbf{V}_{\mathrm{mech}} + \tilde{s}^2 \tilde{\mathbf{P}} \hat{\mathbf{G}}_2 \tilde{\mathbf{P}}$ in place of $\mathbf{V}$. It is valid only under the working model, with homoskedastic clipped errors; it is not guaranteed when clipping binds. The two intervals differ only in the sampling term, so they nearly coincide where the mechanism noise or the ridge bias dominates, and the guaranteed one is wider by up to $\sqrt{\bar{\sigma}^2/\tilde{s}^2}$ where the sampling variance dominates: on the synthetic population ($B_y^2 = 8.34$, $\tilde{s}^2 \approx 0.50$) the median ratio of half-widths is 1.26–1.46 at $\varepsilon = 1$ and 2.3–3.7 at $\varepsilon = 8$ once the ridge is off (Section 11.3).

Remark 9.2 (How accurate is first order?)**.** The validity ratio controls convergence of the Neumann series, but the relative size of the neglected mechanism bias in the Wald interval is of order $\sigma_2 \,\mathrm{tr}(\mathbf{P}_\lambda) \leq \sqrt{p}/(2\vartheta_0)$, which at $\vartheta^\star = 2.5$ and $p = 11$ (setting I) can be 0.66. Coverage claims are therefore asymptotic as $\vartheta_0 \to \infty$ at fixed p. Algorithm 6 reports $\sigma_2 \,\mathrm{tr}(\tilde{\mathbf{P}})$ as a diagnostic.

The intervals are centred at $\tilde{\boldsymbol{\theta}}$, not at the bootstrap bias-corrected estimate $2\tilde{\boldsymbol{\theta}} - \overline{\boldsymbol{\theta}^*}$, because the squared mechanism bias is at most 0.19% of the mean squared error under setting I (cell average; Section 11.8).

9.5 When can the ridge be dropped?

The stage-1 rule is explicit, so the intersection size from which it stops adding a ridge can be read off the protocol parameters and one property of the data.

The threshold. Let $\Sigma = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$ be the per-record second-moment matrix of the preprocessed, clipped design, intercept included, and $\ell = \lambda_{\min}(\Sigma)$ its weakest direction. With n matched records $\mathbf{G} \approx n\Sigma$ and $\lambda_{\min}(\mathbf{G}) \approx n\ell$, up to a relative fluctuation of order $n^{-1/2}$. Ignoring the stage-1 noise in $\lambda_{\min}(\hat{\mathbf{G}}_1)$, the full ridge is $\lambda_I(n) \approx \max\{0,\ g^\star + r_1 - n\ell\}$, which falls linearly in n and vanishes at

$$n_{\text{off}} = \frac{g^\star + r_1}{\ell} = \frac{\sqrt{p}}{\ell\sqrt{2\rho_{\text{run}}}} \left[\frac{2\vartheta^\star \Delta}{\sqrt{1 - f_1}} + \frac{q}{\sqrt{p}} \frac{\Delta_G}{\sqrt{f_1}} \right], \tag{9}$$

by (7), where $q = r_1/\sigma_1$ is $2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)}$ for the union bound and $q_{1-\alpha_L}$ for the exact radius (Section 9.2). With $p = 11$, setting I ($\vartheta^\star = 2.5$, $f_1 = 0.25$, exact radius, $q = 7.42$) gives

$$n_{\text{off}} \approx \frac{19.1\,\Delta + 14.8\,\Delta_G}{\ell\sqrt{2\rho_{\text{run}}}},$$

against $(17.5\,\Delta + 36.5\,\Delta_G)/(\ell\sqrt{2\rho_{\text{run}}})$ at the previous setting ($f_1 = 0.1$, union bound) and $(11.5\,\Delta + 8.9\,\Delta_G)/(\ell\sqrt{2\rho_{\text{run}}})$ at setting P ($\vartheta^\star = 1.5$, $f_1 = 0.25$, $q_{0.25} = 4.44$). Here $\sqrt{2\rho_{\text{run}}} =$ 0.089, 0.173, 0.335, 0.642 and 1.21 for $\varepsilon = 0.5$, 1, 2, 4 and 8 at $\delta = 10^{-9}$. Since $\sqrt{\rho_{\text{run}}}$ grows almost linearly in ε at fixed δ, n_{off} scales like the sensitivity divided by $\ell\varepsilon$. The two terms have different origins. $g^\star/\ell$ is the threshold if ℓ were known in advance; r_1/ℓ is the price of learning it from stage 1. With $\Delta_G \approx 0.9\Delta$, as for the simulation population, $r_1 \approx 0.69\,g^\star$ under setting I, so learning ℓ multiplies the threshold by about 1.7; at the previous setting $r_1 \approx 1.9\,g^\star$ and the factor was 2.9. The private bound $\hat{L}$ on $\lambda_{\min}(\mathbf{G})$, which makes the interval for $\boldsymbol{\beta}$ finite without a public bound on $\|\boldsymbol{\beta}\|$, becomes positive earlier, from $n \approx r_1/\ell$: about $0.41\,n_{\text{off}}$ under setting I and $0.65\,n_{\text{off}}$ at the previous setting.

Below the threshold. With $\Psi = \mathbf{I}$, the component of $\boldsymbol{\theta}_\lambda$ along an eigenvector of $\mathbf{G}$ with eigenvalue $n\ell_k$ keeps the fraction $n\ell_k/(n\ell_k + \lambda)$ of the least-squares coefficient. In the weakest direction $n\ell + \lambda \approx g^\star + r_1 = n_{\text{off}}\ell$, so the fraction kept is n/n_{off}: the shrinkage there falls linearly in n and is gone at n_{off}. Directions with $\ell_k \gg \ell$ are barely shrunk. A weak direction carries little of the variation in $\mathbf{x}$, so shrinking it costs little in prediction error but a lot in the coefficients that load on it; this is why prediction recovers before the coefficients do (Section 11.5).

The transition. The noise lowers $\lambda_{\min}(\hat{\mathbf{G}}_1)$ on average, the more so when several eigenvalues of Σ are close to ℓ, so the rule chooses a larger λ than the formula and switches off after n_{off}. Table 5 gives the threshold for the simulation population of Section 11.7 ($\ell = 0.599$, $\Delta = 86.2$, $\Delta_G = 77.2$, $p = 11$), and Table 22 simulates the stage-1 rule at multiples of it.

- Setting I: the ridge is never off at $0.8\,n_{\text{off}}$, off in 3–4% of releases at n_{off}, and off in 95–99.5% at $1.2\,n_{\text{off}}$. The previous setting gave 3–5% and 94–99%. We take $1.2\,n_{\text{off}}$ as the size from which the ridge is off with high probability.
- Setting P: the transition is wider. At most 1% of releases are off at n_{off}, 46–63% at $1.2\,n_{\text{off}}$ and 98–99% at $1.4\,n_{\text{off}}$. Its $g^\star + r_1$ is about 0.6 times that of setting I in units of σ_2 (smaller $\vartheta^\star$, radius $q_{0.25} = 4.44$), so the downward bias of $\lambda_{\min}(\hat{\mathbf{G}}_1)$, which the formula ignores, is a larger share of it. The first choice P_0 ($\vartheta^\star = 0.5$, Section 10) switched off later still: no release off at n_{off} and 56–70% at $1.4\,n_{\text{off}}$.

Table 14 uses the committed bound $\ell = 0.494$ instead of the population value, so its thresholds are 21% higher. Under the earlier mechanism, which used that committed bound instead of a stage-1 release, the ridge switched off at exactly $g^\star/\ell$ for that mechanism's σ and $\vartheta^\star = 2$.

Table 5: Ridge threshold for the simulation population, $\delta = 10^{-9}$, setting I. n_{off} from (9); $g^\star/\ell$ is the threshold with ℓ known in advance. Last two columns: n_{off} at the previous setting and at setting P. Table 22 gives the simulated switch-off.

	setting I						previous	P
ε	σ_1	σ_2	$g^\star$	r_1	n_{off}	$g^\star/\ell$	n_{off}	n_{off}
0.5	1737	1119	18,562	12,878	52,531	31,014	81,208	31,498
1	892	575	9,538	6,617	26,993	15,937	41,729	16,185
2	461	297	4,927	3,419	13,945	8,233	21,557	8,361
4	240	155	2,569	1,783	7,272	4,293	11,241	4,360
8	128	82	1,364	946	3,860	2,279	5,967	2,314

Check on real data. On protein (Section 11.5, 19,206 matched records, $p = 10$), ℓ computed from the matched design predicts the ridge the encrypted runs chose (Table 6). Those runs used the previous setting ($f_1 = 0.1$, union-bound radius). The runs chose 1–19% more than predicted, always more, as the downward bias of $\lambda_{\min}(\hat{\mathbf{G}}_1)$ implies; the gap is largest where σ_1 is largest.

Table 6: Protein, 19,206 matched records, $\delta = 10^{-9}$, previous setting: ℓ of the matched design, n_{off} from (9), the ridge $g^\star + r_1 - n\ell$ it predicts, and the median ridge of the five encrypted runs of Table 20.

Preprocessing	Δ	Δ_G	ℓ	ε	n_{off}	predicted λ	encrypted λ
standardize	87.3	80.9	0.00191	1	13.1 M	24,953	29,678
standardize	87.3	80.9	0.00191	4	3.53 M	6,695	7,321
winsorize $\lvert z\rvert \le 2$	40.5	34.2	0.00125	1	8.74 M	10,902	12,552
winsorize $\lvert z\rvert \le 2$	40.5	34.2	0.00125	4	2.35 M	2,919	3,088
block unit norm	8.4	2.83	0.00201	1	691 k	1,353	1,385
block unit norm	8.4	2.83	0.00201	4	186 k	336	338

What sets ℓ. ℓ is a property of the joint design after preprocessing, so near-collinearity decides n_{off}, and it may lie within either party's block or across them. On protein, ℓ is about 0.002 under every preprocessing, which keeps the ridge on to between 186k and 13M records at the previous setting. Setting I lowers these thresholds by a factor of 0.63 (standardized) to 0.80 (block-normalized), for example to 8.3M and 555k at $\varepsilon = 1$. The source of the small ℓ differs.

- Standardized: the weakest direction is $0.77\,x_{R,1} - 0.62\,x_{O,5}$; the two features have correlation 0.997, so the two parties hold almost the same variable. R's own block is much better conditioned, $\lambda_{\min}(\mathbf{A}/n) = 0.010$. Neither party can see this collinearity in its own data.
- Block-normalized: the weakest direction lies inside R's block, $-0.66\,x_{R,1} + 0.69\,x_{R,2} - 0.27\,x_{R,3}$, and $\ell \approx \lambda_{\min}(\mathbf{A}/n) = 0.0021$: normalizing a block of strongly correlated features leaves it close to degenerate.

Table 7 repeats the full preprocessing with one feature removed. Removing $x_{R,1}$ removes both weak directions: with block normalization ℓ rises to 0.013 and n_{off} falls from 691k to 102k at $\varepsilon = 1$ and from 186k to 27k at $\varepsilon = 4$. Removing $x_{O,5}$ instead helps only the standardized design.

Table 7: Protein: ℓ and n_{off} ($\delta = 10^{-9}$, previous setting) with one feature removed before preprocessing; bounds recomputed from the reference split.

Features	Preprocessing	Δ	ℓ	n_{off}, $\varepsilon = 1$	n_{off}, $\varepsilon = 4$
all 9	standardize	87.3	0.00191	13.1 M	3.53 M
all 9	block unit norm	8.4	0.00201	691 k	186 k
without $x_{R,1}$	standardize	79.6	0.0130	1.69 M	455 k
without $x_{R,1}$	block unit norm	8.4	0.0131	102 k	27 k
without $x_{O,5}$	standardize	74.4	0.0100	2.03 M	547 k
without $x_{O,5}$	block unit norm	8.4	0.0021	647 k	174 k

Two diagnostics cost no budget. First, $\mathbf{A}$ is a principal submatrix of $\mathbf{G}$, so $\lambda_{\min}(\mathbf{G}) \le \lambda_{\min}(\mathbf{A})$ by Cauchy interlacing, and after linkage R obtains from its exact $\mathbf{A}$ the lower bound $n_{\text{off}} \ge n(g^\star + r_1)/\lambda_{\min}(\mathbf{A})$ before any release; O can bound ℓ the same way from its own block on the register. Second, a collinearity across the parties is invisible to both but visible on a public reference set that carries both schemas, where one exists; screening features on it is the optional Setup module of Section 6.1.

Levers. Table 8 lists what lowers n_{off} at $\varepsilon = 1$ on the simulation population, where the previous setting gives 41.7k. Setting I combines the exact radius with $f_1 = 0.25$, the value the pre-registered rule of Section 10 chose. n_{off} alone would favour $f_1 \approx 0.35$, but it is flat over 0.25–0.45, and the larger σ_2 of a larger f_1 costs accuracy once the ridge is off.

Table 8: Levers on the ridge threshold, simulation population, $\varepsilon = 1$, $\delta = 10^{-9}$ (at the previous setting, $f_1 = 0.1$ with the union-bound radius, $n_{\text{off}} = 41.7$k).

Lever	n_{off}	Cost
Smaller Δ, larger ℓ: per-party transforms (Section 6.1); feature screening on public reference data	$\propto \Delta/\ell$	none in privacy; on protein a factor 6.8 from one feature
Exact $(1-\alpha_L)$ quantile for r_1 ($7.42\sigma_1$ instead of $11.53\sigma_1$)	32.0k (−23%)	none
$f_1 = 0.25$ instead of 0.1	33.1k (−21%)	σ_2 up 9.5% once the ridge is off
Both of the above: **setting I**	27.0k (−35%)	σ_2 up 9.5%
Exact radius and $f_1 = 0.35$	26.5k (−37%)	σ_2 up 18%
$\vartheta^\star = 2$ instead of 2.5 (previous setting)	38.8k (−7%)	thinner validity margin (Remark 9.2)
Setting P: $\vartheta^\star = 1.5$, $\alpha_L = 0.75$, $f_1 = 0.25$, exact radius	16.2k (−61%)	no interval guarantee; the ridge is off with high probability only from about $1.4\,n_{\text{off}}$; a smaller $\vartheta^\star$ (P_0: 0.5, 11.3k) is refused by the gate where $\Delta_G \ll \Delta$ (Section 11.11)
A public lower bound on ℓ instead of stage 1	at best 14.6k (−65%)	validity rests on the reference data being representative; the gate still guards each release
Larger ε	$\propto 1/\sqrt{\rho_{\text{run}}} \approx 1/\varepsilon$	budget

Below n_{off} the ridge is not a tuning choice: it is what certifies the validity ratio $\vartheta^\star$ on which the expansion and the intervals rest (Remark A.2). The levers lower the threshold; none of

them makes the ridge unnecessary below it while the intervals are wanted. Setting P lowers the target itself and gives up the intervals. A study can be planned by evaluating (9) at the intended ε, with ℓ estimated on public reference data or bounded through $\lambda_{\min}(\mathbf{A})$.

A rigorous bracket. Two results bound the transition without simulation.

- *The noise biases $\lambda_{\min}(\hat{\mathbf{G}}_1)$ downward.* For a symmetric $\mathbf{G}$ and a zero-mean symmetric noise matrix $\mathbf{E}$ with $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} < \infty$, $\mathbb{E}\lambda_{\min}(\mathbf{G}+\mathbf{E}) \le \lambda_{\min}(\mathbf{G})$, since $\lambda_{\min}(\mathbf{G}+\mathbf{E}) \le \mathbf{v}^\top(\mathbf{G}+\mathbf{E})\mathbf{v}$ for a unit bottom eigenvector $\mathbf{v}$ of $\mathbf{G}$. The gap grows with the number of eigenvalues of $\mathbf{G}$ near $\lambda_{\min}(\mathbf{G})$. This is the bias that (9) ignores.
- *The switch-off is bracketed.* Condition on the data and take Gaussian slot noise. The rule returns $\lambda = 0$ with probability at least $1-\alpha'$ once $\lambda_{\min}(\mathbf{G}) \ge g^\star + r_1 + \sigma_1 q'_{1-\alpha'}$, where $q'_{1-\alpha'}$ is the $(1-\alpha')$ quantile of $\lambda_{\max}$ of the unit stage-1 noise matrix, by Weyl's inequality. It returns $\lambda > 0$ with probability at least $1-\alpha'$ while $\lambda_{\min}(\mathbf{G}) \le g^\star + r_1 - \sqrt{2}\,\sigma_1 z_{1-\alpha'}$, because $\mathbf{v}^\top\mathbf{E}_1\mathbf{v}$ is Gaussian with variance at most $2\sigma_1^2$. With $\lambda_{\min}(\mathbf{G}) \approx n\ell$, the switch-off therefore lies between $(1-\sqrt{2}\,\sigma_1 z_{1-\alpha'}/(g^\star+r_1))\,n_{\mathrm{off}}$ and $(1+\sigma_1 q'_{1-\alpha'}/(g^\star+r_1))\,n_{\mathrm{off}}$, with probability at least $1-\alpha'$ at each end. The ratio $\sigma_1/(g^\star+r_1)$ does not depend on ε. The upper end cannot be improved in terms of $\lambda_{\min}(\mathbf{G})$ alone: for $\mathbf{G}$ proportional to the identity it is attained exactly (proofs, Part B, Proposition S8).

On the simulation population at $\alpha' = 0.05$ ($q'_{0.95} = 6.303$ at $p = 11$) the bracket is $[0.87, 1.35]\,n_{\mathrm{off}}$ under setting I and $[0.79, 1.58]\,n_{\mathrm{off}}$ under setting P, at every ε. The simulated transitions, from n_{off} to $1.2\,n_{\mathrm{off}}$ under I and to $1.4\,n_{\mathrm{off}}$ under P, lie inside them.

A comparison bound gives the size of the bias for any spectrum: $\mathbb{E}\lambda_{\min}(\hat{\mathbf{G}}_1) \ge \mathbb{E}\min_{\|\mathbf{u}\|=1}(\mathbf{u}^\top\mathbf{G}\mathbf{u} - 2\sigma_1\mathbf{g}^\top\mathbf{u})$ with $\mathbf{g}$ standard normal, by the Sudakov–Fernique inequality, and $\lambda_{\min}(\hat{\mathbf{G}}_1)$ concentrates with variance proxy $2\sigma_1^2$ (proofs, Part B, Lemma S8′). On the simulation population at n_{off} under setting I it bounds the downward bias by $3.2\,\sigma_1$, against $1.7\,\sigma_1$ in simulation, which is too loose to improve on the bracket.

[**Check 9:** The switch-off factors 1.2 (setting I and the previous setting) and 1.4 (setting P) remain empirical. A bound needs a tail bound for lambda-min of the noisy stage-1 Gram that uses the spectrum and is much sharper than the comparison bound; at n-off the three smallest eigenvalues of the simulation population's Gram lie within about 2.4 sigma-1 of each other, too close for second-order perturbation.]

9.6 Mechanism bias

The noise in $\hat{\mathbf{G}}_2$ biases $\tilde{\boldsymbol{\theta}}$ at order σ_2^2. Appendix A gives the leading term, $\sigma_2^2 M(\mathbf{P}_\lambda)\boldsymbol{\theta}_\lambda$ conditionally on a high-probability event, with explicit constants (Theorems A.1 and A.2), and a zero-budget correction that subtracts its plug-in estimate. The correction is an option of the protocol and is off in our evaluation: the squared mechanism bias is at most 0.19% of the mean squared error under setting I (cell average; Section 11.8).

9.7 Status of the statistical guarantees

The companion proofs (Part B) establish: the union-bound radius r_1 for the one-sided event (exact); the independence and closed forms of the two-stage rule; the bound on $\|\boldsymbol{\beta}\|$ for the projection coefficient with no model assumed; the Wald remainder exactly, with the plug-in variance to first order and normality of the sampling term asymptotically; and the bootstrap

RSS correction to first order, without explicit constants. The exact-quantile radius of settings I and P has an exact level for Gaussian slot noise with fresh Monte Carlo draws, and the coupling of Section 9.2 transfers it to the lattice noise at a cost of about 10^{-37} in probability and a relative 10^{-8} in the radius; the total-variation constant of the coupling and the decoding offset have proved bounds (Part B, Lemma S0 and Proposition S7; Part A, Proposition 4.5). The bracket on the switch-off of Section 9.5 is proved for Gaussian slot noise and transfers to the protocol by the same coupling; its upper end is attained by Gram matrices proportional to the identity, and the size of the downward bias of $\lambda_{\min}(\hat{\mathbf{G}}_1)$ is bounded only loosely (Part B, Section S8). The mechanism bias of Appendix A is proved with explicit constants (Part B, Theorem S9). The bias-aware interval uses a public bound on the conditional variance of the clipped response, with which it is valid whether or not clipping binds (Part B, Proposition S10b); a guarantee for the working-model interval, with $\tilde{s}^2$, is open when clipping binds.

10 Parameters

The protocol is defined for a parameter vector $\mathcal{P}$. Table 10 lists every parameter: its role, the constraint it must satisfy, and the value used in this paper's evaluation. The last column is an *evaluation setting*, one admissible choice among many; it is not part of the protocol.

The constraints are of three kinds. Some are hypotheses that $\mathcal{P}$ itself must meet, such as the ranges of the shares and the fact that the bounds and the preprocessing constants are public and fixed in advance; the parties check them in Setup. Some involve quantities that CALIBRATE derives: the existence of κ, $\delta_{\mathrm{DP}} > 0$, a scale that floods, the size of q_0 and the security bound on the chain. CALIBRATE checks these and refuses the run otherwise (Section 7.3). The rest, such as f_1, $\vartheta^\star$ and κ_Ψ, affect only utility or cost and need only lie in their ranges. We call $\mathcal{P}$ *admissible* if all constraints of Table 10 hold. Theorem 7.2, Proposition 7.2 and the input-privacy and correctness statements of Section 8 hold for every admissible $\mathcal{P}$, with the constants they state; none of them uses a particular value. The statistical results of Section 9 need in addition $\vartheta^\star > 1$ and the intervals on ($\alpha_{\mathrm{CI}} > 0$).

Two evaluation settings. Sections 11.7–11.12 use two named settings of the same protocol. They differ only in the statistical parameters (Table 9).

- **Setting I (inference)** keeps the interval guarantee. It uses the exact-quantile radius (Section 9.2), $\vartheta^\star = 2.5$, $\kappa_\Psi = 2$, $\alpha = 0.05$ with $\alpha_L = \alpha_U = \alpha/20$, and $f_1 = 0.25$.
- **Setting P (prediction)** is tuned for test error only. It uses the exact-quantile radius, $\vartheta^\star = 1.5$, $\alpha_L = 0.75$, $\kappa_\Psi = 2$ and $f_1 = 0.25$. Its intervals are off and $\alpha_L > \alpha$, so the statistical results of Section 9 do not apply to it, and R reports the estimate without intervals. The interval guarantee holds only under setting I. P replaces a first choice, P_0 ($\vartheta^\star = 0.5$, $\alpha_L = 0.5$, $f_1 = 0.2$), whose gate refused most releases on block-normalized real data. P_0's results are kept for reference.

Both keep the gate $\hat{\vartheta} \geq 1$, which is part of the protocol, and both have the same privacy guarantee: the noise depends only on $(\varepsilon, \delta, f_1)$, while r_1, $\vartheta^\star$, κ_Ψ and α_L act on released statistics. The previous setting ($f_1 = 0.1$, union-bound radius, $\vartheta^\star = 2.5$, $\kappa_\Psi = 2$, $\alpha_L = \alpha_U = \alpha/20$), which Sections 11.1–11.5 use, is kept as a reference.

How the settings were chosen. On the synthetic population of Section 11.7 only, by a rule written down before any selection or real-data run (`reports/v31_settings/settings_choice.md` in the repository): $\delta = 10^{-9}$, $\varepsilon \in \{0.5, 1, 2, 4, 8\}$, 14 values of n from 100 to 100,000 with 200 cohorts each (seeds independent of the validation runs), $R^2 = 0.5$, and common random numbers across candidates.

- I takes the $f_1 \in \{0.20, 0.25, \ldots, 0.50\}$ that minimizes the mean, over the 70 (n, ε) cells, of $\log(\mathrm{RMSE_{DP}}/\mathrm{RMSE_{OLS}})$ of the slopes against the true coefficients. $f_1 = 0.25$ wins. The criterion is flat between 0.2 and 0.3, and $R^2 = 0.3$ and 0.9 give the same choice.
- $\mathrm{P_0}$, the first choice of P, took the candidate in $\vartheta^\star \in \{0.5, 1, 1.5, 2.5\} \times \alpha_L \in \{0.0025, 0.25, 0.5\} \times f_1 \in \{0.1, 0.2, 0.35\}$ that minimized the mean ratio of test MSE to that of non-private least squares on the same cohorts, with $\boldsymbol{\theta} = 0$ for a refused release. $(0.5, 0.5, 0.2)$ won. It lay on the edge of the grid in $\vartheta^\star$ and α_L, and as pre-registered the grid was not extended. No candidate's gate refused more than 2% of releases in any cell on this population. On block-normalized real data, however, $\mathrm{P_0}$'s gate refused most releases (Section 11.9).
- P was then re-chosen by a second rule, also written down before any run of it (`reports/v31_P2/settings_choice_P2.md`). It uses two populations: the one above, and the same cohorts with each party's block of every row scaled to unit norm ($B_R^2 = 2$, $B_O = 1$, $B_y = 2.89$). At $f_1 = 0.25$ these have $\sigma_1 = 1.55\,\sigma_2$ and $0.46\,\sigma_2$. On Wang's 29 datasets block normalization gives 0.25–0.83 (median 0.56), with 8 below 0.46. The candidates are $\vartheta^\star \in \{0.5, 0.75, 1, 1.25, 1.5, 2, 2.5\} \times \alpha_L \in \{0.0025, 0.05, 0.25, 0.5\} \times f_1 \in \{0.1, 0.2, 0.25, 0.35\}$, with the exact radius and the intervals off. P minimizes the same criterion as $\mathrm{P_0}$, averaged over 140 cells (both populations, five ε, 14 values of n, 200 cohorts each, $R^2 = 0.5$), among the candidates whose gate refuses at most 1% of releases in every cell. Ties within 0.1% go to the larger $\vartheta^\star$. The grid choice $(1.5, 0.5, 0.25)$ lay on the edge in α_L, so as pre-registered $\alpha_L = 0.75$ was added once. $(1.5, 0.75, 0.25)$ wins, with a criterion of 1.241 against 1.274 for I and 1.286 for $\mathrm{P_0}$. It lies on the edge of the extended grid and, as pre-registered, was kept. The constraint binds: six candidates with $\vartheta^\star = 1.25$ do up to 0.3% better but refuse 2–15% of releases in some block-normalized cells, and no candidate with $\vartheta^\star = 0.5$ meets it. At $R^2 = 0.3$ and 0.9 P is also first by the criterion.

Table 9: The evaluation settings. Only the statistical parameters differ. σ_1, σ_2 and n_{off} for the simulation population at $\varepsilon = 1$, $\delta = 10^{-9}$. $\mathrm{P_0}$ is the first choice of P, replaced because its gate refused most releases on block-normalized data.

	previous (reference)	I (inference)	P (prediction)	$\mathrm{P_0}$ (replaced)
radius r_1	union bound	exact quantile	exact quantile	exact quantile
r_1/σ_1 at $p = 11$	11.53	7.42	4.44	4.94
f_1	0.1	0.25	0.25	0.2
$\vartheta^\star$	2.5	2.5	1.5	0.5
κ_Ψ	2	2	2	2
α_L, α_U	0.0025, 0.0025	0.0025, 0.0025	0.75, unused	0.5, unused
intervals	yes	yes	no	no
σ_1, σ_2	1411, 525	892, 575	892, 575	998, 557
n_{off}	41.7k	27.0k	16.2k	11.3k
used in	Sections 11.1–11.5	Sections 11.7–11.12	Sections 11.7, 11.9–11.12	as P, for reference

Limitations of the settings. Four limitations apply.

- The selection re-chose only f_1 for setting I, and $(\vartheta^\star, \alpha_L, f_1)$ for setting P. Setting I keeps $\vartheta^\star = 2.5$ and $\kappa_\Psi = 2$, and setting P keeps $\kappa_\Psi = 2$; these values come from simulations of earlier variants of the rule and were not re-tuned under the two-stage rule with the exact radius.
- Setting I's criterion averages over n, so it trades accuracy after the switch-off for accuracy before it. Once the ridge is off, $f_1 = 0.25$ leaves 9.5% more noise s.d. in stage 2 than the previous setting, and the coefficient error reaches twice the sampling floor 9–13% later at $R^2 = 0.5$ (Section 11.7). A criterion that weights the ridge-off regime more, or an f_1 that depends on the planned n against n_{off}, was not evaluated.
- Setting P lies on the edge of its extended grid, at $\alpha_L = 0.75$, so its stage-1 radius is the 25% quantile of $\lambda_{\max}$ of the unit noise. Its refusal constraint binds: the six candidates with $\vartheta^\star = 1.25$ that do up to 0.3% better on the selection populations refuse 2–15% of releases in some block-normalized cells. P is therefore the best candidate of this grid under this bound, not an optimum; a wider grid or another bound could choose differently, and as pre-registered the grid was extended only once.
- Setting P was chosen after its first choice P_0 failed on block-normalized real data, and its selection populations cover σ_1/σ_2 down to 0.46 at $f_1 = 0.25$ only. Under block normalization, 8 of Wang's 29 datasets have a lower ratio, down to 0.25 (solar), and there its gate refuses more often than the 1% it was chosen under: up to 7.2% of releases on solar at $\varepsilon = 1$ (Section 11.9). P is not validated below the selection's noise ratio or on response distributions heavier-tailed than those of the 29 datasets. The ratio is public before any release, so a prediction setting that depends on it is possible; it was not evaluated.

Table 10: Parameters of the protocol. "Derived" marks quantities that CALIBRATE computes from the others. The last column is the evaluation setting, not part of the protocol.

Parameter	Role	Constraint	Evaluation setting
Security			
N	CKKS ring dimension; $N_s = N/2$ slots	power of two; with the chain, $\log_2(QP) \le L_{\max}(N, \lambda_{\text{sec}})$	2^{14}
λ_{sec}	computational security level	sets $L_{\max}$ [1]: 438 for $N = 2^{14}$ at 128 bits, ternary secrets	128 bits (classical)
chain q_0, q_1, q_2; P; d_{num}	last-level modulus; primes removed by the two rescales; special modulus and digits of key switching	$\log_2(QP) \le L_{\max}$; $q_0 > 2(\Omega_f S_{\max} + 2^\kappa B_{\text{tot},\max} + t_\eta \sigma_{\text{c},\max})$ (checked by CALIBRATE)	q_0: primes of 60 and 59 bits; q_1, q_2: 59 bits; P: two 60-bit primes; $d_{\text{num}} = 2$; $\log_2(QP) = 357$
secret and error distributions	RLWE assumptions of Theorem 7.2 and G1	uniform ternary secrets; discrete Gaussian errors, mean zero and σ_e^2-subgaussian (model (M5)), sampled honestly by both parties	$\sigma_e = 3.19$ (bounds use 3.2); OpenFHE's table sampler, subgaussianity checked numerically
δ_B	failure probability of $B_{\text{tot},s}$	$\delta_B \in (0,1)$; $n_{\text{rel}}(1+e^{\varepsilon_{\text{run}}})(\delta_B + 2^{-97}) < \zeta\delta_{\text{run}}$, and $n_{\text{rel}}(\delta_B + \delta')$ below the run's ledger share	2^{-50}

continued on the next page

Table 10, continued

Parameter	Role	Constraint	Evaluation setting
$\kappa_{\min}$	floor on the flooding parameter (statistical security)	integer ≥ 1	40
κ	flooding parameter	derived by (8); if fixed instead, $\delta_{\mathrm{DP}} > 0$	derived (Table 4)
t_η	noise margin in q_0, in units of $\sigma_{c,s}$	$t_\eta > 0$; correctness failure $2Ne^{-t_\eta^2/2}$ per release	10
Privacy of a run			
$(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$	budget of a run	$\varepsilon_{\mathrm{run}} > 0$, $\delta_{\mathrm{run}} \in (0,1)$; admitted by the ledger	$\varepsilon \in \{0.5, 1, 2, 4, 8\}$, $\delta = 10^{-9}$
f_1	share of ρ_{run} for stage 1	$f_1 \in (0,1)$, $f_1 + f_y < 1$	I and P: 0.25 (previous: 0.1)
f_y	share of a separate $\mathbf{y}^\top\mathbf{y}$ release (separate mode only)	$f_y \in (0, 1 - f_1)$; no protocol default	0.1 in the implementation; every evaluation run is in joint mode
ζ	largest share of δ_{run} the slack may take	$\zeta \in (0,1)$	0.1
mode, n_{rel}	$\mathbf{y}^\top\mathbf{y}$ in stage 2 or apart; number of releases	derived: joint ($n_{\mathrm{rel}} = 2$) iff $\mathbf{y}^\top\mathbf{y}$ is free (Remark 7.1)	derived
Ledger of a base register			
$(\varepsilon_L, \delta_L)$	cap over all runs on the register	$\varepsilon_L \geq \varepsilon_{\mathrm{run}}$, $\delta_L \geq \delta_{\mathrm{run}}$ for every run; public	one run per register: cap $= (\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$; example $(10, 10^{-6})$ in Table 4
ζ_L	share of δ_L reserved for slack	$\zeta_L \in (0,1)$	0.1
ξ_L	share of ε_L reserved for smudging	$\xi_L \in (0,1)$	10^{-3}
δ'	per-release hockey-stick slack of smudging	$\delta' \in (0,1)$	2^{-100}
admission rule	runs fixed in advance, or chosen adaptively; a run's full cost is charged at admission, atomically, without refund	the filter rule unless every run is fixed in advance (Section 7.4)	fixed (one run)
OPRF quota	membership look-ups per researcher and project	policy	none
Statistics			
$\vartheta^\star$	validity target of the ridge	$\vartheta^\star > 0$; the statistical results need $\vartheta^\star > 1$, the refusal bound $\vartheta^\star > 2$; coverage is asymptotic in $\vartheta^\star$ (Remark 9.2)	I: 2.5; P: 1.5
r_1	stage-1 radius (Section 9.2)	union bound, or an upper confidence bound on the exact $(1 - \alpha_L)$ quantile of the noise; both from public quantities	exact quantile, 99% upper bound (I and P); union bound in the previous setting
κ_Ψ	when to shrink only O's block	$\kappa_\Psi \geq 0$	2 (all settings)
$\alpha = \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U$; intervals	error level and its split; intervals on or off	with intervals on, each part > 0 and $\alpha < 1$; with them off, only $\alpha_L \in (0,1)$, which sizes r_1	I: $\alpha = 0.05$, $\alpha_L = \alpha_U = 0.0025$; P: intervals off, $\alpha_L = 0.75$

continued on the next page

Table 10, continued

Parameter	Role	Constraint	Evaluation setting
η_B, B	bootstrap truncation and number of draws	$\eta_B \in (0,1)$; Monte Carlo error of relative order $B^{-1/2}$	1/2; 2000
B_β^{pub}	public a-priori bound on $\|\boldsymbol{\beta}\|$	> 0, or ∞	∞ unless stated
$\bar{\sigma}^2$	public bound on $\mathrm{Var}(\tilde{y}_i \mid \mathbf{X})$ for the bias-aware interval	$0 < \bar{\sigma}^2 \le B_y^2$, fixed before the data are used; below B_y^2 only as a declared bound on the error variance before clipping when only $\mathbf{y}$ is clipped	B_y^2
bias correction	optional correction (16)	on or off	off
Data			
B_R, B_O, B_y	clipping bounds on the preprocessed data	public and fixed before the data are used (reference data or domain knowledge); $B_R \ge 1$ covers the intercept	99.5% quantiles on a public reference split; simulation population $(5.59, 6.24, 2.89)$; block normalization: $B_R^2 = 2$, $B_O = 1$
preprocessing	per-party transforms (Section 6.1)	fixed functions of one record and public constants (Proposition 6.1)	standardize; optionally winsorize at $\|z\| \le 2$ or block unit norm
Φ, $\bar{n}_O$	study scope; padded register size	Φ on public attributes only; $\bar{n}_O \ge \|D_\Phi\|$, public	whole register; $\bar{n}_O$ from 4,096 to 65,536
$n_R^{\max}$	certified cohort cap	$n_R \le n_R^{\max}$, public	cohort size plus a margin
Labeled PSI			
N_B, t, F	BFV ring dimension; plaintext modulus; field elements per item (the item length is $F\log_2(t-1)$)	t prime, $t \equiv 1 \bmod 2N_B$; $p_{\mathrm{pass}} \le \delta_{\mathrm{psi}}$ for false matches (Appendix C), checked at Setup	2^{14}; $t \approx 2^{32}$; $F = 2$ (64-bit items)
h, N_{bins}, d_{cap}, δ_{psi}	cuckoo hashing, bins, largest partition degree, failure probability	sizes are public functions of these; failures are correctness events, not charged to δ	3; 2^{14}; 64; 2^{-40}
κ_ψ, B_ψ, $\delta_{B,\psi}$, q'	flooding of the PSI replies (SanitizeBFV)	$\sigma_\psi \ge \max\{2^{\kappa_\psi} B_\psi/\sqrt{2\pi}, 2^{42}\}$; $q'/(2t) > t_\eta \sigma_\psi + B_\psi + t$; B_ψ a worst-case bound (Section 8)	40; 2^{21} (evaluation noise measured, not bounded); 0; two 60-bit limbs
Encoding			
Ω_U	scale of R's vectors	exactly representable; $\Omega_U \ge \Omega_f$ keeps B_{tot} independent of $\bar{n}_O$	2^{68}
Ω_u	scale of the one-hot masks	exactly representable	q_1
Ω_V, Ω_f	scale of O's data; final scale	derived: smallest $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1 q_2)$ that floods, $\sigma_{c,s} \ge \max\{2^{\kappa} B_{\mathrm{tot},s}/\sqrt{2\pi}, 2^{42}\}$, tracked exactly	derived (Table 4)

Derived quantities. Table 11 follows Calibrate through one run at the previous setting ($f_1 = 0.1$). At setting I ($f_1 = 0.25$) the split gives $\rho_1 = 0.00374$ and $\rho_2 = 0.01123$, so $\sigma_1 = 892.4$ and $\sigma_2 = 575.2$, and Ω_f is 0.13 bits lower ($2^{62.62}$); setting P has the same f_1 and so the same

noise (P_0, with $f_1 = 0.2$: $\sigma_1 = 997.8$ and $\sigma_2 = 556.9$). κ, γ, δ_{DP} and ρ_{run} do not depend on f_1.

Table 11: Example: the derived quantities at the previous setting ($f_1 = 0.1$) for the simulation population ($d_R = 4$, $d_O = 6$, $p = 11$), $\varepsilon_{\mathrm{run}} = 1$, $\delta_{\mathrm{run}} = 10^{-9}$, a single-run ledger and $\bar{n}_O = 4{,}096$, as computed by the implementation.

Quantity	How	Value
Δ, Δ_G	DeltaRep (Proposition 7.1)	86.15, 77.16
mode, n_{rel}	$\mathbf{y}^\top\mathbf{y}$ free (Remark 7.1)	joint, 2
κ	(8); $\kappa_{\mathrm{run}} = 37$, so the floor binds	40
γ; slack $(1 + e^{\varepsilon})\gamma$	step 2	1.82×10^{-12}; 0.68% of δ_{run}
δ_{DP}; ρ_{run}	step 3, tight conversion	9.93×10^{-10}; 0.01497
ρ_1, ρ_2	split by f_1	0.00150, 0.01347
σ_1, σ_2	(7)	1410.3, 524.9
Ω_V; Ω_f	step 4	$2^{53.75}$; $2^{62.75}$
$B_{\mathrm{tot},1}$, $B_{\mathrm{tot},2}$	Theorem 7.1, OpenFHE's gadget	$2^{26.43}$, $2^{26.59}$
$\Delta_{x,1}$, $\Delta_{x,2}$	(6)	$2^{27.46}$, $2^{27.57}$
$\sigma_{c,1}$, $\sigma_{c,2}$	$\lceil \widehat{\Delta}^2_{\mathrm{enc},s}/(2\rho_s) \rceil^{1/2}$	$2^{66.71}$, $2^{65.29}$
q_0 needed	step 5	$2^{81.0}$ ($2^{89.0}$ at $\bar{n}_O = 10^6$); available 2^{119}
$\log_2(QP)$	chain	$357 \leq 438$

The chain in the evaluation. The implementation uses $N = 2^{14}$, a ciphertext modulus of four primes of 60, 59, 59 and 59 bits (q_0 is the product of the first two, about 2^{119}; q_1 and q_2 are the others), two 60-bit special primes and decomposition number 2, for $\log_2(QP) = 357$, below the 128-bit limit of 438 for $N = 2^{14}$. A 64-bit build of OpenFHE keeps every prime below 2^{60}. Honest inputs need q_0 of 80–92 bits over $\bar{n}_O \in [4{,}096, 10^6]$ and $\varepsilon \leq 4$, and 98 bits at $\varepsilon = 8$ (Table 4); the specification's "70–73 bits" is too small. $N = 2^{15}$ brings no gain: it makes the aggregation at 10^6 rows about 40% slower.

11 Empirical evaluation

All results in this section are for protocol v3.1: its calibration, the full OpenFHE implementation (Sections 11.1 and 11.5), and plaintext simulations of its release and solve steps, which reproduce the encrypted runs (Sections 11.5 and 11.7). Each subsection states its settings. They are evaluation settings (Section 10), not part of the protocol. Sections 11.1–11.5 use the previous setting: $f_1 = 0.1$, the union-bound radius, $\vartheta^\star = 2.5$, $\kappa_\Psi = 2$ and $\alpha = 0.05$ with $\alpha_L = \alpha_U = \alpha/20$. Sections 11.7–11.12 use the two named settings of Table 9: I (inference) and P (prediction), with the previous setting as a reference. Unless stated otherwise $\delta = 10^{-9}$ and the bias correction is off. Every study fixed the flooding parameter in advance instead of deriving it by (8); each subsection states its values. The simulation studies, the benchmarks and the encrypted checks of settings I and P use $\kappa = 40$, 40, 40, 44 and 48 for $\varepsilon = 0.5$, 1, 2, 4 and 8. The rule gives 40, 40, 40, 41 and 46 at $\delta = 10^{-9}$ and 40 at every ε at $\delta = 10^{-6}$. Using it would change σ_1 and σ_2 by less than 0.2% (at most 0.15%, at $\varepsilon = 8$ and $\delta = 10^{-9}$), because κ enters the noise only through the slack. Where a subsection compares with the earlier single-release mechanism of the previous version of this paper (one analytic-Gaussian release, a ridge set from a committed eigenvalue bound, and the bias correction applied), it says so. The scripts that produce every number are in the `experiments/` directory of the project repository.

11.1 The cryptographic layer is exact

A prototype of the linkage and aggregation steps was run end to end over sockets, on six cohorts ($n = 200$ to 64,000 matched records out of $n_O = 2{,}000$ to 80,000) with a plaintext backend, and on one cohort ($n = 400$ out of $n_O = 4{,}000$) with OpenFHE 1.5.1 (BFV, $N = 16{,}384$, for the linkage; CKKS with scale 2^{50} for the aggregation). Labeled PSI recovered exactly the true match set in every run, with no misaligned rows (count, set and row-by-row alignment checked). With the noise switched off, the regression agreed with its plaintext target to 2.9×10^{-10} with the plaintext backend and 1.7×10^{-10} with OpenFHE. The OpenFHE run took about 13 s end to end. The prototype hashes identifiers with SHA-256 instead of an OPRF, uses a single CKKS key, and has no release step.

Full implementation of v3.1. The `implementation/` directory of the repository implements every phase of v3.1 on OpenFHE 1.5.1, with R and O as separate processes and a transcript of every message: RFC 9497 OPRF (ristretto255), labeled PSI under R's BFV key with flooded replies, threshold key generation, the seeded upload and key conversion, the aggregation circuit, and the release with exact discrete-Gaussian noise inside O's share. A small C++ layer provides the polynomial-level access that the release needs. On the toy example and on a 4,096-row synthetic register it recovers exactly the true match set and alignment; with the noise switched off by a test-only flag the decrypted statistics match their plaintext values to 10^{-11} relative; with the noise on, the measured noise s.d. is $0.996\,\sigma_2$ and $0.976\,\sigma_1$; the randomness-dependent encryption error stays about 8 bits below the worst-case B_{tot}; and R never receives c_0' while O never receives anything that could serve as a decryption share of R. Its deviations from the specification are listed in `implementation/DEVIATIONS.md`.

11.2 Calibration under v3.1

Table 12 gives σ_2/Δ for two releases with $f_1 = 0.1$ (the previous setting) and $\mathbf{y}^\top\mathbf{y}$ free; σ_2 scales as $1/\sqrt{1-f_1}$, so it is 1.095 times larger at setting I ($f_1 = 0.25$) and at setting P, which has the same f_1; Table 13 compares it with the single analytic-Gaussian release the earlier experiments simulated. The planning formula of (7) is conservative by 6.1–33.4% over $\varepsilon \in [0.1, 8]$ and $\delta \in [10^{-9}, 10^{-6}]$: 6.1% at $(\varepsilon, \delta) = (8, 10^{-9})$ and 33.4% at $(0.1, 10^{-6})$.

Table 12: Stage-2 noise σ_2/Δ under v3.1 (two releases, $f_1 = 0.1$ as in the previous setting, tight conversion).

$\delta \setminus \varepsilon$	0.1	0.25	0.5	1	2	4	8
10^{-6}	41.6	17.6	9.15	4.78	2.51	1.34	0.727
10^{-8}	51.6	21.4	11.0	5.68	2.95	1.55	0.825
10^{-9}	56.0	23.1	11.9	6.09	3.15	1.64	0.871

Table 13: σ_2 of v3.1 divided by the single-release analytic-Gaussian σ at the same (ε, δ).

$\delta \setminus \varepsilon$	0.1	0.25	0.5	1	2	4	8
10^{-5}	1.16	1.16	1.15	1.14	1.14	1.13	1.12
10^{-6}	1.15	1.14	1.14	1.13	1.13	1.12	1.11

Table 14 works the calibration through for the simulation population of Section 11.7 at $\delta = 10^{-9}$. Learning the ridge from stage 1 costs data: at $\varepsilon = 1$ the ridge stays on until about 51k matched records at the previous setting (33k at setting I), against 16k for a single release with the eigenvalue bound known in advance.

Table 14: Worked calibration at the previous setting ($f_1 = 0.1$, union-bound radius), simulation population ($B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$, $p = 11$, $\Delta = 86.15$, $\Delta_G = 77.16$, population eigenvalue bound $\ell = 0.494$), $\delta = 10^{-9}$. "Ridge off from": the n above which the stage-1 rule gives $\lambda = 0$ (typically), and for comparison a single release with ℓ known.

ε	ρ_{run}	σ_1	σ_2	r_1	λ at $n = 10$k	ridge off from	one release, ℓ known
0.5	0.0040	2744	1021	31637	43633	98k	31k
1	0.0150	1410	525	16256	20018	51k	16k
2	0.0561	728	271	8396	7951	26k	8.2k
4	0.2063	380	141	4379	1784	14k	4.3k
8	0.7328	202	75	2324	0	7.2k	2.3k

11.3 Interval coverage under v3.1

Coverage here is unconditional (Section 9.3): it is counted over all releases, and a refused release makes no claim, while it still spends the budget. In the simulations of setting I (Section 11.7) and in the real-data check of Table 17, the gate refused no release, so conditional and unconditional coverage coincide there. Table 15 simulates the release and solve steps of Algorithm 6 (2,000 replications per scenario, $p = 11$), in scenarios chosen to exercise each branch of the ridge rule. The Wald interval hits its target with or without the ridge. The bias-aware interval for $\boldsymbol{\beta}$ in Table 15 is the working-model interval, with $\tilde{s}^2$; it is conservative, because it uses the worst-case bias, and it is unbounded when the ridge is on and no public bound on $\|\boldsymbol{\beta}\|$ is given. Table 16 compares it with the guaranteed interval, with $\bar{\sigma}^2$, including populations where clipping binds. For contrast, a naive Wald interval around a ridge estimate, read as an interval for $\boldsymbol{\beta}$, covered only 42–81% in the earlier simulations when no eigenvalue bound was available.

Table 15: Coverage of both intervals over the 11 coefficients (simulation of the release and solve steps; noise levels set directly).

Scenario	n	Ridge	Wald for $\boldsymbol{\theta}_\lambda$ (target 0.955)	Bias-aware for $\boldsymbol{\beta}$, working model (target ≥ 0.95)
Ridge off	4,000	$\lambda = 0$	0.946–0.958	0.945–0.968
Ridge on, whole matrix	1,500	$\Psi = \mathbf{I}$, median $\lambda = 477$	0.948–0.960	1.000 with a public bound on $\|\boldsymbol{\beta}\|$; unbounded without
Ridge on, O's block only	3,000	$\Psi = \Pi_O$, median $\lambda = 439$	0.944–0.962	unbounded without a public bound
Small n, public bound	200	$\Psi = \mathbf{I}$, median $\lambda = 1{,}021$	0.946–0.962	1.000

The guaranteed interval when clipping binds. Table 16 compares the two bias-aware intervals for $\boldsymbol{\beta}$ under setting I with the calibrated noise ($\delta = 10^{-9}$; 1,000 cohorts per cell, a fresh pair of releases per cohort, the implementation's bootstrap), on three populations with the features, feature bounds and slopes of the synthetic population ($R^2 = 0.5$ before clipping). *Base* is the synthetic population ($B_y = 2.89$; y clipped in 0.5% of rows). *y-clip* has the same homoskedastic errors and $B_y = 1.27$, the 80% quantile of $|y|$, so that 20% of the responses are clipped. *Heteroskedastic* has errors whose standard deviation grows with $|x|$ in O's first (lognormal) feature, with mean variance 0.5, and $B_y = 1.51$ (10% clipped); the noisier records are those of high leverage. Each population is calibrated with its own B_y. Coverage is of the projection coefficient $\boldsymbol{\beta}^* = \mathbf{G}^{-1}\tilde{\mathbf{X}}^\top \mathbb{E}[\tilde{\mathbf{y}} \mid \mathbf{X}]$ of each cohort, the estimand of Proposition S10b, computed in closed form from the Gaussian errors; coverage of the population coefficient of the clipped data differs from it by at most 0.04 (reports/v31_sigmabar).

- The guaranteed interval, with $\bar{\sigma}^2 = B_y^2$, covers every slope in every cell: 0.956–1.000 for single slopes, 0.967–1.000 pooled. Where the sampling variance dominates it is close to 1 and wide: on the synthetic population at $\varepsilon = 8$ its median half-width is 2.3–3.7 times that of the working-model interval, close to $\sqrt{B_y^2/\tilde{s}^2} = 4.1$ at $n = 10^5$. With a smaller B_y the factor is smaller: 1.4–1.9 for y-clip and 1.6–2.4 for the heteroskedastic population at $\varepsilon = 8$, and at most 1.5 at $\varepsilon = 1$.
- The working-model interval covers the slopes as well as the Wald interval covers $\boldsymbol{\theta}_\lambda$ on the base and y-clip populations (0.936–0.977 per slope). With homoskedastic errors before clipping, clipping lowers the variance of the records near the bound, which are not the ones of high leverage, so $\tilde{s}^2$ is not too small there. On the heteroskedastic population it under-covers the slope of the feature that drives the variance at $\varepsilon = 8$, increasingly with n: 0.923, 0.894 and 0.829 at $n = 10$k, 25k and 10^5 (the population coefficient: 0.908, 0.867 and 0.788), while it is still 0.945–0.967 at $\varepsilon = 1$, where the mechanism noise dominates.
- With the declared bound $\bar{\sigma}^2 = 0.5$, the error variance before clipping, which is admissible for the base and y-clip populations, the interval covers 0.936–0.985 per slope and is as wide as the working-model interval on the base population and at most 14% wider on y-clip.

Where the ridge is on, both intervals are unbounded while $\hat{L} = 0$, and dominated by the bias bound otherwise (half-widths within 3% of each other at $n = 25$k, $\varepsilon = 1$).

Real data with calibrated noise. Table 15 sets the noise levels directly. Table 17 and Figure 2 repeat the check for the Wald interval on real data with the noise of a real budget: setting I, σ_1 and σ_2 from Calibrate at $\delta = 10^{-9}$ and $\varepsilon \in \{1, 4, 8\}$, and block normalization with the reference rows of fold 0 (Section 11.9). The data are fixed. The cohort is the first n matched rows of fold 0, or all of them, so $\boldsymbol{\theta}_\lambda$ is a fixed function of the data and of each release's λ. Each cell has 1,000 independent releases of the whole mechanism, with the implementation's bootstrap (2,000 antithetic pairs). Coefficients of one release share its noise, so each interval in the table is clustered by release.

- On protein, bike, elevators and pol, block normalization keeps the ridge on in every release at every size these datasets allow. Protein's n_{off} is about 80k to 560k for $\varepsilon = 8$ to 1, and the block-normalized designs of bike and elevators are singular. We add 3droad ($p = 4$), where the ridge is off from $n = 1{,}000$ at $\varepsilon \geq 4$ and from 5,000 at $\varepsilon = 1$, so that both regimes are checked on real data.

Table 16: The two bias-aware intervals for $\boldsymbol{\beta}$, setting I, $\delta = 10^{-9}$, 1,000 cohorts per cell, ridge off in every release: coverage of the projection coefficient $\boldsymbol{\beta}^*$ (pooled over the ten slopes, and the lowest slope) and median half-width. Guaranteed: $\bar{\sigma}^2 = B_y^2$; working model: $\tilde{s}^2$. Target 0.955 per slope.

			guaranteed ($\bar{\sigma}^2 = B_y^2$)			working model ($\tilde{s}^2$)		
population (B_y^2)	ε	n	pooled	lowest	half-width	pooled	lowest	half-width
base (8.34)	1	100,000	0.996	0.994	0.026	0.948	0.936	0.018
	8	25,000	1.000	1.000	0.040	0.952	0.944	0.014
	8	100,000	1.000	1.000	0.020	0.953	0.937	0.0054
y-clip (1.61)	1	100,000	0.977	0.971	0.017	0.957	0.947	0.015
	8	25,000	0.999	0.997	0.019	0.961	0.948	0.012
	8	100,000	1.000	1.000	0.0088	0.963	0.950	0.0046
heteroskedastic (2.28)	1	100,000	0.981	0.974	0.018	0.956	0.945	0.015
	8	25,000	1.000	0.998	0.022	0.953	0.894	0.011
	8	100,000	1.000	0.999	0.010	0.950	0.829	0.0043

- The Wald interval covers $\boldsymbol{\theta}_\lambda$ 0.945–0.964 of the time over the 48 cells, and single coefficients 0.935–0.979. On protein it covers 0.952–0.963 in all twelve cells, and every protein interval contains the target or lies above it. The gate refused no release.
- While the ridge is on, the interval leans conservative: 26 of the 40 ridge-on cells lie above the target (elevators 0.959–0.964). One cell lies below: 3droad at $n = 25{,}000$, $\varepsilon = 1$ (0.945). Rerun with 5,000 fresh releases, it gives 0.955 (0.952–0.959).
- Coverage of $\boldsymbol{\beta}$ cannot be checked this way. With a fixed cohort, coverage of the full-data least-squares fit (the reference of Section 11.4) is conditional on one sample, whereas the bias-aware interval's guarantee holds over the sampling and the noise together. Where the mechanism noise is small against the sampling error, each coefficient is covered in almost every release or in almost none (3droad at $n = 25{,}000$: 0.67–0.85 over three slopes). While $\hat{L} = 0$ the interval is unbounded anyway. Section 11.4 checks the intervals for $\boldsymbol{\beta}$ on real data with resampled cohorts.

11.4 Interval methods: the parametric bootstrap of Ferrando, Wang and Sheldon

Our Wald interval borrows the parametric bootstrap of Ferrando, Wang and Sheldon [28] (FWS). Their interval for differentially private least squares is also a competitor to our interval for $\boldsymbol{\beta}$. This subsection runs it on the releases of setting I and compares coverage and width, estimand by estimand.

The FWS interval. From released statistics $\hat{\mathbf{G}}$, $\hat{\mathbf{c}}$ and $\widehat{yy}$, FWS take the unregularized estimate $\hat{\boldsymbol{\beta}} = \hat{\mathbf{G}}^{-1}\hat{\mathbf{c}}$, the scaled design $\hat{Q} = \hat{\mathbf{G}}/n$, replaced by its projection with eigenvalues at least 0.1 if it is not positive definite, and a residual variance $\hat{s}^2$. Each bootstrap draw simulates the sampling noise of $\mathbf{y}$ given the design and a fresh draw $(\mathbf{E}_b^*, \mathbf{f}_b^*)$ of the mechanism noise,

$$\boldsymbol{\beta}_b^* = \big(\hat{Q} + \mathbf{E}_b^*/n\big)^{-1}\big(\hat{Q}\hat{\boldsymbol{\beta}} + \mathbf{Z}_b^*/\sqrt{n} + \mathbf{f}_b^*/n\big), \qquad \mathbf{Z}_b^* \sim \mathcal{N}(\mathbf{0}, \hat{s}^2\hat{Q}),$$

and the interval is the percentile interval of the draws, coordinate by coordinate. We reimplemented it from their code (`DPCIs-OLS.py` of `github.com/ceciliaferrando/PB-DP-CIs`,

Table 17: Real data, block normalization, setting I, calibrated noise ($\delta = 10^{-9}$), fixed cohort of the first n matched rows of fold 0 (the last n of each dataset but 3droad is all of them): coverage of $\boldsymbol{\theta}_\lambda$ by the Wald interval over all coefficients, 1,000 releases per cell, with a release-clustered 95% interval. Target 0.955. Ridge: on or off in every release of the cell (3droad at $n = 1{,}000$: on at $\varepsilon = 1$, off at 4 and 8). [a]Rerun with 5,000 fresh releases: 0.955 (0.952–0.959).

dataset	p	n	ridge	$\varepsilon = 1$	$\varepsilon = 4$	$\varepsilon = 8$
protein	10	1,000	on	0.962 (0.958–0.966)	0.960 (0.955–0.964)	0.959 (0.954–0.963)
		5,000	on	0.963 (0.960–0.967)	0.961 (0.957–0.965)	0.955 (0.950–0.960)
		20,000	on	0.958 (0.953–0.962)	0.959 (0.954–0.964)	0.952 (0.947–0.958)
		36,584	on	0.958 (0.953–0.962)	0.958 (0.952–0.963)	0.953 (0.947–0.959)
bike	18	1,000	on	0.962 (0.959–0.965)	0.961 (0.958–0.964)	0.959 (0.956–0.962)
		5,000	on	0.960 (0.957–0.963)	0.959 (0.955–0.962)	0.956 (0.952–0.959)
		13,903	on	0.958 (0.955–0.962)	0.959 (0.955–0.962)	0.956 (0.952–0.959)
elevators	19	1,000	on	0.963 (0.960–0.966)	0.963 (0.960–0.966)	0.962 (0.959–0.965)
		5,000	on	0.961 (0.958–0.964)	0.961 (0.958–0.964)	0.961 (0.958–0.964)
		13,279	on	0.964 (0.961–0.967)	0.962 (0.959–0.965)	0.959 (0.956–0.962)
pol	27	1,000	on	0.961 (0.958–0.963)	0.960 (0.958–0.962)	0.958 (0.956–0.961)
		5,000	on	0.958 (0.956–0.961)	0.960 (0.958–0.963)	0.956 (0.953–0.959)
		12,000	on	0.960 (0.958–0.963)	0.957 (0.954–0.960)	0.959 (0.956–0.962)
3droad	4	1,000	on/off	0.955 (0.949–0.961)	0.961 (0.955–0.968)	0.951 (0.944–0.959)
		5,000	off	0.959 (0.953–0.966)	0.960 (0.953–0.967)	0.954 (0.947–0.961)
		25,000	off	0.945[a] (0.937–0.952)	0.956 (0.949–0.963)	0.956 (0.949–0.963)

commit `f0d2bc6`). Fed the same inputs and the same random numbers, the two give the same bootstrap draws to 10^{-15} (relative). On FWS's own simulation ($N = 1{,}000$, $d = 4$, $\varepsilon = 5$, Laplace noise, 400 trials) their pipeline covers 0.958 and ours 0.950, with median widths 0.581 and 0.585. We run two versions, both with 2,000 draws at level $1 - \alpha_{\mathrm{CI}} = 0.955$, the per-coordinate level of our intervals under setting I.

- *FWS on the stage-2 release:* the same $\hat{\mathbf{G}}_2$ (with R's exact $\mathbf{A}$), $\hat{\mathbf{c}}_2$ and $\widehat{yy}_2$ from which our intervals are computed, and our noise law in the bootstrap: s.d. σ_2 on the O-dependent entries and zero on $\mathbf{A}$. FWS release s^2 through a separate mechanism; here $\hat{s}^2 = (\widehat{yy}_2 - \hat{\mathbf{c}}_2^\top \hat{\boldsymbol{\beta}})/(n-p)$, set to 0.1 when negative, as FWS do with theirs. Stage 1 is not used.
- *FWS with one release:* the whole budget on a single release of the stage-2 statistics, with no stage 1. This is the natural FWS mechanism under our adjacency and accountant. Its noise s.d. is $\sqrt{1 - f_1} = 0.87$ times σ_2 (for example 498 against 575 at $\varepsilon = 1$ on the synthetic population).

Both versions use only post-processing of statistics that the vertical protocol releases, so both could run in the vertical setting.

Estimands. The methods target different quantities, so we report every method's coverage of every estimand.

- The ridge target $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$ is a function of the matched data and of λ, and only the mechanism noise enters its uncertainty. Our Wald interval targets it. It is defined

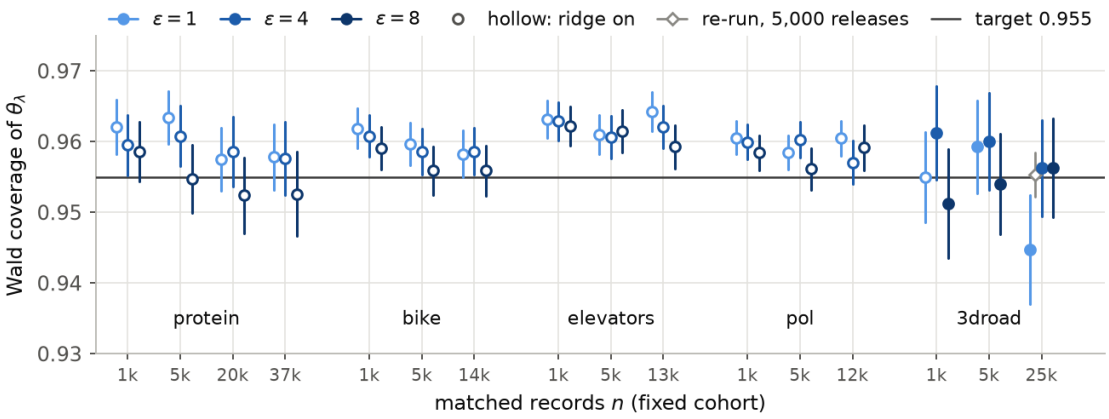


Figure 2: The coverage of Table 17: Wald coverage of $\boldsymbol{\theta}_\lambda$ under setting I with calibrated noise on fixed real cohorts, with release-clustered 95% intervals; colour: ε; hollow: the ridge is on. Grey diamond: the rerun of the one cell whose interval lies below the target. Horizontal line: the target 0.955.

only for a release that chooses a ridge; with $\lambda = 0$ it is the sample least-squares coefficient $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$.

- The regression coefficient $\boldsymbol{\beta}$ of the working model for the clipped data (Section 9.1). FWS target it, and so does our bias-aware interval; both intervals include the sampling variance. On the synthetic population we take the population least-squares coefficient of the clipped data, from 4 million rows. On real data we take the least-squares fit on all rows of the dataset outside the reference set.
- On the synthetic population, also the generating coefficients before clipping.

FWS's draws resample $\mathbf{y}$ given the design, so its interval is too wide for the sample coefficient $\hat{\boldsymbol{\beta}}$. It has no ridge, so it has no $\boldsymbol{\theta}_\lambda$ of its own.

Synthetic population. The cohorts, releases and intervals of setting I are those of Section 11.7 ($\delta = 10^{-9}$, $R^2 = 0.5$, 200 cohorts per n from 100 to 10^5, five values of ε, a fresh pair of releases per cohort and ε); our half-widths reproduce the published ones to 10^{-14}. The ridge is on in every release in 41 of the 70 (n, ε) cells, off in every release in 27, and mixed in 2. Table 18 and Figure 3 give coverage of the ten slopes and median half-widths.

Results on the synthetic population.

- *The Wald interval and $\boldsymbol{\theta}_\lambda$.* The Wald interval covers $\boldsymbol{\theta}_\lambda$ 0.946–0.967 in every cell, with the ridge on or off, and its half-width is 0.056–0.058 while the ridge is on. FWS's interval is not centred on $\boldsymbol{\theta}_\lambda$: in the ridge-on cells where it is narrow enough to be informative it covers $\boldsymbol{\theta}_\lambda$ only 0.57–0.86.
- *Ridge off: FWS on our release is our working-model bias-aware interval.* In the 27 cells where the ridge is off in every release, the two intervals for $\boldsymbol{\beta}$ have the same median half-width to within 1% (0.0176 against 0.0176 at $n = 10^5$, $\varepsilon = 1$; 0.0053 against 0.0054 at $\varepsilon = 8$) and cover 0.943–0.958 (FWS) and 0.946–0.962 (bias-aware). With $\lambda = 0$ both

Table 18: Synthetic population, setting I, $\delta = 10^{-9}$, $R^2 = 0.5$, 200 cohorts per n: coverage of the ten slopes and, in brackets, median half-width (standardized scale; the slopes lie between -0.27 and 0.45). $\boldsymbol{\theta}_\lambda$: ridge target; $\boldsymbol{\beta}$: population least-squares coefficient of the clipped data. "ridge": share of releases with $\lambda > 0$. Bias-aware: the working-model interval, with $\tilde{s}^2$ (the guaranteed interval, with $\bar{\sigma}^2 = B_y^2$, is compared in Table 16); "unbounded" when no interval is finite; square brackets give the share of finite intervals and their median half-width when only some are; an unbounded interval counts as covering. Target 0.955 for every interval.

			v3.1			FWS	
ε	n	ridge	Wald for $\boldsymbol{\theta}_\lambda$	Wald read for $\boldsymbol{\beta}$	bias-aware for $\boldsymbol{\beta}$	stage-2 release, $\boldsymbol{\beta}$	one release, $\boldsymbol{\beta}$
1	1,000	1.00	0.955 (0.057)	0.102	unbounded	1.000 (10.5)	1.000 (10.6)
	5,000	1.00	0.967 (0.057)	0.164	unbounded	1.000 (5.33)	0.999 (2.43)
	7,500	1.00	0.959 (0.056)	0.192	unbounded	0.972 (0.404)	0.965 (0.278)
	10,000	1.00	0.957 (0.056)	0.223	unbounded	0.960 (0.212)	0.951 (0.172)
	25,000	1.00	0.960 (0.057)	0.798	1.000 (0.299)	0.957 (0.070)	0.949 (0.061)
	50,000	0.00	0.963 (0.034)	0.956	0.960 (0.035)	0.958 (0.035)	0.953 (0.030)
	100,000	0.00	0.958 (0.017)	0.949	0.959 (0.018)	0.958 (0.018)	0.958 (0.015)
8	500	1.00	0.963 (0.057)	0.140	unbounded	1.000 (8.79)	1.000 (8.11)
	1,000	1.00	0.953 (0.056)	0.188	unbounded	0.980 (0.582)	0.972 (0.342)
	2,500	1.00	0.954 (0.057)	0.401	1.000 [98% fin.; 1.21]	0.951 (0.107)	0.948 (0.093)
	10,000	0.00	0.956 (0.024)	0.921	0.960 (0.029)	0.958 (0.029)	0.956 (0.026)
	100,000	0.00	0.955 (0.0024)	0.617	0.958 (0.0054)	0.956 (0.0053)	0.949 (0.0052)

add the same two variances, the mechanism's and the sampling variance, to the same estimate; one by a percentile bootstrap and the other by a Wald formula. FWS with one release covers 0.947–0.961 and is 3–13% narrower (0.0154 at $n = 10^5$, $\varepsilon = 1$), because it does not spend a quarter of the budget on stage 1; the difference shrinks as the sampling variance takes over (2.6% at $n = 10^5$, $\varepsilon = 8$). All three are wider than the Wald interval, which has no sampling term: read as an interval for $\boldsymbol{\beta}$, the Wald interval covers only 0.62 at $n = 10^5$, $\varepsilon = 8$.

- *Ridge on: FWS is finite where the bias-aware interval is not.* While the private bound $\hat{L}$ is zero the bias-aware interval is unbounded: in every release at $n \leq 10$k for $\varepsilon = 1$ and at $n \leq 1$k for $\varepsilon = 8$. FWS uses no ridge and stays finite, and it covers $\boldsymbol{\beta}$ in all 41 ridge-on cells (0.951–1.000). It is informative only once $\lambda_{\min}(\mathbf{G})$ is large against the noise. At $\varepsilon = 1$ its median half-width is 9.5–11.8 for $n \leq 2.5$k (where $\hat{Q}$ had to be projected in 93–100% of releases), 5.3 at 5k, 0.40 at 7.5k, 0.21 at 10k, 0.12 at 15k and 0.070 at 25k ($n_{\text{off}} = 27$k). At $\varepsilon = 8$ it is 8.8–11.1 for $n \leq 500$, 0.58 at 1k and 0.11 at 2.5k ($n_{\text{off}} = 3.9$k). The change sits where $\vartheta_0 = \lambda_{\min}(\mathbf{G})/(2\sigma_2\sqrt{p})$ passes 1, near 6.4k records at $\varepsilon = 1$ and 0.9k at $\varepsilon = 8$. Between there and n_{off}, FWS gives a valid finite interval for $\boldsymbol{\beta}$ while ours is unbounded or, where $\hat{L} > 0$, wider (median 2.3 at 15k and 0.30 at 25k for $\varepsilon = 1$, finite ones only).
- *FWS with one release* behaves like FWS on our release, with half-widths 13–41% smaller in the ridge-on cells where both are informative (0.28 against 0.40 at 7.5k, 0.061 against 0.070 at 25k for $\varepsilon = 1$), and coverage 0.948–1.000.

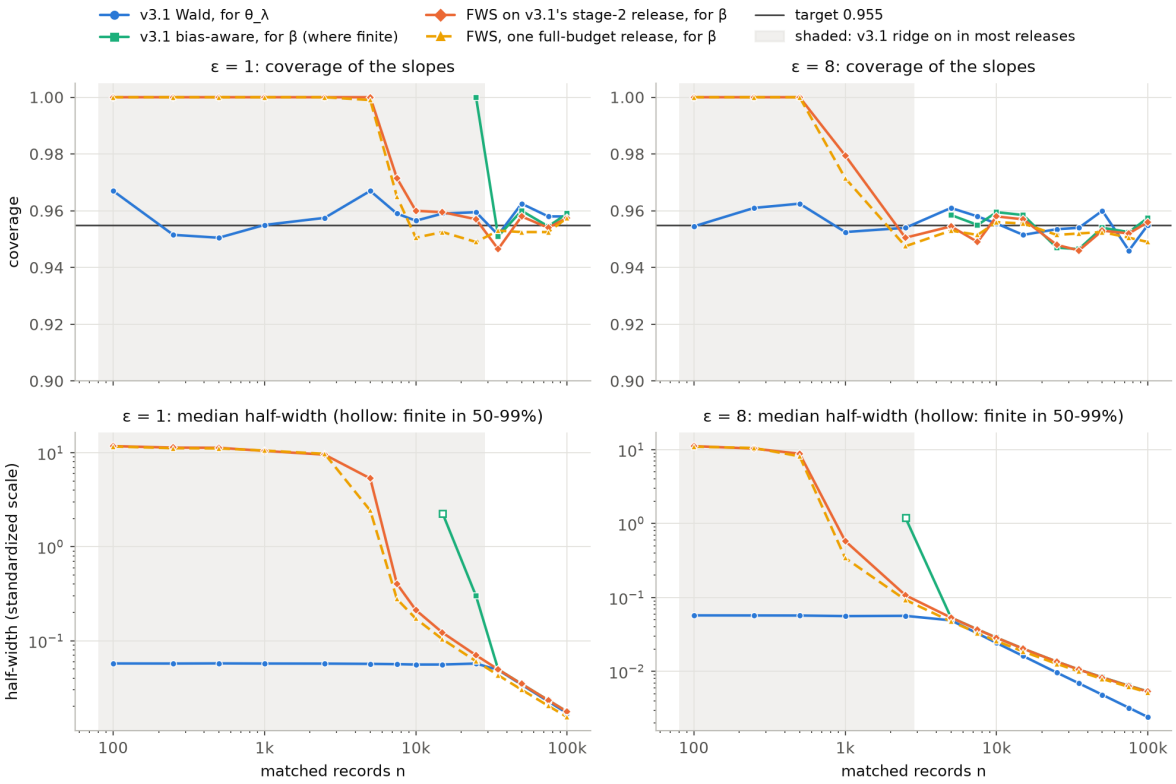


Figure 3: Synthetic population, setting I, $\delta = 10^{-9}$, $R^2 = 0.5$: coverage of the ten slopes (top) and median half-width (bottom, log scale) against n, for $\varepsilon = 1$ (left) and 8 (right). Each interval is shown for its own target: the Wald interval for $\boldsymbol{\theta}_\lambda$, the others for $\boldsymbol{\beta}$. The bias-aware interval is drawn only where at least half of its intervals are finite. Shaded: the stage-1 ridge is on in most releases.

Real data. We repeat the comparison on three of Wang's datasets with many rows (3droad, protein and bike), with block normalization and the reference rows of fold 0 as in Section 11.9. The rows outside the reference set form a finite population, and the least-squares fit of the clipped y on the clipped design over all of them is the reference coefficient $\boldsymbol{\beta}$. Cohorts of n rows are drawn from it without replacement, 200 per n, and on each the release, the intervals and both FWS versions are computed as above ($\delta = 10^{-9}$). A cohort is at most a sixth of the population, so the finite-population correction, which makes every interval for $\boldsymbol{\beta}$ slightly conservative, is at most 17% of the sampling variance. We also ran elevators but leave it out here: its block-normalized design is nearly singular, with a population coefficient of 5.4×10^4, so its least-squares fit is not a meaningful reference. Table 19 gives the results.

- The Wald interval covers $\boldsymbol{\theta}_\lambda$ 0.938–0.978 in every cell of the four datasets (elevators included), with the ridge on or off.
- On 3droad ($p = 4$) the ridge is off in 11 of the 15 cells. There FWS on our release and the bias-aware interval coincide again (for example half-widths 0.088 and 0.089 at $n = 5$k, $\varepsilon = 1$), and both cover the population coefficient 0.945–0.970, although the real errors need not be homoskedastic. In the four ridge-on cells $\hat{L}$ is often positive and the

Table 19: Real data, block normalization, setting I, $\delta = 10^{-9}$, 200 cohorts per n drawn from the rows outside the reference set: coverage of the slopes and, in brackets, median half-width. $\boldsymbol{\beta}$: least-squares fit on all those rows (3droad 391,387, protein 41,157, bike 15,641). Target 0.955.

				v3.1		FWS	
dataset	ε	n	ridge	Wald for $\boldsymbol{\theta}_\lambda$	bias-aware for $\boldsymbol{\beta}$	stage-2 release, $\boldsymbol{\beta}$	one release, $\boldsymbol{\beta}$
3droad	1	1,000	1.00	0.960 (0.148)	1.000 (2.17)	0.987 (0.606)	0.968 (0.458)
	1	5,000	0.00	0.938 (0.079)	0.948 (0.089)	0.948 (0.088)	0.958 (0.078)
	1	25,000	0.00	0.948 (0.016)	0.952 (0.024)	0.957 (0.024)	0.968 (0.022)
	8	1,000	0.00	0.953 (0.057)	0.945 (0.107)	0.950 (0.107)	0.953 (0.102)
	8	5,000	0.00	0.968 (0.011)	0.970 (0.042)	0.970 (0.042)	0.968 (0.041)
	8	25,000	0.00	0.945 (0.0022)	0.962 (0.018)	0.958 (0.018)	0.955 (0.018)
protein	1	1,000	1.00	0.962 (0.089)	unbounded	1.000 (16.2)	1.000 (15.9)
	1	5,000	1.00	0.954 (0.087)	unbounded	0.998 (12.4)	0.999 (11.4)
	8	1,000	1.00	0.958 (0.086)	unbounded	0.992 (10.4)	0.992 (9.54)
	8	5,000	1.00	0.948 (0.074)	unbounded	0.994 (2.11)	0.983 (1.00)
bike	1	500	1.00	0.960 (0.071)	unbounded	1.000 (16.3)	1.000 (16.7)
	1	2,000	1.00	0.953 (0.071)	unbounded	1.000 (28.7)	1.000 (28.0)
	8	500	1.00	0.967 (0.071)	unbounded	1.000 (24.4)	1.000 (21.5)
	8	2,000	1.00	0.957 (0.078)	unbounded	0.940 (6.00)	0.943 (5.54)

bias-aware interval is finite but wider than FWS's (2.17 against 0.61 at $n = 1$k, $\varepsilon = 1$).

- On protein and bike the ridge is on in every cell (n_{off} is 555k on protein at $\varepsilon = 1$, Section 9.5), $\hat{L} = 0$, and the bias-aware interval is unbounded. FWS is finite but not informative: its median half-width is 2.1–29 on coefficients whose population values lie within ± 2, because $\lambda_{\min}(\mathbf{G})$ is far below the noise ($\vartheta_0 \ll 1$); $\hat{Q}$ had to be projected in every bike release and in most protein releases at $\varepsilon \leq 2$. It covers 0.93–1.00; the two cells below target are protein at $n = 2.5$k and bike at $n = 2$k, both at $\varepsilon = 8$ (0.929 and 0.940).

Why the estimands matter. The Wald interval is narrow because $\boldsymbol{\theta}_\lambda$ is a fixed function of the matched records: it answers what the protocol would have output without the privacy noise. FWS and the bias-aware interval answer a different question, what the coefficient of the working model is, and must add the sampling variance; with the ridge off they are the same interval. They differ in how they handle a poorly conditioned design. The bias-aware interval keeps the ridge and pays for its bias with a bound on $\|\boldsymbol{\beta}\|$, which is infinite while $\hat{L} = 0$. FWS drops the ridge and lets the bootstrap measure the instability of $\hat{\mathbf{G}}_2^{-1}$, which costs variance but no bound. Once $\vartheta_0 \gtrsim 1$ that variance is moderate, and FWS is the better interval for $\boldsymbol{\beta}$; below it, neither gives a useful interval for $\boldsymbol{\beta}$, and only $\boldsymbol{\theta}_\lambda$ has one.

Remark 11.1 (Proposal, not proved: an FWS-type interval for $\boldsymbol{\beta}$ while the ridge is on)**.** While the ridge is on, R could also report an FWS-type percentile interval for $\boldsymbol{\beta}$ from the unregularized stage-2 solve $\hat{\mathbf{G}}_2^{-1}\hat{\mathbf{c}}_2$, offered only when $\lambda_{\min}(\hat{\mathbf{G}}_2) \geq 2\sigma_2\sqrt{p}$, the gate with $\lambda = 0$. On the synthetic population it covers where the bias-aware interval is unbounded, from about $\vartheta_0 = 1$, roughly $0.25\, n_{\text{off}}$ (setting I). Its validity rests on the parametric bootstrap and is *not proved* for the noise structure of v3.1, and its gate would use the same release as the interval. It is not part of the protocol.

FWS release s^2 through their own mechanism; here it is the plug-in from the released $\mathbf{y}^\top\mathbf{y}$. Its residual sum of squares is biased downward by $\sigma_2^2 \operatorname{tr}\mathbf{P} + \hat{\boldsymbol{\beta}}^\top \mathbb{E}[\mathbf{EPE}]\hat{\boldsymbol{\beta}}$ to second order, with $\mathbf{P} = \mathbf{G}^{-1}$ and an explicit remainder (proofs, Part B, Proposition S10a). Correcting it by that term changes FWS's coverage of the ten slopes by at most 0.0005 in each of the 41 ridge-on cells of Table 18, and its median half-width by at most 3%: where the ridge is on, the mechanism noise dominates FWS's width, and the plug-in is floored at 0.1 in up to half of the releases.

11.5 End-to-end runs of the implementation

These runs use the full implementation: OPRF and labeled PSI, threshold CKKS upload and aggregation, two releases with exact discrete-Gaussian noise inside O's share, and the solver, with $\delta = 10^{-9}$ and about 40–45 s per run on one core per party. They use the previous statistical setting ($f_1 = 0.1$, union-bound radius, $\vartheta^\star = 2.5$). They were made with a fixed flooding parameter, $\kappa = 40$ at $\varepsilon \in \{1, 4\}$ and $\kappa = 48$ at $\varepsilon = 8$, before the rule (8), which gives 40, 41 and 46 at the evaluation settings. The difference changes Ω_f but σ_1 and σ_2 by less than 0.2%.

Real data. The protein dataset (45,730 rows, 9 features) is split into a public reference set of 10% (standardization and transform constants, 99.5% bounds), a test set of 20%, and O's register (32,011 rows, padded to 32,768). R holds 4 features and O the other 5 and y. R's cohort is 19,206 register members plus 1,000 people O does not hold. Table 20 gives five encrypted runs per setting. Every median lies inside the 95% range of a plaintext simulation of the same mechanism, so the encryption does not change what the mechanism produces. Pooled over all 30 runs, the Wald interval covers $\boldsymbol{\theta}_\lambda$ for 286 of 300 coefficients (target 0.955). A plaintext simulation of the same six configurations, with the noise levels of the encrypted runs and 10,000 releases each, covers 0.955–0.958, with release-clustered 95% intervals inside 0.953–0.960. The encrypted counts, 46 to 50 of 50 per configuration, are what batches of five runs give: a plaintext batch covers no more in 21–100% of draws (Figure 4).

Table 20: Protein, 19,206 matched records: test MSE of the private estimate (median of 5 encrypted runs), the 95% range of a plaintext simulation of the same mechanism (2,000 draws), non-private least squares under the same preprocessing, their ratio, and the median ridge. Predicting zero gives 1.010.

Preprocessing	Δ	ε	Encrypted	Simulated range	Non-private	Ratio	λ
standardize	87.3	1	0.883	0.871–0.896	0.728	1.21	29,678
standardize	87.3	4	0.815	0.807–0.821	0.728	1.12	7,321
winsorize $\lvert z\rvert \le 2$	40.5	1	0.850	0.838–0.855	0.726	1.17	12,552
winsorize $\lvert z\rvert \le 2$	40.5	4	0.788	0.784–0.795	0.726	1.09	3,088
block unit norm	8.4	1	0.793	0.786–0.795	0.745	1.06	1,385
block unit norm	8.4	4	0.759	0.757–0.760	0.745	1.02	338

Block normalization costs a little in the non-private fit (0.745 against 0.728) and gains much more in the private one: at $\varepsilon = 4$ its private error, 0.759, is within 4% of the best non-private model. The ridge is on in every setting, so the coefficients remain shrunk (63–82% of the way from least squares to zero with block normalization); prediction recovers faster than the coefficients do. Section 9.5 predicts these ridges to within 1–19% and traces them to a

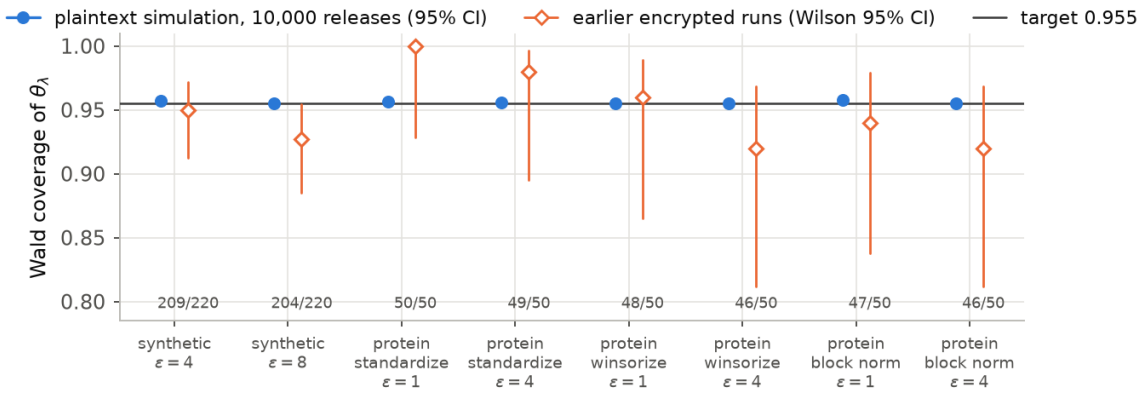


Figure 4: Wald coverage of $\boldsymbol{\theta}_\lambda$ in the configurations of the end-to-end runs of this section (previous setting): a plaintext simulation of the same mechanism on the same data with the noise levels of the encrypted runs (10,000 releases, release-clustered 95% interval) against the encrypted runs (20 per synthetic and 5 per protein configuration; Wilson 95% interval, which ignores the clustering and is therefore slightly too narrow). Horizontal line: the target 0.955.

near-collinearity that keeps the ridge on until 186k–13M matched records.

Synthetic data with the ridge off. With the simulation population of Section 11.7, 20,000 matched records in a 24,000-row register and fixed data, 20 encrypted runs per setting gave $\lambda = 0$ every time. The estimate was within 4.0% ($\varepsilon = 4$, $\kappa = 40$) and 2.5% ($\varepsilon = 8$, $\kappa = 48$) of least squares (median relative distance), and the Wald interval covered $\boldsymbol{\theta}_\lambda$ for 0.950 (209/220) and 0.927 (204/220) of the coefficients, with standardized errors of mean +0.04 and +0.03 and s.d. 1.02 and 1.05.

A plaintext simulation of these two configurations on the same data, with the noise levels of the encrypted runs and 10,000 releases each, covers 0.957 (0.956–0.958, release-clustered) at $\varepsilon = 4$ and 0.955 (0.954–0.957) at $\varepsilon = 8$. Single coefficients cover 0.951–0.961, and the standardized errors have s.d. 0.993 and 0.999. So the coverage at $\varepsilon = 8$ is on target, and 204 of 220 is a low draw: a plaintext batch of 20 runs covers no more with probability 0.05. Pooled over the eight configurations of this section, the encrypted runs covered 699 of 740 coefficients (0.945) against 707 expected from the simulations (s.d. 6.1; one-sided $p = 0.08$). The runs of Section 11.7 test the encrypted coverage with many more runs.

11.6 Large-scale runs

We ran the full implementation 78 times at register sizes from 10^5 to 10^6 rows, in three regimes, each cell three times on one fixed dataset ($\delta = 10^{-9}$, κ derived, a fresh single-run ledger per run, two felts per PSI item; Section 12 gives the cost). (A) The register grows and the cohort is fixed: $\bar{n}_O$ = 106,496, 303,104 and 1,007,616 (10^5, 3×10^5 and 10^6 rows of the synthetic population, padded), $n_R^{\max} = 20{,}000$ with 12,000 matched records, settings I and P on the standardized population and P on the block-normalized one, $\varepsilon \in \{1, 4\}$. (B) A large matched set: $\bar{n}_O$ = 1,007,616, $n_R^{\max} = 100{,}000$ with 60,000 or 95,000 matched records, setting I, $\varepsilon \in \{0.5, 1\}$. (C) 3droad: O's register is its 304,413 rows outside a 10% reference set and a 20% test set

(padded to 311,296), R holds the first feature, O the other two and y, block normalization, 12,000 matched records, settings I and P, $\varepsilon \in \{1, 4\}$. Before running we required, at each size of (A), exact linkage, the ridge and the estimates inside the 95% range of a plaintext simulation of the same mechanism on the same data, a noise s.d. within 3% of its calibrated value, and every phase within 25% of the cost model; the last failed, as Section 12 describes.

- *Linkage.* In all 78 runs R's match set and every position agreed with the plaintext join. No felt-wise false match occurred, and the cuckoo insertion never needed its fallback.
- *Noise.* Pooled over the 78 runs, the 3.04 million decrypted slot values outside the packed entries have mean -0.0007 (standard error 0.0006) and s.d. 0.9997 (Figure 6). The released entries have s.d. 1.000 (stage 1, 3,450 values, Kolmogorov–Smirnov $p = 0.83$) and 1.012 (stage 2, 4,224 values, $p = 0.41$); per cell and release the s.d. lies within 0.994–1.007.
- *Against simulation.* The ridge lay inside the simulated 95% range in all 78 runs (under I at 12,000 matched and $\varepsilon = 1$: 11.3k–11.8k against 9.9k–12.9k), and the coefficients in 0.939–0.975 of cases per size of (A), with $|z| \leq 3.4$ against the simulated law given the run's ridge. Under I the Wald interval covered $\boldsymbol{\theta}_\lambda$ for 344 of 354 coefficients (0.972); 400 plaintext releases per cell on the same data cover 0.941–0.968.
- *Accuracy at* 12,000 *matched.* The register size does not change the release, so (A) repeats the accuracy of Section 11.7 at this n: the population test MSE relative to least squares on the same cohort is 1.26–1.33 under I and 1.11–1.13 under P at $\varepsilon = 1$ (standardized, ridge on; simulated medians 1.29 and 1.13–1.14), 1.008–1.018 under P block-normalized, and at most 1.006 at $\varepsilon = 4$.
- *Regime B.* The ridge was off in all 12 runs, as predicted: n_{off} is 52.5k at $\varepsilon = 0.5$ and 27k at $\varepsilon = 1$, and in simulation the ridge is on in 17% of releases at 60k and $\varepsilon = 0.5$, within the switch-off factor 1.2, and in none elsewhere. The bias-aware interval for $\boldsymbol{\beta}$ was finite in every run, with median half-widths 0.055 and 0.029 at 60k ($\varepsilon = 0.5$, 1) and 0.034 and 0.018 at 95k, and in simulation it covers the true slopes 0.941–0.969 of the time. Prediction is within 1.2% of least squares; the coefficient RMSE is 9.1, 4.7, 4.7 and 2.6 times that of least squares.
- *3droad.* The ridge was off in every release, and the test MSE lay within 0.9995–1.0014 of the block-normalized non-private fit in every run. Block normalization reduces R's single feature to its sign, so that fit predicts little better than the mean (0.973 against 0.994 on standardized y). The Wald interval covered 23 of 24 coefficients.

Figure 5 extends the simulation of Section 11.7 to 10^6 matched records, with the encrypted runs as spot checks. From 10^5 matched records prediction is within 0.5% of least squares at every $\varepsilon \geq 0.5$ in each setting, the ridge is off, the Wald interval covers 0.945–0.966, and the coefficient RMSE relative to least squares falls to 1.45 ($\varepsilon = 0.5$) and 1.15 ($\varepsilon = 1$) at 10^6 under I.

11.7 Coefficient accuracy against intersection size

Setup. The population has $d_R = 4$ and $d_O = 6$ features on realistic raw scales (a score, a binary indicator, lognormal income and assets, a count and an age), Toeplitz-correlated with coefficient 0.3. They are standardized with public population constants and clipped to the population's 99.5% bounds ($B_R = 5.59$ with the intercept, $B_O = 6.24$, $B_y = 2.89$ at $R^2 = 0.5$). The generating slopes give $R^2 \in \{0.3, 0.5, 0.9\}$. The weakest direction of the clipped design is $\ell = \lambda_{\min}(\Sigma) = 0.599$, estimated from 4 million population rows, and $\lambda_{\min}(\mathbf{A})/n = 0.656$.

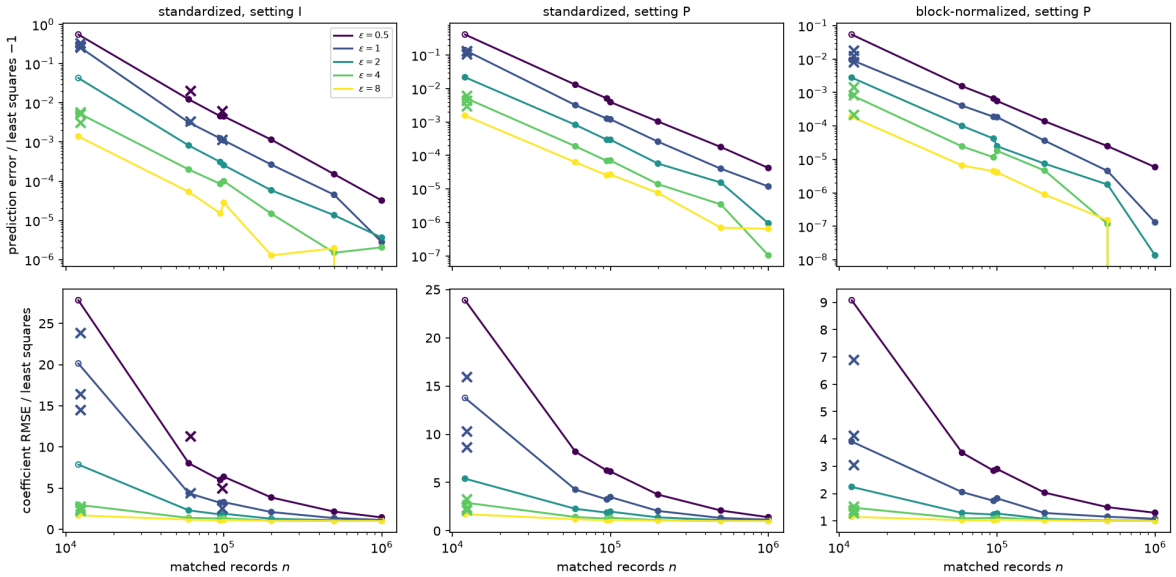


Figure 5: Plaintext simulation at 12k to 10^6 matched records (100 cohorts per n, $\delta = 10^{-9}$, $R^2 = 0.5$): population test MSE relative to least squares on the same cohort minus one (top) and coefficient RMSE relative to least squares (bottom). Hollow markers: the ridge is on in most releases. Crosses: the encrypted runs of regimes A (12k) and B (60k, 95k).

The mechanism is v3.1, simulated in plaintext with the implementation's own calibration and solver: a stage-1 release of (triu $\mathbf{B}, \mathbf{C}$) with s.d. σ_1, the stage-1 rule of Section 9.2, a stage-2 release of every O-dependent entry with s.d. σ_2, the gate $\hat{\vartheta} \geq 1$, and the solve, with the bias correction off. The simulated noise is Gaussian in the slot domain with the calibrated s.d. The encrypted implementation's measured noise is $0.996\,\sigma_2$ and $0.976\,\sigma_1$ (Section 11.1), and its estimates on protein lie inside the simulated range (Section 11.5). A further encrypted check in the ridge-on regime is reported at the end of this subsection.

This subsection uses setting I (Table 9) for the coefficients and the intervals, and adds setting P for prediction, with its first choice P_0 for reference. For P it also uses a block-normalized version of the population, on the same cohorts and noise draws. The previous setting is quoted for reference. Its runs used the same cohorts and the same unit noise draws, so the comparison is paired. All runs use $\delta = 10^{-9}$, $\mathbf{y}^\top \mathbf{y}$ in stage 2 (it is free for these bounds), and the fixed flooding parameter $\kappa = 40$ for $\varepsilon \leq 2$, 44 at $\varepsilon = 4$ and 48 at $\varepsilon = 8$ (the rule (8) would give 41 and 46 at $\varepsilon = 4$ and 8, and change σ_1 and σ_2 by less than 0.2%). With $\Delta = 86.2$ and $\Delta_G = 77.2$, setting I gives $\sigma_1 = 1737$, 892, 461, 240, 128 and $\sigma_2 = 1119$, 575, 297, 155, 82 for $\varepsilon = 0.5$, 1, 2, 4, 8 (Table 5). For each n from 10^2 to 10^5 we draw 200 cohorts. For each cohort and each ε we run an independent pair of releases, and we report the RMSE of the slopes against the true coefficients.

Results. Figure 7 and Table 21 show two regimes. Setting I moves the boundary between them and trades accuracy in one regime against the other.

- *Ridge on.* While the stage-1 ridge is on, the estimate is shrunk towards zero, and its RMSE stays close to that of the all-zero estimate (0.25). The smaller ridge of setting I

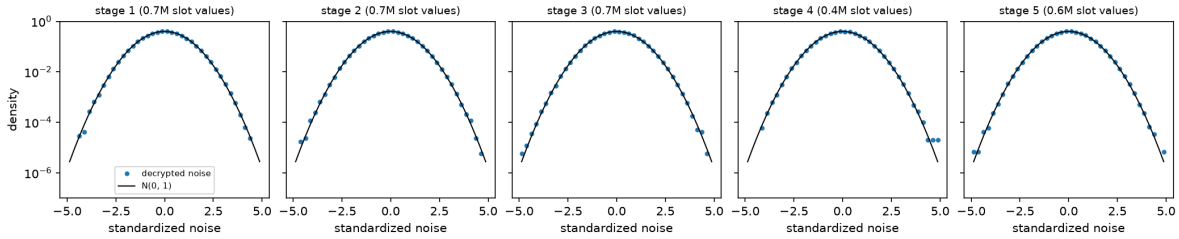


Figure 6: Decrypted noise of every release of the 78 large runs, standardized by the calibrated σ (all slots outside the packed entries, real and imaginary parts), against $\mathcal{N}(0,1)$; one panel per stage of the runs (10^5, 3×10^5, 10^6 rows, regime B, 3droad).

Table 21: $\delta = 10^{-9}$, $R^2 = 0.5$: median RMSE of the standardized slopes against the true coefficients, with the ratio to the non-private floor in brackets, under setting I and at the previous setting (same cohorts and unit noise). *: the stage-1 ridge is on in most releases.

n	setting	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	OLS floor
10^3	I	0.241 (10.7)*	0.237 (10.5)*	0.226 (10.1)*	0.211 (9.4)*	0.182 (8.1)*	0.0224
	previous	0.242 (10.8)*	0.239 (10.7)*	0.232 (10.4)*	0.222 (9.9)*	0.203 (9.0)*	
10^4	I	0.197 (27.5)*	0.154 (21.5)*	0.079 (11.0)*	0.023 (3.2)	0.013 (1.8)	0.0072
	previous	0.213 (29.8)*	0.184 (25.7)*	0.131 (18.3)*	0.043 (6.0)*	0.012 (1.7)	
10^5	I	0.015 (5.8)	0.0083 (3.2)	0.0048 (1.9)	0.0034 (1.3)	0.0028 (1.1)	0.0026
	previous	0.014 (5.3)	0.0076 (2.9)	0.0045 (1.7)	0.0033 (1.3)	0.0028 (1.1)	

lowers the error here. At $n = 10^4$ the median RMSE is 0.197, 0.154 and 0.079 for $\varepsilon = 0.5$, 1 and 2, against 0.213, 0.184 and 0.131 at the previous setting. At $\varepsilon = 4$ the ridge is already off under I, and the RMSE is 0.023 against 0.043.

- *Ridge off.* Once the ridge switches off, the DP error falls like $1/n$, faster than the sampling error, and merges into the floor. Here setting I is slightly worse, because $f_1 = 0.25$ leaves 9.5% more noise s.d. in stage 2. The error reaches twice the floor at $n \approx 8.0$k ($\varepsilon = 8$), 27k ($\varepsilon = 4$) and 88k ($\varepsilon = 2$), against 7.3k, 23k and 79k at the previous setting. At $n = 10^5$ the ratio is 3.2 and 5.8 for $\varepsilon = 1$ and 0.5, against 2.9 and 5.3.

A stronger signal lowers the floor but not the DP error. At $R^2 = 0.9$ the crossings under I are at 25k and 56k ($\varepsilon = 8$, 4; previous setting 23k and 51k), and none is reached by 10^5 for $\varepsilon \le 2$. At $R^2 = 0.3$ they are at 5.9k, 17k and 67k ($\varepsilon = 8$, 4, 2; previous setting 7.0k, 15k and 56k). Only where the crossing lies close to n_{off}, as at $R^2 = 0.3$ and $\varepsilon = 8$, does setting I reach it sooner.

The earlier single-release mechanism ($\delta = 10^{-5}$, committed eigenvalue bound) reached twice the floor at 3.2k, 9.4k and 34k for $\varepsilon = 8$, 4 and 2. Under setting I these crossings are 2.5–2.9 times larger (2.3–2.5 at the previous setting), close to the ratio of the noise variances, $(\sigma_2/\sigma)^2 = 2.5$–$3.0$ under I and 2.1–2.5 at the previous setting.

When the ridge switches off. Section 9.5 predicts the switch-off under setting I at $n_{\text{off}} = 53$k, 27k, 14k, 7.3k and 3.9k for $\varepsilon = 0.5$ to 8, 35% below the 81k, 42k, 22k, 11k and 6.0k of the previous setting. The simulation agrees.

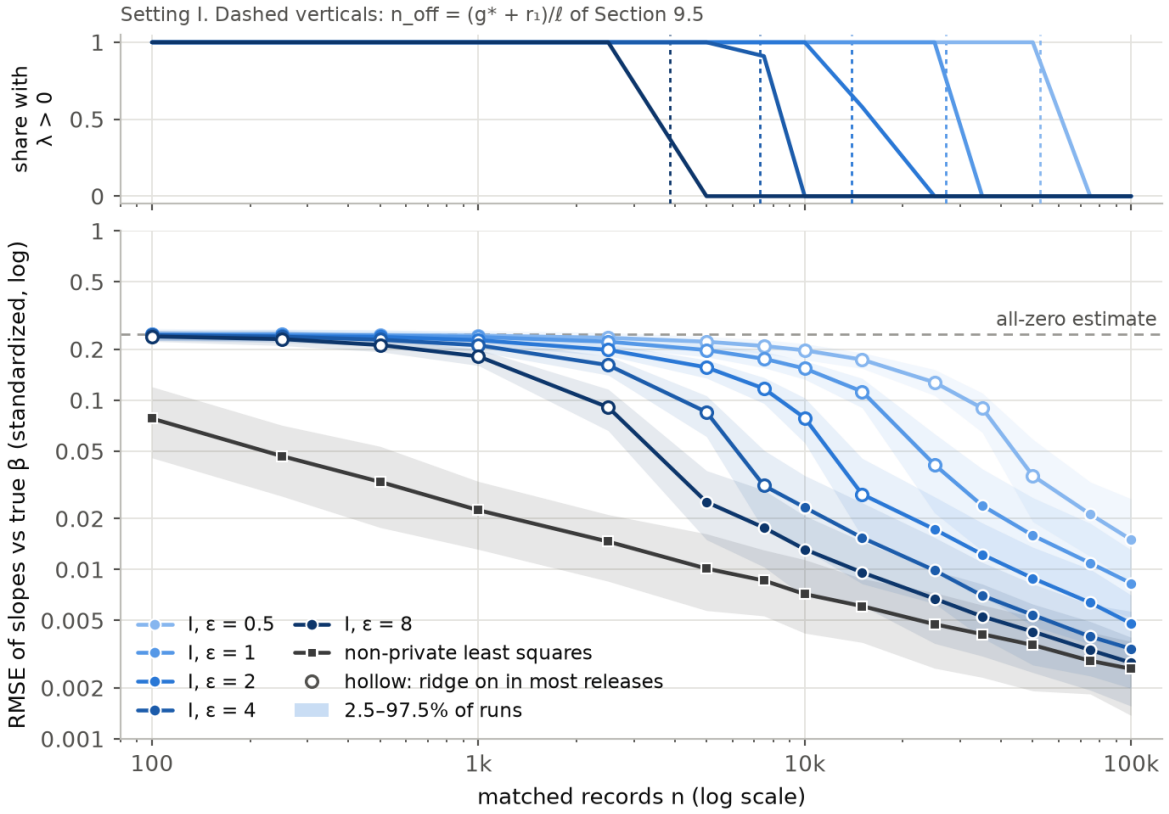


Figure 7: Setting I, $\delta = 10^{-9}$, $R^2 = 0.5$. Bottom: RMSE of the slopes against the true coefficients (standardized scale) against the intersection size n. Lines: median over 200 runs, each with a new cohort and a new pair of releases. Bands: 2.5–97.5% of runs. Hollow markers: the stage-1 ridge is on in most releases. Grey: non-private least squares on the true matches. Top: share of releases with $\lambda > 0$; dashed verticals at n_{off} of (9).

- On the grid of Figure 7, every point below n_{off} has the ridge on in every release, and every point from $1.3\,n_{\text{off}}$ has it off in every release. The two points just above n_{off} (1.03 and $1.08\,n_{\text{off}}$) have it on in 91% and 58% of releases.
- Table 22 puts the transition between n_{off} (3–4% of releases off) and $1.2\,n_{\text{off}}$ (95–99.5% off), as at the previous setting (3–5% and 94–99%).
- Below the threshold the rule chooses 1.5–1.6 times the linear prediction, because $\lambda_{\min}(\hat{\mathbf{G}}_1)$ is biased downward.
- Near n_{off} the rule takes the O-block ridge $\Psi = \Pi_O$ in about half of the releases with $\lambda > 0$ (51–54% at n_{off}), because $\lambda_{\min}(\mathbf{A})/n > \ell$. Below $0.8\,n_{\text{off}}$ it never does.
- Under setting P the switch-off is wider relative to its own n_{off} (Section 9.5): at most 1% of releases are off at n_{off}, 46–63% at $1.2\,n_{\text{off}}$ and 98–99% at $1.4\,n_{\text{off}}$, and at $0.8\,n_{\text{off}}$ the rule chooses 2.0–2.3 times the linear prediction. Under P_0 only 56–70% were off at $1.4\,n_{\text{off}}$, and the factor at $0.8\,n_{\text{off}}$ was 3.0–3.4.

While the ridge is on, setting I holds the validity ratio $\vartheta_\lambda = \lambda_{\min}(\mathbf{G} + \lambda\mathbf{I})/(2\sigma_2\sqrt{p})$ at a median of 5.0–5.2 (10–90%: 4.6–5.5), against 7.9–9.2 at the previous setting and $\vartheta^\star = 2.5$. Setting

Table 22: Stage-1 rule under setting I on the simulation population, $\delta = 10^{-9}$: share of 200 releases with $\lambda = 0$ at $n = m\,n_{\text{off}}$, each on a fresh cohort, and the median λ at $0.8\,n_{\text{off}}$ against the linear prediction $g^\star + r_1 - n\ell$ of Section 9.5.

		share with $\lambda = 0$ at $m =$					λ at $0.8\,n_{\text{off}}$	
ε	n_{off}	0.6	0.8	1.0	1.2	1.4	median	predicted
0.5	52,531	0.00	0.00	0.03	0.99	1.00	9,186	6,288
1	26,993	0.00	0.00	0.03	0.97	1.00	4,784	3,231
2	13,945	0.00	0.00	0.04	0.99	1.00	2,618	1,669
4	7,272	0.00	0.00	0.04	0.95	1.00	1,376	871
8	3,860	0.00	0.00	0.04	0.95	1.00	744	462

P holds it at 3.4–3.5, and at 2.0 on the block-normalized population (I: 3.2; P_0: 2.9–3.0 and 1.1–1.2).

Prediction recovers first, and fastest under setting P. On a held-out sample of the population (clipped design, unclipped y), setting I brings the test MSE within 10% of that of non-private least squares at $n \approx$ 38k, 20k, 10k, 5.3k and 2.9k for $\varepsilon = 0.5$ to 8 (0.72–$0.76\,n_{\text{off}}$, while the ridge is still on). The previous setting needed 58k, 30k, 15k, 8.0k and 4.2k. Setting P, which was chosen for this, needs 26k, 14k, 7.1k, 4.0k and 2.1k (Figure 8). That is 1.3–1.5 times fewer records than I and 2.0–2.2 times fewer than the previous setting. On this population the first choice P_0 needed 5–13% fewer records than P (23k, 12k, 6.2k, 3.6k and 2.0k). Within 1% the settings are close: 70k to 5.0k for P, 72k to 5.0k for I, and 91k to 7.0k for the previous setting. P's advantage over I shrinks with a stronger signal. At $R^2 = 0.9$ P needs 1.04–1.20 times fewer records than I ($\varepsilon = 1$: 28k against 32k); at $R^2 = 0.3$ it needs 1.3–1.4 times fewer. On this population the gate refused no release under I or P, at any R^2 and n. Under setting I, twice-the-floor accuracy on the coefficients needs 2.8, 5.0 and 8.8 times as many records as prediction within 10%, for $\varepsilon = 8$, 4 and 2.

The block-normalized population. The bottom row of Figure 8 uses the same cohorts with each party's block of every row scaled to unit norm ($B_R^2 = 2$, $B_O = 1$, $B_y = 2.89$; $\Delta = 10.2$, $\Delta_G = 2.83$, $\ell = 0.105$). There $\sigma_1 = 0.46\,\sigma_2$ at $f_1 = 0.25$, against $1.55\,\sigma_2$ above, and P's n_{off} is 16k to 1.2k for $\varepsilon = 0.5$ to 8.

- Setting P comes within 10% at 9.4k, 4.8k, 2.5k, 1.5k and 729 matched records, against 15k, 7.5k, 4.0k, 2.1k and 1.1k under I (1.4–1.6 times fewer).
- Its gate refused at most 1% of releases in any cell, 0.06% on average, and also at most 1% at $R^2 = 0.3$ and 0.9.
- P_0's gate refused 98–100% of releases wherever its ridge was on, so its test MSE stays near that of predicting the mean until the ridge switches off.

Coefficients on different scales. Raw-scale errors cannot be compared across coefficients, so Figure 9 shows the *efficiency ratio* $\text{RMSE}_{\text{DP},j}/\text{SE}_{\text{OLS},j}$ under setting I. It does not change when x_j or y is rescaled. A ratio r means the release is about as accurate as least squares on n/r^2 records. While the ridge is on, the ratio grows with n and is largest for the largest coefficients, which the ridge shrinks most. Once $\lambda = 0$ in every release, the ten slopes' ratios lie

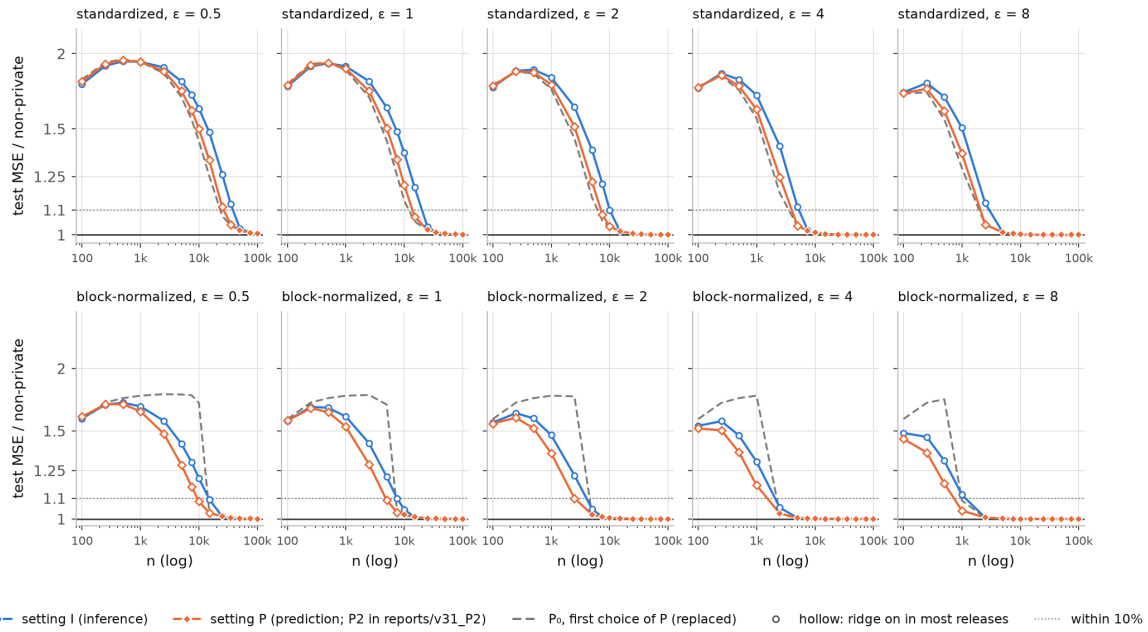


Figure 8: Test MSE on the population divided by that of non-private least squares on the same cohort, against n, $\delta = 10^{-9}$, $R^2 = 0.5$. Top row: the standardized population. Bottom row: the same cohorts with block-normalized features. Mean over 200 cohorts, a refused release counted as $\boldsymbol{\theta} = 0$. Blue: setting I. Orange: setting P. Grey dashed: P_0, the first choice of P, whose gate refuses the block-normalized releases while its ridge is on. Hollow markers: the stage-1 ridge is on in most releases. Dotted horizontal: within 10%.

within -21% and $+25\%$ of their median, except at $\varepsilon = 8$ and $n \geq 50$k, where one coefficient reaches $+34\%$ (previous setting: -20% to $+26\%$, and $+36\%$). Neither block is consistently harder: the O-block average is 0.90–1.19 times the R-block average.

Inference under setting I. At $R^2 = 0.5$ every release also runs the intervals of Section 9.4, with a bootstrap of 2,000 antithetic pairs and no public bound on $\|\boldsymbol{\beta}\|$. Coverage holds in every regime we simulated.

- *Wald interval for $\boldsymbol{\theta}_\lambda$.* It covers 0.945–0.966 of the coefficients in each of the 70 (n, ε) cells, pooled over the 200 releases and 11 coefficients, with the ridge on or off (target 0.955). The previous setting gave 0.945–0.965. The lowest single coefficient in any cell covers 0.91 under both settings, which is what 770 coefficient-cells of 200 releases produce by chance.
- *Bias-aware interval for $\boldsymbol{\beta}$.* When the ridge is off in every release (27 cells under I, 20 at the previous setting), it is finite and covers the true slopes 0.930–0.964 of the time and the clipped-population coefficients 0.946–0.962 (previous setting: 0.930–0.964 and 0.945–0.962); at large n clipping moves the target. While the ridge is on, $\hat{L} = 0$ in most releases and the interval is unbounded. Where $\hat{L} > 0$ it is finite but wide: median half-width up to 3.2, against 0.058 for the Wald interval.
- *Naive reading.* Read as an interval for $\boldsymbol{\beta}$, the Wald interval covers only 9–85% in the 41 cells where the ridge is on in every release (previous setting: 2–60% in 47 cells). The reports also give 0.088–0.950 over the 43 cells where the ridge is on in most releases.

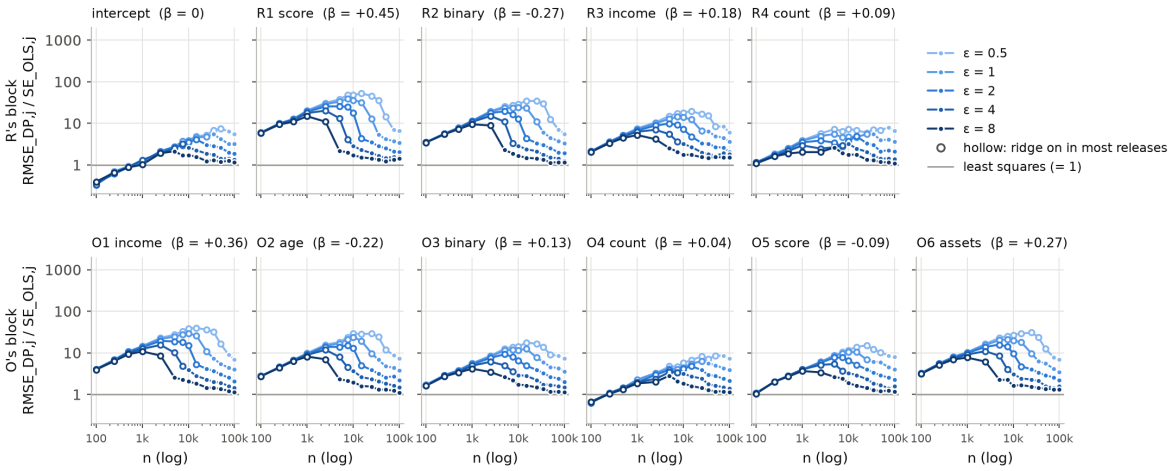


Figure 9: Efficiency ratio $\mathrm{RMSE}_{\mathrm{DP},j}/\mathrm{SE}_{\mathrm{OLS},j}$ per coefficient against n, setting I, $\delta = 10^{-9}$, $R^2 = 0.5$. Top row: R's block, with the intercept. Bottom row: O's block. Hollow markers: the stage-1 ridge is on in most releases.

These add two cells in which the ridge is off in 9% and 42% of releases ($n = 7{,}500$ at $\varepsilon = 4$, and $n = 15{,}000$ at $\varepsilon = 2$), which cover 0.90 and 0.95. The smaller ridge of setting I shrinks less, so the naive reading is less wrong, but it is still wrong.

The gate refused no release in any cell, at any R^2.

Encrypted check of settings I and P. We ran the full OpenFHE implementation 450 times on one synthetic dataset from this population: $n = 10{,}000$ matched records in a register of 12,000 rows padded to 16,384, $\delta = 10^{-9}$. There were three configurations. In two of them the ridge is on: setting I at $\varepsilon = 2$ and setting P at $\varepsilon = 1$, where the matched design's n_{off} is 14.1k and 16.4k. In the third it is off: setting I at $\varepsilon = 8$, where n_{off} is 3.9k. The flooding parameter was $\kappa = 40$, 40 and 48. Each configuration got 100 runs. Because I at $\varepsilon = 2$ came out low (below), 150 more runs of it followed with fresh seeds. A run took 21–23 s (median; three runs at a time, one core per party). R read the exact radius from the cached quantile: $r_1 = 3{,}419$ under I at $\varepsilon = 2$ and 3,964 under P.

- No release was refused, and **A** was exact. We rebuilt each run's decrypted releases from its recorded noise and fed them to the plaintext solver with the run's seed. This gave the run's λ, estimate and half-widths to within 10^{-11} (relative), so an encrypted run differs from a plaintext one only through its noise.
- The ridge lay inside the 95% range of a plaintext simulation of the same mechanism on the same data (4,000 draws). Under I at $\varepsilon = 2$ it did so in 95 of 100 runs and 144 of 150 runs of the repeat (range 2,578–4,232). Under P it did so in 92 of 100 runs (4,837–7,855). Under I at $\varepsilon = 8$ it was 0 in every run, as simulated. The encrypted coefficients lay inside the simulated 95% ranges in 0.940, 0.955, 0.944 and 0.957 of cases (I at $\varepsilon = 2$, its repeat, P, I at $\varepsilon = 8$).
- Under I at $\varepsilon = 2$, the Wald interval covered $\boldsymbol{\theta}_\lambda$ for 1,037 of 1,100 coefficients (0.943; release-clustered 95% interval 0.928–0.958). The repeat covered 1,586 of 1,650 (0.961; 0.953–0.970), and the two together 0.954 (0.946–0.962). Under I at $\varepsilon = 8$ it covered 1,054

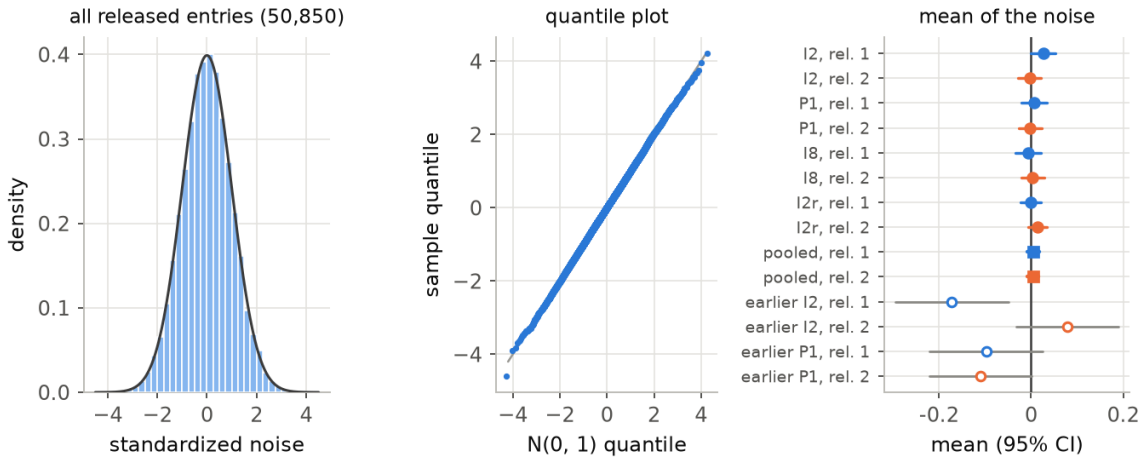


Figure 10: Raw decrypted noise of 450 encrypted runs (settings I and P, $n = 10{,}000$, register 16,384), standardized by the calibrated σ_1 and σ_2. Left: histogram of all released entries against $\mathcal{N}(0, 1)$. Middle: quantile plot. Right: mean with 95% interval by configuration (I2: setting I at $\varepsilon = 2$; I2r: its repeat; P1: setting P at $\varepsilon = 1$; I8: setting I at $\varepsilon = 8$) and release, pooled, and for the earlier five-run checks (hollow).

of 1,100 (0.958; 0.945–0.971). The plaintext simulation of the same configurations covers 0.9585 and 0.9555 (10,000 releases each). The first 100 runs at $\varepsilon = 2$ lie just below the 2.5% quantile of the plaintext simulation for 100 runs (1,040). The repeat and the noise checks below put this down to chance.

- The raw decrypted noise was standardized by the calibrated σ_1 and σ_2: in stage 1, the 51 O-dependent entries of $\mathbf{G}$; in stage 2, those, $\mathbf{c}$ and $y^\top y$. Pooled over all 450 runs, the stage-1 noise has mean +0.006 (standard error 0.007) and s.d. 0.995 (Kolmogorov–Smirnov $p = 0.13$ against the calibrated law, $\mathcal{N}(0, 1)$ after standardization). The stage-2 noise has mean +0.004 (0.006) and s.d. 1.006 ($p = 0.48$). For each configuration and release, every mean lies within 1.9 standard errors of zero. The slots outside the packed entries carry the same noise on a plaintext zero. There are more than 1.6 million of them per configuration and release. Their mean lies within 0.0011 of zero (standard error 0.0006–0.0008), and their s.d. is 0.999–1.001, release by release. Over all 14.7 million slot values the mean is −0.0004 (standard error 0.0003).
- The negative means of the earlier five-run checks are not reproduced: −0.17 on the stage-1 noise under I, and −0.10 under P (Figure 10). The same configurations now give +0.03 and −0.001 on the stage-1 noise under I (5,100 and 7,650 entries), and +0.006 and −0.003 under P.

11.8 Does the bias correction matter?

Under v3.1 the ridge comes from stage 1, so it is independent of the stage-2 noise (Proposition 9.1), and Theorem A.2 applies conditionally on (λ, Ψ). We ask the question again with the earlier conditional design at the previous setting ($\delta = 10^{-9}$, $R^2 = 0.5$, fixed $\kappa = 40$, 40, 40, 44 and 48 for $\varepsilon = 0.5$ to 8), and repeat it more briefly under setting I, with the same κ.

Table 23: Mechanism bias and the optional correction (16) under v3.1 at the previous setting, $\delta = 10^{-9}$, $R^2 = 0.5$; 45 cells (n, ε), binned by $\vartheta_0 = \lambda_{\min}(\mathbf{G})/(2\sigma_2\sqrt{p})$. Bias share and largest standardized bias: maxima over the cells of a bin. MSE ratio: corrected over uncorrected. K: smallest number of averaged releases of the same data at which the squared bias equals the variance of the average.

			bias^2/MSE	$\max_j \lvert b_j\rvert/\text{sd}_j$	MSE ratio against		
ϑ_0	cells	ridge on	(uncorrected)	(uncorrected)	$\boldsymbol{\theta}_\lambda$	$\hat{\boldsymbol{\beta}}$	K
0–1	15	15	0.001%	0.009	0.9951–0.9954	1.0000–1.0005	81,616
1–2.5	6	6	0.012%	0.024	0.9952–0.9954	1.0009–1.0021	9,069
2.5–7	9	9	0.048%	0.044	0.9945–0.9952	1.0025–1.0109	2,118
7–15	5	0	0.048%	0.037	0.9960–0.9972	0.9960–0.9972	2,103
≥ 15	10	0	0.011%	0.018	0.9990–1.0000	0.9990–1.0000	8,782

- For each $n \in \{500, 1{,}000, \ldots, 128{,}000\}$ (doubling) and each ε we draw five datasets.
- For each dataset we draw four stage-1 releases, each of which fixes (λ, Ψ).
- For each of these we draw 10,000 antithetic pairs $(\mathbf{E}, \mathbf{f})$, $(-\mathbf{E}, -\mathbf{f})$ of stage-2 releases.

The comparison against the true coefficients uses the cohort-level runs of Section 11.7.

No, not even as a variance reduction (Table 23 and Figure 11 for the previous setting; setting I below).

- *The bias is negligible.* The squared mechanism bias of Theorem A.2, averaged over a cell's datasets and stage-1 releases, is at most 0.048% of the mean squared error around $\boldsymbol{\theta}_\lambda$ in any of the 45 cells (0.071% for a single dataset and stage-1 release). The largest coefficient bias is at most 0.044 of its standard deviation. The earlier mechanism gave 0.75% and 0.14. The reason is the stage-1 rule: while the ridge is on it holds ϑ_λ at 7.9–9.2, well above $\vartheta^\star = 2.5$, because of the radius r_1; in these cells the ridge is off only from $\vartheta_0 = 9.6$. The bias peaks around the switch-off and decays like $1/\vartheta_0^2$ after it.
- *The theorem holds.* Wherever $\vartheta_0 \geq 1$, the simulated bias is 0.997–1.010 times the leading term $\sigma_2^2 M(\mathbf{P}_\lambda)\boldsymbol{\theta}_\lambda$.
- *Against* $\boldsymbol{\theta}_\lambda$ the correction lowers the mean squared error by at most 0.55%: by about 0.5% with the ridge on and by 0.4% or less once it is off. At most 0.05 percentage points of this come from removing bias; the rest is a variance reduction. The earlier mechanism gave 7–12% with the ridge on, when ϑ_λ was near 2.
- *Against the sample least-squares estimate*, with the ridge on, the correction raises the mean squared error by up to 1.1%. Deterministic shrinkage (Proposition A.1) is 92–99.5% of that error, and the correction leaves it alone by design.
- *Against the true coefficients* it changes the mean squared error by -0.4% to $+1.7\%$. The largest increase is with a moderate ridge just below the switch-off ($n = 75$k at $\varepsilon = 0.5$).
- *Averaging.* The bias would matter only if at least about 2,100 releases of the same data were averaged. The earlier mechanism gave 130.
- *Real data.* On the 29 datasets of Section 11.9 the correction changed the test MSE of v3.1 at the previous setting by -0.01% to $+0.39\%$, against -3.7% to $+2.3\%$ under the earlier mechanism.

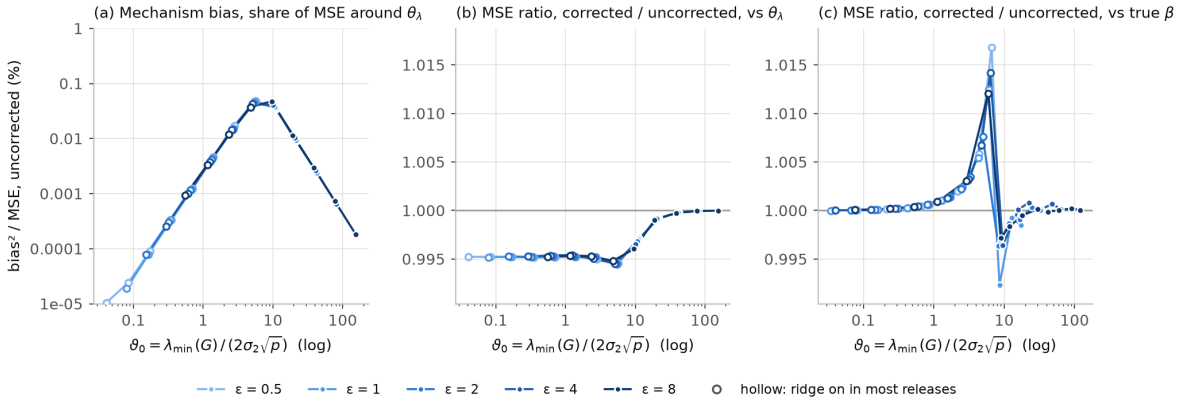


Figure 11: The optional correction under v3.1 at the previous setting, $\delta = 10^{-9}$, $R^2 = 0.5$, against $\vartheta_0 = \lambda_{\min}(\mathbf{G})/(2\sigma_2\sqrt{p})$, one line per ε; hollow markers: the stage-1 ridge is on in most releases. (a) Squared mechanism bias as a share of the mean squared error around $\boldsymbol{\theta}_\lambda$ (uncorrected estimate). (b) Mean squared error of the corrected estimate divided by that of the uncorrected one, against $\boldsymbol{\theta}_\lambda$. (c) The same against the true coefficients, from the cohort-level runs of Section 11.7. Values below 1 favour the correction.

Setting I. The same design with 5,000 instead of 10,000 antithetic pairs per (dataset, λ) gives the same answer, with a larger but still negligible bias.

- The squared mechanism bias, averaged over a cell's datasets and stage-1 releases, is at most 0.19% of the mean squared error around $\boldsymbol{\theta}_\lambda$ in any of the 45 cells (0.048% at the previous setting). The largest share for a single dataset and stage-1 release in any cell is 0.24% (0.071%). The largest coefficient bias is 0.078 of its standard deviation (0.044). The bias peaks around the switch-off, where $\vartheta_0 \approx 4.4$–5.0.
- The bias is larger because setting I holds ϑ_λ at 4.6–5.3 while the ridge is on, against 7.9–9.2 at the previous setting; the leading term scales like $1/\vartheta_\lambda^2$.
- Against $\boldsymbol{\theta}_\lambda$ the correction lowers the mean squared error by 1.4–1.7% with the ridge on and by at most 1.4% once it is off (previous setting: about 0.5%).
- Theorem A.2 holds: wherever $\vartheta_0 \geq 1$, the simulated bias is 0.994–1.017 times the leading term.
- The bias would matter only if at least about 490 releases of the same data were averaged (2,100 at the previous setting).
- The gate refused no release.

Leaving the correction off therefore costs nothing that matters under either setting.

11.9 Benchmark against AdaSSP

AdaSSP [75] perturbs the same sufficient statistics and solves a ridge whose penalty adapts to a private estimate of $\lambda_{\min}(\mathbf{X}^\top\mathbf{X})$. It assumes one curator holding the joined table, which our setting rules out. The comparison therefore measures the utility cost of never pooling the data.

Setup. We use the 29 UCI datasets [23] of Wang's published evaluation, with 10-fold cross-validation on the folds of the earlier comparison. This subsection uses $\delta = 10^{-6}$, as Wang's

published evaluation does. There are 50 noise draws per fold and $\varepsilon \in \{0.5, 1, 2, 4, 8\}$. The v3.1 arms use the fixed $\kappa = 40$, 40, 40, 44 and 48; the rule (8) would give 40 at every ε at this δ and change σ_1 and σ_2 by less than 0.02%.

- *Rows.* In each fold, 10% of all rows, taken from the training folds, form a public reference set. Only the per-party preprocessing uses it, to fix its constants and bounds. Every arm trains on the remaining training rows (80% of the data) and is tested on the held-out fold.
- *Split.* R holds the first $\lfloor d/2 \rfloor$ features and O the rest and y. Every training row counts as matched.

We compare seven arms.

(a) *AdaSSP (published):* Wang's algorithm and noise constants, with his preprocessing: z-scored features, each whole row scaled to unit norm, $y/\max|y|$. Its guarantee covers the whole record $(\mathbf{x}, y)$, not only O's part, so it is nominally stronger than that of v3.1. Its published constants, however, are calibrated for add/remove adjacency, and under our replace-one adjacency they give an ε up to 28% above the nominal one (end of this subsection).

(b) *AdaSSP (matched):* AdaSSP's adaptive ridge under our adjacency and our accountant. $\mathbf{A}$ is exact; $\rho_{\text{run}}/3$ goes to $\lambda_{\min}$ and $2\rho_{\text{run}}/3$ to one joint release of the O-dependent entries with $\Delta = \sqrt{10}$, with the tight conversion. It is idealized, because its private $\lambda_{\min}$ step needs $\mathbf{X}^\top\mathbf{X}$ in the clear.

(c) *v3.1 with joint row normalization, under settings I and P:* v3.1 on Wang's preprocessed data, without an intercept, with $B_R = B_O = B_y = 1$. It is idealized too. Wang's normalization scales each row $(\mathbf{x}_R, \mathbf{x}_O)$ by its joint norm, which needs both parties' features in one row, and no party has them in the vertical setting. The arm isolates the change of mechanism from the change of preprocessing.

(d) *v3.1 with block unit norm, under settings I and P:* each party normalizes its own block of every row (Section 6.1), so $B_R^2 = 2$ with R's intercept and $B_O = 1$. B_y is the 99.5% quantile on the reference set.

(e) *v3.1 with standardization, under setting I:* each party standardizes its own columns with reference-set constants; the bounds are the 99.5% quantiles on the reference set.

The previous setting's arms (c)–(e) are taken from their earlier run on the same folds and matched rows, and are not rerun. AdaSSP was rerun with new noise draws, which changed its test MSE by a median of 0.0% (at most 4.7% on any dataset). The v3.1 arms run both releases, the stage-1 rule, the gate and the solve, with the bias correction off. A refused release outputs $\boldsymbol{\theta} = 0$. Settings I, P and P_0 of one preprocessing share their unit noise draws. P was run after the other two, on the same draws; rerunning I on them reproduced its test MSE to within 10^{-14} (relative). Arms with different preprocessing have different non-private baselines. Each arm is therefore reported in two ways:

- as a ratio of its test MSE to that of its own non-private fit, $(\mathbf{Z}^\top\mathbf{Z} + \mathbf{I})^{-1}\mathbf{Z}^\top\mathbf{y}$ (Wang's non-private estimator), on the same matched rows with the same preprocessing; arms (a)–(c) share this fit;
- as an absolute test MSE on one common scale, Wang's $y/\max|y|$, which adds the cost of the preprocessing.

Methods are compared by paired tests over folds (95% CI of the mean difference of absolute

Table 24: Test MSE at $\varepsilon = 1$, $\delta = 10^{-6}$, as a ratio to each arm's own non-private fit (Table 25 gives the non-private test MSE). Last column: block unit norm at the previous setting, from its earlier run.

			AdaSSP		(c) joint norm		(d) block norm		(e) std.	(d)
Dataset	n	d	(a)	(b)	I	P	I	P	I	previous
housing	506	13	1.83	2.05	2.35	2.22	2.44	2.28	3.13	2.50
wine	1,599	11	1.75	1.94	2.21	2.06	1.87	1.73	3.02	1.92
airfoil	1,503	5	1.10	1.16	1.34	1.19	1.43	1.27	1.76	1.51
concrete	1,030	8	1.53	1.73	2.18	1.94	2.56	2.26	2.31	2.69
bike	17,379	17	1.04	1.07	1.22	1.13	1.38	1.22	1.47	1.43
elevators	16,599	18	1.57	1.77	2.26	2.07	1.13	1.10	2.68	1.13
pol	15,000	26	1.05	1.08	1.26	1.16	1.53	1.38	1.66	1.63
kin40k	40,000	8	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
protein	45,730	9	1.01	1.01	1.03	1.02	1.02	1.01	1.12	1.03

Table 25: Absolute test MSE at $\varepsilon = 1$, $\delta = 10^{-6}$, on Wang's $y/\max|y|$ scale. The first two columns are the non-private fits under Wang's preprocessing (shared by (a)–(c)) and under block unit norm (d). Bold: lowest private arm in the row among (a)–(d).

	non-private		AdaSSP		(c) joint norm		(d) block norm		(d)
Dataset	Wang	block	(a)	(b)	I	P	I	P	previous
housing	0.0400	0.0396	**0.0732**	0.0821	0.0941	0.0889	0.0966	0.0902	0.0990
wine	0.0201	0.0252	**0.0350**	0.0390	0.0444	0.0413	0.0472	0.0438	0.0485
airfoil	0.0534	0.0521	**0.0586**	0.0622	0.0715	0.0638	0.0746	0.0661	0.0787
concrete	0.0445	0.0381	**0.0681**	0.0770	0.0968	0.0865	0.0977	0.0861	0.103
bike	0.0279	0.0270	**0.0290**	0.0299	0.0341	0.0315	0.0372	0.0329	0.0388
elevators	0.00872	0.0229	**0.0136**	0.0154	0.0197	0.0180	0.0259	0.0253	0.0260
pol	0.135	0.123	**0.141**	0.146	0.170	0.157	0.188	0.170	0.201
kin40k	0.0632	0.0632	0.0633	0.0633	0.0633	0.0633	**0.0632**	**0.0632**	0.0632
protein	0.119	0.121	**0.120**	0.120	0.122	0.121	0.124	0.122	0.124

test MSE, t with 9 degrees of freedom).

Section 11.10 adds two more baselines on the same folds and rows: vertical full-record SSP, which noises R's block too, and TukeyEM, which needs no bounds.

Results. Tables 24–26 and Figure 12 separate the mechanism, the setting and the preprocessing.

- *Setting I narrows the gap, but not the count.* Arm (c) has the same preprocessing as AdaSSP. Under setting I, at $\varepsilon = 1$, it has significantly higher test MSE than AdaSSP (published) on 23 of 29 datasets and lower on 5, with a median gap of +19%; at the previous setting the counts were 20 and 6 and the gap +28%. Against AdaSSP (matched), which has our guarantee, it is higher on 18 and lower on 6, with a median gap of +12% (18 and 8, +17% at the previous setting). At $\varepsilon = 8$ the median gaps are +11% against (a) and +6% against (b). The ridge is still the cause. Setting I's median ridge is 2.8 times that of AdaSSP (matched) (range 2.4–3.9 over 26 datasets) and 4.1 times that of AdaSSP

Table 26: Paired comparisons over all 29 datasets, $\delta = 10^{-6}$: number of datasets on which the first arm has significantly lower / higher absolute test MSE (95% CI over folds). The remaining datasets show no significant difference. Rows marked "previous" are from the earlier run of the previous setting.

Comparison	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$
(d) block, I vs (a) AdaSSP, published	6 / 19	5 / 20	4 / 21	3 / 22	1 / 21
(d) block, P vs (a)	6 / 19	5 / 20	5 / 18	2 / 20	3 / 15
(d) block, P_0 vs (a)	6 / 19	5 / 20	3 / 21	2 / 24	2 / 22
(d) block, previous vs (a)	5 / 19	5 / 19	4 / 21	3 / 23	1 / 21
(d) block, P vs (b) AdaSSP, matched	9 / 16	5 / 19	6 / 18	5 / 14	9 / 15
(c) joint, I vs (a) AdaSSP, published	7 / 19	5 / 23	4 / 23	1 / 21	0 / 22
(c) joint, P vs (a)	6 / 20	5 / 22	3 / 24	0 / 23	0 / 25
(c) joint, P_0 vs (a)	6 / 20	4 / 22	1 / 25	0 / 24	0 / 26
(c) joint, previous vs (a)	7 / 19	6 / 20	3 / 22	2 / 22	0 / 23
(c) joint, I vs (b) AdaSSP, matched	8 / 15	6 / 18	6 / 20	5 / 19	2 / 20
(c) joint, P vs (b)	8 / 15	6 / 18	6 / 18	4 / 18	1 / 18
(c) joint, P_0 vs (b)	11 / 10	6 / 11	6 / 12	5 / 12	3 / 12
(c) joint, previous vs (b)	10 / 17	8 / 18	5 / 20	5 / 21	3 / 20
(d) block, I vs (d) block, previous	18 / 4	19 / 1	21 / 1	23 / 1	24 / 0
(d) block, P vs (d) block, I	18 / 2	21 / 1	23 / 1	24 / 1	23 / 0
(d) block, P vs (d) block, P_0	20 / 5	19 / 4	19 / 4	20 / 3	18 / 4
(c) joint, P vs (c) joint, I	18 / 7	19 / 6	21 / 5	20 / 3	20 / 2
(c) joint, P vs (c) joint, P_0	7 / 18	7 / 20	8 / 19	6 / 20	7 / 19
(e) std., I vs (d) block, I	2 / 24	2 / 24	2 / 25	2 / 23	2 / 18

(published), against 4.2 times AdaSSP (matched) at the previous setting; for example 536, 188 and 127 on housing and 372, 150 and 101 on protein (786 and 559 before). Setting I still certifies $\vartheta^\star = 2.5$ at confidence $1 - \alpha_L$ from a quarter of the budget.

- *Setting P narrows the gap in the implementable arm.* Under block normalization, (d) under setting P has higher test MSE than AdaSSP (published) on 20 of 29 datasets at $\varepsilon = 1$ and lower on 5, the same counts as setting I, but the median gap is +14% instead of +25%. At $\varepsilon = 8$ it is higher on 15 and lower on 3 (bike, kin40k and pol), with a median gap of +4.3% (+10.7% under I). Against AdaSSP (matched) the median gap is +6.3% at $\varepsilon = 1$ and +1.4% at $\varepsilon = 8$. Setting P has lower test MSE than setting I on 18–24 datasets and higher on at most 2 at every ε, by a median of 3–7%. Its ridge is 0.6 times that of I. On protein the gap to AdaSSP (published) is 2.0% (0.122 against 0.120), and on bike it is 13%. Its gate refused 0.2% of releases on average over the 29 datasets and all ε, more than 1% on at most 3 datasets at any ε, and at most 7.2% (solar at $\varepsilon = 1$, where $\sigma_1 = 0.25\,\sigma_2$).
- *Setting P under joint normalization.* Arm (c) under P has a median gap of +4.9% against AdaSSP (matched) at $\varepsilon = 1$ (lower test MSE on 6 datasets, higher on 18) and +2.6% at $\varepsilon = 8$ (1 and 18). Against AdaSSP (published) the gaps are +8.8% and +7.8%. Its ridge is 1.6–1.9 times that of AdaSSP (matched), against 2.6–2.9 times under I, and it beats (c) under I on 18–21 datasets at every ε. Its gate refused 0.03% of releases on average, and at most 2.4% (servo, $\varepsilon = 8$).

- *The first choice P_0 failed under block normalization.* P_0 ($\vartheta^\star = 0.5$, $\alpha_L = 0.5$, $f_1 = 0.2$) matched AdaSSP (matched) to within about 1% under joint normalization (a median gap of +0.2% to +1.1% at every ε, with a ridge 1.1–1.4 times AdaSSP's). Under block normalization, however, its gate refused 65% of releases on average over the 29 datasets and all ε: at least a quarter on 25 of them, and all of them on elevators, gas and solar. A refused release outputs $\boldsymbol{\theta} = 0$, so (d) under P_0 was worse than AdaSSP (published) by a median of +36% at $\varepsilon = 1$ and +81% at $\varepsilon = 8$. Under joint normalization setting P is worse than P_0 on 18–20 datasets, by a median of 0.3–2.9%; that is the price of a gate that passes under block normalization, where P beats P_0 on 18–20 datasets. Section 11.11 explains why block normalization made P_0's gate fail.
- *The preprocessing.* Under setting I, block normalization (d) and joint normalization (c) differ significantly on 17 of 29 datasets in absolute test MSE ($\varepsilon = 1$; 5 lower, 12 higher), and their ratios to their own non-private fits are similar. Their differences come mostly from the non-private fits: on elevators, block normalization's non-private test MSE is 2.6 times that of Wang's preprocessing, while on concrete it is 0.86 times. Plain standardization (e) is worse than block normalization on 18–25 datasets, because heavy tails inflate the bounds.

AdaSSP's published noise constants. AdaSSP adds noise of s.d. $s = \sqrt{\log(6/\delta)}/(\varepsilon/3)$ to each of three statistics: $\lambda_{\min}$, $\mathbf{X}^\top\mathbf{X}$ (symmetrized, so off-diagonal s.d. $s/\sqrt{2}$) and $\mathbf{X}^\top\mathbf{y}$.

- Under add/remove adjacency each of the three has sensitivity 1 for unit-norm rows. The three releases then cost $\rho = 3/(2s^2)$, which the tight conversion turns into $(0.31, 0.64, 1.35, 2.87, 6.24)$-DP at $\delta = 10^{-6}$ for nominal $\varepsilon = (0.5, 1, 2, 4, 8)$, which is valid, with room.
- Under replace-one adjacency, which is ours, the sensitivities of $\mathbf{X}^\top\mathbf{X}$ (the Frobenius norm of $\mathbf{x}\mathbf{x}^\top - \mathbf{x}'\mathbf{x}'^\top$, which matches the symmetrized noise) and of $\mathbf{X}^\top\mathbf{y}$ are $\sqrt{2}$ and 2. The same constants then give $(0.48, 1.01, 2.14, 4.60, 10.2)$-DP: above the nominal ε for $\varepsilon \geq 1$, by up to 28% (at $\varepsilon = 8$).
- Recalibrated to exactly (ε, δ) under replace-one adjacency, with the tight conversion and an equal split, AdaSSP's test MSE rises by a median 2–3% (range −5% to +28%).

The published column therefore overstates AdaSSP's advantage only slightly, and its stronger guarantee holds only nominally: it protects the whole record, but at our adjacency it exceeds its nominal ε for $\varepsilon \geq 1$. The gap to v3.1 lies in the ridge rule.

[**Check 10:** AdaSSP's adjacency: the published constants are consistent with add/remove adjacency (sensitivity 1 for each statistic); confirm against the statement in Wang (2018) before citing it.]

11.10 Two further baselines: full-record SSP and TukeyEM

Two more baselines run on the folds, reference rows and matched rows of Section 11.9, at $\delta = 10^{-6}$ with 50 noise draws per fold, $\varepsilon \in \{0.5, 1, 2, 4, 8\}$ and the same fixed κ, and are reported in the same two ways. They answer two questions that AdaSSP does not: what keeping R's block exact buys, and what the clipping bounds and the per-party preprocessing cost. They differ in what they need.

- *Full-record SSP* could run in the vertical setting: it needs a secure aggregation of the whole

Gram matrix, as in the protocols below, and output noise. With block normalization it is implementable; with joint normalization it is idealized, like arm (c).

- *TukeyEM* needs the joined table: each of its least-squares fits uses whole rows. It is an idealized central baseline, like AdaSSP.
- The FWS intervals of Section 11.4 post-process our own release, so they could run in the vertical setting.

Full-record SSP. Karr et al. [45], Hall et al. [38] and Gascón et al. [35] compute the whole Gram matrix securely, $\mathbf{A}$ included. Adding noise to every released entry gives the natural competitor that combines secure regression with output privacy. We keep everything else of v3.1: the two releases with the split f_1, the tight zCDP conversion with the same slack, the stage-1 rule, the gate, and settings I and P. The releases now also contain $\operatorname{triu}\mathbf{A}$ (every entry but the constant n), and the adjacency replaces a whole record $(\mathbf{x}_R, \mathbf{x}_O, y)$. Two things change. The sensitivity is that of Proposition 11.1. The exact radius is the quantile of $\lambda_{\max}$ of a unit noise matrix without the zero block on $\mathbf{A}$, computed by the rule of Section 9.2: at $p = 11$, $q_{0.9975} = 7.87$ instead of 7.42 (setting I) and $q_{0.25} = 4.87$ instead of 4.44 (setting P). Each full-record release adds noise on $\mathbf{A}$, from a separate stream, to the same unit noise on the O-dependent entries as the v3.1 release it is compared with, so the comparison is paired draw for draw. The v3.1 arms recomputed on these streams reproduce the published fold values to 10^{-14}. The question is therefore only what the exact $\mathbf{A}$ buys.

Proposition 11.1 (Sensitivity of the full-record release)**.** *Let a record be $(\dot{\mathbf{x}}, \mathbf{x}_O, y)$ with $\dot{\mathbf{x}} = (1, \mathbf{u})$, $\|\dot{\mathbf{x}}\| \le B_R$, $\|\mathbf{x}_O\| \le B_O$ and $|y| \le B_y$, and let the release be the upper triangle of $\sum_i \mathbf{w}_i \mathbf{w}_i^\top$ with $\mathbf{w}_i = (\dot{\mathbf{x}}_i, \mathbf{x}_{O,i}, y_i)$, without its constant entry n, every entry with the same noise. For the blocks $k \in \{\mathbf{u}, \mathbf{x}_O, y\}$ of a record and of its replacement write n_k and n'_k for their squared norms and g_k for their inner product. Then the replace-one ℓ_2 sensitivity is*

$$\Delta^2_{\text{full}} = \sup\Big[\sum_k (n_k + n'_k - 2g_k) + \sum_k (n_k^2 + n_k'^2 - 2g_k^2) + \sum_{k<l} (n_k n_l + n'_k n'_l - 2g_k g_l)\Big], \qquad (10)$$

over the norms allowed by the bounds and $|g_k| \le \sqrt{n_k n'_k}$, with $|g_k| = \sqrt{n_k n'_k}$ for a block of dimension one (y among them). For a block of dimension at least two the supremum is attained with both norms at the bound. Without an intercept the first sum is dropped; without $\mathbf{y}^\top\mathbf{y}$ (separate mode) the term $(n_y - n'_y)^2$ is dropped; for stage 1 every term with y is dropped.

Proof sketch. The first sum is the change of the first moments (the intercept row), the others that of $\operatorname{triu}(\mathbf{w}\mathbf{w}^\top - \mathbf{w}'\mathbf{w}'^\top)$ block by block. For each diagonal block $\|\operatorname{triu}\mathbf{S}\|^2 \le \|\mathbf{S}\|_F^2$, which gives the bound. In two coordinates, for any norms and inner product, the two vectors can be chosen so that $\mathbf{x}\mathbf{x}^\top - \mathbf{x}'\mathbf{x}'^\top$ is diagonal, so the bound is attained for blocks of dimension at least two. The bracket is a concave quadratic in the g_k, every other coefficient is nonnegative, and the admissible set of g_k grows with the norms, so its maximum is nondecreasing in n_k and n'_k. For blocks of dimension at least two what remains is a maximization over $t = yy' \in [-B_y^2, B_y^2]$ of an explicit piecewise quartic, attained at $t = \pm B_y^2$, at a breakpoint of the pieces or at a real root of a cubic, so the supremum has a closed form computed by exact enumeration, as for Proposition 7.1 (proofs, Part B, Theorem S10c); for a block of dimension one that value is a valid, conservative bound. The values below use the earlier numerical maximization (a dense grid refined by local search, rounded up by a relative 10^{-12}), which agrees with the enumeration to 10^{-12} over 19 bound vectors. With R's row held fixed the same computation reproduces

Table 27: Full-record SSP against v3.1 on the simulation population ($\delta = 10^{-9}$, $R^2 = 0.5$, the cohorts and unit noise of Section 11.7): matched records at which the stage-1 ridge switches off (n_{off} of (9) with the full-record noise and radius), at which test MSE comes within 10% of non-private least squares, and at which the median slope RMSE reaches twice the OLS floor. Each cell: v3.1 / full-record. ">100k": not reached.

	setting I			setting P		
ε	n_{off}	prediction ≤ 1.1	slopes $\leq 2\times$	n_{off}	prediction ≤ 1.1	slopes $\leq 2\times$
0.5	52.5k / 58.7k	38.1k / 42.8k	>100k / >100k	31.5k / 35.7k	26.5k / 30.5k	>100k / >100k
1	27.0k / 30.2k	20.1k / 22.1k	>100k / >100k	16.2k / 18.4k	13.7k / 15.8k	>100k / >100k
2	13.9k / 15.6k	10.0k / 11.6k	88.2k / >100k	8.4k / 9.5k	7.1k / 8.2k	88.2k / >100k
4	7.3k / 8.1k	5.3k / 6.0k	26.6k / 31.6k	4.4k / 4.9k	4.0k / 4.3k	26.6k / 31.6k
8	3.9k / 4.3k	2.9k / 3.4k	8.0k / 9.4k	2.3k / 2.6k	2.1k / 2.2k	8.0k / 9.4k
block-normalized population: prediction ≤ 1.1						
0.5		14.7k / 15.4k			9.4k / 9.8k	
1		7.5k / 7.9k			4.8k / 5.1k	
2		4.0k / 4.1k			2.5k / 2.7k	
4		2.1k / 2.1k			1.5k / 1.6k	
8		1.1k / 1.2k			729 / 767	

Proposition 7.1 to 10^{-12}, and a direct maximization over vectors with $d_R, d_O \in \{1, 2, 3\}$ from many starting points never exceeds (10) and attains it. □

Without an intercept and at unit bounds (arm (c), $d_R, d_O \geq 2$), $\Delta^2_{\text{full}} = 32/3$ (at $\cos\angle(\mathbf{u}, \mathbf{u}') = \cos\angle(\mathbf{x}_O, \mathbf{x}'_O) = 1/3$ and $y' = -y$) against $\Delta^2 = 10$, and $\Delta^2_{G,\text{full}} = 6$ against $\Delta^2_G = 4.5$. On the simulation population (standardized, $B_R = 5.59$ with the intercept, $B_O = 6.24$, $B_y = 2.89$), Δ rises from 86.2 to 92.7 (+7.5%) and Δ_G from 77.2 to 85.8 (+11%). Under block normalization the sensitivity of the stage-2 release without $\mathbf{y}^\top\mathbf{y}$, which sets σ_2 there because $\mathbf{y}^\top\mathbf{y}$ is released separately, does not change at all (10.00 at $B_y = 2.89$), and Δ_G rises from 2.83 to 3.27 (+15%). On the 29 datasets (fold 0, $\varepsilon = 1$), σ_1 rises by 12–16% and σ_2 by 0–7% under block normalization, and by 0–16% and 0–3% under joint normalization.

Results for full-record SSP.

- *Synthetic population, standardized* (Table 27, Figure 13). Noising $\mathbf{A}$ moves every threshold up. The ridge switches off 12% later under setting I and 13% later under P (30.2k against 27.0k records at $\varepsilon = 1$ under I). Prediction within 10% of non-private needs 7–16% more records (22.1k against 20.1k under I and 15.8k against 13.7k under P at $\varepsilon = 1$). Twice the OLS floor on the slopes needs 17–18% more (31.6k against 26.6k at $\varepsilon = 4$, 9.4k against 8.0k at $\varepsilon = 8$), and at $\varepsilon = 2$ it is not reached by 10^5 records (88k for v3.1). Once the ridge is off the slopes stay less accurate: at $n = 10^5$ their median RMSE is 3.54 against 3.20 times the floor at $\varepsilon = 1$, and 1.40 against 1.31 at $\varepsilon = 4$.
- *Synthetic population, block-normalized.* Here σ_2 is unchanged and only stage 1 and the noise on $\mathbf{A}$ cost. Prediction within 10% needs 3–7% more records.
- *The 29 datasets* (Table 28, Figure 14). Full-record SSP has significantly higher test MSE than v3.1 on most datasets and lower on few, but the gap is small. Under block

normalization and setting I it is higher on 21–24 of 29 datasets and lower on at most 2, at every ε, with a median gap of +0.4% to +1.0%; under setting P, higher on 19–23 and lower on at most 2, by +0.6% to +1.2%. The largest gaps are on energy and sml (+5.1% and +4.1% under I at $\varepsilon = 1$, +7.3% and +6.3% under P). Under joint normalization the median gap is +1.0% to +1.7%; full-record SSP is lower on 7 datasets at $\varepsilon = 0.5$ and 5 at $\varepsilon = 1$ under setting I. Six of the seven have little linear signal (non-private test MSE within 25% of that of predicting the mean: fertility, forest, pendulum, pumadyn32nm, solar and tamielectric), where its larger ridge shrinks a noisy estimate further; the seventh is concreteslump (103 rows). Its median ridge is 8–10% larger than v3.1's under block normalization and 14–17% under joint normalization. Its gate refused 0.14% of releases on average under block normalization and setting P at $\varepsilon = 1$ (at most 2.4%, solar).

- *Against AdaSSP* the verdicts do not change: under block normalization and setting I full-record SSP is worse than AdaSSP (published) on 20 of 29 datasets at $\varepsilon = 1$ with a median gap of +26%, against +25% for v3.1.

- *Cryptographic cost* (measured; single thread, $d_R = 4$, $d_O = 6$, $n_R^{\max} = 2{,}000$). The releases also hold triu $\mathbf{A}$ without n: $(d_R + 1)(d_R + 2)/2 - 1 = 14$ more entries. The four sums $\sum_t b_t x_{R,j,t}$ come from R's existing uploads. The ten products need ten more encrypted upload vectors $b \cdot x_j x_k$ from R, on top of the six of v3.1. We implemented this extra work as a benchmark path of the implementation, not as a protocol mode: R encrypts and O converts the ten vectors, O aggregates all 77 entries under encryption (the new ones against a vector of ones), and each packed release is made and decoded; a noiseless test decryption returns triu $\mathbf{A}$ to 10^{-11}. At 16k and 64k register rows, O's aggregation grows by a factor 1.21 and 1.15 (1.5 s more than v3.1's 7.4 and 10.2 s; the cost model's estimate was $77/63 = 1.22$), and a whole run by 1.08 and 1.09 over a measured v3.1 run of 21 and 33 s (estimate: 1.16). The upload grows by 10 and 40 MB, exactly as estimated, which multiplies the traffic by 1.15 and 1.46 (estimate: 1.19 and 1.55, against a v3.1 baseline with one felt per PSI item). The releases cost the same, since each is one ciphertext whose noise is sampled per ring coefficient. With the larger entry counts, Calibrate raises $\log_2 B_{\text{tot}}$ by 0.15–0.18, with the same κ and σ.

Keeping R's block exact therefore buys about 1% of test MSE on real data with per-party preprocessing, consistently but little, and 7–18% of the records needed on standardized synthetic data, where R's bound is large ($B_R = 5.59$). The gain grows with R's share of the sensitivity: under block normalization B_y dominates Δ, so the exact $\mathbf{A}$ saves only the noise on $\mathbf{A}$ itself and part of Δ_G. Its other benefits are that no budget and no circuit work are spent on statistics that R already holds, and that R can use the exact $\mathbf{A}$ for diagnostics that cost no budget (Section 9.5).

TukeyEM. Amin et al. [2] split the rows at random into m disjoint groups, fit least squares on each, and release a point of high approximate Tukey depth among the m fits. A propose-test-release (PTR) step first checks, with half the budget, that the regions of high depth are stable; if the check fails, nothing is released, which we score as $\boldsymbol{\theta} = 0$, as a refused release of v3.1. The exponential mechanism on depth uses the other half. A record changes one fit, so the guarantee is (ε, δ)-DP for a change of one whole record, at the nominal $(\varepsilon, \delta) = (\varepsilon, 10^{-6})$ and with no zCDP conversion. TukeyEM needs no bound on the features or the labels, so it runs on the z-scored features (reference-set constants, with an intercept column) and the standardized y, without

Table 28: Full-record SSP on the 29 datasets, $\delta = 10^{-6}$: number of datasets on which it has significantly lower / higher absolute test MSE than the arm it is compared with (95% CI over folds, t with 9 degrees of freedom), and in brackets the median over datasets of the ratio of their test MSE, minus one, in percent. First four rows: against v3.1 with the same preprocessing and setting. The verdicts on the ratio to the own non-private fit differ by at most one dataset.

Full-record SSP	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$
(d) block norm, I	1/21 (+0.4)	1/23 (+0.7)	2/23 (+1.0)	1/24 (+0.8)	0/24 (+0.6)
(d) block norm, P	2/19 (+0.6)	1/20 (+0.9)	2/22 (+1.2)	0/23 (+0.8)	0/23 (+0.7)
(c) joint norm, I	7/21 (+1.0)	5/21 (+1.6)	3/22 (+1.7)	2/23 (+1.1)	0/24 (+1.1)
(c) joint norm, P	7/20 (+1.3)	5/21 (+1.5)	5/22 (+1.7)	3/21 (+1.0)	1/23 (+1.0)
(d), P vs v3.1 (d), I	18/2 (−2.7)	22/1 (−3.8)	24/1 (−5.5)	24/1 (−3.8)	23/0 (−3.7)
(d), I vs (a) AdaSSP, published	6/19 (+15.6)	5/20 (+25.9)	4/21 (+20.2)	3/22 (+13.8)	1/21 (+11.2)
(d), P vs (a)	6/19 (+10.3)	5/20 (+15.4)	5/19 (+11.7)	2/20 (+9.0)	3/16 (+5.3)
(c), P vs (b) AdaSSP, matched	7/15 (+1.3)	6/18 (+6.8)	6/19 (+4.8)	4/19 (+3.3)	1/20 (+4.6)

clipping; its own non-private fit is the ridge fit on that design. We reimplemented it from the reference code (`dp_regression/tukey.py` of `github.com/google-research/google-research`, commit `ba728e2`). Replaying the reference code's random numbers, the two give the same fits, PTR distances and outputs in 60 of 60 trials. We changed three details of the reference code:

- the Laplace noise of the PTR test has scale $1/(\varepsilon/2)$, which its threshold $\ln(1/(2\delta))/(\varepsilon/2)$ requires, instead of $\varepsilon/2$;
- the released point is drawn from the depth level that the exponential mechanism chose, not from the next deeper one;
- an empty set of depths in the PTR check counts as passing instead of raising an error.

On the reference code's own example ($n = 22{,}000$, $d = 10$) the first two leave the pass rates unchanged and raise the median error of the released model by 0.1–5%. The number of groups trades the PTR test against the accuracy of each fit. We fixed it in advance to groups of $2p$ rows, $m = \lfloor n/(2p) \rfloor$, close to the reference example ($m = 1{,}000$ to $2{,}000$ at $n = 22{,}000$, $p = 11$). We also ran $m = \lfloor n/(kp) \rfloor$ for $k \in \{1.5, 3, 5, 10, 25\}$; choosing the best k per dataset and ε on the test folds, an optimistic oracle, changes the counts against v3.1 below by at most one dataset.

Where TukeyEM fails. The PTR distance is a lower bound on the number of fits that would have to change to make the release unsafe. It cannot exceed $t - 1$ with $t = \lfloor \lfloor m/2 \rfloor / 2 \rfloor$, and the test passes only if the distance plus Laplace noise exceeds $\ln(1/(2\delta))/(\varepsilon/2)$, which is 26.2 at $\varepsilon = 1$ and 3.3 at $\varepsilon = 8$ for $\delta = 10^{-6}$. So TukeyEM needs about $m \geq 110$ groups at $\varepsilon = 1$ and $m \geq 20$ at $\varepsilon = 8$, that is $n \geq 220p$ and $n \geq 40p$ matched rows with groups of $2p$. With our split, 21 of the 29 datasets have $m < 110$ and 12 have $m < 20$. Enough groups are necessary but not sufficient: the regions of high depth are stable only if the fits agree, which fits on $2p$ rows rarely do. Table 29 and Figure 15 give the results.

- *The test fails on every dataset with fewer than about* 3,500 *rows per coefficient, unless* ε *is large.* At $\varepsilon \leq 2$ it passes in every run on kin40k, protein, tamielectric and 3droad ($n/p \geq 3{,}556$, $m \geq 1{,}777$) and in no run on the other 25 datasets. There its distance bound takes its smallest value, -1, even where m would allow more (pol, elevators and bike, $m = 222$–386). At $\varepsilon = 8$ it also passes on pol, elevators, bike and airfoil (98%), and

Table 29: TukeyEM on the datasets where its PTR test passes at some ε ($\delta = 10^{-6}$, $m = \lfloor n/(2p) \rfloor$ groups, 10 folds, 50 runs per fold): share of runs that pass, test MSE divided by its own non-private fit, and absolute test MSE divided by that of v3.1 with block normalization under setting P, at $\varepsilon = 1$ / 8. On the other 21 datasets the test fails in every run at every ε.

Dataset	n	p	m	PTR passes	ratio to own fit	ratio to v3.1 (d), P
airfoil	1,203	6	100	0.00 / 0.98	2.041 / 1.059	1.565 / 1.026
pol	12,000	27	222	0.00 / 1.00	1.872 / 5.335	2.027 / 7.440
elevators	13,280	19	349	0.00 / 1.00	4.170 / 1.147	1.448 / 0.420
bike	13,904	18	386	0.00 / 1.00	3.721 / 1.012	3.264 / 1.066
kin40k	32,000	9	1,777	1.00 / 1.00	1.004 / 1.003	1.007 / 1.007
protein	36,584	10	1,829	1.00 / 1.00	1.116 / 1.083	1.092 / 1.074
tamielectric	36,625	4	4,578	1.00 / 1.00	1.001 / 1.001	1.001 / 1.001
3droad	347,900	4	43,487	1.00 / 1.00	1.002 / 1.002	0.988 / 0.988

fails on the other 21 datasets in every run.

- *Where it passes, it is accurate on the large datasets.* Its test MSE is within 0.4% of its own non-private fit on kin40k, tamielectric and 3droad, and 12% ($\varepsilon = 1$) and 8% ($\varepsilon = 8$) above it on protein. At $\varepsilon = 8$ it is 1.2% above on bike, 6% on airfoil and 15% on elevators. On pol it passes at $\varepsilon = 8$ but releases a poor model: 5.3 times its non-private test MSE and 7.4 times that of v3.1. Its fits on groups of 54 rows vary widely (pol's features are discrete), and the released model predicts worse than the mean (0.983 against 0.345 on Wang's scale).
- *Against v3.1.* TukeyEM has significantly higher absolute test MSE than v3.1 with block normalization under setting P on 25 of 29 datasets at $\varepsilon = 1$ and lower on 2 (median gap +26%), and higher on 24 and lower on 3 at $\varepsilon = 8$ (+31%). Against setting I the counts are 25 and 1 (+18%) and 24 and 3 (+30%). It is lower on 3droad, where the unclipped z-scored fit beats block normalization (0.02649 against 0.02689 non-private), on pumadyn32nm, where the test fails and predicting the mean beats any fit of these noise-only data, and at $\varepsilon = 8$ on elevators, where block normalization's own non-private fit is 2.6 times worse than the z-scored one.

TukeyEM therefore answers the question about bounds only where it passes. There, the bounds and the clipping of v3.1 cost little: on protein, kin40k and tamielectric v3.1 is as accurate or more so. Per-party block normalization can cost more than the privacy noise (elevators), and a method without bounds avoids that cost, but at the sizes of most of Wang's datasets TukeyEM releases nothing.

[**Check 11:** TukeyEM: the adjacency (a change of one record changes one of the m fits; the PTR distance is a Hamming distance between the sets of fits) and the PTR noise scale 2/epsilon are taken from the reference code and the PTR threshold, because the paper of Amin et al. was not accessible when this was written. Confirm both against the paper, and the choice of m against its recommendation.]

[**Check 12:** Full-record SSP: the measured cost assumes that R uploads the products b x-j x-k. The U-side bound of those vectors (at most B-R squared) enters the derivation of B-tot, which was only re-evaluated with the larger entry counts, not re-derived. A secure-aggregation protocol of the kind of Karr et al. or Gascon et al. would have its own cost.]

Table 30: Matched records needed to come within 10% of the arm's own non-private test MSE, $\delta = 10^{-9}$: v3.1 with block unit norm under settings I and P, under P_0 and at the previous setting, and AdaSSP (published). ">" means not reached at the sizes available.

Dataset (n available)	Arm	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$
bike (13,903)	v3.1, I	>14k	>14k	>14k	13.6k	7.3k
	v3.1, P	>14k	>14k	>14k	8.6k	4.5k
	v3.1, P_0	>14k	>14k	>14k	>14k	>14k
	v3.1, previous	>14k	>14k	>14k	>14k	8.3k
	AdaSSP	>14k	9.1k	4.5k	2.3k	1.2k
elevators (13,279)	v3.1, I	>13k	>13k	>13k	8.0k	4.1k
	v3.1, P	>13k	>13k	9.5k	5.1k	2.5k
	v3.1, P_0	>13k	>13k	>13k	>13k	>13k
	v3.1, previous	>13k	>13k	>13k	8.9k	4.7k
	AdaSSP	>13k	>13k	>13k	>13k	8.8k
pol (12,000)	v3.1, I	>12k	>12k	>12k	>12k	>12k
	v3.1, P	>12k	>12k	>12k	>12k	10.5k
	v3.1, P_0	>12k	>12k	>12k	12.0k	6.4k
	v3.1, previous	>12k	>12k	>12k	>12k	>12k
	AdaSSP	>12k	8.5k	4.3k	2.2k	1.0k
protein (36,584)	v3.1, I	16k	8.2k	4.1k	2.1k	985
	v3.1, P	10k	5.0k	2.5k	1.2k	599
	v3.1, P_0	>37k	>37k	>37k	>37k	32k
	v3.1, previous	20k	10k	5.2k	2.6k	1.3k
	AdaSSP	4.5k	2.2k	992	427	166

[**Check 13:** Full-record SSP: the closed form of the proofs (Part B, Theorem S10c) covers blocks of dimension at least two; for tamielectric, whose block for R has dimension one, it is a conservative bound and the reported value comes from the numerical maximization. Decide whether the baseline code should switch to the enumeration.]

11.11 Intersection size on real data

Figure 16 repeats the experiment of Figure 7 on the four datasets with at least 15,000 rows and a real linear signal: bike, elevators, pol and protein. It runs v3.1 with block normalization under settings I and P, and AdaSSP (published), all at $\delta = 10^{-9}$. The previous setting is taken from its earlier run on the same folds and matched sets.

- In each fold the reference set is set aside as in Section 11.9. The remaining training rows are shuffled once, and the first n form the matched set, nested across n. The held-out fold is the test set.
- There are 40 noise draws per fold and n. The v3.1 runs of settings I and P share their unit noise draws and use the fixed $\kappa = 40$, 40, 40, 44 and 48 for $\varepsilon = 0.5$ to 8.
- Each arm is compared with its own non-private fit at the same n.

Table 30 gives the n at which each arm comes within 10% of its own non-private test MSE.

- *Protein, setting I.* v3.1 comes within 10% of non-private at 16k, 8.2k, 4.1k, 2.1k and 985 matched records for $\varepsilon = 0.5$ to 8, 18–24% fewer than the previous setting (20k, 10k, 5.2k,

2.6k, 1.3k). AdaSSP needs 3.6–5.9 times fewer records than setting I (4.4–7.8 times fewer than the previous setting). The earlier single-release mechanism needed 5.4k, 2.8k, 1.4k and 619 for $\varepsilon = 0.5$ to 4.

- *Bike, elevators and pol, setting I.* Bike now comes within 10% at $\varepsilon = 4$ (13.6k) and $\varepsilon = 8$ (7.3k); the previous setting did so only at $\varepsilon = 8$ (8.3k). Elevators does at 8.0k and 4.1k for $\varepsilon = 4$ and 8 (previous: 8.9k and 4.7k), relative to block normalization's own non-private fit, which is 2.8 times worse than that on standardized features. Pol does not, at the 12k records available: at that size its test MSE is 1.63 and 1.22 times non-private at $\varepsilon = 1$ and 4 (previous: 1.74 and 1.27).

- *Setting P.* Its gate refused at most 1% of releases at every n and ε on all four datasets: at most 0.3% on bike, 1.0% on elevators, and none on pol and protein. P comes within 10% sooner than setting I wherever either does. On protein it needs 10k, 5.0k, 2.5k, 1.2k and 599 matched records for $\varepsilon = 0.5$ to 8, 1.6–1.7 times fewer than I. On bike it needs 8.6k and 4.5k for $\varepsilon = 4$ and 8 (I: 13.6k and 7.3k). On elevators it needs 9.5k, 5.1k and 2.5k for $\varepsilon = 2$, 4 and 8, where I needed 8.0k and 4.1k for $\varepsilon = 4$ and 8 only. On pol it gets there at $\varepsilon = 8$ only (10.5k). AdaSSP still needs 2.2–3.6 times fewer records than P on protein. At the largest n and $\varepsilon = 1$ the median stage-1 ridge under P is 600 on protein and 1,204 on bike, against 1,056 and 1,936 under I.

- *Why the first choice P_0 failed.* While the ridge was on, the gate refused 96–100% of P_0's releases on bike, 100% on elevators and 82–97% on protein, at every n and ε. A refused release outputs $\boldsymbol{\theta} = 0$ and still spends the budget, so P_0's test MSE stayed near that of predicting the mean: at the largest n it was 3.9, 1.6 and 1.34 times non-private. With $\vartheta^\star = 0.5$ the gate passes only if r_1 exceeds $\sigma_2\sqrt{p}$ by more than the extra downward shift that the stage-2 noise causes in $\lambda_{\min}$ (Section 9.3). On the standardized synthetic population, where $\Delta_G \approx 0.9\Delta$, P_0 had $\sigma_1 = 1.79\,\sigma_2$ and $r_1 = 8.9\,\sigma_2$, far above $\sigma_2\sqrt{11} = 3.3\,\sigma_2$. Block normalization bounds $\mathbf{x}$ tightly but leaves B_y as large as before, so $\Delta_G/\Delta \approx 0.34$ on protein. Then $\sigma_1 \approx 0.67\,\sigma_2$ and $r_1 = q_{0.5}\sigma_1 \approx 4.53 \times 0.67\,\sigma_2 \approx 3.1\,\sigma_2$ ($p = 10$), below $\sigma_2\sqrt{10} = 3.2\,\sigma_2$ even before the stage-2 noise, the larger of the two, lowers $\lambda_{\min}$ further. P_0's selection population never entered this regime. Setting P's selection included a block-normalized population with $\sigma_1 = 0.46\,\sigma_2$, and with $\vartheta^\star = 1.5$ its gate does not depend on r_1 to pass. On pol, where $\sigma_1 = 0.83\,\sigma_2$ under P, the gate refused at most 2% of P_0's releases, and there P_0 remains the best v3.1 arm: within 10% at 12.0k ($\varepsilon = 4$) and 6.4k ($\varepsilon = 8$).

- *The ridge stays on throughout.* At the largest n and $\varepsilon = 1$ the median stage-1 ridge under setting I is 1,056 on protein and 1,936 on bike (previous setting: 1,334 and 2,226). Section 9.5 puts the switch-off for protein at 555k matched records at $\varepsilon = 1$ under setting I.

Setting P was chosen on synthetic populations whose noise ratio does not go below $\sigma_1 = 0.46\,\sigma_2$; on the 29 benchmark datasets its gate refused up to 7.2% of releases, on solar ($\sigma_1 = 0.25\,\sigma_2$). Both are limitations of the setting (Section 10).

11.12 Where the utility goes

Prediction is not interpretation. Test MSE includes the irreducible noise, so it is forgiving. On the simulation population, setting I brings prediction within 10% of non-private at about $0.73\,n_{\text{off}}$, while the ridge is still on; coefficients within twice the non-private error take 2.8–8.8 times as many records (Section 11.7). Intervals for $\boldsymbol{\beta}$ are informative only once the ridge is off or a public bound on $\|\boldsymbol{\beta}\|$ is available.

The gap to AdaSSP is the ridge; the two settings shrink it only in part. At the same (ε, δ) and unit bounds, the stage-2 noise of v3.1 is lower than that of AdaSSP under our guarantee: $\sigma_2 = 16.5$ at settings I and P (both $f_1 = 0.25$) and 15.1 at the previous setting, against 17.6, at $\varepsilon = 1$ and $\delta = 10^{-6}$. The difference lies in the ridge, which the stage-1 rule sizes to certify $\vartheta^\star$ at confidence $1 - \alpha_L$.

- *Setting I keeps the interval guarantee.* The exact radius and $f_1 = 0.25$ cut the ridge by about a third (2.8 instead of 4.2 times that of AdaSSP (matched)), hold ϑ_λ near 5 instead of 8–9, and lower n_{off} by 35% and the records needed for prediction within 10% by a third on the simulation population; its intervals cover in every simulated regime. On real data it improves on the previous setting on 18–24 of 29 datasets, but it does not change the verdict against AdaSSP: with block normalization it is still worse on 20 of 29 at $\varepsilon = 1$, and the median gap falls only from 28% to 25%. Once the ridge is off, its larger σ_2 costs 9.5% in noise s.d.
- *Setting P gives up the interval guarantee.* It holds the validity ratio at about 3.5 instead of 5 (2.0 on the block-normalized population), its ridge is about 0.6 times that of setting I, and its gate still passes: it was chosen under a bound on refusals on standardized and block-normalized synthetic data, and it refused 0.2% of block-normalized releases on average over the 29 datasets. With block normalization, the preprocessing a vertical deployment can use, it is the best v3.1 arm. It is better than setting I on 18–24 datasets at every ε, and the median gap to AdaSSP (published) falls from +25% to +14% at $\varepsilon = 1$ and from +11% to +4% at $\varepsilon = 8$. It is still worse than AdaSSP on 20 of 29 datasets at $\varepsilon = 1$. Under joint normalization its ridge is 1.8 times that of AdaSSP (matched). The first choice P_0 ($\vartheta^\star = 0.5$) came within about 1% of AdaSSP (matched) there, with a ridge 1.3 times AdaSSP's, but was refused on most block-normalized releases. With $\vartheta^\star < 1$ the gate passes only when the stage-1 noise is not much smaller than the stage-2 noise ($\sigma_1 = 1.34\,\sigma_2$ for unit bounds, about $0.67\,\sigma_2$ for block-normalized protein), and block normalization makes Δ_G about a third of Δ.

Per-party block normalization costs little against Wang's joint normalization, which no party can apply in the vertical setting; under setting I the two differ significantly on 17 of 29 datasets, mostly through their non-private fits.

What the other baselines add. Three comparisons bound the cost of the design choices (Sections 11.4 and 11.10).

- *The exact* $\mathbf{A}$ *is a modest accuracy gain.* Noising R's block as well, with the protocol otherwise unchanged, raises test MSE by a median of 0.4–1.7% on the 29 datasets (significantly on 19–24 of them, at every ε, setting and preprocessing), and the records needed on the standardized synthetic population by 7–18%. The exact $\mathbf{A}$ also means that

no budget and no circuit work are spent on statistics that R already holds.

- *Dropping the bounds does not help at these sizes.* TukeyEM, on the joined table and without bounds, fails its stability test on 25 of 29 datasets at $\varepsilon \leq 2$ and has higher test MSE than v3.1 under setting P on 24–25 of them. Where it passes, it shows that block normalization can cost more than the privacy noise (elevators), and that clipping costs little (protein, kin40k).
- *For $\boldsymbol{\beta}$, a bootstrap without the ridge helps where ours gives no interval.* The interval of Ferrando et al. on our stage-2 release equals our bias-aware interval once the ridge is off, and stays finite and covers while the ridge is on, from about $\vartheta_0 = 1$, roughly a quarter of n_{off} on the synthetic population. Below that, only $\boldsymbol{\theta}_\lambda$ has a useful interval.

How well the data fill their bounds matters as much as n. The noise is sized for the committed bounds, so headroom the data do not use is lost signal. After scaling by $\max |y|$, the typical $|y|$ is only 0.19 of its bound on elevators and 0.24 on wine. Choosing bounds that typical records come close to, and units or transforms that make this possible (for example log income), is part of using the protocol well.

Conditioning decides when the coefficients become usable. The ridge is off from about $(g^\star + r_1)/\ell$ matched records (Section 9.5). On protein a near-collinearity involving a single feature keeps it on until 555k records with block normalization and 8.3M without, at $\varepsilon = 1$ under setting I (691k and 13M at the previous setting). Removing that feature, a decision that public reference data can inform, lowers the block-normalized threshold about sevenfold (102k at the previous setting).

Budget. Every release, including a refused one, is charged to O's ledger. Model search and exploratory re-runs spend the budget quickly; the protocol suits a pre-specified model per research question.

12 Cost

Table 31: Cost per run, with $m = \lceil \bar{n}_O / N_s \rceil$ chunks and $|\mathcal{E}|$ released entries. O's computation is $\Omega(n_{\text{Sc}})$ in the number n_{Sc} of non-dummy rows in scope, which is inherent for oblivious selection (Proposition 12.1); the public scope Φ is the main lever on it.

Step	Communication	O's computation	Rounds
Link	$n_R^{\max}$ OPRF calls; PSI query and response (sublinear in $\bar{n}_O$)	per-run preprocessing $O(\bar{n}_O)$; per-query evaluation	2–3
Upload	$(d_R + 2)\, m$ seeded ciphertexts (half size)	$(d_R + 2)\, m$ key conversions	1
Aggregate	–	$\lvert\mathcal{E}\rvert\, m$ ciphertext–plaintext products; $\lvert\mathcal{E}\rvert \log_2 N_s$ rotations; $\lvert\mathcal{E}\rvert$ mask products	–
Release (×2 or 3)	2 ring elements each	1 re-randomization, 1 partial decryption, N exact samples	2–3

Proposition 12.1 (Work linear in the scope)**.** *Consider any two-party protocol in which O reads its register row by row and whose decrypted statistic is within $\tau_{\text{dec}} < B_O^2/2$ of $\mathbf{z}(D)$ except*

with probability p_{err}, for every input in the clipped domain. Suppose that O's views for any two cohorts of the same size are distinguishable with advantage at most adv*. Let $n_{\mathrm{Sc}} \le \bar{n}_O$ be the number of non-dummy rows in scope. Then O reads at least $(1 - 2p_{\mathrm{err}} - \mathrm{adv})\, n_{\mathrm{Sc}}$ rows in expectation. In v3.1,* adv *is the slack of G2, so O's work is $\Omega(n_{\mathrm{Sc}})$.*

Proof sketch. The proofs (Part A, Proposition 9.5) prove this in a cell-probe model, following the companion book on the protocol (Chapter 20 there). If O never reads a matched row, changing that row's $\mathbf{x}_O$ from $\mathbf{0}$ to $B_O\mathbf{e}_1$ does not change O's computation but moves B_{11} by $B_O^2 > 2\tau_{\mathrm{dec}}$, so a matched row is skipped with probability at most $2p_{\mathrm{err}}$. Whether a row is skipped is a function of O's view, so an unmatched row is skipped with probability at most adv more: otherwise O could tell a cohort that matches the row from one that does not. Summing over the non-dummy rows gives the bound. O knows its dummy rows and may skip them, so the bound is in n_{Sc}, not $\bar{n}_O$; $\bar{n}_O$ is chosen close to n_{Sc}. □

The measured runs use the chain of Section 10, with $\Omega_U = q_2$ and $\kappa = 40$; the operations, and so the timings, are the same at the evaluation settings.

Measured cost. We ran the full implementation at $\bar{n}_O$ = 106,496, 303,104 and 1,007,616 register rows with $n_R^{\max}$ = 20,000 padded queries ($p = 11$, two felts per PSI item, four OpenMP threads per party and four aggregation entries in parallel at O, on a 4-core 2.8 GHz Xeon with 15 GB of memory), and at 10^6 rows with $n_R^{\max}$ = 100,000 (Section 11.6). Table 32 gives the cost of one run; the communication equals the model's exact count. In these runs, at 10^6 rows, O spent 80 of its 152 s on the OPRF evaluation of its whole register (fresh key every run), which was then single-threaded, 19 s on PSI binning and interpolation, 18 s on the CKKS aggregation, 15 s on the PSI evaluation and 12 s on SanitizeBFV; 369 of the 475 MB of traffic is R's upload. With 100,000 queries the PSI evaluation and sanitization grow with the ten cuckoo tables (74 and 62 s) and the traffic to 756 MB.

Parallel OPRF. O's OPRF evaluation now runs in parallel over the rows, with the same key and the same group operations, and with outputs bit-identical to the single-threaded evaluation. On four threads it takes 1.8 instead of 7.8 s at 10^5 rows and 17.7 instead of 72.7 s at 10^6 rows (4.1×). We re-measured O's total by full runs, not by substituting the OPRF time: one run each at 10^5 and 10^6 rows (setting I, $\varepsilon = 1$, $n_R^{\max}$ = 20,000, release padding on). At 10^6 rows O took 91 s instead of 152 s, of which 18.6 s was the OPRF, and the run took 120 s end to end instead of 179 s, with the same peak memory (2.3 GB) and traffic; at 10^5 rows, 28.6 and 36 s (Table 32). The OPRF is now as costly as the PSI preprocessing (18.5 s) and the aggregation (16.8 s). Both runs linked exactly, and in each the second release reached R 2 ms after its deadline of 0.49 and 0.51 s (the first also waits for the aggregation).

The cost model, built from per-operation microbenchmarks and exact operation counts, failed with the inputs published before the runs (Figure 17): it overestimated O's total by 25–52% at every size, and single phases by up to a factor 2.8, which fails the 25% agreement we had required before running. Three causes were found. The microbenchmarks behind it ran 1.4–2.2 times faster on the idle machine than in the published files. Its interpolation term assumed full partitions, whereas the fill is 0.15 to 0.58. Its aggregation speedup was measured with one thread per entry, not with the four threads per entry of these runs. We refreshed the model accordingly. All microbenchmarks were re-measured on the idle machine. The interpolation is

Table 32: Measured cost of one run (setting I, $\varepsilon = 1$, mean of 3 runs; four threads per party), against the cost model with the inputs published before the runs and with the refreshed model, both for the code that was run (single-threaded OPRF). The last three rows: one run each with the parallel OPRF and release padding, and the refreshed model for it. Wall time in seconds.

$\bar{n}_O$	106,496	303,104	1,007,616	1,007,616
$n_R^{\max}$	20,000	20,000	20,000	100,000
O, measured	34.5	56.6	151.9	270.5
O, model (published inputs)	50.2	77.0	193.9	357.3
O, refreshed model	32.2	55.0	146.4	252.2
run, end to end	42	67	179	313
O's CPU time, all threads	78	109	256	550
peak memory O / R (GB)	0.76 / 0.36	0.99 / 0.66	2.33 / 1.73	2.34 / 1.79
traffic (MB)	133	205	475	756
O, measured, parallel OPRF	28.6	–	91.1	–
O, refreshed model, parallel OPRF	25.9	–	85.3	–
run, end to end, parallel OPRF	36	–	120	–

split into the membership polynomials, whose degree is always D, and the label interpolation, whose cost is scaled by the expected sum of squared partition counts under the binomial bin loads; on four threads the model also accounts for the static schedule, which gives the fullest partitions to one thread. It now predicts the measured interpolation to within 0.88–0.98 at 10^5 to 10^6 rows. The aggregation uses the speedup of the configuration that is run (2.83 for four threads $\times$ four entries, against 3.51 for one thread per entry). Evaluated for the code of the 78 runs, the refreshed model gives O's total to within a factor 0.95–1.12 in every run, against 0.94–1.25 with the earlier re-measured inputs. Every phase of at least 2 s at O lies within 25% in every run, except the upload conversion at 10^6 rows (1.11–1.40: the isolated microbenchmark misses the memory traffic of 738 MB of ciphertexts). On R's side, its OPRF takes 1.12–1.31 times the model. For the runs with the parallel OPRF the model gives 1.07 (10^6 rows) and 1.10 (10^5 rows) of O's total. The earlier extrapolation (260 s of O's CPU on one core and 76 s on four cores at 10^6) assumed one felt and 2,000 queries; with two felts and 20,000 queries the refreshed model predicts 85 s on four cores, and 91 s were measured.

13 Extensions and limitations

A deviating researcher. v3.1 makes no claim against a malicious R. Making output privacy hold against one requires (i) zero-knowledge proofs at upload that the ciphertexts are well formed with short errors, that $b_t \in \{0,1\}$, $\|\dot{\mathbf{x}}_t\| \le B_R$, $(1-b_t)\dot{\mathbf{x}}_t = \mathbf{0}$ and $\sum_t b_t \le n_R^{\max}$, together with a coarse norm bound that rules out wraparound (candidates include lattice-based and VOLE-based proofs); (ii) key generation with R committing first, plus proofs that its shares are short; and (iii) OPRF inputs bound to a certified cohort. A homomorphic check that relies on R's own decryption share cannot do this job.

Remark 13.1 (Hiding membership)**.** Replacing labeled PSI with circuit-PSI (for example VOLE-PSI with payloads [68]) gives both parties shares of $\mathbf{b}$ and $\dot{\mathbf{X}}_R$ in τ-order, so R no longer learns I. The aggregation would move to BFV or use wide statistical masking, because shares modulo 2^ℓ do not reduce inside CKKS. $\mathbf{A}$ must then be computed under encryption and noised,

and the adjacency extends to changing a record's match status, which raises the sensitivity.

Prediction after training (future work). After a run R holds $\tilde{\boldsymbol{\theta}} = (\tilde{\boldsymbol{\theta}}_R; \tilde{\boldsymbol{\theta}}_O)$. Predicting y_i for a new individual needs $\mathbf{x}_{O,i}^\top \tilde{\boldsymbol{\theta}}_O$, and only O holds $\mathbf{x}_{O,i}$. Split inference, in which O evaluates $\mathbf{x}_{O,i}^\top \tilde{\boldsymbol{\theta}}_O$ under encryption and releases it as in Algorithm 5, is not part of v3.1. Three things are unspecified: the sensitivity of a prediction release, its noise, and its charge to the ledger. Each decrypted prediction is a projection of $\mathbf{x}_{O,i}$ in a direction R knows, so repeated queries or model versions reconstruct $\mathbf{x}_{O,i}$ unless every one is charged.

What v3.1 supports is distillation to an R-only model. The student coefficient that regresses the joint model's fitted values on R's features is $\hat{\boldsymbol{\beta}}_R^{\mathrm{st}} = \mathbf{A}^{-1}\mathbf{c}_R = \boldsymbol{\beta}_R + \mathbf{A}^{-1}\mathbf{C}\boldsymbol{\beta}_O$. R computes $\mathbf{A}^{-1}\hat{\mathbf{c}}_R$ from its exact $\mathbf{A}$ and the released $\hat{\mathbf{c}}_R$; this is post-processing, costs no budget and needs no interaction at test time. Its loss against the joint fitted values is $\boldsymbol{\beta}_O^\top \mathbf{S} \boldsymbol{\beta}_O$, with $\mathbf{S} = \mathbf{B} - \mathbf{C}^\top \mathbf{A}^{-1}\mathbf{C}$ the Schur complement of $\mathbf{A}$ in $\mathbf{G}$: when it is small, distillation suffices; when it is large, O's features carry signal that R's cannot proxy.

Further limitations.

- The two evaluation settings were not fully re-tuned: setting I keeps $\vartheta^\star = 2.5$ and $\kappa_\Psi = 2$, and setting P keeps $\kappa_\Psi = 2$, from earlier variants of the rule, and setting I's selection criterion averages over n (Section 10).
- Setting P is the best candidate of a grid on whose edge it lies, under a refusal bound that binds, and it was chosen after its first choice failed. Its selection covered $\sigma_1/\sigma_2 \geq 0.46$; 8 of the 29 benchmark datasets lie below, and on solar its gate refused up to 7.2% of releases (Section 10).
- Exact-identifier linkage only.
- Linear models only; generalized linear models via approximate sufficient statistics are future work.
- The variance is model-based. A sandwich variance would need a third release of $\sum_i r_i^2 \mathbf{x}_i \mathbf{x}_i^\top$, a ciphertext–ciphertext computation with its own sensitivity.
- No verifiability of O's computation.

14 Summary and open items

Protocol v3.1 links records with an OPRF and labeled PSI, aggregates under two-party threshold CKKS, and makes each release a threshold decryption in which O's share carries the DP noise and floods the encryption error. O never decrypts, R never sees a noiseless statistic that depends on O's data, and the budget is enforced by O's control of its share. Two releases separate the choice of the ridge from the estimate, which makes the bias theory and the intervals valid. The protocol and its guarantees are stated for any admissible parameter vector (Section 10); Calibrate derives the flooding parameter, the scale and the noise from it, and the ledger composes runs, fixed or adaptive, under a per-register cap: it charges a run's full cost at admission and accounts the smudging step in hockey-stick divergence, so that across runs it no longer raises the flooding parameter. The evaluation uses two settings of the same protocol, chosen on synthetic data by a pre-registered rule. Setting I keeps the interval guarantee: with an exact Monte Carlo quantile for the stage-1 radius and $f_1 = 0.25$ it lowers the ridge threshold by 35%, and its intervals cover in every simulated regime. Setting P gives up the intervals for a

smaller ridge. It was chosen subject to a bound on gate refusals, after a first choice failed on block-normalized real data. With block normalization it beats setting I on 18–24 of 29 datasets at every ε and nearly halves the median gap to AdaSSP at $\varepsilon = 1$ (+14% against +25%), but AdaSSP on the joined table still predicts better on 20 of them.

Remaining obligations.

O1. The worst-case error bound: proven for the circuit as written, as an explicit function of the parameters, for centred key switching ($B_{\text{tot}} \approx 2^{25.3}$ at the proofs' example) and for OpenFHE's uncentred key switching in the model it defines ($2^{26.6}$ at the evaluation settings), with decoupled scales; that OpenFHE 1.5.1 realizes the model rests on reading its source, and (M5) for its sampler on a numerical check. Proofs for the restructured circuit, which is not used, are open.

O2. Smudging for the discrete Gaussian, for every $\sigma_c \geq 2^{42}$ in statistical distance and with no floor in hockey-stick divergence (proven), and a concrete constant-time exact sampler (open). Releases are now padded to a deadline fixed from public parameters, which removes the dependence of their latency on η except in deadline overruns, whose probability is measured, not bounded.

O3. Simulation security of threshold key generation against a semi-honest R: proven, by a reduction to plain RLWE, since O sends only its public-key share (hybrid H4 of Theorem 7.2).

O4. The two-stage rule, the RSS correction and the coverage of both intervals, checked by simulation (Sections 11.3 and 11.7, the latter at setting I) and, for the Wald interval, also on fixed cohorts of five real datasets with calibrated noise (Section 11.3); on real data the intervals for $\boldsymbol{\beta}$ are checked only against a full-data reference (Section 11.4).

O5. PSI parameters with explicit failure probabilities (correctness only, not charged to δ). The item length now meets them at $\bar{n}_O = 10^6$: two felts per item and a label-consistency test give $p_{\text{pass}} = 2^{-67.5}$ there with $n_R^{\max} = 20{,}000$ (Appendix C), and SETUP checks the bound. The cuckoo-hashing failure probability is now bounded for an insertion that finds an assignment whenever one exists, which the implementation's augmenting-path fallback provides, and the worst-case bound B_ψ on the evaluation noise of the PSI replies, which SANITIZEBFV floods, is proved for the textbook BFV model; checking OpenFHE's BFVrns against that model remains.

Other open items. Whether the rule for runs fixed in advance stays valid when requests are chosen adaptively (Proposition 7.2); a proof of the subgaussianity of OpenFHE's error sampler, which is checked only numerically, and of the match between OpenFHE's key switching and the model in which the error bound is proved, which rests on reading its source; the free-set boundary for $\mathbf{y}^\top \mathbf{y}$ and a closed form for $d_O = 1$; explicit constants for the bootstrap RSS correction; a guarantee for the working-model interval, with the data-driven variance, when clipping binds (the reported bias-aware interval uses a public variance bound and is valid); a bound on the downward bias of $\lambda_{\min}(\hat{\mathbf{G}}_1)$ sharp enough to turn the switch-off factors 1.2 and 1.4 into bounds; and sixteen citation details flagged for verification in the proofs. The no-wraparound condition at the earlier levels, which earlier drafts listed here, is proved, and CALIBRATE, which sizes the modulus chain, is proved sound.

A Mechanism bias and the optional correction

Throughout this section σ denotes the stage-2 slot s.d. σ_2, and (λ, Ψ) is fixed independently of the stage-2 noise (Proposition 9.1).

A.1 Noise model

Write the stage-2 statistics as $\hat{\mathbf{G}}_2 = \mathbf{G} + \mathbf{E}$ and $\hat{\mathbf{c}}_2 = \mathbf{c} + \mathbf{f}$, with

$$\mathbf{E} = \begin{bmatrix} \mathbf{0} & \mathbf{E}_C \\ \mathbf{E}_C^\top & \mathbf{E}_B \end{bmatrix}, \qquad \mathbf{f} = \begin{bmatrix} \mathbf{f}_R \\ \mathbf{f}_O \end{bmatrix}. \tag{11}$$

The top-left block of $\mathbf{E}$ is identically zero, since $\mathbf{A}$ is exact. Each independent entry on and above the diagonal of the O-dependent blocks is drawn with variance σ^2 and mirrored, so every entry, the diagonal included, has variance σ^2. With $\chi_{ij} = 0$ iff i, j both lie in R's block and $\chi_{ij} = 1$ otherwise,

$$\mathbb{E}[E_{ij}E_{kl}] = \sigma^2 \chi_{ij}\big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \delta_{ij}\delta_{ik}\delta_{il}\big). \tag{12}$$

The last term removes the diagonal double-count of the GOE convention. $\mathbf{E}$ and $\mathbf{f}$ occupy disjoint entries of the release, are independent, and are symmetric about zero. (The lattice noise induces slot-domain noise with these first and second moments; the CKKS error is flooded and of lower order.)

Condition on the realized sample, so $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$ is fixed, and set $\mathbf{P} = \mathbf{G}^{-1}$,

$$\vartheta_0 = \frac{\lambda_{\min}(\mathbf{G})}{2\sigma\sqrt{p}}, \qquad \mathcal{E}_\eta = \big\{\|\mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}\|_{\mathrm{op}} \le \eta\big\} \quad (0 < \eta < 1). \tag{13}$$

$\mathcal{E}_\eta$ is invariant under $\mathbf{E} \mapsto -\mathbf{E}$. Define

$$M(\mathbf{P}) = \big(\mathbf{P}^2 + \mathrm{tr}(\mathbf{P})\mathbf{P} - \mathbf{P}\,\mathrm{diag}(\mathbf{P})\big)\Pi_O + \big(\mathbf{P}\Pi_O\mathbf{P} + \mathrm{tr}(\Pi_O\mathbf{P})\mathbf{P}\big)\Pi_R, \tag{14}$$

where $\mathrm{diag}(\mathbf{P})$ is the diagonal matrix formed from the diagonal of $\mathbf{P}$.

Remark A.1 (The unconditional mean does not exist)**.** With Gaussian $\mathbf{E}$, $(\mathbf{G}+\mathbf{E})^{-1}(\mathbf{c}+\mathbf{f})$ has no finite mean: already for $p = 1$, $\mathbb{E}[1/(a+e)]$ diverges. Statements about its bias must therefore be conditional, or about a truncated estimator. We condition on $\mathcal{E}_\eta$, whose complement has exponentially small probability.

A.2 Result

Theorem A.1 (Leading-order conditional bias)**.** *Fix $\eta \in (0,1)$ with $\eta\vartheta_0 > 1$. Then, for $\tilde{\boldsymbol{\beta}} = (\mathbf{G} + \mathbf{E})^{-1}(\mathbf{c} + \mathbf{f})$,*

$$\mathbb{E}\big[\tilde{\boldsymbol{\beta}} \mid \mathcal{E}_\eta\big] - \hat{\boldsymbol{\beta}} = \sigma^2 M(\mathbf{P})\,\hat{\boldsymbol{\beta}} + O(\sigma^4) + O\big(e^{-p(\eta\vartheta_0 - 1)^2/2}\big),$$

where the constants are explicit: in the norm $\|\mathbf{v}\|_{\mathbf{G}} = (\mathbf{v}^\top\mathbf{G}\mathbf{v})^{1/2}$ the remainder is at most $\Pr(\mathcal{E}_\eta)^{-1}\big[\kappa_p^4\vartheta_0^{-4}(1-\eta)^{-1}(\|\hat{\boldsymbol{\beta}}\|_{\mathbf{G}} + \sigma\sqrt{\mathrm{tr}\,\mathbf{P}}) + 2\sqrt{2}\,\kappa_p^2\vartheta_0^{-2}e^{-p(\eta\vartheta_0-1)^2/2}\|\hat{\boldsymbol{\beta}}\|_{\mathbf{G}}\big]$, with $\kappa_p = (2\sqrt{p} + 2.512 + 2\sqrt{2})/(2\sqrt{p})$, which is 1.81 at $p = 11$ (proofs, Part B, Theorem S9). The moment-vector noise $\mathbf{f}$ contributes nothing at this order; it is variance-only. The term $-\mathbf{P}\,\mathrm{diag}(\mathbf{P})\Pi_O$ is the single-draw correction to the GOE contraction.

The proof is given in Appendix B. Setting $\Pi_O \to \mathbf{I}$ and $\Pi_R \to \mathbf{0}$ recovers the homogeneous-Gram bias, the correct check when every block is noised.

Remark A.2 (When the expansion is valid)**.** $\mathbf{PE}$ is similar to $\mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}$, so the Neumann series for $(\mathbf{I}+\mathbf{PE})^{-1}$ converges iff $-\mathbf{G} \prec \mathbf{E} \prec \mathbf{G}$; on $\mathcal{E}_\eta$ it converges geometrically with ratio η. For the largest eigenvalue, $\mathbb{E}\lambda_{\max}(\mathbf{E}) \le 2\sigma\sqrt{p}$, and $\lambda_{\max}(\mathbf{E})$ has sub-Gaussian tails above its mean; $\lambda_{\max}(-\mathbf{E})$ has the same law. Bounding the two one-sided tails of $\|\mathbf{E}\|_{\mathrm{op}} = \max\{\lambda_{\max}(\mathbf{E}), \lambda_{\max}(-\mathbf{E})\}$ separately gives $\Pr(\mathcal{E}_\eta^c) \le 2\exp(-p(\eta\vartheta_0 - 1)^2)$ for $\eta\vartheta_0 > 1$. Whether $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \le 2\sigma\sqrt{p}$ also holds is not proved. Since $\lambda_{\min}(\mathbf{G})$ grows linearly in n while σ does not depend on n, the condition holds with high probability once n is large enough; below that, a ridge is required.

A.3 Schur-complement reading

Let $\mathbf{S} = \mathbf{B} - \mathbf{C}^\top\mathbf{A}^{-1}\mathbf{C}$ be the Schur complement of $\mathbf{A}$ in $\mathbf{G}$. Block inversion gives $\Pi_O\mathbf{P}\Pi_O = \mathbf{S}^{-1}$, hence

$$\mathrm{tr}(\Pi_O\mathbf{P}) = \mathrm{tr}(\mathbf{S}^{-1}), \qquad \mathrm{tr}(\mathbf{P}) = \mathrm{tr}(\mathbf{A}^{-1}) + \mathrm{tr}\big(\mathbf{A}^{-1}\mathbf{C}\mathbf{S}^{-1}\mathbf{C}^\top\mathbf{A}^{-1}\big) + \mathrm{tr}(\mathbf{S}^{-1}). \tag{15}$$

The two trace weights governing the leading bias are $\mathrm{tr}(\mathbf{G}^{-1})$ for the O-coefficients and $\mathrm{tr}(\mathbf{S}^{-1})$ for the R-coefficients. When O's features carry little signal beyond R's, $\mathbf{S}$ is near-singular and the private estimator is worst-biased in R's own coefficients.

A.4 Ridge

Let $\mathbf{G}_\lambda = \mathbf{G} + \lambda\Psi$, $\mathbf{P}_\lambda = \mathbf{G}_\lambda^{-1}$, and let ϑ_λ and $\mathcal{E}_{\eta,\lambda}$ be as in (13) with $\mathbf{G}$ replaced by $\mathbf{G}_\lambda$.

Theorem A.2 (Ridge mechanism bias)**.** *Suppose (λ, Ψ) is independent of $(\mathbf{E}, \mathbf{f})$ and $\eta\vartheta_\lambda > 1$. Then*

$$\mathbb{E}\big[\tilde{\boldsymbol{\theta}} \mid \mathcal{E}_{\eta,\lambda}\big] - \boldsymbol{\theta}_\lambda = \sigma^2 M(\mathbf{P}_\lambda)\,\boldsymbol{\theta}_\lambda + O(\sigma^4) + O\big(e^{-p(\eta\vartheta_\lambda - 1)^2/2}\big),$$

with the moment-vector noise $\mathbf{f}$ again variance-only.

The proof is that of Theorem A.1 under $\mathbf{G} \mapsto \mathbf{G}_\lambda$, which leaves $\mathbf{E}$, its covariance (12) and $\mathbf{f}$ untouched.

Proposition A.1 (Total bias against least squares: shrinkage split)**.** *If the estimand is $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$, the total bias of the private ridge splits as*

$$\mathbb{E}[\tilde{\boldsymbol{\theta}} \mid \mathcal{E}_{\eta,\lambda}] - \hat{\boldsymbol{\beta}} = \underbrace{\boldsymbol{\theta}_\lambda - \hat{\boldsymbol{\beta}}}_{\textit{deterministic}} + \underbrace{\mathbb{E}[\tilde{\boldsymbol{\theta}} \mid \mathcal{E}_{\eta,\lambda}] - \boldsymbol{\theta}_\lambda}_{\textit{mechanism}}, \qquad \boldsymbol{\theta}_\lambda - \hat{\boldsymbol{\beta}} = -\lambda\mathbf{P}_\lambda\Psi\hat{\boldsymbol{\beta}}.$$

The deterministic term grows in λ. The leading mechanism term decays in λ: for $\Psi = \mathbf{I}$, $\|\sigma^2 M(\mathbf{P}_\lambda)\boldsymbol{\theta}_\lambda\| \le (3 + p + d_O)\,\sigma^2\|\mathbf{c}\|/(\lambda_{\min}(\mathbf{G}) + \lambda)^3$, which is $O(\sigma^2/\lambda^3)$; for $\Psi = \Pi_O$ with $\mathbf{A}$ invertible it is $O(\sigma^2/\lambda)$.

For $\Psi = \mathbf{I}$, $\lambda_{\min}(\mathbf{G}_\lambda) = \lambda_{\min}(\mathbf{G}) + \lambda$ grows without bound. For $\Psi = \Pi_O$, $\lambda_{\min}(\mathbf{G} + \lambda\Pi_O) \to \lambda_{\min}(\mathbf{A})$ as $\lambda \to \infty$, so the O-block ridge cannot validate the expansion when $\lambda_{\min}(\mathbf{A})$ is too small, which is why the stage-1 rule falls back to $\Psi = \mathbf{I}$.

A.5 Why the ridge must not depend on the estimation noise

Earlier drafts chose (λ, Ψ) from the same noisy Gram matrix used for estimation. That choice is post-processing and costs no privacy, but it makes λ a function of $\mathbf{E}$ at first order, so Theorem A.2 no longer applies. In simulation, with λ fixed independently the correction below removed 87–98% of the mechanism bias, whereas under the adaptive rule it removed essentially none: the residual after correction was 1.0–1.6 times the uncorrected bias. The two-stage rule of Section 9.2 keeps the adaptivity and restores independence.

A.6 Optional zero-budget correction

Since σ, d_R, d_O and Π_O are public and $\hat{\mathbf{G}}_2$ is already private, the bias is estimable by post-processing:

$$\tilde{\boldsymbol{\theta}}_{\mathrm{bc}} = \tilde{\boldsymbol{\theta}} - \sigma^2 M(\tilde{\mathbf{P}})\,\tilde{\boldsymbol{\theta}}. \tag{16}$$

Substituting $\hat{\mathbf{G}}_2$ for $\mathbf{G}$ inside a term already of order σ^2 injects only $O(\sigma^3)$ fluctuation and $O(\sigma^4)$ bias. One must not additionally invert the deterministic shrinkage, which would re-amplify precisely the directions the ridge protects. In practice the bias removed is small (Section 11.8). The squared mechanism bias of one run is at most 0.05% of the mean squared error around $\boldsymbol{\theta}_\lambda$ at the previous setting and 0.19% under setting I (cell averages). The correction lowers that error by about 0.5% at the previous setting and by 1.4–1.7% under setting I while the ridge is on, mostly by reducing variance. The correction is therefore an option of the protocol, off in our evaluation.

B Proof of Theorem A.1

Throughout, condition on the sample so that $\mathbf{G}$, $\mathbf{c}$ and $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$ are fixed, and write $\mathbf{P} = \mathbf{G}^{-1}$ and $\mathbf{H} = \mathbf{P}\mathbf{E}$.

Step 1: Neumann expansion on $\mathcal{E}_\eta$. On $\mathcal{E}_\eta$ the spectral radius of $\mathbf{H}$ is at most $\eta < 1$ (Remark A.2), so

$$\tilde{\boldsymbol{\beta}} = (\mathbf{G}+\mathbf{E})^{-1}(\mathbf{c}+\mathbf{f}) = (\mathbf{I}+\mathbf{H})^{-1}\mathbf{P}(\mathbf{c}+\mathbf{f}) = \big(\mathbf{I}-\mathbf{H}+\mathbf{H}^2-\mathbf{H}^3+\mathbf{H}^4(\mathbf{I}+\mathbf{H})^{-1}\big)\big(\hat{\boldsymbol{\beta}}+\mathbf{P}\mathbf{f}\big),$$

and $\|\mathbf{H}^4(\mathbf{I}+\mathbf{H})^{-1}\|$ is bounded by a constant times $\|\mathbf{E}\|_{\mathrm{op}}^4/(1-\eta)$ in the $\mathbf{G}$-norm.

Step 2: odd terms and the moment noise vanish. $\mathcal{E}_\eta$ is invariant under $\mathbf{E} \mapsto -\mathbf{E}$, $\mathbf{E}$ is symmetric about zero, and $\mathbf{f}$ is independent of $\mathbf{E}$ and symmetric about zero. Hence $\mathbb{E}[\mathbf{H}^m \mid \mathcal{E}_\eta] = 0$ for odd m, and every term containing $\mathbf{P}\mathbf{f}$ has zero conditional expectation. Therefore

$$\mathbb{E}[\tilde{\boldsymbol{\beta}} \mid \mathcal{E}_\eta] = \hat{\boldsymbol{\beta}} + \mathbb{E}[\mathbf{H}^2 \mid \mathcal{E}_\eta]\,\hat{\boldsymbol{\beta}} + O(\sigma^4).$$

Step 3: removing the conditioning.

$$\mathbb{E}[\mathbf{H}^2 \mid \mathcal{E}_\eta] - \mathbb{E}[\mathbf{H}^2] = \big(\mathbb{E}[\mathbf{H}^2\mathbf{1}_{\mathcal{E}_\eta}] - \mathbb{E}[\mathbf{H}^2]\Pr(\mathcal{E}_\eta)\big)/\Pr(\mathcal{E}_\eta),$$

whose norm is at most $2\big(\mathbb{E}\|\mathbf{H}\|^4\big)^{1/2}\Pr(\mathcal{E}_\eta^c)^{1/2}/\Pr(\mathcal{E}_\eta) = O\big(\sigma^2 e^{-p(\eta\vartheta_0-1)^2/2}\big)$ by Cauchy–Schwarz and Remark A.2.

Step 4: the second-moment contraction. Write $\mathbf{p} = (P_{11}, \ldots, P_{pp})^\top$ for the diagonal of $\mathbf{P}$ and $\mathbf{X} = (\chi_{ij})$.

Lemma B.1. *For any fixed symmetric* $\mathbf{P}$*, with* $\circ$ *the Hadamard product,* $\mathbb{E}[\mathbf{EPE}] = \sigma^2\big(\mathbf{X} \circ \mathbf{P} + \operatorname{diag}(\mathbf{Xp}) - \Pi_O \operatorname{diag}(\mathbf{p})\big)$.

Proof. Entrywise, $\mathbb{E}[(\mathbf{EPE})_{ab}] = \sum_{i,j} P_{ij}\mathbb{E}[E_{ai}E_{jb}]$, and (12) gives $\mathbb{E}[E_{ai}E_{jb}] = \sigma^2\chi_{ai}(\delta_{aj}\delta_{ib} + \delta_{ab}\delta_{ij} - \delta_{ai}\delta_{aj}\delta_{ab})$. Summing against the three terms gives $\sigma^2\chi_{ab}P_{ab}$, $\sigma^2\delta_{ab}\sum_i \chi_{ai}P_{ii}$, and $-\sigma^2\delta_{ab}\chi_{aa}P_{aa}$. With $\chi_{aa} = \mathbf{1}[a \in O]$ these are the (a, b) entries of $\mathbf{X} \circ \mathbf{P}$, $\operatorname{diag}(\mathbf{Xp})$ and $-\Pi_O \operatorname{diag}(\mathbf{p})$. □

Writing $\chi_{ij} = 1 - (\Pi_R)_{ii}(\Pi_R)_{jj}$ turns the first two terms into projector form: $\mathbf{X} \circ \mathbf{P} = \mathbf{P} - \Pi_R\mathbf{P}\Pi_R$ and $\operatorname{diag}(\mathbf{Xp}) = \operatorname{tr}(\mathbf{P})\mathbf{I} - \operatorname{tr}(\Pi_R\mathbf{P})\Pi_R$. Left-multiplying by $\mathbf{P}$ and using $\Pi_O \operatorname{diag}(\mathbf{p}) = \operatorname{diag}(\mathbf{P})\Pi_O$,

$$\mathbb{E}[\mathbf{PEPE}] = \sigma^2\big[\mathbf{P}^2 - \mathbf{P}\Pi_R\mathbf{P}\Pi_R + \operatorname{tr}(\mathbf{P})\mathbf{P} - \operatorname{tr}(\Pi_R\mathbf{P})\mathbf{P}\Pi_R - \mathbf{P}\operatorname{diag}(\mathbf{P})\Pi_O\big].$$

Step 5: regrouping. Substituting $\Pi_O = \mathbf{I} - \Pi_R$, the first four terms collapse to $(\mathbf{P}^2 + \operatorname{tr}(\mathbf{P})\mathbf{P})\Pi_O + (\mathbf{P}\Pi_O\mathbf{P} + \operatorname{tr}(\Pi_O\mathbf{P})\mathbf{P})\Pi_R$, which gives $\mathbb{E}[\mathbf{H}^2] = \sigma^2 M(\mathbf{P})$. Combining with Steps 2 and 3 proves the theorem. □

Degenerate check. With $\Pi_O = \mathbf{I}$ and $\Pi_R = \mathbf{0}$ the formula gives $\sigma^2(\mathbf{P}^2 + \operatorname{tr}(\mathbf{P})\mathbf{P} - \mathbf{P}\operatorname{diag}(\mathbf{P}))$; for $p = 1$ this is $\sigma^2 a^{-2}\hat{\boldsymbol{\beta}}$, matching a direct expansion of $c/(a + e)$ with $\operatorname{Var}(e) = \sigma^2$.

Extension to ridge (Theorem A.2). Replacing $\mathbf{G}$ by $\mathbf{G}_\lambda = \mathbf{G} + \lambda\Psi$ with (λ, Ψ) independent of $(\mathbf{E}, \mathbf{f})$ leaves $\mathbf{E}$, its covariance and $\mathbf{f}$ untouched, and Steps 1–5 hold verbatim with $\mathbf{P} \mapsto \mathbf{P}_\lambda$. If λ depended on $\mathbf{E}$, Step 2 would fail: $\mathbf{P}_\lambda$ would no longer be fixed under the conditional expectation.

C Labeled PSI with an OPRF and record alignment

R must learn I and the alignment π, while O learns only public sizes. We use labeled PSI in the style of Chen, Laine and Rindal [16, 17, 19] on OPRF values. The construction runs under BFV with R's own key, whose exact arithmetic modulo t is needed for the zero test and label recovery. Algorithm 2 gives the steps; the toy example deck works them through with numbers.

Pre-processing at O (fresh every run). O pads its scope register to $\bar{n}_O$, draws a uniformly random permutation τ of its rows and a fresh OPRF key k. Each row j becomes the item $F_k(\mathrm{id}_j)$, mapped into $\mathbb{Z}_t^F$ as F field elements (*felts*, as in APSI) that each avoid the padding root $t - 1$, with label $\tau(j)$ encrypted under F independent keys derived from $F'_k(\mathrm{id}_j)$, one per felt ($F = 2$ in the evaluation, so items have 64 bits). Felt f of a bin occupies slot $fN_{\mathrm{bins}} + b$, that is, ciphertext f; each slot has its own membership and label polynomial, and the items of a partition differ in every felt. Each item is hashed into all h candidate bins; each bin is split into α partitions of at most D items, where D and α are public functions of $(\bar{n}_O, h, N_{\mathrm{bins}}, d_{\mathrm{cap}})$ chosen so that overflow has probability at most δ_{psi}. Every partition's membership polynomial is padded with the root $t - 1$ to degree exactly D, and its label polynomial interpolates (item, encrypted label) pairs.

Query at R. R pads its identifiers to $n_R^{\max}$, obtains F_k and F'_k of each through the OPRF, cuckoo-hashes [64] the items into N_{bins} bins with the same h hash functions (at most one item per bin), and sends BFV encryptions of the slot-wise powers $\mathbf{w}^{2^i}$, $i < W$, with $W = \lceil \log_2 D \rceil + 1$.

Evaluation and reply at O. O forms the encryptions of $\mathbf{w}^d$, $d \le D$, as products of window ciphertexts, evaluates the membership and label polynomials of each partition as plaintext-coefficient linear combinations, masks with $\mathbf{r}_1, \mathbf{r}_2$ uniform in $(\mathbb{Z}_t^*)^{N_{\text{bins}}}$ from a CSPRNG, and returns

$$\textsc{SanitizeBFV}\big(\mathbf{r}_1 \odot P(\mathbf{w})\big), \qquad \textsc{SanitizeBFV}\big(\mathbf{r}_2 \odot P(\mathbf{w}) + L(\mathbf{w})\big).$$

The first is 0 in a bin iff R's item there is in O's set, and the second is then that item's encrypted label; otherwise both are uniform. The mask $\mathbf{r}_1$ hides $P(\mathbf{w})$ at non-matches, which would otherwise carry information about O's items. SanitizeBFV and its parameters $(\kappa_\psi, B_\psi, q')$ are discussed under G4 in Section 8.

Recovery at R. For each occupied bin and partition, R declares a match only if the flag is zero in all F felt slots of the bin *and* the F label felts, each opened under its own key derived from F'_k of R's item, give the same position; that position is $\pi(i) = \tau(j)$. A true match always passes both tests. Flags that are zero in every felt but whose labels disagree come from felts matched by different items of the partition, which the zero test alone allows, and R treats them as no match, without an abort. R aborts the run if an identifier gets two different positions (ambiguous), if a position lies outside $[1, \bar{n}_O]$, or if two identifiers get the same position; an honest O assigns distinct positions, so only a false match causes any of these. When a bin has more than one partition ($\alpha > 1$; $\alpha = 5$ at $\bar{n}_O = 10^6$ in the evaluation parameters), R also learns which partition flags its match. That index depends on the loads and insertion order of O's items in the bin, which are pseudorandom to R outside I, and the simulator of G4 reproduces it (Section 8).

Remark C.1 (Collision probability)**.** A false match requires a query and the items of one partition of its bin to share all F felt values, possibly from different items. Felts are uniform on $t-1$ values (up to the bias of reducing a 256-bit hash, which the implementation includes), independent across felts, and the items of a partition differ in every felt, so with partition loads $c_a \le D$ summing to at most the public load bound B,

$$p_{\text{felt}} = \frac{n_R^{\max} S_F}{(t-1)^F}, \qquad S_F = \lfloor B/D \rfloor D^F + (B \bmod D)^F,$$

bounds the probability that some query has all F flags zero in some partition. Such an event passes the label test only if $F-1$ independent keys of the query hit given values, and only an event that passes can abort the run (ambiguous, out of range, or repeated position) or change the output:

$$p_{\text{pass}} = p_{\text{felt}}\, t^{-(F-1)}, \qquad p_{\text{silent}} = p_{\text{pass}}\, \bar{n}_O / t.$$

p_{pass} is the false-match term p_{false} of the slack of G1 and G2 (Section 8), and Setup refuses unless $p_{\text{pass}} \le \delta_{\text{psi}}$. With $F = 1$ there is no label test, $p_{\text{pass}} = p_{\text{felt}}$, and $t \approx 2^{32}$ fails the constraint at every evaluation size ($p_{\text{pass}} = 2^{-18.8}$ at $\bar{n}_O = 4{,}096$, $n_R^{\max} = 600$; the earlier per-item estimate $n_R^{\max} \bar{n}_O h/(N_{\text{bins}}(t-1))$ gave 8.5×10^{-5} at 10^6 and $n_R^{\max} = 2{,}000$). The evaluation uses $F = 2$, $N_{\text{bins}} = 2^{14}$, $h = 3$, $D = 64$. At $\bar{n}_O = 10^6$ ($B = 308$, $\alpha = 5$) and $n_R^{\max} = 20{,}000$: $p_{\text{felt}} = 2^{-35.5}$, which alone exceeds $\delta_{\text{psi}} = 2^{-40}$ and is why the label test is

part of the match rule; $p_{\text{pass}} = 2^{-67.5}$ and $p_{\text{silent}} = 2^{-79.6}$. The per-item formula with F felts, $n_R^{\max}\bar{n}_O h/(N_{\text{bins}}(t-1)^2)$, which counts only collisions of a whole item, gives $2^{-42.2}$. At $n_R^{\max} = 100{,}000$, $p_{\text{pass}} = 2^{-65.2}$. Two felts double the query, the powers and the replies per cuckoo table at the same N_{bins}; SanitizeBFV's conditions are unchanged, since every reply ciphertext is as before.

Remark C.2 (Pre-shuffle necessity)**.** Without τ, the positions $\pi(i)$ recovered by R would reflect O's original database ordering (e.g. by region or registration date), leaking attributes outside the DP guarantee. The fresh shuffle makes positions uniformly random. It must be applied by O inside the protocol, not assumed of the input files.

Remark C.3 (Linkage privacy)**.** O sees $n_R^{\max}$ uniformly random group elements in the OPRF and BFV ciphertexts under R's key in the PSI, so it learns neither I nor n_R.

D Status of the proofs

The companion proofs document [83] (Part A: cryptography and DP [84]; Part B: sensitivity and statistics [85]; and a composition appendix on composition across runs and the hockey-stick ledger [86]) labels every step as exact, first order, asymptotic, numerical or open. Several results were first written in the companion book on the protocol and added to the proofs after a second reading that checked every step and constant (Part A, Sections 4, 7 and 9; Part B, Sections S7–S10). Table 33 summarizes it.

Table 33: Status of the claims.

Claim	Status	Notes
Encoding sensitivity (Lemma 7.1)	proven	$\sqrt{2/N}$ and the $\sqrt{N}$ rounding term
Sensitivity of the centre, eq. (6)	proven	any encoding scales, any changed row; $\Delta_{x,2} \approx 2^{27.6}$ at the evaluation settings
Smudging (Lemma 7.2)	proven	explicit constants; the factor $1 + 2^{-80}$ needs $\sigma_c \geq 2^{42}$ (earlier stated for 2^{20}); hockey-stick form (Remark 7.5) proven with no floor
Worst-case B_{tot} (Theorem 7.1)	proven, explicit	$\approx 2^{25.3}$ for the circuit as written with centred key switching; $\approx 2^{26.6}$ for OpenFHE's key switching, proven for the model it defines, whose match with OpenFHE 1.5.1 rests on reading its source; assumes (M5) and honestly sampled errors; measured errors 7.8–8.7 bits below it
Decoding offset	proven	public bound, 10^{-8} to $5 \cdot 10^{-6}$ in the slots at the evaluation settings
zCDP of the ideal mechanism; tight conversion	proven	$\rho_{\text{run}} = 0.01496$ at $\varepsilon = 1$, $\delta = 10^{-9}$
Output privacy (Theorem 7.2)	proven (hybrids)	RLWE for the re-randomizer and plain RLWE for O's public-key share; R's errors honestly sampled; slack γ statistical, not negligible in λ_{sec}; standard simulation jointly with R's output
Calibrate	proven	its outputs meet the hypotheses of Theorem 7.2 and the no-wraparound condition

Table 33, continued

Claim	Status	Notes
Input and cohort privacy (G1, G2)	proven	O's view only; simulators, up to the slack $p_{\text{cuckoo}} + p_{\text{false}}$ of R's observable aborts; $p_{\text{cuckoo}} \le 2^{-49.1}$ at $n_R^{\max} = 2{,}000$ and $\le 2^{-41.1}$ up to 100,000 (balanced tables) for an insertion that finds an assignment whenever one exists; $p_{\text{false}} = p_{\text{pass}} \le 2^{-65}$ with two felts
Correctness (no wraparound)	proven	failure probability $\le \delta_B + 2Ne^{-t_\eta^2/2}$ per release; no condition at the earlier levels
Composition across runs (Proposition 7.2)	proven, fixed and adaptive (filter)	composition appendix, Theorem 4.2; hockey-stick ledger, Corollary 5.4; the rule for fixed runs with adaptively chosen requests is not proved
G4 with SanitizeBFV	proven (random oracles, CDH)	first layer without circuit privacy; second layer proven for the textbook BFV model (evaluation noise $\le 2^{-31}$ after the switch), with OpenFHE's BFVrns not checked against the model
O's work (Proposition 12.1)	proven	cell-probe model, in n_{Sc}
Replace-one sensitivity (Proposition 7.1)	proven, exact for $d_O \ge 2$	regime 3 reduces to three cubics; $d_O = 1$ valid but conservative
Full-record sensitivity (Proposition 11.1)	proven, closed form for blocks of dimension ≥ 2	exact enumeration; conservative for a block of dimension one
When $\mathbf{y}^\top\mathbf{y}$ is free	partly proven	boundary between $2(r+o)$ and $3(r+o)$ numerical only
Operator-norm radius r_1	proven (one-sided), both radii	union bound about 1.55× conservative at $p = 11$; the exact quantile (settings I and P) has an exact level with fresh draws, transferred to the lattice noise by a coupling (failure about 10^{-37}) whose constants are proved
Two-stage (λ, Ψ)	proven	independent of the stage-2 noise; closed form for λ_O
Switch-off bracket	proven	tight among Gram matrices with a given $\lambda_{\min}$; the factors 1.2 and 1.4 are empirical
Bound on $\|\boldsymbol{\beta}\|$	proven	for the projection coefficient, no model assumed
Mechanism bias (Theorems A.1 and A.2)	proven, explicit constants	remainder $O(\vartheta_0^{-4})$ relative
Coverage	partly first order	Wald remainder exact; plug-in variance first order; normality asymptotic; bias-aware interval with the public variance bound $\bar{\sigma}^2$ valid under binding clipping; working-model interval (with $\tilde{s}^2$) open under binding clipping
Bootstrap RSS correction	first order	constants not explicit; the plug-in RSS of the FWS comparison has an explicit second-order bias

References

[1] M. Albrecht, M. Chase, H. Chen, J. Ding, S. Goldwasser, S. Gorbunov, S. Halevi, J. Hoffstein, K. Laine, K. Lauter, S. Lokam, D. Micciancio, D. Moody, T. Morrison, A. Sahai, and V. Vaikuntanathan, "Homomorphic encryption security standard," HomomorphicEncryption.org, Toronto, Canada, November 2018.

[2] K. Amin, M. Joseph, M. Ribero, and S. Vassilvitskii, "Easy differentially private linear regression," in *Proc. ICLR*, 2023. arXiv:2208.07353.

[3] T. B. Armstrong, M. Kolesár, and S. Kwon, "Bias-aware inference in regularized regression models," arXiv:2012.14823, 2020 (v2, 2023). Working paper.

[4] G. Asharov, A. Jain, A. López-Alt, E. Tromer, V. Vaikuntanathan, and D. Wichs, "Multiparty computation with low communication, computation and interaction via threshold FHE," in *Proc. EUROCRYPT*, LNCS vol. 7237, 2012.

[5] A. F. Barrientos, J. P. Reiter, A. Machanavajjhala, and Y. Chen, "Differentially private significance tests for regression coefficients," *J. Comput. Graph. Statist.*, pp. 440–453, 2019.

[6] A. F. Barrientos, A. R. Williams, J. Snoke, and C. M. Bowen, "A feasibility study of differentially private summary statistics and regression analyses with evaluations on administrative and survey data," *J. Amer. Statist. Assoc.*, vol. 119, no. 545, pp. 52–65, 2024.

[7] D. Bogdanov, L. Kamm, B. Kubo, R. Rebane, V. Sokk, and R. Talviste, "Students and taxes: a privacy-preserving study using secure computation," *Proc. Privacy Enhancing Technologies*, vol. 2016, no. 3, pp. 117–135, 2016.

[8] Boneh, Gennaro, Goldfeder, Jain, Kim, Rasmussen, and Sahai, "Threshold cryptosystems from threshold fully homomorphic encryption," in *Proc. CRYPTO*, 2018, pp. 565–596.

[9] G. Brown, J. Hayase, S. Hopkins, W. Kong, X. Liu, S. Oh, J. C. Perdomo, and A. Smith, "Insufficient statistics perturbation: Stable estimators for private least squares (extended abstract)," in *Proc. COLT*, PMLR vol. 247, 2024, pp. 750–751. Full version: arXiv:2404.15409.

[10] Brown, Dvijotham, G. Evans, Liu, Smith, and A. G. Thakurta, "Private gradient descent for linear regression: Tighter error bounds and instance-specific uncertainty estimation," in *Proc. ICML*, PMLR vol. 235, 2024, pp. 4561–4584.

[11] P. Buddhavarapu, A. Knox, P. Mohassel, S. Sengupta, E. Taubeneck, and V. Vlaskin, "Private matching for compute," Cryptology ePrint Archive, Paper 2020/599, 2020.

[12] M. Bun and T. Steinke, "Concentrated differential privacy: Simplifications, extensions, and lower bounds," in *Proc. TCC*, 2016.

[13] C. L. Canonne, G. Kamath, and T. Steinke, "The discrete Gaussian for differential privacy," in *Proc. NeurIPS*, 2020.

[14] Chaudhuri, Monteleoni, and Sarwate, "Differentially private empirical risk minimization," *J. Mach. Learn. Res.*, vol. 12, pp. 1069–1109, 2011.

[15] Checri, Sirdey, Boudguiga, and Bultel, "On the practical CPAD security of 'exact' and threshold FHE schemes and libraries," in *Advances in Cryptology – CRYPTO 2024*, 2024. Cryptology ePrint Archive, Paper 2024/116.

[16] H. Chen, K. Laine, and P. Rindal, "Fast private set intersection from homomorphic encryption," in *Proc. ACM CCS*, 2017, pp. 1243–1255.

[17] H. Chen, Z. Huang, K. Laine, and P. Rindal, "Labeled PSI from fully homomorphic encryption with malicious security," in *Proc. ACM CCS*, 2018, pp. 1223–1237.

[18] J.-H. Cheon, A. Kim, M. Kim, and Y. Song, "Homomorphic encryption for arithmetic of approximate numbers," in *Proc. ASIACRYPT*, 2017, pp. 409–437.

[19] Cong, Cruz Moreno, da Gama, Dai, Iliashenko, Laine, and Rosenberg, "Labeled PSI from homomorphic encryption with reduced computation and communication," in *Proc. ACM CCS*, 2021, pp. 1135–1150. Page range from an institutional (imec) record.

[20] A. Dasylva, M. De Cubellis, F. De Fausti, and L. Franssen, "Linking trade data from different national statistical offices through a private set intersection," *J. Official Statist.*, vol. 41, no. 2, pp. 569–597, 2025.

[21] J. de Jong, B. Kamphorst, and S. Kroes, "Differentially private block coordinate descent for linear regression on vertically partitioned data," *J. Cybersecurity and Privacy*, vol. 2, no. 4, pp. 862–881, 2022.

[22] W. Du, Y. S. Han, and S. Chen, "Privacy-preserving multivariate statistical analysis: Linear regression and classification," in *Proc. SIAM SDM*, 2004, pp. 222–233.

[23] D. Dua and C. Graff, *UCI Machine Learning Repository*. University of California, Irvine, School of Information and Computer Sciences, 2019.

[24] Dwork and Nissim, "Privacy-preserving datamining on vertically partitioned databases," in *Proc. CRYPTO*, LNCS vol. 3152, 2004, pp. 528–544.

[25] C. Dwork, K. Kenthapadi, F. McSherry, I. Mironov, and M. Naor, "Our data, ourselves: Privacy via distributed noise generation," in *Proc. EUROCRYPT*, LNCS vol. 4004, 2006, pp. 486–503.

[26] G. Evans, G. King, M. Schwenzfeier, and A. Thakurta, "Statistically valid inferences from privacy-protected data," *American Political Science Review*, vol. 117, no. 4, pp. 1275–1290, 2023.

[27] V. Feldman and T. Zrnic, "Individual privacy accounting via a Rényi filter," in *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021. arXiv:2008.11193.

[28] C. Ferrando, S. Wang, and D. Sheldon, "Parametric bootstrap for differentially private confidence intervals," in *Proc. AISTATS*, PMLR vol. 151, 2022, pp. 1598–1618.

[29] J. Foulds, J. Geumlek, M. Welling, and K. Chaudhuri, "On the theory and practice of privacy-preserving Bayesian data analysis," in *Proc. UAI*, 2016.

[30] Freedman, Nissim, and Pinkas, "Efficient private matching and set intersection," in *Proc. EUROCRYPT*, LNCS vol. 3027, 2004, pp. 1–19.

[31] D. Froelicher, P. Egger, J. S. Sousa, J. L. Raisaro, Z. Huang, C. Mouchet, B. Ford, and J.-P. Hubaux, "UnLynx: A decentralized system for privacy-conscious data sharing," *Proc. Privacy Enhancing Technologies*, vol. 2017, no. 4, pp. 232–250, 2017.

[32] D. Froelicher, J. R. Troncoso-Pastoriza, J. S. Sousa, and J.-P. Hubaux, "Drynx: Decentralized, secure, verifiable system for statistical queries and machine learning on distributed datasets," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3035–3050, 2020.

[33] D. Froelicher, J. R. Troncoso-Pastoriza, A. Pyrgelis, S. Sav, J. S. Sousa, J.-P. Bossuat, and J.-P. Hubaux, "Scalable privacy-preserving distributed learning," *Proc. Privacy Enhancing Technologies*, vol. 2021, no. 2, pp. 323–347, 2021.

[34] D. Froelicher, J. R. Troncoso-Pastoriza, J. L. Raisaro, M. A. Cuendet, J. S. Sousa, H. Cho, B. Berger, J. Fellay, and J.-P. Hubaux, "Truly privacy-preserving federated analytics for precision medicine with multiparty homomorphic encryption," *Nature Communications*, vol. 12, article 5910, 2021.

[35] A. Gascón, P. Schoppmann, B. Balle, M. Raykova, J. Doerner, S. Zahur, and D. Evans, "Privacy-preserving distributed linear regression on high-dimensional data," *Proc. Privacy Enhancing Technologies*, vol. 2017, no. 4, pp. 345–364, 2017.

[36] I. Giacomelli, S. Jha, M. Joye, C. D. Page, and K. Yoon, "Privacy-preserving ridge regression with only linearly-homomorphic encryption," in *Proc. ACNS*, LNCS vol. 10892, 2018, pp. 243–261.

[37] Q. Guo, D. Nabokov, E. Suvanto, and T. Johansson, "Key recovery attacks on approximate homomorphic encryption with non-worst-case noise flooding countermeasures," in *Proc. 33rd USENIX Security Symposium*, 2024.

[38] R. Hall, S. E. Fienberg, and Y. Nardi, "Secure multiple linear regression based on homomorphic encryption," *J. Official Statist.*, vol. 27, no. 4, pp. 669–691, 2011.

[39] S. Hardy, W. Henecka, H. Ivey-Law, R. Nock, G. Patrini, G. Smith, and B. Thorne, "Private federated learning on vertically partitioned data via entity resolution and additively homomorphic encryption," arXiv:1711.10677, 2017.

[40] C. Heinze-Deml, B. McWilliams, and N. Meinshausen, "Preserving differential privacy between features in distributed estimation," arXiv:1703.00403, 2017.

[41] I. Hwang, Y. Hwang, M. Kim, D. Lee, and Y. Song, "On the (in)security of approximate computation protocols from CKKS," Cryptology ePrint Archive, Paper 2025/395, 2025. Earlier title: "Provably secure approximate computation protocols from CKKS."

[42] M. Ion, B. Kreuter, et al., "On deploying secure computing: Private intersection-sum-with-cardinality," in *Proc. IEEE EuroS&P*, 2020, pp. 370–389. Cryptology ePrint Archive, Paper 2019/723.

[43] S. Jarecki, A. Kiayias, and H. Krawczyk, "Round-optimal password-protected secret sharing and T-PAKE in the password-only model," in *Proc. ASIACRYPT*, 2014.

[44] P. Kairouz, Z. Liu, and T. Steinke, "The distributed discrete Gaussian mechanism for federated learning with secure aggregation," in *Proc. ICML*, PMLR vol. 139, 2021, pp. 5201–5212.

[45] A. F. Karr, X. Lin, A. P. Sanil, and J. P. Reiter, "Privacy-preserving analysis of vertically partitioned data using secure matrix products," *J. Official Statist.*, vol. 25, no. 1, pp. 125–138, 2009.

[46] D. Kifer, A. Smith, and A. Thakurta, "Private convex empirical risk minimization and high-dimensional regression," in *Proc. COLT*, PMLR vol. 23, 2012, pp. 25.1–25.40.

[47] K. Kluczniak and G. Santato, "On circuit private, multikey and threshold approximate homomorphic encryption," *IACR Communications in Cryptology*, vol. 2, no. 1, 2025. Cryptology ePrint Archive, Paper 2023/301.

[48] W. Krichene, N. E. Mayoraz, S. Rendle, S. Song, A. Thakurta, and L. Zhang, "Private learning with public features," in *Proc. AISTATS*, PMLR vol. 238, 2024, pp. 4150–4158.

[49] T. Lepoint, S. Patel, M. Raykova, K. Seth, and N. Trieu, "Private join and compute from PIR with default," in *Proc. ASIACRYPT*, 2021.

[50] B. Li and D. Micciancio, "On the security of homomorphic encryption on approximate numbers," in *Proc. EUROCRYPT*, LNCS vol. 12696, 2021, pp. 648–677.

[51] B. Li, D. Micciancio, M. Schultz, and J. Sorrell, "Securing approximate homomorphic encryption using differential privacy," in *Proc. CRYPTO*, LNCS vol. 13507, 2022, pp. 560–589.

[52] S. Lin, E. Paquette, and E. D. Kolaczyk, "Differentially private linear regression with linked data," *Harvard Data Science Review*, vol. 6, no. 3, 2024. arXiv:2308.00836.

[53] S. Lin, A. Slavković, and D. R. Bhoomireddy, "Differentially private linear regression and synthetic data generation with statistical guarantees," in *Proc. AISTATS*, PMLR vol. 300, 2026. arXiv:2510.16974.

[54] Y. Liu, T. Fan, T. Chen, Q. Xu, and Q. Yang, "FATE: An industrial grade platform for collaborative learning with data protection," *J. Mach. Learn. Res.*, vol. 22, no. 226, pp. 1–6, 2021.

[55] Y. Liu, Y. Kang, T. Zou, Y. Pu, Y. He, X. Ye, Y. Ouyang, Y.-Q. Zhang, and Q. Yang, "Vertical federated learning: Concepts, advances, and challenges," *IEEE Trans. Knowl. Data Eng.*, 2024.

[56] McGregor, Mironov, Pitassi, Reingold, Talwar, and Vadhan, "The limits of two-party differential privacy," in *Proc. IEEE FOCS*, 2010, pp. 81–90.

[57] Mironov, Pandey, Reingold, and Vadhan, "Computational differential privacy," in *Proc. CRYPTO*, LNCS vol. 5677, 2009, pp. 126–142.

[58] I. Mironov, "On significance of the least significant bits for differential privacy," in *Proc. ACM CCS*, 2012, pp. 650–661.

[59] Mohammed, D. Alhadidi, B. C. M. Fung, and M. Debbabi, "Secure two-party differentially private data release for vertically partitioned data," *IEEE Trans. Dependable Secure Comput.*, vol. 11, no. 1, pp. 59–71, 2014.

[60] P. Mohassel and Y. Zhang, "SecureML: A system for scalable privacy-preserving machine learning," in *Proc. IEEE S&P*, 2017.

[61] C. Mouchet, J. Troncoso-Pastoriza, J.-P. Bossuat, and J.-P. Hubaux, "Multiparty homomorphic encryption from ring-learning-with-errors," *Proc. Privacy Enhancing Technologies*, vol. 2021, no. 4, pp. 291–311, 2021.

[62] V. Nikolaenko, U. Weinsberg, S. Ioannidis, M. Joye, D. Boneh, and N. Taft, "Privacy-preserving ridge regression on hundreds of millions of records," in *Proc. IEEE S&P*, 2013, pp. 334–348.

[63] T. Ogilvie, "Differential privacy for free? Harnessing the noise in approximate homomorphic encryption," in *Proc. CT-RSA*, 2024, pp. 292–315. Cryptology ePrint Archive, Paper 2023/701.

[64] R. Pagh and F. F. Rodler, "Cuckoo hashing," *J. Algorithms*, vol. 51, no. 2, pp. 122–144, 2004.

[65] B. Pinkas, T. Schneider, O. Tkachenko, and A. Yanai, "Efficient circuit-based PSI with linear communication," in *Proc. EUROCRYPT*, LNCS vol. 11478, 2019, pp. 122–153.

[66] J. P. Reiter, A. Oganian, and A. F. Karr, "Verification servers: Enabling analysts to assess the quality of inferences from public use data," *Comput. Statist. Data Anal.*, vol. 53, no. 4, pp. 1475–1482, 2009.

[67] A. Davidson, A. Faz-Hernández, N. Sullivan, and C. A. Wood, "Oblivious pseudorandom functions (OPRFs) using prime-order groups," RFC 9497 (Informational, IRTF CFRG stream), December 2023.

[68] P. Rindal and P. Schoppmann, "VOLE-PSI: fast OPRF and circuit-PSI from vector-OLE," in *Proc. EUROCRYPT*, 2021.

[69] R. Rogers, A. Roth, J. Ullman, and S. Vadhan, "Privacy odometers and filters: Pay-as-you-go composition," in *Advances in Neural Information Processing Systems 29 (NIPS)*, 2016, pp. 1921–1929.

[70] A. P. Sanil, A. F. Karr, X. Lin, and J. P. Reiter, "Privacy preserving regression modelling via distributed computation," in *Proc. ACM SIGKDD*, 2004, pp. 677–682.

[71] O. Sheffet, "Differentially private ordinary least squares," in *Proc. ICML*, PMLR vol. 70, 2017, pp. 3105–3114.

[72] E. Shi, T.-H. H. Chan, E. Rieffel, R. Chow, and D. Song, "Privacy-preserving aggregation of time-series data," in *Proc. NDSS*, 2011.

[73] T. Stevens, C. Skalka, C. Vincent, J. Ring, S. Clark, and J. Near, "Efficient differentially private secure aggregation for federated learning via hardness of learning with errors," in *Proc. USENIX Security*, 2022, pp. 1379–1395.

[74] D. Vu and A. Slavković, "Differential privacy for clinical trial data: Preliminary evaluations," in *Proc. IEEE ICDM Workshops*, 2009, pp. 138–143.

[75] Y.-X. Wang, "Revisiting differentially private linear regression: optimal and adaptive prediction & estimation in unbounded domain," in *Proc. UAI*, 2018, pp. 93–103. Code and data: https://github.com/yuxiangw/optimal_dp_linear_regression.

[76] C. Wang, J. Liang, M. Huang, B. Bai, K. Bai, and H. Li, "Hybrid differentially private federated learning on vertically partitioned data," arXiv:2009.02763, 2020.

[77] J. Whitehouse, A. Ramdas, R. Rogers, and Z. S. Wu, "Fully-adaptive composition in differential privacy," in *Proc. ICML*, PMLR vol. 202, 2023, pp. 36990–37007.

[78] R. Wu, X. Yang, Y. Yao, J. Sun, T. Liu, K. Q. Weinberger, and C. Wang, "Differentially private multi-party data release for linear regression," in *Proc. UAI*, PMLR vol. 180, 2022, pp. 2128–2137.

[79] D. Xu, S. Yuan, and X. Wu, "Achieving differential privacy in vertically partitioned multiparty learning," arXiv:1911.04587, 2019.

[80] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 12:1–12:19, 2019.

[81] J. Zhang, Z. Zhang, X. Xiao, Y. Yang, and M. Winslett, "Functional mechanism: Regression analysis under differential privacy," *Proc. VLDB Endow.*, vol. 5, no. 11, 2012.

[82] Protocol v3.1: specification (honest-but-curious model). Repository file `docs/protocol_v31.pdf` (source `docs/protocol_v31.tex`), 2026.

[83] Protocol v3.1: proofs appendix. Part A: cryptographic and DP guarantees; Part B: sensitivity and statistical guarantees; Part C: composition across runs and the hockey-stick ledger. Repository file `docs/proofs/protocol_v31_proofs.pdf`, 2026.

[84] Protocol v3.1, proofs Part A: cryptographic and DP guarantees. Repository file `docs/proofs/crypto_proofs.pdf`, 2026.

[85] Protocol v3.1, proofs Part B: sensitivity and statistical guarantees. Repository file `docs/proofs/stats_proofs.pdf`, 2026.

[86] Protocol v3.1, proofs, composition appendix: composition across runs and the hockey-stick ledger. Repository file `docs/proofs/composition/composition_proof.pdf`, 2026.

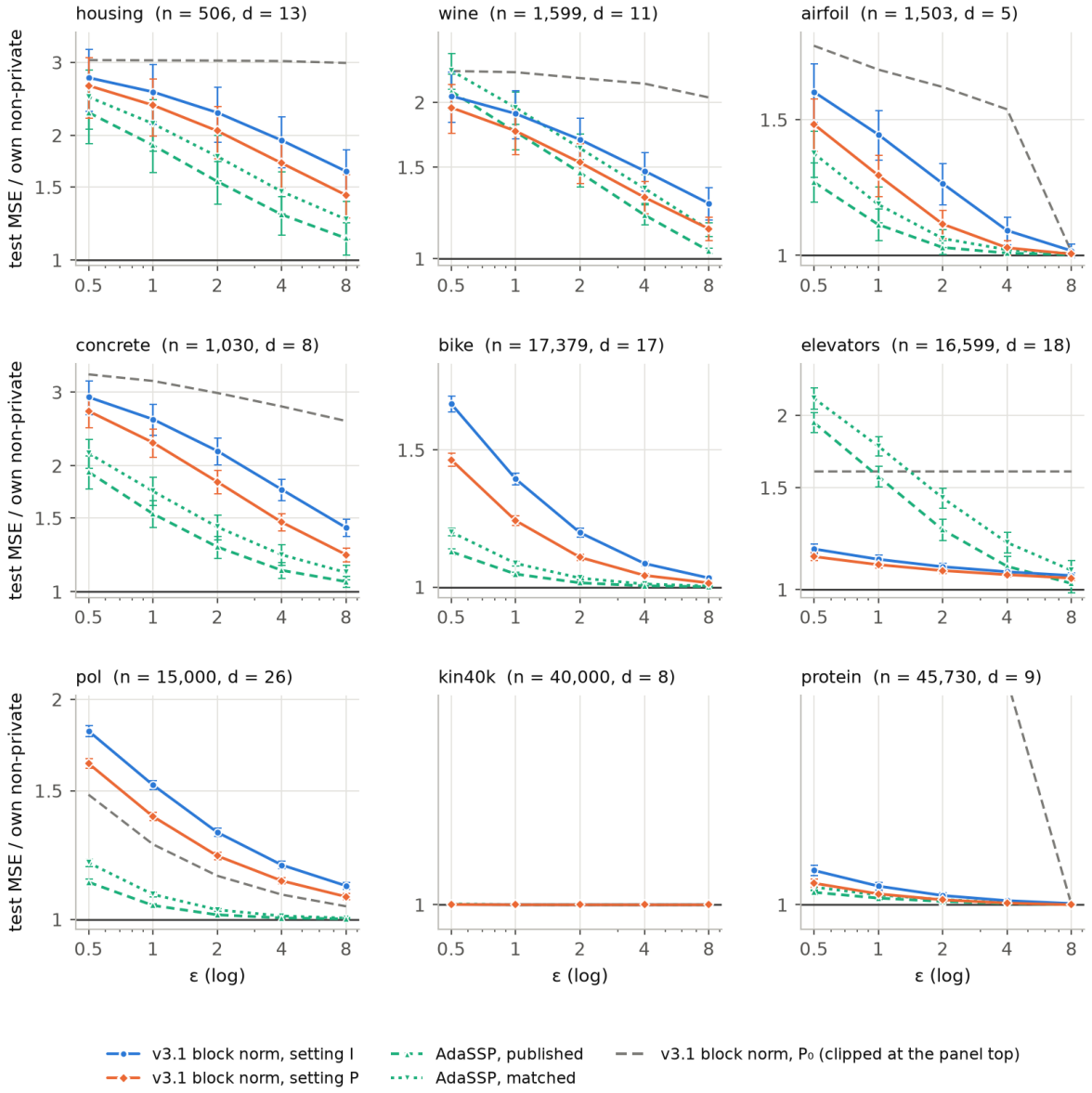


Figure 12: Test MSE divided by each arm's own non-private test MSE against ε, $\delta = 10^{-6}$, on nine of the datasets. Blue: v3.1 with block unit norm, setting I. Orange: the same, setting P. Aqua: AdaSSP, published (dashed) and matched (dotted). Grey dashed: v3.1 with block unit norm under P_0, the first choice of P, clipped at the top of each panel. Points: mean over 10 folds and 50 noise draws. Error bars: 95% CI of the mean (t over folds).

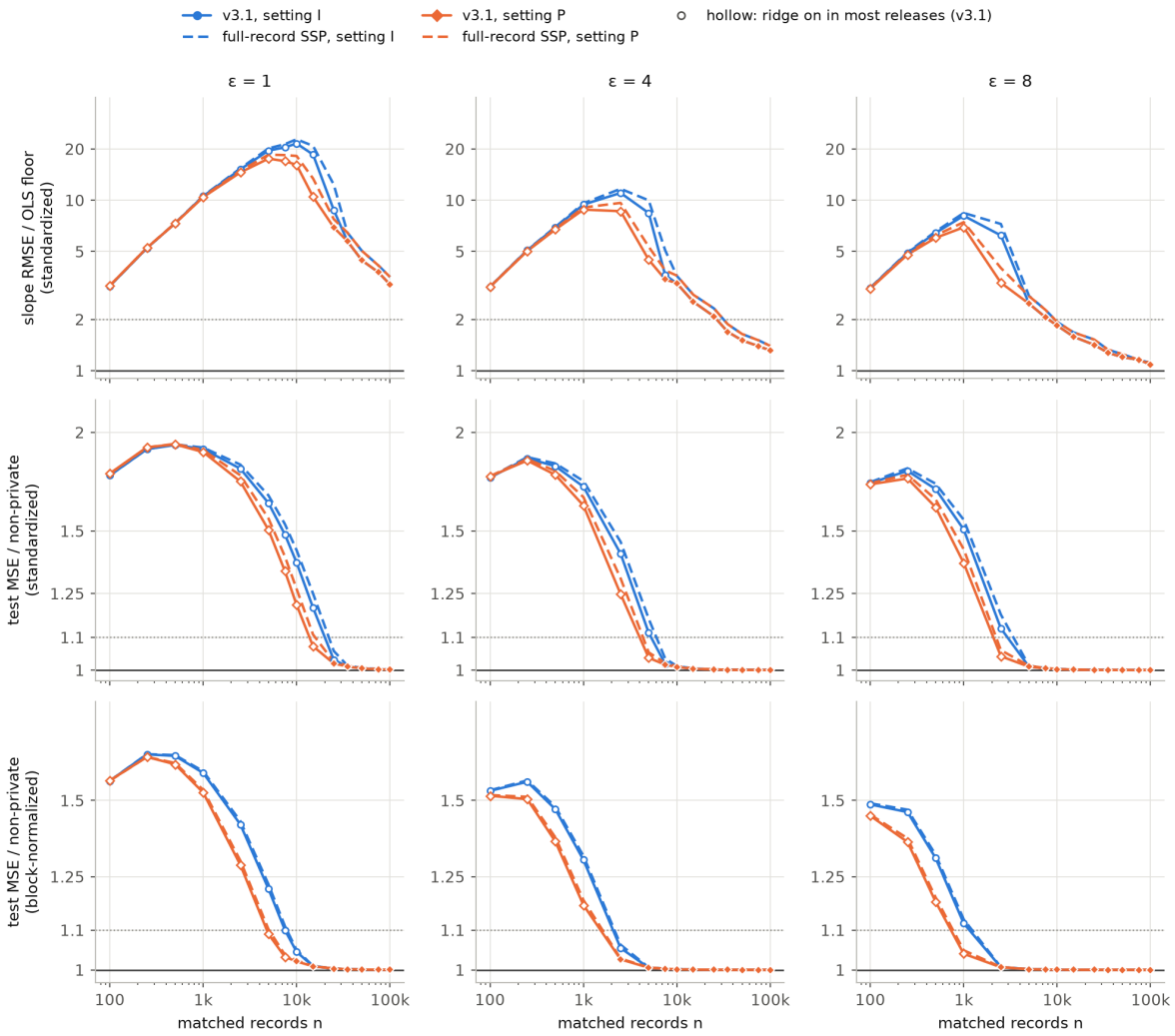


Figure 13: Full-record SSP (dashed) against v3.1 (solid) on the simulation population, $\delta = 10^{-9}$, $R^2 = 0.5$; blue: setting I, orange: setting P; columns $\varepsilon = 1$, 4, 8. Top: median slope RMSE divided by that of non-private least squares (standardized population). Middle and bottom: population test MSE divided by that of non-private least squares on the same cohort, for the standardized and the block-normalized population. Hollow markers: the v3.1 ridge is on in most releases.

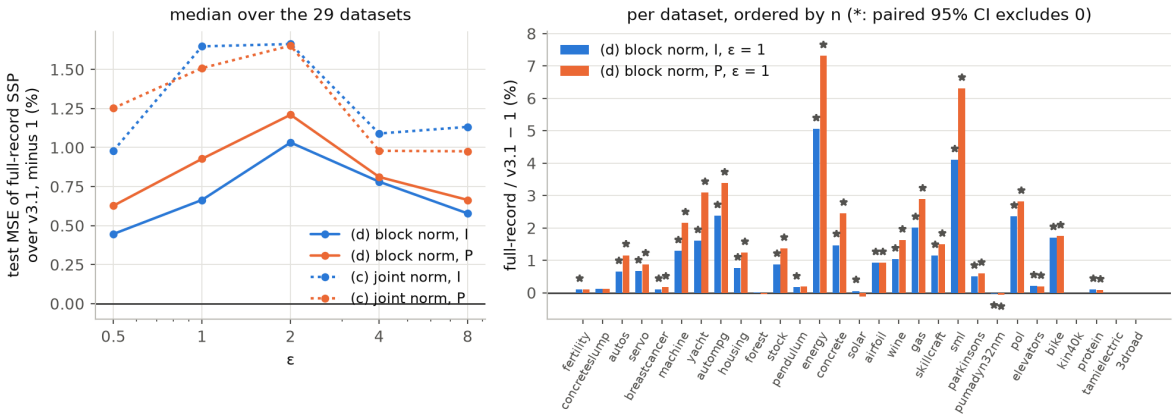


Figure 14: Full-record SSP against v3.1 on the 29 datasets, $\delta = 10^{-6}$. Left: median over datasets of the absolute test MSE of full-record SSP divided by that of v3.1 with the same preprocessing and setting, minus one, against ε. Right: the same per dataset at $\varepsilon = 1$, block normalization; a star marks a paired 95% CI over folds that excludes zero.

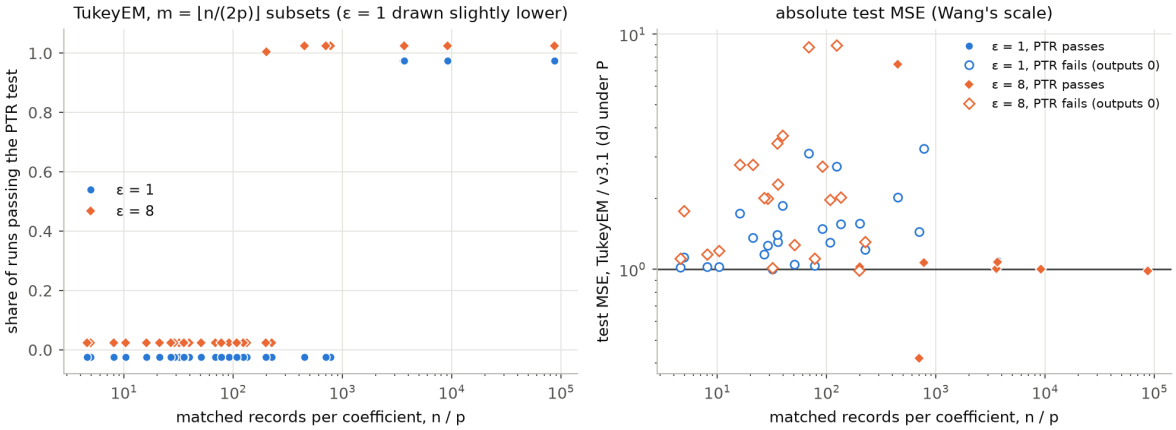


Figure 15: TukeyEM on the 29 datasets, $\delta = 10^{-6}$, $m = \lfloor n/(2p) \rfloor$ groups, against matched rows per coefficient. Left: share of runs that pass the PTR test, at $\varepsilon = 1$ (circles, drawn slightly lower) and 8 (diamonds). Right: absolute test MSE divided by that of v3.1 with block normalization under setting P; hollow: the test fails in most runs, so TukeyEM outputs $\boldsymbol{\theta} = 0$.

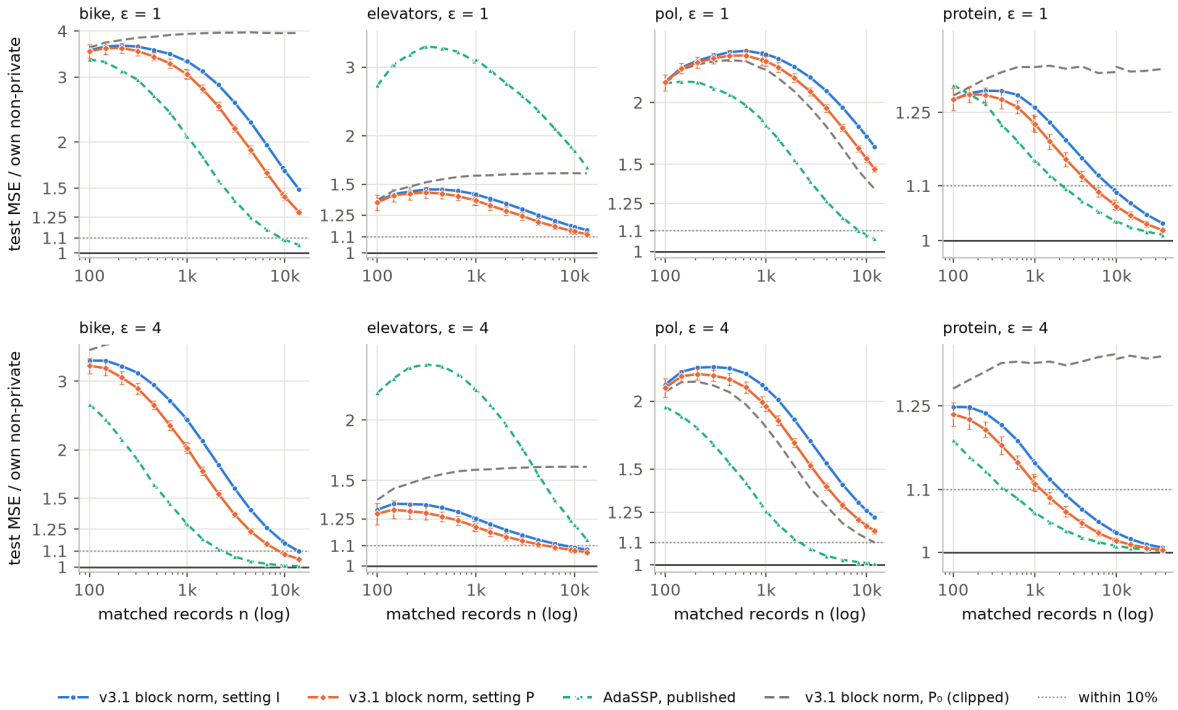


Figure 16: Test MSE divided by the arm's own non-private test MSE at the same n, against the number of matched records, $\delta = 10^{-9}$; top row $\varepsilon = 1$, bottom row $\varepsilon = 4$. Blue: v3.1 with block unit norm, setting I. Orange: the same, setting P. Aqua dashed: AdaSSP (published). Grey dashed: v3.1 with block unit norm under P_0, the first choice of P (clipped). Error bars: 95% CI over folds (setting P).

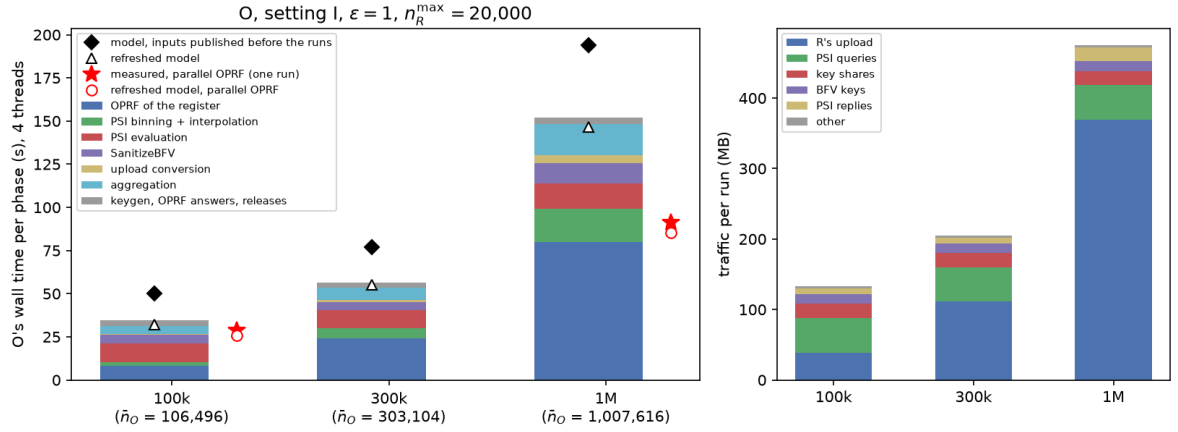


Figure 17: Left: O's wall time per phase at 10^5, 3×10^5 and 10^6 register rows ($n_R^{\max} = 20{,}000$, setting I, $\varepsilon = 1$, four threads, single-threaded OPRF), against the cost model with its published inputs (filled diamonds) and the refreshed model (open triangles). Stars: O's total in one run with the parallel OPRF; open circles: the refreshed model for it. Right: traffic per run.